

ABSTRACTS COLLECTION

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AUDIT AND CLINICAL GOVERNANCE

P001. Abstract not included

P002

WAITING TIME FOR LASER TREATMENT AMONG
PROLIFERATIVE DIABETIC RETINOPATHY REFERRALS IN A
HOSPITAL-BASED DIABETIC RETINOPATHY SERVICE – A
QUALITY IMPROVEMENT PROJECT

Corresponding Author: Nadir A M Ali, orcid.org/0000-0001-9718-5571

Nadir Ali¹, Stacey Law¹, Ashish Patwardhan¹

¹Royal Cornwall Hospitals NHS Trust, Truro, UK

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Introduction: A quality improvement project was conducted at Royal Cornwall Hospital, Truro, in response to multiple DATIX reports of delayed laser panretinal photocoagulation treatment (PRP) to new referrals with proliferative diabetic retinopathy (PDR).

Method: The project involved an initial audit of the newly referred PDR patients with delayed PRP treatment in the period from 1st January 2019 to 31st December 2019. This was followed by a change of practice, which included a new 3-tier clinic system that isolates urgent cases and fast-track them to have treatment in a dedicated Laser clinic, while other non-urgent patients are seen in either a virtual or a face-to-face clinic based on the gradeability of their fundus photographs.

The duration from the referral day to the first PRP day, the duration between first and second laser treatments, and the delay reason (if any) were identified. Any wait of >6 weeks was defined as 'delayed first laser session'. The service was re-audited after 2 years from change implementation to measure the change's outcome.

Results: In 2022, 620 laser procedures were done for 428 patients. Of those, 71 patients were new referrals from DESP. In contrast, 255 laser procedures were done for 188 patients in the year 2019. Of those 79 patients were new patients referred from DESP.

Of the new referrals, 21% had delayed 1st session in 2022 as compared to 42% in 2019, indicating over 50% reduction in response to the change implemented.

Conclusion: Considering the limited NHS resources and the high load on this service, a 3-tier service delivery (virtual, face-to-face and Laser clinics) seems to be the best mode to maximize the hospital-based diabetic retinopathy capacity using the available manpower and infrastructure.

P003

IMPROVING TIMELINESS OF ANGIOGRAPHY AND ANTI-VEGF
INJECTION ADMINISTRATION IN PATIENTS WITH WET
ACTIVE AMD

Corresponding author: Ruqaiyah Behranwala, orcid.org/0000-0001-9914-6090

Ruqaiyah Behranwala¹, Theodora Gkika¹, Susan Downes¹

¹Oxford University Hospitals NHS Foundation Trust, Oxford Eye Hospital, Oxford, UK

Disclosures: No funding and no conflicts of interest

Introduction: Wet active AMD is characterised by CNV. OCT is the first diagnostic step and FFA is offered to people with suspected wet active AMD to confirm the diagnosis. Wet AMD can deteriorate rapidly. Minimising delays in starting treatment increases the chances of preserving vision and quality of life.

Method: All patients that attended NEW AMD clinic were identified. Only patients with a diagnosis of wet AMD were included. Time (in days) was recorded for each patient from date of clinic assessment to 1) FFA being carried out 2) FFA being read 3) Decision to book injections 4) Injections received.

Results: There were fewer breaches after the protocol was implemented. FFA results were being reviewed and acted upon faster after the protocol was implemented. The most common reason for delay of injections was lack of capacity before both before and after protocol implementation.

Conclusion: Address capacity issues by running extra injection clinics. Implement treat and extend protocols to help increase capacity for injections. Increase use of OCT-A with associated teaching for workforce. Ensure practitioners are aware that intravenous angiography needs only be used in cases where there is doubt over the diagnosis.

P004

FORECASTING THE SURGE IN DEMAND FOR EYE CARE
SERVICES IN WALES

Corresponding Author: Ahmed Bilal, orcid.org/0000-0002-6846-5731

Ahmed Bilal¹, Muslim Bilal², Elizabeth Williams³, Mark Tuson³, Sartaj Ahmed⁴, Yih-Horng Tham⁴, Catrin Bertalot⁵, Rhianon Reynolds⁶, James E Morgan^{1,7}

¹Cardiff & Vale University Health Board, Cardiff, UK; ²School of Medicine, Cardiff University, Cardiff, UK; ³School of Mathematics,

Cardiff University, Cardiff, UK; ⁴Swansea Bay University Health Board, Swansea, UK; ⁵Betsi Cadwaladr University Health Board, Abergele, UK; ⁶Aneurin Bevan University Health Board, Newport, UK; ⁷School of Optometry and Vision Science, Cardiff University, Cardiff, UK

Disclosures: no conflicts of interest and no applicable funding

Introduction: The increasing ageing population projections for Wales over the next two decades will result in increased prevalence of age-related ophthalmic diseases. Current lack of UK population-based surveys, confounded by poor referral and diagnosis data coding, absence of a unified patient electronic record for ophthalmology causes difficulties for service planning. Necessitating epidemiological modelling to predict future demand and aid in planning for capacity shortfalls in an already strained service, as highlighted by the Pyott Report.

Method: Literature review within Ovid-MEDLINE for studies of comparable population demographics to Wales with relevant incidence and prevalence data of the major eye diseases, cataract, glaucoma and retinal diseases (diabetic retinopathy and age related macular degeneration). A mean age categorised incidence and prevalence was derived from these studies and extrapolated against age categorised Welsh population projections obtained from the office for national statistics (StatsWales).

Results: Between 2023 - 2033, the number of those aged ≥ 70 and aged ≥ 80 years is projected to increase by 16.0% (505 to 586K) and 31.3% (182 to 239K) respectively. With projected increase in incidence of cataract, glaucoma and retinal diseases by 11%, 11.7%, 13.0% respectively. With projected increase in prevalence of cataract, glaucoma and retinal diseases by 12.5%, 12.9%, 12.6% respectively.

Conclusion: Demand is projected to increase at an alarming rate over the next nine years. Continuing at the current trajectory, without urgent action to plan for the foreseeable rise in demand for eye care, will result in an unsustainable model of eye care provision in Wales. A disaster of longer waiting lists and patients suffering from untreated, treatable ophthalmic diseases.

P005

SINGLE-CENTRE AUDIT OF SECONDARY IOL INSERTION IN EYES WITH NO ZONULAR SUPPORT

Corresponding author: Miranda Buckle, orcid.org/0000-0002-6722-7125

Miranda Buckle¹, Johannes Keller¹

¹Bristol Eye Hospital, Bristol, UK

Disclosures: No conflicts of interest and no applicable funding

Introduction: A single-centre service evaluation was conducted to audit outcomes for eyes undergoing secondary intraocular lens (IOL) implantation with no zonular support.

Method: An electronic patient record (Medisoft Ltd) was used to identify cases using the terms "secondary IOL", "scleral-fixed IOL", and "IOL exchange". Data were extracted on 28th November 2023 for the period 1st Jan 2019 to 27th Nov 2023. All cases of sulcus IOL implantation were excluded.

Results: 273 eyes without zonular support underwent secondary IOL implantation during the audit period: 120 had iris-claw IOL, 45 had intrascleral haptic-fixated IOL (Yamane technique), 36 had sutured scleral-fixated, and 72 had sutureless scleral-fixated IOL (Carlevalle IOL).

76% had pre-operative acuity documented and 91% had post-operative acuity documented. Only 1.5% (n = 4) had pre-operative refractive target documented, of which one case achieved a post-operative refractive outcome within 1D of target.

Complications included lens dislocation (7.3%), cystoid macular oedema (20.7%), uveitis (3.7%), glaucoma (9.3%), retinal detachment (1.9%), and endophthalmitis (0.2%). There was no significant difference in the incidence of each complication between the four surgical strategies (repeated measures ANOVA, $p > 0.05$). Sub-group analysis was carried out to interrogate the incidence of CMO: IOL type was strongly associated with CMO occurrence in eyes with sutured IOL ($p < 0.0005$), but not in the Yamane group ($p > 0.5$). The most frequent indications for surgery in eyes which developed CMO following sutureless IOL implantation were complicated cataract extraction and dislocated IOL.

Conclusion: Improved recording of target refraction is needed to enable better evaluation of final refractive outcomes. For sutured IOLs, IOL type may influence post-operative CMO occurrence. Complicated cataract extraction and dislocated IOL may be risk factors for post-operative CMO occurrence after sutureless secondary IOL implantation.

P006

CERTIFICATION OF VISUAL IMPAIRMENT IN NORTHERN IRELAND: 10 YEARS OF DATA

Corresponding author: Laura Cushley, orcid.org/0000-0003-0697-8854

Laura Cushley¹, Tunde Peto^{1,2}, Roseleen McCann², Tanya Moutray², A. Jonathan Jackson²

¹Centre for Public Health, Queen's University Belfast, Belfast, UK;

²Belfast Health and Social Care Trust / Northern Ireland Clinical Research Network, Belfast, UK

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Introduction: Collection of certifications of visual impairment data (CVI) in the UK dates back to 1948 and publication of this data in Northern Ireland dates back to 1984. In 2012, there was a comprehensive review of the process and annual reports have been published since 2013 with information on causes, levels of visual impairment and other factors.

Method: Since 2014 annual clinical and demographic data were collected and recorded in an Excel Spreadsheet. Double entry checks were completed on 10% of the data to ensure validation. SPSS statistical package was used to analyse all data over the 10-year period.

Results: Over the 10-year period there was an 80% increase in certification. The exception to year-on-year increase occurred in 2020 due to the impact of COVID-19 on service delivery. The most common cause of severe sight impairment (SSI) and Sight Impairment (SI) in adults was age-related macular degeneration (AMD), accounting for approximately 50% of certifications annually. The second and third biggest causes of certification were glaucoma and diabetic eye disease, followed by hereditary retinal diseases. While adult certifications have increased year on year, children's CVI has fluctuated throughout the years. The biggest cause of childhood visual impairment remains cerebral visual impairment followed by Albinism, Optic Atrophy and hereditary retinal disease.

Conclusion: Annual certification of sight loss in Northern Ireland has increased by 80% since 2013. The leading cause of both SSI & SI CVI in adults remains AMD followed by Glaucoma and Diabetic Eye Disease. CVI in children has not increased but the biggest cause of visual impairment in children remains cerebral visual impairment.

P007

MEASURING LEVEL OF HARM IN OPHTHALMIC SETTING - AN AUDIT

Corresponding author: Anna Grabowska, orcid.org/0000-0002-3053-2142

Anna Grabowska¹, Eoin O'Sullivan¹

¹King's College Hospital, London, UK

Disclosures: No conflicts of interest and no applicable funding

Introduction: In 2021 The Royal College of Ophthalmologists published guidance for measuring level of harm in Ophthalmic setting.

The aim of this audit was to compare our measurement of harm level against the criteria proposed by RCOphth post external harm review. Furthermore, we compared the level of harm to the proposed level of investigation e.g. a serious event can occur but with little harm.

Method: This was a retrospective review of consecutive minor, moderate and severe incidents which were reported between June 2020 and June 2021 in Ophthalmology at King's College Hospital (KCH). The measure of agreement used was Cohen's kappa along with the Confidence Interval 95%. Cohen's kappa (κ) of: 0.40 - 0.59 is considered moderate agreement, 0.60 - 0.79 is considered substantial agreement, 0.80 - 0.99 is considered outstanding agreement.

Results: Forty-four incidents were recorded from four clinical sites. There was a moderate agreement between the level of harm by RCOphth and KCH (Cohen's kappa = 0.553, $p < 0.001$, Confidence Interval 95% = [0.46, 0.64]). There was a low agreement between the level of investigation by RCOphth and KCH (Cohen's kappa = 0.349, $p = 0.029$, Confidence Interval 95% = [0.28, 0.41]). Further analysis showed that the low agreement was a result of KCH team tending to perform higher investigation level than that suggested by the level of harm.

Conclusion: RCOphth guidance is a useful tool when measuring level of harm in Ophthalmic setting but clinical judgement must still be used by the clinician undertaking any investigation into an incident of harm.

P008

ASSESSING THE EFFECTIVENESS OF AN MSK RHEUMATOLOGY BACK PAIN QUESTIONNAIRE IN EARLY REFERRAL AND DIAGNOSIS OF ANKYLOSING SPONDYLITIS IN UVEITIS PATIENTS

Corresponding author: Netra Kalla, orcid.org/0009-0002-3162-5137

Netra Kalla¹, Arsalan Bangash, Faaraan Bangash¹

¹Northampton General Hospital NHS Trust, Northampton, UK

Disclosures: No conflicts of interest and no funding

Introduction: Uveitis is the most common initial extra-articular manifestation of axial spondylo-arthritis (SpA), occurring in up to 1/3 of patients. A considerable 40% of uveitis patients have undiagnosed spondylarthritis due to inadequate referrals (Judith Rademacher, et al). Our study aimed to address the diagnostic delay in axial spondylo-arthritis (SpA) by evaluating the effectiveness of implementing the MSK Rheumatology Back Pain Questionnaire in carrying out appropriate rheumatology referrals for uveitis patients.

Method: The patients presenting to the uveitis clinic at Northampton General Hospital were reviewed prospectively from May 2023-July 2023. A total of 40 patients' data was collected and analysed. All patients completed the 12-point questionnaire. (A score of ≥ 4 indicating referral). Symphony/Medi viewer/ICE were used to view any previous patient notes, previous investigations and whether the patients are already under rheumatology or not.

Results: Previously, 25% of patients that were clinically indicated for referral, were not referred to rheumatology appropriately. Implementing the MSK Rheumatology Back Pain Questionnaire improved these statistics significantly – 96% of appropriate patients were referred. In addition, it resulted in a substantial decrease in the number of missed/delayed referrals - (from 25% to 4%).

Conclusion: Our study showed evidence that implementation of the rheumatology questionnaire can enhance the referral process of ophthalmology patients. Thereby, potentially leading to earlier diagnoses and improved outcomes for patients with axial SpA. Despite this substantial improvement, it is also worth noting that consistent documentation and good clinic attendance is still quintessential for management of patients with uveitis and suspected axial SpA. Overall, our study emphasises that since conditions can overlap in specialties, there should be greater emphasis on working more coherently and with greater engagement to achieve better patient outcomes.

P009

CHANGING TO ELECTRONIC PATIENT RECORDS: HOW CAN WE CLICK AWAY FROM PAPER?

Corresponding author: Janvi Karia, orcid.org/0000-0002-6897-7411

Janvi Karia¹, Swapnil Parmar¹, Mya Ohn¹

¹Mid and South Essex NHS Foundation Trust, UK

Disclosures: no conflicts of interest and no applicable funding

Introduction: The NHS Long Term Plan targets a core level of digitalisation by 2024 with a key aim to have complete migration to electronic paper records (EPR). We audited the ophthalmology departments within a large NHS trust to assess clinical usage of EPR; there is often a reliance on paper records. Problems with this include difficulty in accessing previous records which disrupts continuity of care and reducing care quality and can cause patient harm.

Method: Members of the eye department including nurses, technicians and clinicians completed an anonymous, online questionnaire regarding their documentation habits and preferences. Responses were quantitatively and qualitatively analysed and a departmental meeting held where the results were disseminated and shared with the relevant staff groups. Based on initial results, changes were implemented, and the questionnaire sent out again for re-audit.

Results: Less than 50% of the department uses EPR for the majority of documentation. The major barriers to improving EPR use included (1) poor quality technology (2) difficulty in using the software. Changes have been implemented including efforts with Medisight (EPR company) to ensure software is up-to-date and bug free. Liaison with IT department to ensure the department has adequate number of functioning PCs available for use. EPR teaching sessions were set up for staff. The same questionnaire was then sent to department members post changes. The results are being compared with pre-intervention data and statistically analysed for significant change. Anticipated results are a positive change via increase in % of documentation with EPR. Final results to be shared at conference.

Conclusion: Better utilisation of EPR can cause improvements within ophthalmology departments by reducing duplication of workload, increasing patient flow and therefore decreasing ophthalmic waitlists.

P010

USING MICROSOFT SHAREPOINT LISTS FOR COLLABORATIVE ELECTRONIC WAITING LISTS

Corresponding author: Ahmed Khamis, orcid.org/0009-0006-6789-5154

Ahmed Khamis¹, Ian De Silva¹

¹University Hospitals of Leicester, Leicester, UK

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Introduction: Waiting list forms are an essential way of communicating information to the surgical booking teams to ensure that cases can be appropriately prepared and booked prior to the day of surgery. Conventional paper or other versions of electronic forms have faced numerous issues such as lost forms, illegible forms, incomplete forms and the poor interactivity or static nature as they may not be a live database. Inadequately filled forms or poor communication may cause inappropriate preparation of cases and cancellations on day of surgery.

Method: A quality improvement study at University Hospitals of Leicester of 75 paper waiting list forms were compared to the same quantity of waiting list forms submitted electronically. Data on completion of essential and cataract specific domain questions were collated.

Results: The paper waiting list showed that essential information was a third incomplete and in cases of cataract waiting lists, completion of questions on biometry, medical health, prognostic and complexity questions were only 42% complete.

Utilising the Microsoft SharePoint Lists enabled mandating of the completion of questions, and this allowed a 100% completion of both essential information and cataract specific questions on biometry, medical health, prognostic and complexity factors.

Conclusion: The electronic system utilises a unique set of list permissions and data environments that fits within the criteria of the trust privacy team. It also allows simultaneous multiuser actions and ability to view with specific filters to prioritise your workflow. An automated risk assessment score is also integrated within the system to facilitate recognition of appropriate cases based on surgical expertise. The integration with the Microsoft outlook emails allows automated email reminders, reports and requests which have helped the whole team work more collaboratively.

P011

DIGITAL AND SOCIAL BARRIERS TO REMOTE CONSULTATION IN OPHTHALMOLOGY CLINICS

Corresponding author: Henry Rocha, orcid.org/0000-0001-6669-0831

Henry Rocha^{1*}, Jen Lim^{1*}, Louise Allen²

¹University of Cambridge School of Clinical Medicine, Cambridge, UK; ²Cambridge University Hospitals NHS Foundation Trust, Cambridge, UK

*These authors contributed equally

Disclosures: no conflicts of interest and no applicable funding

Introduction: Ten million ophthalmology appointments take place in the UK annually. The NHS Long Term Plan (NHS LTP) and Greener NHS programme promote the use of remote consultation (RC) to reduce waiting times and carbon emissions. This study investigates patient opinions about RC, home vision testing and digital access among ophthalmic patients.

Method: Patients in Addenbrooke's Hospital eye clinics were surveyed by students using a validated eHealth innovation acceptability questionnaire. Respondents were chosen to give diversity in age, ethnicity, sex, and specialty clinics. Statistical significance of associations was calculated using chi-squared tests with Bonferroni correction.

Results: One hundred and one patients were surveyed. Glaucoma, retinal and emergency eye clinic patients were the most frequent respondents (29%, 23% and 22% respectively). Ninety-six percent of households owned at least one smartphone and 95% had internet access. Ninety percent had two smart communication devices, including at least one large-screened device (required for some vision self-testing applications). Digital access was associated with neither age nor ethnicity. Respondents younger than 55 and 65 were significantly more likely to have favourable views about RC and home vision testing respectively ($p < 0.05$). Reasons for reluctance towards RC included a lack of the human touch and receiving inadequate reassurance. Sixty-seven percent of respondents had travelled by car or bus, resulting in 0.69 tonnes of CO₂-equivalent emissions.

Conclusion: Our study found no association between digital access and age but patients over 65 years were significantly less likely to favour RC and home vision testing, indicating a need to educate, support, and reassure older patients when suggesting RC. Reaching the NHS LTP target of 30% RC in the surveyed population could have resulted in a reduction of 0.21 tonnes of CO₂-equivalent emissions.

P012

IMPROVING SAFETY AND EFFICIENCY IN EYE CASUALTY

Corresponding Author: Shi Pei Loo, orcid.org/0009-0003-8680-7218

Shi Pei Loo¹, Kaleena Michael¹

¹University Hospital Ayr, NHS Ayrshire and Arran, Ayr, UK

Disclosures: no conflicts of interest and no applicable funding

Introduction: In recent years, there has been a rise in demand for emergency eye care services due to an ageing population and more health seeking behaviour. There is an important need to streamline acute services to maximise the use of resources and ensure safe patient care. Hence, we audited referral patterns and patient outcomes of our Eye Casualty service with the aim to improve the efficiency of the service and patient safety.

Method: Prospective data collection was carried out over two weeks. Clinicians and nurses running the Eye Casualty service were asked to fill in pre-designed audit forms. The data obtained was analysed and presented at the ophthalmology departmental meeting.

Results: A total of 120 forms were filled in (96.0% response rate) with 119 completed forms. 74.2% were new patients and 25.0% were review patients. 47.5% of referrals were from optometrists, 18.3% from A&E, and the remaining from Other Specialties and General Practitioners. 9.2% cases were self-referrals. Referral letters were present for only 29.2% of new patients. The most common diagnosis was under the category Cornea (36.7%), followed by Neurology (15.0%). 28.3% of patients were discharged. The most common outcome was Others (Follow up clinic) (45.8%) and 3.3% were listed for urgent surgery.

Conclusion: There is room for improvement for referral patterns. Our eye casualty service should only be reserved for new cases, hence there was inappropriate slot utilisation for 25% of patients. Our data has also identified a need to liaise with different referral sources to ensure that all patients have a referral letter for assessment. These recommendations along with information on management outcomes will be used to inform resource utilisation and further service designs.

P013. Abstract not included

P014

AUDIT OF DIABETES CARE PROCESSES IN PATIENTS WITH DIABETES ATTENDING MEDICAL RETINA (MR) CLINICS

Corresponding Author: Duncan J Maclean, orcid.org/0009-0004-2419-8338

Duncan J Maclean¹, Sam Philip¹, Cynthia Santiago¹

¹NHS Grampian, Aberdeen, UK

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Introduction: Diabetic Retinopathy (DR) is a leading cause of blindness and visual impairment around the world. It is well known that effectively managing glycaemic control, blood pressure, and regular screening are protective against the progression of DR. However, there is a subsection of the diabetic community who despite best efforts continues to have progression of sight-threatening DR.

We conducted a rapid evidence review to synthesise available evidence and formulate audit standards for Diabetes care processes and risk factor management for DR.

Method: Evidence synthesis included Embase search and review of guidance from the American Diabetes Association, Diabetes Canada, the Royal College of Ophthalmologists, and NICE. We then assessed the attainment of these in patients attending our regional tertiary care MR Clinics.

Results: Key agreed audit standards included the following; HbA1C <53mmol, Blood Pressure <140/90mmHg, cholesterol <5mmol/l.

Of 1700 patients with diabetes, only 18% had HbA1C <53mmol/l, 60.5% had BP of < 140/90 mmHg and 76.8% had cholesterol level of <5 mmol/l. 36% had HbA1c above 75.

Conclusion: These findings show that the control of risk factors especially glycaemic control in our population is sub-optimal. There is a strong need for quality improvement initiatives to address this important clinical problem as this strongly correlates with progression of DR and avoidable visual impairment.

P015

AUDIT OF NOVEL HYBRID MODEL OF CARE IN GENERAL OPHTHALMOLOGY USING DIAGNOSTICS TESTS FIRST, THEN REMOTE CONSULTATION BY VIDEO OR PHONE WITHIN 2 WEEKS

Corresponding author: Anca Marinescu, orcid.org/0000-0002-6794-8539

Anca I Marinescu¹, Caterina M Hall²

¹Moorfields Eye Hospital, London, UK; ² University Hospitals Sussex NHS Foundation Trust, Worthing, UK

Disclosures: no conflicts of interest and no applicable funding

Introduction: Trialling a more efficient care model whereby patients attend for diagnostic tests and have a follow up consultation by phone or video. This addresses clinic space constraints, improves productivity and reduces waiting time in clinic. We audit 111 patients.

Normally patients attend the clinic, have diagnostic checks, then wait to see a consultant. Consultants can see limited numbers and often the patient may spend hours waiting. The new model requires suitable patients to attend for diagnostics, leave immediately and have remote consultation within a few days. The technicians use a structured questionnaire for basic information but the remote consultation asks more probing questions to correlate findings with symptoms.

Method: Suitable conditions identified during triage:

Suspicious discs

Disc drusen

Reduced vision

Non specific blurry vision

Macula changes seen (normal VA)

Visual field defects with normal IOP's and CDR

Neurological conditions such as IIH can be monitored

The technicians undertake the same suite of tests for all patients for simplicity.

Results: Stratford - 56 patients reviewed. 46.5% discharged, 28.5% referred, 12.5% follow ups, 9% DNA and 3.5% cancelled by patient.

Brent Cross - 55 patients reviewed. 24% discharged, 24% referred, 20% follow ups, 32% DNA

Conclusion: At Stratford we are approaching the 50% discharge rate.

Brent Cross had a high DNA rate which we are addressing with better patient communication.

Administratively patients get both appointments when the diagnostics are booked, so they know when they will be called or have video – this is also a failsafe.

The patient can have diagnostics any weekday reducing the constraint of limited clinic space.

Video/phone is often more efficient so the consultant can increase the number of patients seen per session.

P016

EVALUATING OPHTHALMOLOGY REFERRAL PRACTICES IN CANDIDEMIA PATIENTS

Corresponding author: Sara F Memon, orcid.org/0000-0003-0444-2925

Sara F Memon¹, Melissa Hutchings^{1,2}, Sarah Bourne¹

¹University Hospitals Dorset NHS Foundation Trust, Dorset, UK;

²University Hospital Southampton NHS Foundation Trust, Southampton, UK

Disclosures: No conflicts of interest and no funding

Introduction: Current guidelines from the Infectious Diseases Society of America (IDSA) recommend ophthalmic examinations for candidaemia patients, but challenges have arisen due to declining rates of ocular candida involvement. Guidelines from RCOphth suggest a case-by-case approach based on the presence of specific risk factors. However, the decision to refer may be at the discretion of the ward doctor who may have little experience of the condition.

Method: We examined the records of 12 inpatients with positive candida blood cultures, spanning ages 2 to 80 (median age 65). We noted choice of empirical antifungal agent, decision to refer to ophthalmology, and subsequent timing of eye examination and follow up. Additionally, we identified key risk factors for developing candidaemia in our patient group.

Results: 7/12 patients were referred to ophthalmology, with 6/7 receiving eye examination. The majority of patients (75.0%, n = 9) were commenced on empirical caspofungin, with 16.7% (n = 2) commenced on other agents (e.g. fluconazole, amphotericin B). No change in antifungal therapy occurred post-eye examination. Two patients reported visual symptoms (16.7%), of which one had confirmed candida endophthalmitis. Only one patient who had initial unremarkable eye examination had follow up to assess for interval change (16.7%). Challenges included inconsistent advice regarding referral to ophthalmology and arrangements for follow up, oversight of the patient's visual symptoms, and insufficient referrals from patients in the Intensive Care Unit (ITU).

Conclusion: Our findings underscore the need for standardised guidance for ophthalmology referrals for patients with candidaemia, considering the needs of non-verbal and ITU patients and highlighting the prominent risk factors before making a referral. Importantly when ophthalmology referral is deemed appropriate, considering the necessary follow-up, particularly when empirical therapy may lack ocular penetrance, is key.

P017

PROLIFERATIVE DIABETIC RETINOPATHY LASER TREATMENT AUDIT

Corresponding author: Evans Okoh, orcid.org/0009-0000-1383-3181

Evans Okoh¹, Adelehin Ijasan²

¹University of Aberdeen, Aberdeen, UK; ²Aberdeen Royal Infirmary, Aberdeen, UK

Disclosures: No conflicts of interest and no applicable funding

Introduction: Diabetic retinopathy is a sight-threatening disease of the retina. Panretinal photocoagulation (PRP) is a laser therapy for treating retinal angiogenesis, which is common a problem in diabetes.

Aim: The aim of this audit is to describe the short-term outcomes of patients undergoing their first treatment of PRP.

Method: For this retrospective study, 80 patients who had their first PRP treatment between March 2020 to March 2022 were investigated. Patient demographics, laser shots per session, waiting times for laser treatment, co-morbidities, number of laser sessions, visual acuities at the end of study period, and outcome were all considered.

Results: According to the NHS Diabetic Eye Screening Programme (DES) June 2016, standard 6 states that people identified as having active, untreated high-risk proliferative diabetic retinopathy can access nationally approved treatments within 5 days of referral, and 20 days for those with early retinopathy. Only 48.3% of patients in this audit received treatment within 20 days, with a mean waiting time of 27 days. The main reasons for this delay were patient anxiety, discomfort, laser machine breakdown, need for additional treatment e.g. antiVEGF and exacerbation of retinopathy despite significant PRP treatment. The future aim is to reduce the waiting time within two weeks. A major limitation to this audit is that only one senior clinician highlighted the first lasers done on the software, as a result all data used for this audit were from one clinician.

Conclusion: Though the outcomes of our cohort were good, as patients received their PRP on time compared to DES standards; reducing the number of PRP sessions per patient could reduce waiting times and the burden on the service.

P018

DEFINING A STANDARDISED MINIMUM CLINICAL DATASET FOR GLAUCOMA PROCEDURES

Corresponding Author: Ian A S Rodrigues, orcid.org/0000-0002-4913-1306

Ian A S Rodrigues¹, Anthony Khawaja^{2,3}, Gus Gazzard^{2,3}, Ananth Viswanathan^{2,3}, Andrew Tatham⁴

¹Guy's and St Thomas' NHS Foundation Trust, London, UK; ²NIHR Biomedical Research Centre, Moorfields Eye Hospital NHS Foundation Trust, London, UK; ³UCL Institute of Ophthalmology, London, UK; ⁴Princess Alexandra Eye Pavilion, Edinburgh, UK

Disclosures: no conflicts of interest and no applicable funding

Introduction: Standardised routine clinical data collection that enables national audits have furthered knowledge and helped improve outcomes in multiple ophthalmic conditions. However, despite the high volumes and expenditure in glaucoma procedures globally, currently no agreement exists on the data variables that should be collected to compare their outcomes.

Method: A set of defined clinical variables to be collected for patients undergoing any intraocular pressure lowering (IOP) procedure was developed by a RCOphth subcommittee, in collaboration with glaucoma societies in United Kingdom & Ireland (UKEGS) and Europe (EGS). Principles around standardising definitions, minimising data collection burden and electronic medical record implementation were adhered to.

Results: The following data items were deemed mandatory to collect for all purposes:

- Pre-operative: Glaucoma type; Visual acuity; IOP and Topical glaucoma medication number (on same date); Oral acetazolamide use.
- Intraoperative: Procedure date; Surgeon; Procedure name (+/-lens extraction); If revision surgery.
- Post-operative (every 12 months): Complications (with date);

Reoperations for complications; Further IOP lowering procedures (with date); Visual acuity; IOP and Topical glaucoma medication number (on same date); Oral acetazolamide use.

Further desirable items to enable more in-depth audits and analyses, plus those considered optional for collection were defined in the full clinical dataset. It also defines the data type for each clinical variable and references recommended classifications and coding for glaucoma related diagnoses, procedures and post-operative complications.

Conclusion: This dataset represents an agreed set of standardised clinical information for patients undergoing IOP lowering surgical or laser procedures, which can be collected from routine care across different platforms. The large volume of real-world data that could be gathered would enable future national and international audits and research to be conducted in comparing outcomes of glaucoma procedures.

P019

ASSESSING UTILISATION OF FOLLOW-UP APPOINTMENTS IN BACTERIAL CONJUNCTIVITIS

Corresponding Author: Saran Santhosh, orcid.org/0009-0008-6053-0720

Saran Santhosh¹

¹Nottingham University Hospitals Trust, Nottingham, UK

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Introduction: Approximately 1% of all appointments made in British General Practice can be attributed to infective conjunctivitis (Dart, 1986) (McDonnell, 1988). Bacterial conjunctivitis is largely a self-limiting condition, for which antibiotics are seldom indicated. The current NICE guidance recommends opting for management with self-care measures and a follow-up appointment. Arranging follow-up with the patient enables General Practitioners to verify the diagnosis as well as resolution of symptoms (National Institute for Health and Care Excellence, 2022). The aim of this audit was to assess if patients in a GP surgery underwent follow-up subsequent to a bacterial conjunctivitis episode. This was set to a 100% standard.

Methods: SystmOne records were used to quantify bacterial conjunctivitis episodes and consequent follow-up appointments. The first audit loop assessed all episodes (n = 41) between 4 November 2021 to 4 November 2022. After the completion of this loop, the results were shared through a departmental meeting. A second loop was later completed and evaluated all episodes (n = 58) between 19 April 2023 and 19 October 2023.

Results: In the first loop, 34.1% of episodes (n = 14) received follow-up. The remaining 65.9% of episodes (n = 27) were not followed-up. In the second loop, follow-up declined to 12.1% (n = 7). The remaining 87.9% of episodes (n = 51) were not followed-up.

Conclusion: Follow-up had significantly diminished in the second cycle. This could have been attributed to increased management with antibiotics in the second cycle, leading to possible lack of perceived need for follow-up. Additionally, an increased number of telephone consultations were utilised, which would limit the ability to assess for resolution through examination. A third audit loop will be performed following the implementation of poster aids in consultation rooms.

P020

THE RCOPHTH NATIONAL OPHTHALMOLOGY DATABASE STUDY OF CATARACT SURGERY: ASSESSMENT OF POSTERIOR CAPSULE RUPTURE FOR THE GENDER OF THE OPERATING SURGEON

Corresponding author: Peng Yong Sim, orcid.org/0000-0001-5500-2634

Peng Yong Sim¹, Paul H J Donachie^{2,3}, Alexander Day^{1,4}, John Buchan^{2,5,6}

¹Moorfields Eye Hospital NHS Foundation Trust, London, UK; ²The Royal College of Ophthalmologists' National Ophthalmology Database, London, UK; ³Gloucestershire Hospitals NHS Foundation Trust, Cheltenham, UK; ⁴UCL Institute of Ophthalmology, London, UK; ⁵International Centre for Eye Health, London School of Hygiene and Tropical Medicine, London, UK; ⁶Leeds Teaching Hospitals NHS Trust, Leeds, UK

Disclosures: no conflicts of interest and no applicable funding

Introduction: To investigate posterior capsule rupture (PCR) rates for the operating surgeon gender, for surgery performed in the traditional NHS and the Independent Sector (ISTC).

Method: Eligible operations were performed between 01/04/2016 and 31/03/2022 in centres supplying data to the national cataract audit for ≥50 eligible operations with data for patients' gender, age at surgery, anterior chamber depth and preoperative visual acuity. Expected PCR was calculated using the new RCOPHTH NOD PCR model.

Results: 952,044 operations were performed in 75 NHS and 58 ISTC centres by 3066 surgeons, 1901 (62.0%) male and 1,165 (38.0%) female.

Male surgeons performed 733,375 (77.0%) operations (55.6% NHS vs 44.4% ISTC), and 294 (15.5%) male surgeons had ISTC data. For male surgeon operations, 13.3% were performed by trainees. PCR occurred in 6,700 (0.91%) operations with an expected PCR rate of 1.05%. The observed PCR rates were 1.28% for NHS vs 0.45% for ISTC, and the expected PCR rates were 1.31% for NHS vs 0.73% for ISTC.

Female surgeons performed 218,669 (23.0%) operations (82.0% NHS vs 18.0% ISTC), and 63 (5.4%) female surgeons had ISTC data. For female surgeon operations, 29.9% were performed by trainees. PCR occurred in 2,904 (1.33%) operations with an expected PCR rate of 1.32%. The observed PCR rates were 1.50% for NHS vs 0.55% for ISTC, and the expected PCR rates were 1.44% for NHS vs 0.78% for ISTC.

Conclusion: Even though the observed and expected PCR rates were slightly lower for male surgeons, for both the NHS and ISTC, this is plausibly due to a higher proportion male surgeons with ISTC data, a higher proportion of male surgeon operations from the ISTC and a lower proportion performed by trainees.

P021

AN AUDIT VABYSMO VS ONGAVIA: WHO IS THE FAIREST OF THEM ALL?

Corresponding Author: Chanelle L Smith, orcid.org/0009-0008-7325-6257

Chanelle L Smith^{1,2}, Sam Jacobs³, Sheeja Thomas⁴

¹Sandwell and West Birmingham NHS Trust, Birmingham, UK; ²The Dudley Group NHS Foundation Trust, Dudley, UK; ³University of Birmingham, Birmingham, UK; ⁴Wye Valley NHS Trust, Hereford, UK

Disclosures: no conflicts of interest and no applicable funding

Introduction: Vabysmo (Faricimab) is an intra-vitreous (IVT) monoclonal antibody injection only approved by NICE in June 2022. Its loading dose consists of 4 separate injections; one injected each month. Ongavia (Ranibizumab) is another anti-VEGF treatment which consists of three injections; one each month apart. IVT injections have transformed the visual prognosis of various retinal conditions such as age-macular degeneration (AMD) and diabetic macular oedema. The aim of our audit was to compare the efficacy of Vabysmo to Ongavia to ensure these treatments deliver the expected outcomes while maintaining a favourable safety profile.

Method: Utilising a retrospective approach, all patients receiving Vabysmo or Ongavia injections at Hereford County Hospital who had completed the loading doses were identified. Patient demographics, central macular thickness (CMT) reduction, visual acuity gain and side effects were recorded.

Results: Ten patients had Vabysmo including 6 females and 4 males with a mean age of 81. Nine patients had Ongavia including 4 females and 3 males with a mean age of 85. All patients were treated for AMD and all ethnicities recorded were Caucasian. The mean CMT reduction for patients taking Vabysmo was $34\mu\text{m} \pm 27.9$ compared with $8\mu\text{m} \pm 49.3$ with Ongavia. The mean number of letters gained with Vabysmo were 6.4 ± 15.4 compared with 2.9 ± 8.5 with Ongavia.

Conclusion: Vabysmo showed a greater CMT reduction and gain in letters compared to Ongavia suggesting greater efficacy at treating AMD. Vabysmo is the first licensed treatment to combine anti-VEGF and anti-Ang-2 action, which has shown quicker resolution of fluid and larger interims between follow ups. There were few side effects noted for both injections. It would be useful in further studies to compare its use for treating diabetic macular oedema.

P022

A SERVICE EVALUATION OF PATIENT SATISFACTION WITH THE CURRENT STANDARD OF GLAUCOMA CARE PROVISION AT DERRIFORD HOSPITAL

Corresponding Author: Ryan Teo, orcid.org/0009-0006-2380-9912

Ryan Teo¹, Kleonikos Tsakiris¹

¹University Hospitals Plymouth NHS Trust, Plymouth, UK

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Introduction: The social and economic burden of glaucoma are likely to increase in the future because of longer life expectancies and an ageing population.

Healthcare providers must continually monitor and improve their management of glaucoma patients.

The project's purpose was to identify areas of improvement in Derriford Hospital's provision of glaucoma care and thereby, advance the European Glaucoma Society's mission statement to maintain or improve the quality of life in affected patients.

Method: The methodology consists of the following:

- Designing the Glaucoma Service Evaluation (12 questions)
- Administering the Service Evaluation
- Statistical Analysis of the Results

Responses were based on a five-point Likert Scale, with 1 being the most negative patient satisfaction score and 5 being the most positive.

Results: 150 patients, comprising 83 males and 67 females, undertook the questionnaire.

The mean age was 75 years.

Patient response rate to all 12 questions was 100%.

The average five-point Likert Scale Score across all 12 questions was 4.37.

Conclusion: The results suggest that overall patient satisfaction with the standard of glaucoma care at Derriford Hospital is positive to a large extent.

Patients are happy with evolutions in glaucoma care such as Virtual Clinics, and the role of allied health professionals.

However, it identified significant areas of improvement - enhancing patient education through the use of leaflets.

This service evaluation demonstrated that patients are largely satisfied with the current level of glaucoma care at Derriford. Most patients are open to novel initiatives, which would minimise patient waiting time at clinics.

More measures, such as patient counselling of medication side-effects and distribution of informative leaflets, should be implemented to improve the standard of glaucoma care and hence, raise patient satisfaction at Derriford Hospital.

P023

CLINIC CANCELLATION QIPP

Corresponding Author: Clara Hoi Ka Wu, orcid.org/0009-0008-7325-6257

Clara Hoi Ka Wu¹, Heidi Laviers², Rathie Rajendram¹

¹Moorfields Eye Hospital, London, UK; ²King's College Hospital NHS Foundation Trust, London, UK

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Introduction: The Patient Access Policy describes the way in which the Trust manages patient referrals, waiting lists, appointments and admissions. There are currently no specific standards for cancelling and rebooking appointments.

This project aims to reduce postponed or cancelled outpatient appointments. We evaluate reasons for cancellations, the amount of notice given and if appointments were subsequently rebooked appropriately.

Method: The following data was collected from all outpatient appointments at Moorfields Eye Centre, St George's Hospital between 1st to 28th February 2023:

- Total number of clinic appointments
- Number of clinic appointments postponed/ cancelled by the patient
- Number of clinic appointments postponed/ cancelled by the hospital

Reasons for cancellation

Amount of notice given to patients

Number of times an individual patient had their appointment postponed/ cancelled

The timeframe in which appointments were subsequently rebooked

Results: The total number of appointments in February 2023 was 8539, of which 2556 (29.9%) were cancelled. 1130 (13.2%) appointments were cancelled by the patient and 1426 (16.7%) by the hospital. 450 (31.5%) cancellations were due to insufficient clinic capacity, 272 (19.1%) had no reason documented and 231 (16.2%) were booked into an incorrect clinic. On average, patients were given 21.7 days' notice for the cancellation, 111 patients had their appointments cancelled on the day. 461 cases had already been rebooked and it took 40 days on average to rebook them.

Conclusion: Cancelled appointments have a significant financial impact on the NHS as payments are generated from the National Tariff Payment System by patients attending appointments. Effective coordination of staff leave and taking into account emergency cases by having one or two buffer appointments can prevent last-minute cancellations.

P024

TRANSFORMING CATARACT REFERRAL PATHWAY

Corresponding Author: Soon Chee Yap, orcid.org/0009-0001-3067-4343

Soon Chee Yap¹, Yan Lynn Ng¹, John Ellis¹

¹Ninewells Hospital, NHS Tayside, Dundee, UK

Disclosures: no conflicts of interest and no applicable funding

Introduction: Increasing demand and backlog for cataract surgery has become a public health concern. Besides maximizing surgical capacity, the next bottleneck in the system is our capacity to see patients in the cataract assessment clinics. We have proposed a solution by introducing an enhanced cataract referral pathway. EnVoi Lutrahealth is an app/web-based system that aims to improve quality of referrals from community optometrists, ensuring sufficient patient information is available at referral to list patient for cataract surgery. This aims to reduce numbers of cataract outpatient visits and to shorten waiting times for cataract surgery.

Method: A prospective study was carried out involving 150 patient referrals over a span of 5 months. We have piloted the enhanced cataract referral pathway at three community optometrists in NHS Tayside – Specsavers Dundee, Specsavers Arbroath, and Optix Angus with cataracts referrals made to Ninewells Hospital, Dundee. We have compared number of patients listed straight for surgery and patients requiring further phone consult or face-to-face consultation. We have also measured referral to listing/booking waiting time and outpatient savings. Lastly, we evaluated each users' satisfaction using validated questionnaires.

Results: We are currently at final stages of data collection, with 131 referrals completed. Preliminary data analysis showed 94.5% patients referred listed straight for surgery. This has resulted in a total of £15,240 in outpatient savings. Average time from referral to booking has been shortened to only 11 days. 7 patients required additional input from secondary care in the form of telephone or face-to-face consultation, which demonstrates safety of the enhanced pathway. Users' satisfaction to follow.

Conclusion: The enhanced cataract referral pathway effectively and safely reduces cataract outpatient visits required and subsequently reduces waiting time and outpatient costs.

CATARACT AND REFRACTIVE SURGERY

P025

USING AN ARTIFICIAL INTELLIGENCE TELEPHONE CALL TO AUTOMATE CATARACT PRE-ASSESSMENT APPOINTMENTS

Corresponding Author: Aisling Higham, orcid.org/0000-0002-8901-0007

Aisling Higham^{1,2}, Ernest Lim², Daiserie Vinluan¹, Kanmin Xue¹, Kikkeri Arun¹, Vera He¹, Rory McKinnon², Sarah Khavandi², James Godwin², Rebecca Turner¹

¹Oxford University Hospitals NHS Foundation Trust, Oxford, UK;

²Ufonia Ltd, Oxford, UK

Disclosures: Aisling Higham, Ernest Lim, Vera He, Rory McKinnon, Sarah Khavandi and James Godwin are employees of Ufonia Ltd.

Funding: This evaluation was funded by an SBRI grant, awarded to Ufonia.

Introduction: Pre-operative assessment is often a bottleneck for surgical waiting list throughput. We evaluate the real-world effectiveness of an artificial intelligence powered telephone conversation agent (Dora R3.9, Ufonia Limited, Oxford, UK) for performing automated cataract surgery pre-operative assessment history taking prior to proceeding to an in-person cataract one-stop assessment.

Method: All patients referred for routine cataract surgery at a large teaching hospital had a call from Dora before their one-stop cataract assessment. Dora assesses 18 domains to determine case complexity (for example medications and ability to lie flat) and undertakes a shared decision making discussion. A Net-Promoter-Score (NPS) of the likelihood of recommending Dora was captured (score 1-10). Effectiveness was assessed through time in motion studies. Accuracy of clinical assessment was compared to an expert-human listening to the calls and ground-truth from face-to-face assessments.

This evaluation was funded by a Small Business Research Initiative grant.

Results: Between March-December 2023, 1243 patients completed pre-assessment telephone calls with Dora. Patients had a mean age of 75 years and gave Dora a mean NPS of 8/10. When comparing Dora to the expert call reviewer, mean agreement was 90% across 29,184 questions. Comparing Dora decisions to face-to-face cataract assessment showed agreement of 86% across 654 values collected over a 3-week audit. Time in motion studies suggested 12 minutes time saving per consultation. 5.4% of patients were unsure or did not want cataract surgery, thus unnecessary in-person appointments were avoided.

Conclusion: In this population, automated cataract pre-operative assessment reliably collects key clinical information from patients. This information can improve surgery conversion rates, shorten appointments, and free clinician time within the eye service.

P026

THE OPPORTUNITY OF INDEPENDENT SECTOR TREATMENT CENTRES FOR CATARACT TRAINING

Corresponding Author: Ernest Iakovlev, orcid.org/0000-0002-5512-5703

Ernest Iakovlev¹, John Buchan², Alexander C Day¹

¹Moorfields Eye Hospital NHS Trust, London, UK; ²Leeds Teaching Hospitals NHS Trust, Leeds, UK

Disclosures: no conflicts of interest and no applicable funding

Introduction: Cataract surgery is the most performed surgical procedure in the NHS. Being able to perform cataract surgery independently is also a core requisite of Ophthalmology training. However, the proportion of NHS cataract surgeries performed in the independent sector has increased from approximately 20% in 2017 to almost 50% in 2021, and the trend is showing no sign of reducing.

Method: The increase in the number of cases performed in the independent sector, in addition to the case complexity remaining low in the independent sector while rising in NHS training hospitals, could be linked to a reduction in the number of cases trainees are doing. As per the GMC's 2022 National Training Survey, the proportion of trainees who had not completed 50 or more cases by end of their ST2 year has increased from 12% in 2019 to 35% in 2022.

Results: The 2021-2022 National Ophthalmology Database audit showed that only 1/96 independent providers had training opportunities. The 2023 GMC training survey also showed that only 8% of trainees were able to access training opportunities in the independent sector, while 60% felt more training was required.

Conclusion: The independent sector is an ideal training location due to the high number of routine cataract surgery cases, whilst NHS hospitals are more suited to trainees later in their training when exposure to more complex cases is required. Closer collaboration is needed between all NHS cataract surgery providers to deliver appropriate training for future generations of ophthalmic surgeons.

P027

EVALUATING THE CATARACT PATHWAY AND SURGERY OUTCOMES IN PATIENTS WITH ADVANCED DEMENTIA AND LEARNING DISABILITY IN A TERTIARY REFERRAL CENTRE

Corresponding Author: Mumta Kanda, orcid.org/0000-0001-6122-1104

Mumta Kanda^{1,2}, Alasdair Warwick¹, Giulio Pocobelli¹, Lucy Barker¹, Vincenzo Maurino¹, Rajesh Deshmukh¹

¹Moorfield's Eye Hospital NHS Foundation Trust, London, UK;

²Queen's Hospital, Romford, UK

Disclosures: no conflicts of interest and no funding

Introduction: Patients with advanced dementia and learning disability who lack capacity pose a unique challenge for cataract surgeons. We analysed the cataract pathway and outcomes for such patients in our tertiary referral centre (Moorfield's Eye Hospital, London).

Method: We conducted a retrospective search of our patient database (OpenEyes™) for cataract surgery between 1st January 2021 and 31st May 2023 in those over 18 with learning disability or dementia and lacking capacity. We collected information on pre- and post-operative clinic assessment and intraoperative details.

Results: Our study included 51 patients (67 eyes); median age 81 (70, 86), 39 had dementia, 9 had learning disability. Median pre-operative logMar best-corrected visual acuity (BCVA) was 1.00 (0.48, 1.81). 24 (36%) eyes had hypermature cataracts. BCVA was missing for 7 (10%) eyes. Difficulty assessing BCVA was reported in 15 (29%) patients. 18 (27%) had no documentation of subjective symptoms. 60 (91%) had a best interests meeting (BIM) and 7 (11%) were done on the day of surgery. 30 (45%) eyes were operated under general anaesthetic, 22 (33%) under local anaesthetic, and 15 (22%) under local anaesthetic with sedation. There were no intra-operative complications. 6 (9%) did not attend post-operative review. Median post-operative BCVA was 0.2 (0.12, 0.48). 4 (6%) patients developed cystoid macular oedema, 6 (9%) developed rebound uveitis, and 1 (1%) had raised intraocular pressure.

Conclusion: Improvement areas include better documentation of pre-operative symptoms as examination can be difficult and unreliable, conducting BIMs in clinics to avoid delays on the day of surgery, and identifying patients needing surgery earlier before they develop advanced cataract. Many patients tolerate local anaesthetic and visual outcomes are promising. We propose a new cataract pathway for these patients.

P028

A COMPARATIVE STUDY OF THE ENVIRONMENTAL AND ECONOMIC IMPACT OF TOPICAL/INTRACAMERAL VERSUS SUB-TENON'S ANAESTHESIA FOR CATARACT SURGERY WITHIN NHS INSTITUTIONS

Corresponding Author: Jonathan Malcolm, orcid.org/0000-0003-4942-355X

Jonathan Malcolm¹, Marta Gruszka-Goh^{2,3}, Paul Donachie^{2,3}, John Buchan^{1,3,4}

¹Leeds Teaching Hospitals NHS Trust, Leeds, UK; ²Gloucestershire Hospitals NHS Foundation Trust, Cheltenham, UK; ³The Royal College of Ophthalmologists' National Ophthalmology Database Audit, London, UK; ⁴International Centre for Eye Health, London School of Hygiene and Tropical Medicine, London, UK

Disclosures: no conflicts of interest and no applicable funding

Introduction: The most common local anaesthesia techniques used for UK cataract surgery are topical ($\pm$ intracameral) and sub-Tenon's. National Ophthalmology Database data for the 2021 NHS year indicates that when local anaesthesia was used, the Independent Sector used topical/intracameral anaesthesia in 97.5% of operations, while NHS institutions used topical/intracameral anaesthesia in 57.2% of operations and sub-Tenon anaesthesia in 41.5% of operations. In NHS institutions, 52.5% of surgery performed by a trainee used topical/intracameral anaesthesia, and 20/74 NHS institutions used topical/intracameral in >80% of operations.

Although both anaesthesia techniques are known to be safe and effective, the economic and environmental implications remain relatively unexplored. This study aimed to compare the economic and environmental impact of topical/intracameral and sub-Tenon's anaesthesia for cataract surgery within the NHS.

Method: Cataract surgery lists at Leeds Teaching Hospitals NHS Trust were observed to determine the consumables and human resource requirements for topical/intracameral and sub-Tenon's anaesthesia. A predominantly process-based analysis was used to estimate the carbon dioxide equivalent (CO₂e) associated with consumables. The provider cost of consumables, waste disposal and human resources was calculated using procurement, waste compliance and staffing data.

Results: The mean CO₂e and cost per case was 0.379 kg and £17.14 respectively for topical/intracameral anaesthesia versus 0.860 kg and £27.74 for sub-Tenon's anaesthesia.

Although some NHS institutions predominantly use topical/intracameral anaesthesia, many still have scope to reduce sub-Tenon's anaesthesia usage and increase topical/intracameral usage. For every 10,000 cases of sub-Tenon's anaesthesia converted to topical/intracameral, NHS institutions can expect to save around 4800 kg of CO₂e and £106,000.

Conclusion: Topical/intracameral anaesthesia is a cheaper and more environmentally sustainable alternative to sub-Tenon's anaesthesia for cataract surgery.

P029

ONE-STOP CATARACT CLINIC QUALITY IMPROVEMENT PROJECT: IMPROVEMENT OF CATARACT PATHWAYS AT A TERTIARY OPHTHALMIC HOSPITAL

Corresponding Author: Mana Rahimzadeh, orcid.org/0000-0002-8738-6967

Disclosures: no conflicts of interest and no funding

Introduction: The One-stop cataract clinic provides a pathway for carrying out pre-assessment, biometry and clinical review in one visit as recommended in the RCOphth High Flow Cataract Surgery guidelines (2022). The primary objectives of this quality improvement project were to evaluate the appropriateness of referrals, identify triaging issues and assess clinic outcomes, thereby enhancing productivity by increasing surgical listing and streamlining the service to improve patient experience.

Method: We carried out a retrospective data collection for the period between 01/08/2022 – 31/01/2023. 540 patients were included in the analysis. Data extracted from electronic medical records included: referral source and reason, refraction, past medical history (PMH), social history, clinic diagnosis, clinic investigations and clinic outcome.

Results: Most referrals were made by opticians and GPs (72.6%, n = 392). Only 61–68% of these referrals included PMH or refraction. Of note, 8.1% of patients had no referral letter. In 96.4% of the patients, the reason for referral was cataract. The remainder of referrals were unrelated eye conditions. In 89.3% of patients, the clinic diagnosis was cataract but 14.1% of these patients were asymptomatic. In 7% the diagnosis was not related to cataracts. Overall, 74.3% of cataract patients were listed for surgery and the remaining were discharged, referred to another specialty or placed under monitoring.

Conclusion: This quality improvement project identifies multiple areas for improvement, particularly in triaging and patient selection. To address this, a nurse-led telephone pre-assessment is recommended to ensure only patients symptomatic with cataract are seen in clinic. We developed a patient questionnaire to be provided prior to telephone appointments, alongside a patient information leaflet and new clinic letter to streamline the pre-assessment process.

P030

CORNEAL CROSSLINKING FOR KERATOCONUS AT FRIMLEY PARK HOSPITAL (2018-2021): A CLINICAL AUDIT

Corresponding Author: Sarita Rai, orcid.org/0009-0005-4851-8428

Sarita Rai¹, Thomas Poole¹

¹Frimley Health NHS Foundation Trust, Frimley, UK

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Introduction: Corneal collagen cross-linking (CXL) is a minimally invasive procedure that aims to slow the progression of keratoconus, which is the most prevalent corneal ectatic disorder characterized by progressive, non-inflammatory changes in stromal collagen structure, by strengthening corneal stromal tissue.

Method: Accelerated epithelium-off crosslinking with 10 minutes of UVA radiation of wavelength 370nm and power 9mW/cm² after the initial riboflavin soaking phase. Retrospective data collection of 109 eyes who had CXL from 2018 until 2021 was carried out to determine baseline and post-CXL best corrected visual acuity, pentacam corneal topography data at 6 and subsequent months (maximum up to 3 years followed up) as well as any adverse outcomes.

Results: Of the 109 eyes crosslinked, 30 eyes had no/ lost follow up, most likely due to the COVID pandemic falling in that

period; therefore, a sample of 79 eyes was used for analysis. The mean age of patients at time of procedure was 27.48 years. 15 (19%) were female and 64 (81 %) were male eyes. Mean follow-up was 2 years. 4 (5.06%) eyes demonstrated progression of keratoconus, which we defined as an increase in K2 or Kmax of 1.5 dioptres or more following the CXL procedure. Two of these four eyes had a repeat crosslinking procedure, one ended up having a keratoplasty and the other intrastromal ring. The remaining 75 eyes (95%) showed no progression. There were no adverse events in the follow up period. Mean BCVA also improved from 6/12 to 6/9, and none of the 75 eyes lost vision.

Conclusion: Corneal crosslinking with an accelerated protocol is a safe and effective procedure to stabilise keratoconus in our population.

P031

POSTERIOR CAPSULAR RUPTURE IN CATARACT SURGERY: A MULTIVARIATE ANALYSIS OF RISK FACTORS

Corresponding Author: Rishi Ramessur, orcid.org/0000-0003-3420-6050

Rishi Ramessur^{1,2}, Alexander Ionides¹, Allan Nghiem¹, Derek Kwun-Hong Ho³

¹Moorfields Eye Hospital NHS Foundation Trust, London, UK;

²Imperial College London NHS Trust, London, UK; ³Gloucestershire Hospitals NHS Foundation Trust, Gloucestershire, UK

Disclosures: no conflicts of interest and no funding

Introduction: Posterior capsular rupture (PCR) is the most significant complication associated with cataract surgery. We sought to understand risk factors associated with this serious complication at Moorfields Eye Hospital (MEH). Additionally, we sought to ascertain whether there was any difference in PCR risk following SubTenon's versus topical anaesthetic.

Method: Using Structured Query Language (SQL), we extracted data on 172,803 cataract operations performed between 2012 and 2022 from Electronic Health Records at MEH. Time Series Analysis and multivariate Ordinary Least Squares Regression were used to identify long-term trends and PCR risk factors.

Results: The average monthly PCR rate remained stable (0.75–1.75%) over the 10-year period, with the exception of a spike to 4% during early 2020 – thought to be due to low volume prioritisation of complex cases during the Covid-19 pandemic. Multivariate regression revealed higher PCR rates associated with Asian (adjusted odds ratio (AOR) 1.003, 95% confidence interval (CI) 1.001–1.005), Black (AOR 1.006, 95% CI 1.004–1.008) and Other ethnicities (AOR 1.003, 95% CI 1.001–1.005). Additionally, higher PCR rates were associated with subconjunctival anaesthesia (AOR 1.012, 95% CI 1.003–1.033) and surgeon grades of registrar (AOR 1.011, 95% CI 1.006–1.017) and fellow (AOR 1.005, 95% CI 1.000–1.009). Lower PCR rates were associated with SubTenon's and topical anaesthetic delivery. There was no significant difference between PCR risk with SubTenon's (AOR 0.993, 95% CI 0.987–0.999) and topical anaesthetic delivery (AOR 0.992, 95% CI 0.985–0.998).

Conclusion: A Learning Health System, as postulated by Hardie, describes a cycle of data to knowledge, knowledge to practice and practice to data. Our insights into risk factors for PCR complications will allow us to tailor our practice in order to optimise outcomes during cataract surgery.

P032

ACCURACY OF TORIC INTRAOCULAR LENS FORMULAS WITH MEASURED POSTERIOR CORNEAL ASTIGMATISM OF DIFFERENT ORIENTATIONS

Corresponding Author: Stephen Stewart, orcid.org/0000-0002-8682-825X

Stephen Stewart¹, Tun Kuan Yeo³, Salissou Moutari², Richard McNeely¹, Jonathan Moore¹

¹Cathedral Eye Clinic, Belfast, UK; ²Queen's University Belfast, Belfast, UK; ³Tan Tock Seng Hospital, Singapore

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Funding: none

Introduction: To assess whether the use of measured posterior keratometry values improved the prediction accuracy of toric intraocular lens power formulas, compared to predicted posterior keratometry values.

Method: 418 eyes undergoing phacoemulsification with implantation of a non-toric monofocal IOL were included in the study. Prediction errors for postoperative refractive astigmatism at 4 weeks postoperatively were compared for the following formulas: Barrett Toric with predicted posterior corneal astigmatism (PPCA), Barrett Toric with measured posterior corneal astigmatism (MPCA), EVO Toric PPCA, EVO Toric MPCA and Holladay I with the Abulafia-Koch regression. Subgroup analysis was conducted to compare prediction errors for eyes with a vertically orientated steep axis of PCA (60-120 degrees) to eyes with a non-vertically orientated steep axis of PCA.

Results: Standard keratometry was with-the-rule in 48% of eyes, while the steep PCA axis was vertically orientated in 91% of eyes. For all eyes, EVO PPCA had a smaller mean absolute error than Barrett MPCA, Barrett PPCA and Abulafia-Koch ($p < 0.01$ for all). EVO PPCA had the highest percentage of eyes within 0.50D of predicted postoperative astigmatism for eyes with vertical PCA (61.2%), while EVO MPCA had the highest percentage for eyes with non-vertical PCA (54.1%). EVO MPCA had the smallest centroid error for all eyes, and the subgroups ($p < 0.01$ for all). Eyes with non-vertical PCA have a lower percentage within 0.50D than eyes with vertical PCA when using PPCA (43% vs 61%, $p = 0.027$), but there is no significant difference between these groups when MPCA is used for eyes with non-vertical PCA (54% vs 61%, $p = 0.40$).

Conclusion: When the steep axis of posterior corneal astigmatism is not vertically orientated, the use of measured posterior keratometry values improves prediction accuracy.

P033. Abstract not included

P034. Abstract not included

P035

POPULATION-BASED VERSUS INDIVIDUAL A CONSTANT OPTIMISATION FOR REFINING REFRACTIVE PREDICTION ACCURACY IN SECOND-EYE CATARACT SURGERY

Corresponding Author: Stephen Stewart, orcid.org/0000-0002-8682-825X

Stephen Stewart¹, Salissou Moutari², Richard McNeely¹, Jonathan Moore¹

¹Cathedral Eye Clinic, Belfast, UK; ²School of Mathematics and Physics, Queen's University Belfast, Belfast, UK

Disclosures: no conflicts of interest and no applicable funding

Introduction: This study aimed to assess if A constant optimisation based on the refractive outcome of the first-operated eye for a given patient will improve the refractive prediction accuracy for the second eye.

Method: Consecutive eyes undergoing delayed bilateral sequential phacoemulsification with implantation of an enhanced monofocal intraocular lens were included. Patients were excluded if their interocular axial length difference was greater than 0.3mm. Prediction accuracy for second-operated eyes was compared for two methods: (1) First-operated eyes were used to generate a population average A constant for the Barrett Universal II formula and this constant was then used for all second-operated eye IOL power calculations; (2) First-operated eyes were used to generate an individual A constant for each patient and then this constant was used for the second-operated eye IOL calculation for a given patient.

Results: 106 eyes of 53 patients were included in the study. Mean absolute error was significant smaller when an individual A constant was used for second eye IOL calculations, compared to the population average A constant ($0.26 \pm 0.26D$ vs $0.34 \pm 0.29D$, $p = 0.046$). Mean prediction error and root mean square error were also smaller for the individual A constant method, although this did not reach statistical significance (-0.009 ± 0.37 vs -0.012 ± 0.45 , $p = 0.97$; 0.36 vs 0.45 , $p = 0.19$). A higher proportion of second eyes were within 0.25, 0.50 and 0.75D of target postoperative refraction for the individual A constant method (58% vs 49%, 81% vs 79%, 98% vs 91%).

Conclusion: First eye individual A constant optimisation reduces prediction error when patients are undergoing second eye surgery, assuming the eyes are biometrically similar. Individual A constant optimisation appears to be a superior method to population A constant optimisation.

P036. Abstract not included

P037

EVALUATION OF ONE YEAR OUTCOME OF TOPOGRAPHY-GUIDED PHOTOREFRACTIVE KERATECTOMY AND COLLAGEN CROSS-LINKING FOR POST-LASIK ECTASIA

Corresponding Author: Natalie K H Yang, orcid.org/0009-0000-8458-8507

Simon P Holland¹, David T C Lin¹, Gregory Moloney¹, Natalie K H Yang¹, Derek B Chan¹

¹The University of British Columbia, Department of Ophthalmology & Visual Sciences, Vancouver, Canada

Disclosures: no conflicts of interest and no funding

Introduction: Crosslinking (CXL) is used to treat post-LASIK ectasia. We evaluated Topography-Guided Photorefractive Keratectomy (TG-PRK) with CXL for post-LASIK ectasia using a Schwind Amaris 1050 excimer laser (SA).

Method: Post-LASIK ectatic eyes that underwent treatment with the SA and Athens protocol CXL were evaluated. Preoperative and 12-month postoperative uncorrected distance visual acuity (UDVA), best corrected visual acuity (CDVA), manifest refraction, and topographic cylinder were analysed.

Results: 82 eyes with complete data at 12 months were included. 59 eyes (72%) showed UDVA $\geq 6/12$ post-operatively. 36 eyes (41%) had improved CDVA. Mean astigmatism changed from $-3.31 \pm 1.61D$ to $-1.11 \pm 1.06D$. Mean spherical equivalent improved from $-1.28 \pm 2.68D$ to $-0.45 \pm 1.53D$. 21 eyes (26%) gained 2 or more lines while 1 eye (1.2%) lost 2 lines or more. No cases showed ectatic progression.

Conclusion: One-year results of TG-PRK with CXL as a treatment for post-LASIK ectasia show safety and efficacy as a potential alternative treatment for post-LASIK ectasia. The technique is an alternative treatment for post-LASIK ectasia in patients who are contact lens intolerant.

P038

OUTCOME OF PHOTOREFRACTIVE KERATECTOMY WITH ADVANCED BEAM PROFILE FOR MYOPIA

Corresponding Author: Natalie K H Yang, orcid.org/0009-0000-8458-8507

David T C Lin¹, Simon P Holland¹, Gregory Moloney¹, Natalie K H Yang¹, Derek B Chan¹

¹The University of British Columbia, Department of Ophthalmology & Visual Sciences, Vancouver, Canada

Disclosures: no conflicts of interest and no funding

Introduction: SmartSurFACE combines transepithelial photorefractive keratectomy (TE-PRK) with Schwind Smart Pulse Technology. It has been shown to improve visual acuity in low to high myopia, but there is limited data on its use in correcting extremely myopic eyes. We evaluated the outcomes of using TE-PRK on the Schwind Amaris laser (SA) with SmartSurFACE beam profile for moderate to extreme myopia.

Method: Patients with moderate myopia (0.00D to -6.00D), high myopia (-6.25D to -10.00D), and extreme myopia (Over -10.00D) treated with TE-PRK with SmartSurFACE beam profile at a laser refractive clinic with 12 months of follow-up were included in this retrospective study. Preoperative and 12-month postoperative uncorrected distance visual acuity (UDVA), best corrected visual acuity (CDVA), and manifest refraction were analysed.

Results: Ranging from -0.25D to -17.50D, a total of 1128 eyes were evaluated. The moderate, high, and extreme myopia groups included 680, 372, and 76 eyes respectively. Mean spherical equivalents improved for all eyes from -5.62 ± 2.82 D to 0.01 ± 0.37 D, for moderate group from -3.75 ± 1.42 D to 0.04 ± 0.33 D, for high group from -7.81 ± 1.14 D to 0.01 ± 0.37 D, and for extreme myopia from -11.66 ± 1.53 D to -0.23 ± 0.61 D. UDVA $\geq 6/7.5$ was achieved by 660 (97%), 352 (95%), and 60 (79%) eyes respectively. Only one patient lost ≥ 2 lines of vision.

Conclusion: TE-PRK with SA with SmartSurFACE beam profile showed efficacy and safety in a range of myopic eyes, including extremely myopic patients. This technique provides an alternative to LASIK and SMILE, which may be limited by low pachymetry, and avoids risks such as ectasia.

P039

TOPOGRAPHY-GUIDED PHOTOREFRACTIVE KERATECTOMY FOR CORRECTION OF IRREGULAR ASTIGMATISM FOLLOWING PENETRATING KERATOPLASTY

Corresponding Author: Natalie K H Yang, orcid.org/0009-0000-8458-8507

Simon P Holland¹, David T C Lin¹, Gregory Moloney¹, Natalie K H Yang¹, Derek B Chan¹

¹The University of British Columbia, Department of Ophthalmology & Visual Sciences, Vancouver, Canada

Disclosures: no conflicts of interest and no funding

Introduction: Penetrating keratoplasty (PKP) can be complicated by high and irregular astigmatism postoperatively. Treatment options such as contact lenses may not be well tolerated. We evaluated the efficacy and safety of Topography-Guided Photorefractive Keratectomy (TG-PRK) to correct irregular astigmatism following PKP.

Method: Patients with TG-PRK for irregular astigmatism post-PKP and 12 months of follow-up data were included in this retrospective study at a laser refractive clinic. There was a total of 182 eyes. Preoperative and postoperative uncorrected distance visual acuity (UDVA), best corrected visual acuity (CDVA), manifest refraction, and topographic cylinder were analysed.

Results: Postoperatively, 59 (32%) eyes had UDVA $\geq 6/12$ at 12 months compared to preoperatively where only 3 (1.6%) eyes had a recorded UDVA $\geq 6/12$. One quarter of the patients gained ≥ 2 lines. 8% lost ≥ 2 lines from incomplete treatment, haze, or cataracts. However, no patient subsequently needed specialty contact lenses.

Conclusion: TG-PRK may improve irregular astigmatism postoperatively and improve both UDVA and CDVA with good efficacy and safety. It may be an alternative to current surgical management for post-PKP irregular astigmatism. Trans-epithelial TG-PRK is a surface procedure that avoids more invasive techniques such as wedge resection and relieving incisions for post-penetrating keratoplasty.

P040

TOPOGRAPHY-GUIDED PHOTOREFRACTIVE KERATECTOMY FOR IRREGULAR ASTIGMATISM AFTER RADIAL KERATOTOMY USING A HIGH-SPEED LASER

Corresponding Author: Natalie K H Yang, orcid.org/0009-0000-8458-8507

Simon P Holland¹, David T C Lin¹, Gregory Moloney¹, Natalie K H Yang¹

¹The University of British Columbia, Department of Ophthalmology & Visual Sciences, Vancouver, Canada

Disclosures: no conflicts of interest and no funding

Introduction: To evaluate Topography-guided Photorefractive Keratectomy (TG-PRK) for irregular astigmatism after Radial Keratotomy (RK) with Schwind Amaris 1050 (SA).

Method: 81 eyes treated with Schwind Amaris 1050 excimer laser with CXL with Athens protocol. Preoperative and post-operative uncorrected distance visual acuity (UDVA), corrected distance visual acuity (CDVA), manifest refraction, and topographic cylinder were analysed after 12 months of follow-up.

Results: 46 eyes (57%) showed UDVA $\geq 6/12$ postoperatively. 36 eyes (45%) had improved CDVA and 9 (11%) gained ≥ 2 lines while 3 (4%) lost 2 or more lines. Mean astigmatism was reduced from -2.22 ± 2.04 D to -1.05 ± 1.01 D. Mean spherical equivalent improved from 2.36 ± 2.23 D to -0.36 ± 1.64 D.

Conclusion: Early results of TG-PRK CXL with SA show efficacy and safety in treating post-RK irregular astigmatism. The technique may be an alternative treatment for post-RK patients who have been unsuccessful with other methods of vision correction such as specialty contact lenses.

P041

EFFICACY OF IMPLANTABLE COLLAMER LENSES FOR PATIENTS WITH KERATOCONUS: A SYSTEMATIC REVIEW AND META-ANALYSIS

Corresponding Author: Ali Alkhabbaz, orcid.org/0000-0002-9931-5269

Ali Alkhabbaz¹, Mohammad Karam², Andre Pollmann³, Tsz Hin Alexander Lau², Khaldoun Abbas⁴

¹Kuwait University, Kuwait; ²Department of Ophthalmology and Visual Sciences, McGill University, Montreal, Canada; ³Department of Ophthalmology and Visual Sciences, University of Montreal, Montreal, Canada; ⁴Faculty of Medicine, University of British Columbia, Vancouver, Canada

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Introduction: Purpose: Assess efficacy of implantable collamer lenses for patients with Keratoconus.

Method: A systematic review and meta-analysis were performed per the PRISMA guidelines, and a search of electronic information was conducted to identify all studies of ICL efficacy in keratoconus patients. The primary outcome measures were best-corrected visual acuity (BCVA), corrected distance visual acuity (CDVA), uncorrected distance visual acuity (UDVA), uncorrected close visual acuity (UCVA), spherical equivalent, manifest refractive cylinder, astigmatism, and sphere.

Results: 24 eligible studies enrolling 656 eyes were identified. There was statistically significant improvement in BCVA up to 2 years post-operatively (SMD: 0.037, $P=0.048$). There was a significant improvement in UDVA (SMD: 0.247, $P=0.021$) and CDVA (SMD: 0.319, $P=0.002$) at 1 year postoperatively. There was a significant difference in UCVA up to 1 year post-operatively (SMD: 0.452, $P=0.022$). There was a statistically significant difference in spherical equivalent up to 1 year post-operatively (SMD: -0.404, $P=0.015$). There was no statistically significant difference in manifest refractive cylinder up to 1 year post-operatively (SMD: -0.286, $P=0.515$). There was a statistically significant difference in astigmatism up to 1 year post-operatively (SMD: -1.041, $P=0.013$). There was no statistically significant difference in the sphere at 6 months post-operatively (SMD: -0.507, $P=0.156$).

Conclusion: The results of this meta-analysis confirm that the use of ICL in patients with keratoconus is effective in improving BCVA up to 2 years postoperatively, as well as UDVA and CDVA at 1 year post-operatively. ICL also improves UCVA up to 1 year post-operatively. ICL improves spherical equivalent at 1 year post-operatively, however not refractive cylinder. However, it improves astigmatism up to 1 year post-operatively. ICL does not improve sphere at 6 months postoperatively.

P042

SAFETY AND EFFICACY OF PHOTOTHERAPEUTIC KERATECTOMY FOR TREATMENT OF RECURRENT CORNEAL EROSION SYNDROME: A RETROSPECTIVE SERVICE EVALUATION IN A TERTIARY CENTRE

Corresponding Author: Ahmed Bardan, orcid.org/0000-0002-8632-7041

Ahmed Al-Wizni¹, Hassan Ahmad², Matilda Woodhill², Mousindha Arjunan², Ahmed Bardan²

¹University of Leeds, Leeds, UK; ²St James' University Hospital, Leeds, UK

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Introduction: Recurrent corneal erosion syndrome (RCES), characterised by defects in hemidesmosomes of the corneal epithelial basement membrane, often presents with awakening pain, photophobia, and visual disturbances. Trauma and epithelial basement membrane degeneration (EBMD) are common aetiologies. Treatment options range from conservative approaches to the use of excimer laser-based Phototherapeutic Keratectomy (PTK). This study aimed to evaluate the safety and efficacy outcomes of receiving PTK for the treatment of RCES performed at St James' University Hospital, Leeds, United Kingdom.

Method: A retrospective analysis of 209 patients (218 eyes) diagnosed with RCES treated with PTK between August 2015 and December 2021 was conducted. This included 114 females and 95 males. The extracted data from patient records included the recurrence rate of RCES, the time gap between PTK and recurrence of symptoms, and reported symptoms and complications following PTK.

Results: Our results demonstrated that 21.1% experienced symptoms postoperatively following PTK, of which 77.7% were due to recurrence, and 18.8% being due to unresolved pain. Furthermore, 8.25% of eyes required further laser treatment. The incidence of post-operative infection following PTK in the group was 0.46%. Other post-operative symptoms included visual symptoms such as blurry or reduced vision that persisted for varying periods.

Conclusion: PTK is a valuable and efficacious treatment option for severe RCES, offering sustained relief with a low recurrence rate. However, further analysis is required to investigate factors contributing to the need for repeat PTK in certain patient subgroups.

P043

EXAMINING CORNEAL SURFACE TEMPERATURE VARIATIONS RESULTING FROM FINGERTIP, KNUCKLE, AND FINGERNAIL EYE RUBBING

Corresponding Author: Eray Atalay, orcid.org/0000-0002-2536-4279

Eray Atalay¹, Musacan Yalçın², Zeynep Serra Özler², Mert Egemen Çalıřkan², Taylan Can Özer², Sertac Eroglu³

¹Eskiřehir Osmangazi University Faculty of Medicine, Eskiřehir, Turkey;

²Eskiřehir Osmangazi University Medical School, Eskiřehir, Turkey;

³Eskiřehir Osmangazi University Faculty of Science, Eskiřehir, Turkey

Disclosures: no conflicts of interest and no applicable funding

Introduction: Eye rubbing has been consistently associated with the development and progression of keratoconus across multiple studies. Although the mechanism is mostly attributed to mechanical forces, cyclic temperature increases may contribute to biomechanical weakening of the cornea. The purpose of this study was to comparatively assess the magnitude of corneal temperature variations induced by three different eye rubbing techniques.

Method: A total of 86 subjects (44 male) between 18 and 24 years of age were enrolled into the study. Subjects were requested to rub their eyes for 20 seconds using their fingertip, knuckle, and fingernails in a randomized fashion using their dominant hands. At each session, 20 seconds resting period was

given for determining the baseline corneal temperature and the intervals between each session were set to 10 minutes. The average, minimum and maximum corneal surface temperatures within a 3 mm diameter region-of-interest circle centred on the cornea were measured using a high-resolution thermal camera (FLIR A8200c), immediately after the rubbing procedure had stopped. The variations in corneal temperature across three different rubbing techniques were compared using repeated measures ANOVA, with the statistical significance at $p < 0.05$.

Results: The mean baseline corneal temperature was $34.4 \pm 0.7^\circ\text{C}$ and the increase from baseline was 1.0°C , 0.97°C and 1.03°C for fingertip, knuckle, and fingernail eye rubbing, respectively (all $p < 0.001$). Corneal temperature variations across the three different types of eye rubbing were similar ($p > 0.05$) for both genders.

Conclusion: Eye rubbing causes an approximately 1°C increase in surface temperature, regardless of the specific eye rubbing technique employed. Coupled with the mechanical forces, the cyclic temperature increase induced by habitual eye rubbing may be detrimental for corneal biomechanical integrity, contributing to its association with keratoconus.

P044

IS THE MEIBOPATCH HEATED PAD USEFUL IN TREATING PATIENTS WITH MEIBOMIAN GLAND RELATED DRY EYE DISEASE? ASSESSING TOLERABILITY, ACCEPTANCE, AND EFFECTIVENESS

Corresponding Author: Tasneem Elghazali Bakhiet, orcid.org/0009-0006-4145-5548

Tasneem Elghazali Bakhiet¹, Gwyn Samuel Williams¹, Abdus Samad Ansari¹, Alistair Thomas¹

¹Swansea University Health Board, Swansea, UK

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Introduction: Dry Eye Syndrome (DES) significantly impacts vision-related quality of life and poses a substantial financial burden in ophthalmology practice globally. Projected expenditure on DES treatments is estimated to exceed 6.6 billion USD by 2027. The Meibopatch (VISUfarma International, Valbonne, France) represents one of the latest devices designed to enhance treatment adherence and aid in managing Dry Eye Syndrome (DES). However, its long-term acceptance and efficacy among patients with posterior blepharitis, a condition closely linked with Meibomian gland dysfunction (MGD), has yet to be demonstrated.

Method: Patients with symptomatic nonresponsive blepharitis were enrolled during clinic visits and asked to self-administer Meibopatch treatment for four weeks. The ocular surface disease index (OSDI) questionnaire was utilized to evaluate symptoms at baseline and following the four-week treatment period. Supplementary assessments on acceptability and tolerability were conducted through post-treatment questionnaires. A grant was obtained from the University Health Board research and development funding and Health Care Research Wales.

Results: A total of 43 patients were recruited. The baseline pre-treatment OSDI score $\pm$ SD (Standard Deviation) was 47.9 ± 16.4 , while the post-treatment OSDI value $\pm$ SD was 37.7 ± 14.9 . Analysis revealed a significant mean difference between pre and post-treatment OSDI scores ($p = 0.0005$), indicating notable symptomatic improvement over the assessment duration. However, the findings diverged from evaluations of acceptability and overall patient satisfaction with the Meibopatch.

Conclusion: The results suggest that Meibopatch represents a safe therapeutic option for managing DES. However, the controversy arises from the incongruence between the substantial reduction in symptoms noted from the OSDI scores and the reported patient outcomes regarding satisfaction and effectiveness. Future efforts should focus on validating the OSDI in blepharitis patients and establishing patient-centred outcomes in the realm of DES.

P045. Abstract not included

P046

EVALUATION OF ALLIED HEALTH PROFESSIONAL-LED KERATOCONUS MANAGEMENT SERVICE IN THE UNITED KINGDOM: THE BIRMINGHAM AND MIDLAND EYE CENTRE STUDY

Corresponding Author: Marianthi Bourlaki, orcid.org/0000-0001-7839-2057

Marianthi Bourlaki¹, Murad Khan², Anil Aralikatti¹, Ankur Barua¹, Darren S J Ting^{1,2}

¹Birmingham and Midland Eye Centre, Sandwell and West Birmingham Hospitals NHS Trust, Birmingham, UK; ²Academic Unit of Ophthalmology, Institute of Inflammation and Ageing, University of Birmingham, Birmingham, UK

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Introduction: Keratoconus is the most common corneal ectatic disorder, affecting 1:375 to 1:2000 people globally. In view of improved community screening, ophthalmic units across the UK are seeing a significant increase in community-to-hospital referrals for keratoconus management.

To cope with the increased workload, several innovative pathways/services have been established, including the setup of virtual keratoconus clinics and CXL services led by nurses and optometrists [collectively known as allied health professionals (AHPs)].^{4,5} This cross-sectional study aimed to examine the effectiveness and safety of AHP-led keratoconus service in Birmingham and Midland Eye Centre (BMEC), one of the UK largest ophthalmic tertiary referral centres that provides region-wide CXL service for the entire Birmingham.

Method: We included 170 patients with suspected/confirmed keratoconus from AHP-led keratoconus clinics at BMEC, run by two nurses (SB, RS) and two optometrists (EK, UP) with 3-9 years of corneal experience, in June-October 2023.

Progression of keratoconus was defined by the presence of ≥ 2 following parameters over the past year:

- Subjective/objective decrease in CDVA ≥ 1 Snellen line
- $\geq 1\text{D}$ increase in cylinder on manifest refraction
- $\geq 1\text{D}$ increase in K2
- $\geq 1\text{D}$ increase in Kmax

Progressive thinning of corneal thickness

The decision was first made by AHPs independently and was recorded as either observe, list for CXL, discharge, or uncertain. This was then compared to the decision made independently by the cornea consultant ophthalmologists. The main outcome measure was the concordance of the clinical decision made between AHPs and corneal consultants.

Results: 53 (31.2%) and 117 (68.8%) were of new and follow-up visits, respectively. AHPs made decisions to observe in 141 (82.9%), list for CXL in 11 (6.5%) cases, discharge in 8 (4.7%) cases, and expressed uncertainty in only 10 (5.9%) cases. An excellent agreement (91.8%) was observed between AHPs and consultants. Of the 10 (5.9%) cases with uncertainty, the consultant recommended observation in 5 cases, CXL in 4 cases and discharge in 1

case. For the four remaining cases with disagreement, AHPs made the decision to observe but the consultant made the decision to discharge. Six cases were correctly listed for CXL at the first visit, and no cases were incorrectly listed for CXL or discharge by the AHPs.

Conclusion: This study highlights an excellent concordance in clinical decision-making between AHPs and consultant ophthalmologists for managing new and follow-up cases of suspected keratoconus, highlighting the effectiveness, safety and feasibility of AHP-led keratoconus services (with minimal consultant support).

With a standardised protocol, similar AHP-led keratoconus service models can be replicated in other UK ophthalmic units to help alleviate the pressure of current ophthalmic services.

P047

INCIDENCE OF STEROID RESPONSE IN LOW RISK PATIENTS WHO UNDERWENT DMEK FOR FUCHS ENDOTHELIAL DYSTROPHY BETWEEN 2014 TO 2020 AT MOORFIELDS EYE HOSPITAL

Corresponding Author: Marketa Cilkova, orcid.org/0000-0001-9566-5352

Marketa Cilkova¹

UCL Institute of Ophthalmology, London, UK

Disclosures: no conflicts of interest and no funding

Introduction: The aim of this 7 year study was to identify the incidence and risk factors for steroid response in non glaucomatous patients who underwent routine Descemet's membrane endothelial keratoplasty (DMEK) or combined phacoemulsification/DMEK surgery at Moorfields Eye Hospital between 2014 and 2020.

Method: This retrospective review of 993 eyes excluded eyes with pre-existing OHT, glaucoma or pupillary block and eyes with raised IOP within the early postoperative period (2 weeks). Incidence and time of onset of steroid response was determined, as well as risk factors such as type of steroid, agents using for intraoperative AC tamponade and re-bubbling.

Results: Median time of steroid response diagnosis following DMEK surgery was 4.5 months. In 85 % of total cases the steroid response occurred within the first 9 months.

In 75% of cases with onset after 9 months steroid response was observed shortly after another intraocular surgery either on the same or contralateral eye, or after an increase in topical steroids for other causes.

In 80% of total cases steroid induced ocular hypertension resolved with switching or discontinuing steroids without further ongoing IOP lowering treatment, 20 % of cases with steroid response required ongoing treatment or monitoring in Glaucoma Service and less than 1% required glaucoma surgery.

Conclusion: Incidence of steroid response is higher than expected in low risk patients with Fuchs endothelial dystrophy who underwent DMEK. Switching to a lower potency steroid formulation should be considered at 4-months postoperatively in low risk DMEK surgeries.

P048

KERATOMETRIC OUTCOMES OF CORNEAL COLLAGEN CROSSLINKING FOR KERATOCONUS PATIENTS AT SUSSEX EYE HOSPITAL: 5 YEARS RESULTS

Corresponding Author: Mohamed G Elghobaier, orcid.org/0000-0003-4420-3937

Mohamed G Elghobaier¹, Sue Bennett¹, Mayank A Nanavaty^{1,2}

¹Sussex Eye Hospital, University Hospitals Sussex NHS Trust, Brighton, UK; ² Sussex Medical School, University of Sussex, Brighton, UK

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Introduction: This study aims to report on the long-term keratometric results of epithelium-off corneal collagen cross-linking (CXL) for patients with keratoconus at Sussex Eye Hospital in Brighton, United Kingdom.

Method: Patients with advanced keratoconus were included in the study. Epithelial removal was performed using 70% alcohol and 2% chlorhexidine gluconate wipes (Clinell, GAMA Healthcare, UK) for 10–20 seconds. CXL was completed by applying dextran-free riboflavin (0.1%) for 10 minutes (Vibex Rapid; Avedro, Inc.), then continuous UV-A light (Avedro KXL system; Avedro, Inc.) for 30 minutes at 3 mW/cm² intensity and 5.4 J/cm² energy. Keratometric data on pre and postoperative were collected. The patients were divided into central, paracentral, and peripheral cones.

Results: From 2014 to 2019, 100 eyes underwent treatment with a 5-year follow-up, with central, paracentral, and peripheral cones comprising 63%, 28%, and 9% respectively. Postoperatively, central cones showed postoperative flattening of Kmax (-1.54 ± 1.94 D) and paracentral cones (-0.62 ± 1.59 D), with no changes in peripheral cones. Patients with central cones had a preoperative posterior Kmean of -7.70 ± 0.80 , which was not significantly changed at the 1-year follow-up. In contrast, patients with paracentral and peripheral cones showed steepening. Postoperative flattening negatively correlated with paracentral and peripheral locations, while positively correlated with pre-operative Kmax values in central cones. Comprehensive 5-year follow-up data will be shared upon presentation.

Conclusion: Postoperative flattening in central cones was consistent across all cone locations, with postoperative flattening correlated with preoperative Kmax values in central cones, but not peripheral and paracentral locations.

P049

ASSOCIATION BETWEEN SOCIODEMOGRAPHIC FACTORS, VISUAL ACUITY AND CORNEAL TOPOGRAPHIC OUTCOMES AFTER COLLAGEN CROSS-LINKING IN PATIENTS WITH KERATOCONUS

Corresponding Author: Cara Heppell, orcid.org/0009-0006-2438-5955

Cara Heppell¹, Paul Nderitu¹, Ruchi Gour¹, Sophie Jones¹

King's College Hospital NHS Foundation Trust, London, UK

Disclosures: no conflicts of interest and no funding

Introduction: Corneal cross-linking (CXL) using riboflavin and UV light is an effective treatment for reducing the risk of keratoconus progression. This study aimed to determine the association between sociodemographic factors (age, sex, ethnicity, deprivation) with post-operative visual acuity (VA) and corneal topography (KMax) in an ethnically and socioeconomically diverse patient cohort.

Method: An audit of 297 eyes of 232 keratoconus patients who underwent CXL between June 2016 and January 2023 at King's College Hospital was conducted. Data on age, sex, ethnicity, index of multiple deprivation (IMD) decile, postoperative complications,

pre- and postoperative best VA (BVA), and KMax was extracted. Univariate Kaplan-Meier and multivariate cox-regression survival analyses were performed with four primary outcomes: BVA improvement (≥ 0.1 logMAR decrease), BVA stability (≤ 0.0 logMAR decrease), KMax improvement ($\geq 1.00D$ decrease) and KMax deterioration ($\geq 1.50D$ increase) at 52 weeks.

Results: At 52 weeks, there was a 51.8% (95%CI: 43.8%–58.6%) probability of BVA improvement, 78.5% (71.7%–83.7%) probability of BVA stability, 53.7% (45.0%–61.0%) probability of KMax improvement, and 30.4% (22.7%–37.4%) probability of KMax deterioration. Multivariate cox-regression analyses showed significant associations ($p < 0.05$) between ethnicity, BVA improvement, BVA stability and KMax deterioration outcomes (hazard ratios (HR) ranging between 2.11 and 10.52) and between education (HR 1.13), employment deciles (HR 1.12) and BVA improvement.

Conclusion: CXL resulted in the stabilisation or improvement of VA and KMax in the majority of eyes. Significant associations were observed between employment and education deprivation and BVA improvement, but not KMax suggesting the influence of sociodemographic factors on visual rehabilitation in keratoconus patients following CXL. Further investigation into the nature of these associations is warranted to guide interventions that ensure visual recovery is maximised across all keratoconus patients following CXL.

P050. Abstract not included

P051

A MULTICENTRE STUDY VALIDATES THE WHO 2022 CLASSIFICATION FOR CONJUNCTIVAL MELANOCYTIC INTRAEPITHELIAL LESIONS WITH IMPROVED CLINICAL AND PROGNOSTIC INFORMATION

Corresponding Author: Yamini Krishna, orcid.org/0000-0001-5067-3682

Yamini Krishna¹, Hardeep Mudhar², Hardeep Mudhar², Simon Cross³, Claudia Auw-Haendrich⁴, Raymond Barnhill⁵, Svetlana Cherepanoff⁶, Ralph Eagle⁷, James Farmer⁸, Robert Folberg⁹, Hans Grossniklaus¹⁰, Martina C Herwig-Carl¹¹, Martin Hyrcza¹², Sandra Lassalle¹³, Karin U Loeffler¹¹, Alexandre Moulin¹⁴, Tatyana Milman⁷, Rob M Verdijk¹⁵, Steffen Heegaard¹⁶, Sarah Coupland¹

¹Liverpool University Hospitals NHS Foundation Trust, Liverpool, UK; ²Royal Hallamshire Hospital, Sheffield, UK; ³University of Sheffield, Sheffield, UK; ⁴University of Freiburg, Freiburg, Germany; ⁵Paris Sciences and Lettres Research University, Paris, France; ⁶St Vincent's Hospital, Sydney, Australia; ⁷Wills Eye Hospital, Philadelphia, PA, USA; ⁸Queen's University, Kingston, Canada; ⁹Oakland University William Beaumont School of Medicine, Rochester, MI, USA; ¹⁰Emory Eye Center, Atlanta, GA, USA; ¹¹University Hospital Bonn, Bonn, Germany; ¹²Arnie Charbonneau Cancer Institute, Calgary, Canada; ¹³Pasteur Hospital, Centre Hospitalier Universitaire de Nice, Nice, France; ¹⁴Jules-Gonin Eye Hospital, Lausanne, Switzerland; ¹⁵Erasmus MC University Medical Center, Rotterdam, Netherlands; ¹⁶Rigshospitalet, Copenhagen, Denmark

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Introduction: Several terminologies and grading systems have been proposed for Conjunctival Melanocytic Intraepithelial Lesions (C-MIL). In 2018, the 4th 'WHO Classification of Eye Tumours' proposed a simplified grading but still captured risk of

disease progression to invasive melanoma. This system was validated in 2021; however, some weaknesses led to further revisions for the 5th WHO Classification in 2022. Herein we validate the WHO 2022 Classification of C-MIL.

Method: C-MIL cases were retrieved from 3 supraregional ocular pathology centres. All adequate biopsies were included and stained with H&E, MelanA, SOX10 and PRAME. Digitised slides were uploaded on the SmartZoom platform and independently scored by 4 ocular pathologists, to obtain a consensus score, before circulating to 14 expert eye pathologists from 9 different countries for independent scoring

Results: Evaluated were 105 cases from 97 patients: 28/105 benign melanoses; 13 Low-grade C-MIL; 37 High-grade; and 27 melanoma in-situ. With this 4-tiered grading system, 93% observers showed fair-to-moderate agreement (Kappa-statistics). With a 3-tiered system (combining High-grade and melanoma in-situ), 93% showed moderate-to-substantial agreement. The agreement level was similar across lesion types for both systems, with Low-grade C-MIL being the most diagnostically challenging. Recurrence rates and invasive melanoma development for High-grade C-MIL (3-tiered system): 34% and 47%, respectively versus melanoma in-situ alone (4-tiered): 55% and 74%, respectively.

Conclusion: This large international collaborative study is the first to undertake a comprehensive review of C-MIL to validate the WHO 2022 Classification and was found to have substantial interobserver agreement and good reproducibility. This classification provides good prognostic accuracy (recurrence risk and development of invasive melanoma). Importantly, it facilitates decision-making for the treating clinician as to which patients can be observed versus those requiring active treatment.

P052

TEMPORARY KERATOPROSTHESIS AND PRIMARY CORNEAL GRAFT FOR OCULAR TRAUMA: A SYSTEMATIC REVIEW AND META-ANALYSIS

Corresponding Author: David McMaster, orcid.org/0000-0003-1900-8785

David McMaster¹, Sophia Halliday², James Bapty³, Richard Blanch⁴

¹Imperial College London, London, UK; ²BRAVO VICTOR, London, UK; ³Portsmouth Hospitals University NHS Trust, Portsmouth, UK; ⁴Neuroscience and Ophthalmology, Institute of Inflammation and Ageing, University of Birmingham, Birmingham, UK

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Introduction: In cases of ocular trauma where severe corneal and retinal injury occur together, a temporary keratoprosthesis (TKP) is often a last resort to allow posterior segment visualisation, enabling vitreoretinal surgery which can then be combined with a penetrating keratoplasty (PKP) to restore corneal clarity in a single operation.

Method: A systematic literature review was performed following PRISMA guidelines. CENTRAL, MEDLINE, Embase, ISRCTN registry and ClinicalTrials.gov were searched from inception to 27 April 2023. Randomised and non-randomised studies assessing visual outcomes after combined surgery were eligible for inclusion. Outcomes included change in best-corrected visual acuity, corneal graft survival and retinal reattachment at final follow-up. Proportional meta-analysis was used to estimate the overall rate of the primary outcomes.

Results: A total of 22 studies met inclusion criteria reporting 420 eyes. All studies were retrospective and non-randomised with follow up times ranging from 2 to 91 months, with at least 79% of

outcomes reported after 6 months. After combined surgery the rate of corneal graft survival was 55% (95% CI 0.46-0.66; I^2 28%, $p < 0.11$), successful retinal attachment was 81% (95% CI 0.71-0.92; I^2 0%, $p < 0.99$) and improved visual acuity, when compared to no change or decrease in visual acuity, was 47% (95% CI 0.36-0.61; I^2 40%, $p = 0.04$).

Conclusion: We found that when combined vitreoretinal surgery and PKP with a TKP are performed, anatomically successful retinal reattachment is likely, approximately half of corneal grafts survive and a similar proportion of patients benefit in terms of improved visual acuity, compared to their preoperative function. This systematic review of the available literature may help inform surgeons of the benefits of using a TKP for cases of ocular trauma.

P053. Abstract not included

P054. Abstract not included

EMERGENCY OPHTHALMOLOGY

P055

ASSESSING NURSES' CONFIDENCE IN INTRAVITREAL INJECTION THERAPY AND MANAGING IOP SPIKES: A QUALITY IMPROVEMENT PROJECT

Corresponding Author: Maryam Al Khalifa, Maryam.Alkhalifa@elht.nhs.uk

Maryam Al Khalifa¹, Eesaa Longden¹, Hoozaifa Master¹, Miguel Kurc¹, Hugo Whyte¹

¹East Lancashire Hospitals NHS Trust, Blackburn, UK

Disclosures: no conflicts of interest and no funding

Introduction: Nurses play an important role in Intravitreal Injection Therapy (IVT) and managing complications, such as Intraocular pressure (IOP) spikes. However, their confidence in this aspect has not been widely explored. This quality improvement project aims to examine the confidence of nurses before and after implementing the changes discussed below, to optimize patient care at Royal Blackburn Hospital (RBH).

Method: A questionnaire was developed to assess the confidence of the nurses performing IVT and assessing IOP spikes. It was distributed amongst all the nurses who perform these injections at RBH. Additionally, a flow chart was posted in the injection room as a visual aid to guide nurses on the management of IOP spikes. We will also make sure to have an 'IOP spike box' available in the injection room containing lidopidine 1% and an iCare machine. There are plans to make amendments to patient documents to include C/D ratios to improve identification of high-risk patients prior to injection.

Results: The results of the questionnaire evaluated confidence, experience and training in managing IOP spikes. The findings revealed varying degrees of confidence and experience among nurses. The nurses who had higher numbers of injections performed reported higher confidence. 2/6 nurses have not received training in recognizing and managing IOP spikes, 4/6 nurses expressed requiring additional support such as training on the use of iCare, as well as management of IOP spikes. There was unanimous agreement between all 6 nurses that the changes discussed above would be helpful.

Conclusion: We believe that the results of this project will provide valuable insights to enhance nurses' competence and improve overall patient care. We look forward to sharing the complete findings at the conference.

P056

THE CONFIDENCE AND COMPETENCY OF GENERAL ACCIDENT AND EMERGENCY DOCTORS IN APPROACHING EYE-RELATED EMERGENCY CONDITIONS

Corresponding Author: Mohd Alhalki, orcid.org/0009-0003-7185-6703

Mohd Alhalki¹

¹West Middlesex University Hospital, Isleworth, UK

Disclosures: no conflicts of interest and no funding

Introduction: Ophthalmic-related conditions present 1.5%-6% of general ED attendance according to the Royal College of Ophthalmologists, 2020. Therefore, it's essential for ED doctors to have the basic sufficient skills to approach ocular emergencies. This study aims to evaluate how confident the ED physicians are in assessing emergency eye conditions.

Method: An online survey comprising 8 questions was created and sent to the local doctors working in Accident and Emergency department. Participants in the questionnaire were junior doctors, senior clinical fellows and consultants. The data was then collected and analysed.

Results: A total of 24 doctors working in A&E have completed the survey which consists of 8 questions, amongst which 83% were senior house officers/junior clinical fellows, 8% were more senior doctors, and the remaining 4% were consultants.

A total of 84% of participants have either below average or no confidence at all with assessing and managing emergency eye conditions. An average of 50% and 62% of A&E doctors don't have any confidence using ophthalmoscope and slit lamp, respectively.

Conclusion: This study demonstrates that most of A&E doctors lack the confidence of approaching ocular emergencies as well as using the basic ophthalmic equipment in ED.

More attention should be focused on providing basic training and education to general A&E doctors to help them being more comfortable using the ophthalmic devices and having the basic fundamental knowledge in dealing with common eye-related causalities, which eventually will reflect on the level of care provided to our patients.

P057

A CONFIDENT CUT: A LOW FIDELITY METHOD TO THE TEACHING OF LATERAL CANTHOTOMY/ CANTHOLYSIS

Corresponding Author: Frederick Bailey, orcid.org/0000-0002-7206-3081

Frederick Bailey^{1*}, Benjamin TK Hui^{2*}, Timothy J Patterson², Barry T Cartmill³

¹Belfast Health and Social Care Trust, Belfast, UK; ²Northern Ireland Medical and Dental Training Agency, Belfast, UK; ³Western Health and Social Care Trust, Londonderry, UK

*Joint first authors

Disclosures: no conflicts of interest and no funding

Introduction: Lateral canthotomy/ cantholysis is a procedural skill which is used in cases of orbital compartment syndrome to limit optic nerve damage. It is a skill required of doctors in various medical specialties, including Emergency Medicine. Despite this,

trainee doctors can lack confidence in this procedure.¹ Effective training methods should be identified to improve confidence in performing this procedure.

Method:

Aims: To design and implement a training programme to improve confidence in performing a lateral canthotomy/cantholysis.

To design and implement a training programme which is feasible and acceptable to both trainers and trainees.

Methods: A previously described, low fidelity model eye was designed to practice the principles of lateral canthotomy/cantholysis.² A tutorial on the topic of lateral canthotomy/cantholysis was created and delivered to a group of 6 Emergency Medicine trainees by a Consultant Ophthalmologist. This was then followed by hands on practice by the trainees on the pre-made model eyes. Participants were asked to complete questionnaires before and after the session focussed on confidence and acceptability of teaching.

Results: Prior to the tutorial, participants had variable confidence in the elements of the lateral canthotomy/cantholysis. Almost all participants did not feel confident in performing the procedure. After the tutorial/practical session, all confidence parameters improved. In particular, all participants felt confident in performing a lateral canthotomy/cantholysis. Written responses highlighted the model eye as a useful practice modality.

Conclusion: A simple, reproducible, low-fidelity model eye can be used to improve confidence in the procedure of a lateral canthotomy/cantholysis.

P058

SEVENTEEN YEARS OF BIRD-RELATED EYE INJURIES AT THE ROYAL VICTORIAN EYE AND EAR HOSPITAL, MELBOURNE, 2006–2022

Corresponding Author: Jeremy P M Flanagan, orcid.org/0000-0002-4506-4388

Jeremy P M Flanagan^{1,2}, Melissa Moyes¹, Todd Goodwin³, Carmel Crock¹, Thomas Campbell^{1,2,4,5}

¹The Royal Victorian Eye and Ear Hospital, East Melbourne, Australia;

²Centre for Eye Research Australia, East Melbourne, Australia;

³James Cook University, Townsville, Australia; ⁴Sunshine Coast

University Hospital, Sunshine Coast, Australia; ⁵University of the Sunshine Coast, Sunshine Coast, Australia

Disclosures: no conflicts of interest and no funding

Introduction: Several Australian bird species are known to swoop at perceived threats to nesting offspring, which may result in serious ophthalmic injury. Despite this, the literature on bird-related ocular injuries is limited to 77 published cases, of which just six describe swooping bird attacks. This does not reflect the clinical caseload at the Royal Victorian Eye and Ear Hospital (RVEEH), a quaternary Australian ophthalmic centre where clinicians frequently treat injuries caused by both swooping and non-swooping birds.

Method: A retrospective review of the RVEEH emergency department database identified 687 bird-related ophthalmic injuries presenting between 1st January 2006 and 31st December 2022.

Results: Most eye injuries were caused by swooping birds (magpies and magpie-larks), with the remainder caused by other species or unidentified/unrecorded species. Most primary diagnoses were abrasions/lacerations of the cornea (37.8%) and conjunctiva (37.7%). Only 11 (1.6%) were penetrating eye injuries,

eight of which were caused by magpies. Most injuries caused by birds occurred during the month of May, which lies outside the breeding seasons of several swooping bird species (July to October).

Conclusion: Most bird-related eye injuries are non-threatening to vision. Public education on swooping attacks should occur four months earlier than the magpie and magpie-lark breeding season, and Australians should consider using eye protection when travelling through areas with known swooping birds.

P059

EVALUATING THE REFERRALS TO EYE CASUALTY AT LRI FOLLOWING THE DISSOLVEMENT OF LOTS

Corresponding Author: Nida Jameel, orcid.org/0009-0002-0739-923X

Nida Jameel¹, Ian De Silva²

¹University of Leicester, Leicester, UK; ²University Hospitals of Leicester NHS Trust, Leicester, UK

Disclosures: no conflicts of interest and no funding

Introduction: University Hospitals of Leicester Eye Casualty (EED) is a walk in service operating 7 days a week with over 20,000 annual consultations. UHL serves a population of approximately 1.2 million people within Leicester City, Leicestershire and Rutland and has no other community urgent eye care service.

Method: A prospective study of 219 consecutive new patient attendances were evaluated for their referral source, diagnosis established in EED, urgency, distance travelled by patients and the index of multiple deprivation decile.

Results: The most common sources of referrals were self-referrals (32.4%), referrals from a general practitioner after a face to face (F2F) consultation (21.9%), optometrist after a F2F consultation (15.1%), general emergency department (10.5%) and NHS 111 (6.8%).

Overall, 8% of cases fitted a definition of an emergency (sight or life-threatening presentation), 50% were deemed to require an assessment between 48 hours and a week (urgent) and 42% could wait longer than a week (non-urgent). In subgroup analysis, optometrists F2F had the highest percentage of emergency cases of the cases they referred but over a half of cases referred by them were non urgent. Self-presentation and NHS 111 had the lowest percentage of non-urgent proportions and the highest percentage of these were for GP's following a telephone consultation.

On average, patients travelled 19 miles, with the highest groups of patients travelling 5-10 miles ($n = 69$). The index of multiple deprivation decile showed that there was an imbalance between the general demographics and those that attended EED.

Conclusion: This study showed a need for a community care assessment service such as a minor eye care service prior to referral to EED to ensure that the most appropriate patients are referred to the service.

P060

BILATERAL POSTERIOR SCLERITIS - COULD IT BE GIANT CELL ARTERITIS?

Corresponding Author: Ayesha Karimi, orcid.org/0000-0003-1714-9234

Ayesha Karimi¹, Ishani Rakshit², Hannah Bolton¹, Tsz Lun Ernest Wong³, Guy Negretti³

¹Maidstone and Tunbridge Wells NHS Trust, Maidstone, UK;

²University of Exeter Medical School, Exeter, UK; ³Surrey and Sussex Healthcare NHS Trust, Redhill, UK

Disclosures: no conflicts of interest and no funding

Introduction: A 75 year-old female with learning difficulty (LD) presented to the Ophthalmology department with an eleven-day history of right followed by left eye pain, redness and blurred vision. Systemic review found a four-week history of headache, pain in her shoulders, hands and calves.

Method: Visual acuity was 6/24 in the right eye and hand movements in the left eye. She had bilateral chemosis and redness, shallow retinal and choroidal detachments in the right eye and extensive retinal and choroidal detachments in the left eye. There was also mild restriction of downgaze. Temporal arteries were palpable but not abnormally thickened or tender. B scan showed thickened posterior sclerae bilaterally, with fluid surrounding the optic nerve. ESR was 60mm/hr and CRP was 20mg/l.

Results: Given the polymyalgic-type symptoms as well as the raised inflammatory markers, she was diagnosed with bilateral posterior scleritis secondary to giant cell arteritis and commenced on oral prednisolone at 60 milligrams once a day, with gastric and bone protection. Her binocular visual acuity improved to 6/12 within 7 days of starting treatment.

Conclusion: Posterior scleritis is a potentially blinding condition, with non-specific symptoms and signs, and as a result diagnosis is commonly delayed. This patient was seen in General Practice as well as in A&E, where diagnoses of angle closure glaucoma as well as iritis were rightfully considered. Her symptoms persisted for eleven days before ophthalmic opinion was sought. She had LD, which added to her diagnostic challenges. The RCOphth has published guidance on how to assess and manage visual loss in people with LD, as well as what reasonable adjustments can be considered to make a positive impact on the patient's assessment and overall outcome.

P061

LUBRICATING EYE DROPS IN THE EYE EMERGENCY DEPARTMENT, A QUALITY IMPROVEMENT PROJECT

Corresponding Author: Noorulain Khalid, orcid.org/0000-0002-1548-324X

Noorulain Khalid¹, Unsaar Hayat², Muhammad Chaudhary³, Christos Iosifidis², Felipe Dhawahir-Scala², Professor Fiona Carley²

¹Manchester University NHS Foundation Trust, Manchester, UK;

²Manchester Royal Eye Hospital, Manchester, UK; ³Good Samaritan Hospital, Vincennes, IN, USA

Disclosures: Fiona Carley - 1) Scope – Member of the Advisory Board 2) Funded to attend EUDES 2023 – VISUfarma BV. No other conflicts of interest or funding.

Introduction: Dry Eye Disease (DED) is a multifactorial condition often seen in association with other pathologies in the Emergency Eye Department (EED) at Manchester Royal Eye Hospital (MREH). Although lubricants are one of the most commonly prescribed ophthalmic medications, the drug and formulation used vary largely between prescribers.

Aims: Produce an up to date standardised guideline for MREH EED using international (DEWS II) and national (NICE, BNF)

guidance, taking the hospital formulary into consideration. Measure and reduce the rate of inappropriate lubricant prescription in EED. Assess impact of change.

Method: Data from 845 patients who attended EED over a period of 3 months was analysed. 157 patients met the inclusion and exclusion criteria.

Following thorough literature analysis and collaboration with the corneal team, an algorithm for prescription of lubricants in EED was produced, detailing first, second and third line medications.

3 PDSA cycles were performed, with data analysis following each. Interventions included distribution of emails to all EED prescribers, circulation of virtual posters of algorithm, physical copies displayed in all EED rooms, in-person reminders at handover, teaching incoming trainees new guidance.

Appropriateness of prescriptions was analysed against the guideline and checked by a registrar with experience in Cornea for clinical correlation.

Results: Over the 3 PDSA cycles, there was:

38% improvement in number of appropriate prescriptions (from 55% to 93%)

51% reduction in expenditure from lubricant prescriptions (saving ~£50/week)

Conclusion: DED is often poorly and inconsistently managed in the EED; trainees receive minimal training in the area. Appropriate prescription of lubricating drops results in improved disease management for patients as well as reducing costs to the trust. Providing an easy to follow guideline helps significantly in achieving both goals.

P062

COMPARATIVE ANALYSIS OF LARGE LANGUAGE MODELS IN THE NHS 111 ONLINE TRIAGING FOR EMERGENCY OPHTHALMOLOGY

Corresponding Author: Shaheryar Ahmed Khan, orcid.org/0000-0002-7545-9364

Shaheryar Ahmed Khan¹, Chrishan Gunasekera²

¹Colchester General Hospital, Colchester, UK, ²Norfolk & Norwich University Hospitals NHS Foundation Trust, Norwich, UK

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Introduction: Ophthalmology has recently seen a particular increase in the use of Artificial intelligence

In this study, we assess the performance of different LLM-chatbots in comparison to the NHS 111 triage system for ophthalmology emergency responses

Method: We utilized the NHS 111 online for answering 21 selected ophthalmology emergency question as a patient. The responses generated from NHS 111 online, was compared to the different LLM-chatbots responses. We selected a range of state-of-the-art models including ChatGPT-3.5, Google Bard, Bing Chat, and ChatGPT-4.0. We asked each LLM chatbot the exact same question. To evaluate the accuracy of responses, this was done twice with different prompts

Results: The LLM-chatbots achieved an overall accuracy of 52% compared with responses from the NHS 111 online. For first prompt, the LLM-chatbots exhibited mean accuracies of 49%, while for the second prompt, the accuracy increased to 56%. In breakdown of the individual chatbots, ChatGPT-3.5 and Google Bard demonstrated accuracies of 57% and 53%, respectively, for first prompt. This increased to 67% for both models when

evaluated with second prompt. Bing Chat displayed an accuracy of 38% for first prompt, decreasing to 33% for second prompt. ChatGPT-4.0 exhibited an accuracy of 57% on both prompts

Conclusion: Overall, the results are promising in this study with a moderate mean percentage accuracy of LLM Chatbots compared to the NHS 111. Our study underscores the potential of LLM-chatbots in generating responses for ophthalmology emergencies, albeit with variable accuracy. Further refinement and adaptation of LLM models along with better training of the chatbots can potentially enhance their accuracy and applicability and could potentially replace the triaging for end users like patients, doctors, nurses and other healthcare professionals in the future.

P063

EVALUATION OF EFFICACY OF ARTIFICIAL INTELLIGENCE IN THE TRIAGING OF PATIENTS IN EYE CASUALTY AND IT'S COMPARISON WITH CLINICIANS AT DIFFERENT GRADES

Corresponding Author: Samreen Khanam, orcid.org/0000-0002-6530-3390

Samreen Khanam¹, Hina Kanani¹, Ian DeSilva¹

¹University Hospitals of Leicester NHS Trust, Leicester, UK

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Introduction: The study was conducted to evaluate the efficacy of artificial intelligence in triaging patients in emergency eye care setting and it was compared with triaging by clinicians.

Method: 30 random patients visiting the eye casualty were asked questions based on a questionnaire. Open access Artificial Intelligence platform (ChatGPT) was asked to triage these patients on the basis of urgency to be seen by a clinician, and also to formulate 3 main differential diagnoses on the basis of the answers to this questionnaire. Similarly, clinicians at different grades were asked to triage and formulate differentials based on the same questionnaire. These findings were compared to the definitive diagnosis and the assessment for urgency, as determined by the treating doctor.

Results: 11 out of 30 cases required same day consultation, as per the treating clinician. ChatGPT agreed with this triaging in 7 patients (63.6 % agreement). 21 patients were deemed to be requiring casualty consultation, and ChatGPT agreed with this triaging in all 21 cases (100% agreement). Out of 9 cases that were determined to be non-urgent (requiring consultation within a month or later), only 1 was agreed upon by ChatGPT. Chat GPT was not found to be very effective in identification of non-urgent cases. In 20 out of 30 cases, the definitive diagnosis was mentioned as a top 3 differentials by ChatGPT. This was comparable to experienced clinicians.

Conclusion: Artificial intelligence has the potential to be used for triaging in eye casualty setting. The formulation of differential diagnoses by Chat GPT was comparable to experienced clinicians. However, poor identification of non-urgent cases was found to be comparable to inexperienced doctors with minimal training in ophthalmology.

P064

AN EVALUATION OF THE EFFECTIVENESS AND SAFETY OF THE LEICESTER COMMUNITY URGENT EYE CARE SERVICE

Corresponding Author: Heera Singh Landa, orcid.org/0009-0004-7486-4204

Heera Singh Landa¹, Amy Hitesh Rana¹, Ian De Silva²

¹University of Leicester, Leicester, UK; ²University Hospitals of Leicester NHS Trust, Leicester, UK

Disclosures: no conflicts of interest and no funding

Introduction: Previous studies have shown that the demand for hospital urgent eye care services is estimated to be 31 per 1000 population. Within Leicester City, Leicestershire and Rutland (LLR), a community urgent eye care service was initiated to assess patients with a regional telephone/ video triage service (LOTS) and a local community optometry urgent eye care service (LUECS).

Method: This retrospective cross-sectional study evaluated the performance of LUECS in terms of outcomes, diagnoses, and safety over a period of 2 months. Eye casualty attendances were recorded for all patients referred to LUECS up to 1 month following their LUECS assessment.

Results: Of 1017 patients using the LOTS service, 34% (n = 346) were discharged, 23% (n = 238) were referred to eye casualty and 34% (n = 348) were referred to LUECS. Of the patients referred to LUECS, only 3% (n = 10) were subsequently referred on to eye casualty, meaning total overall referred to eye casualty was 25% (n = 258). Of those referred to eye casualty from LUECS, 2 patients had retinal tears, 2 had acute anterior uveitis and 1 was due to a chemical injury.

The most common conditions that presented to LOTS/LUECS were Blepharitis (15%), Conjunctivitis (14%), and Chalazion (4%). Of those seen and discharged from LUECS, 4 patients re-attended eye casualty and none of these patients had sight threatening disease.

Conclusion: The LUECS service demonstrated an effective way to manage patients with low risk of urgent eye presentations in a community setting, with 96% of these patients not requiring eye casualty referral. These services showed a good safety profile, with no cases of sight loss or sight-threatening conditions identified following assessment and management solely in the LUECS service.

P065

ENDOPHTHALMITIS AFTER INTRAVITREAL INJECTIONS: AN AUDIT

Corresponding Author: Hoa L Ma, orcid.org/0009-0000-8374-7801

Hoa Ma¹, Channele Smith², Sheeja Thomas²

¹University of Birmingham, Birmingham, UK; ²Wye Valley NHS Trust, Hereford, UK

Disclosures: no conflicts of interest and no funding

Introduction: Endophthalmitis is a rare yet visually devastating complication of intra-ocular procedures. A commonly performed procedure with the risk of endophthalmitis is intravitreal injection for retinal diseases (e.g., age-related macular degeneration, diabetic retinopathy). This audit aims to (1) investigate the incidences of endophthalmitis post-intravitreal injections at a secondary care centre and (2) identify potential risk factors for endophthalmitis with the aim to inform further research to support guideline development and enhance patient safety.

Method: Patient records at Hereford County Hospital from January 2011 – November 2023 were reviewed to identify incidences of endophthalmitis post-intravitreal injections. Patient demographics, injection details, culture results, co-morbidities,

history of previous endophthalmitis and treatment given were recorded.

Results: Fourteen incidences of endophthalmitis were identified. The mean age of patients when diagnosed with endophthalmitis was 76 years. The mean number of injections received until endophthalmitis occurred was 17.3, with higher number of injections associated with infective endophthalmitis. 11 out of 14 incidences occurred in patients with either previous endophthalmitis, systemic or local co-morbidities. There were 3 years with repeated endophthalmitis incidences, one of which revealed a cluster of sterile endophthalmitis post-Eylea injections. All patients had anterior chamber tap and intravitreal antibiotics injections followed by oral antibiotics and corticosteroids. Five pars plana vitrectomies were performed later in patients with persistent vitritis. Mean visual acuity gain post-treatment was 48.5 letters (0.97 LogMAR).

Conclusion: The overall rate of endophthalmitis post-intravitreal injection during the audit period at Hereford County Hospital was 0.12 per 1000 - below the national standard of 0.5 per 1000. Increased number of intravitreal injections and immunosuppression appear to be risk factors for endophthalmitis, with greater severity and poorer visual outcomes in several patients with co-morbidities.

P066

EVALUATION OF THE IMPACT OF AN EMERGENCY FOCUSED OPHTHALMOLOGY TEACHING COURSE ON THE CONFIDENCE OF EMERGENCY DOCTORS

Corresponding Author: Elizabeth J E Mahon, elizabeth.mahon7@nhs.net

Elizabeth J E Mahon¹, Dana Ahnood¹

¹Shrewsbury and Telford NHS Trust, Shrewsbury, UK

Disclosures: no conflicts of interest and no funding

Introduction: It has been established that doctors in Accident and Emergency (A&E) departments have minimal confidence in managing ophthalmic presentations (Murray, P., et al. Eye 2016; <https://doi.org/10.1038/eye.2016.99>). The lack of confidence to accurately assess and manage patients presenting to A&E will likely have an impact on the quality of patient management and the quality of referrals made to eye casualties. We hypothesize that a teaching course covering core emergency ophthalmic examination and management skills will improve attendees' confidence in these areas.

Method: Doctors currently or potentially working in A&E were invited to attend a teaching session where they would rotate between three stations which included slit lamp examination, simulation of foreign body removal and managing chemical eye injury. Teaching was delivered in small groups, with experienced staff leading the sessions. Pre- and post-course questionnaires included a 1-5 Likert scale self-assessment confidence rating on the three areas of teaching and overall confidence.

Results: The course was attended by 13 doctors, with 46% of attendees being junior emergency specialty trainees or trainee equivalents, 38% were foundation doctors and 15% were A&E staff grade doctors equivalent to registrar. 92% of those who attended reported less than one hour of slit lamp examination time. Overall, the attendees' overall confidence went from 1.9 to 4.2 on the Likert scale.

Conclusion: Our findings demonstrate a lack of confidence from the A&E doctors in assessing patients presenting with ophthalmic issues, with an overall confidence score of 1.9 and

92% estimating less than one hour of slit lamp use. There was an improvement to a rating of 4.2, which demonstrates the overall success of the course.

P067

CORNEAL PERFORATION: A SIMULATED FORMAL EDUCATIONAL PROGRAMME AT MOORFIELDS EYE HOSPITAL

Corresponding Author: Alice L Milligan, orcid.org/0000-0002-9196-9647

Alice L Milligan¹, Tara Bader¹, Su-Yin Koay¹

¹Moorfields Eye Hospital NHS Foundation Trust, London, UK

Disclosures: no conflicts of interest and no funding

Introduction: Corneal perforations are ocular emergencies. Timely diagnosis and treatment are paramount to avoid sight-threatening complications, yet formal teaching opportunities are rare and are often performed in an emergency and/or out of hours where supervision may be limited. Our aim was to create an educational package to ensure trainees were confident to perform the procedure correctly.

Method: Step-by-step instructions, a demonstration video and simulations were provided. A Phillips Studio SRT-Head and eye, or an Altomed model eye (discarded after use in cataract/laser simulation) was secured upright on a slit-lamp. The anterior chamber was filled with balanced salt solution and punctured to replicate a leak. The perforation was repaired with cyanoacrylate glue, a sterile cotton tip and disc of sterile plastic drape. The simulation was assessed alongside traditional pigs' eyes and feedback compared. The intervention was validated using pre- and post-teaching questionnaires.

Results: Nineteen ophthalmology trainees took part; twelve attended simulation. In the pre-intervention questionnaire, 100% (19/19) stated that step-by-step instructions with a video would be useful and 84% (16/19) declared face-to-face sessions would be useful. In the post-intervention questionnaire, 100% (12/12) had improved confidence; 84% (16/19) declared the package 'very useful' and 67% (8/12) rated it as 'excellent'. The majority, 67% (8/12) preferred the novel simulation to the pig's eye.

Conclusion: Simulation allows trainees to acquire technical skills without risk to patients. Traditional models do not replicate the texture of the human cornea or the upright position of the patient at the slit lamp. This was completed at minimal cost, using discarded model eyes. To our knowledge, it is the first highly realistic, simulated education package to address this key procedure that does not require animal cadavers.

P068

THE UTILITY OF HIGH FIDELITY SURGICAL SIMULATION IN LATERAL CANTHOTOMY/CANTHOLYSIS IN PREPARING ST1 OPHTHALMOLOGY TRAINEES FOR ON-CALL EMERGENCIES

Corresponding Author: Manal I Patel, orcid.org/0000-0001-7438-2468

Manal Patel¹, Abigail Jones¹

¹Queen Alexandra Hospital, Portsmouth, UK

Disclosures: no conflicts of interest and no funding

Introduction: Ophthalmology trainees in their first year of training are often scheduled to be on the on-call rota after two months of clinical exposure. Retrobulbar haemorrhage and associated orbital compartment syndrome is an ophthalmological emergency requiring urgent lateral canthotomy/cantholysis. The aims of this study were to assess ST1 trainees' confidence in performing this task given lack of relevant clinical experience and to evaluate the efficacy of high-fidelity surgical simulation to increase trainees' confidence and competency in this essential skill.

Method: Surgical simulation using a porcine model of retrobulbar haemorrhage was delivered. Trainees were each expected to perform a lateral canthotomy/cantholysis independently as part of this experience. Pre- and post- simulation session surveys were conducted and compared using student t-test analysis to assess prior experience, knowledge and confidence.

Results: None of the ST1 trainees had any prior experience in performing lateral canthotomy/cantholysis despite being on the on-call rota ($n=4$). As a result of the simulation, trainees subjective rating of ocular adnexa anatomy increased from 2 to 4.5 out of 5 ($p=0.017$). Importantly, trainees' subjective confidence in performing lateral canthotomy/cantholysis in the emergency setting increased from 1.5 to 4 out of 5 as a result of the simulation with $p=0.0025$.

Conclusion: Trainees' confidence, knowledge of indications, and technique for performing lateral canthotomy/cantholysis in the emergency setting was greatly improved by a high-fidelity surgical simulation for retrobulbar haemorrhage. The ability to recognise the signs of an orbital compartment syndrome and act urgently are a crucial competency required for independent ophthalmology on-call shifts. As such, we recommend that trainees are offered the opportunity to perform lateral canthotomy/cantholysis in a high-fidelity model to help prepare them to manage retrobulbar haemorrhage in the emergency setting.

P069

ACCEPTABILITY AND FEASIBILITY OF A VIRTUAL REALITY ENABLED EYE EXAM IN THE BASCOM PALMER EYE INSTITUTE EMERGENCY ROOM

Corresponding Author: Joshua M Reyes, orcid.org/0000-0002-4537-5530

Joshua M Reyes¹, Amanda Tran¹, Giselle C Ricur¹, Tamara Juvier¹, Carlos E Mendoza¹

¹Bascom Palmer Eye Institute, Miami, FL, USA

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Introduction: Rapid and reliable testing of visual function is crucial in ophthalmic emergencies. Often, these tests, conducted before seeing an ophthalmologist or optometrist, can cause delays due to staff workload. The VisuALL system, a virtual reality device, measures various aspects of eye exams, including visual acuity, visual field, and more. This study assesses the feasibility and acceptability of using VisuALL in an eye emergency room.

Method: Eligible patients at Bascom Palmer Eye Institute Emergency Room were invited for the study. After consent, they used VisuALL for tests including confrontation visual fields, extraocular motility, colour vision, and pupillometry, followed

by a manual exam for comparison. Participants then completed a paper survey to determine acceptability.

Results: A total of 50 patients participated with the mean (SD) age of 51.1 (15.1) years. 48 (96%) completed the full examination. The mean (SD) time taking the test was 5 minutes 38 seconds (1 minute 34 seconds) and the mean (SD) time wearing the headset, accounting for time required to give instructions and repeat a test, was 10 minutes 56 seconds (1 minutes 57 seconds). In total 38 patients (76%) found the virtual reality exam useful, with 43 (86%) stating they would be willing to use the device again in the future and recommended the exam to a family member or friend.

Conclusion: Integration of the VisuALL device into the emergency room was feasible and patients reported a high level of satisfaction through the posttest survey. Examination results were overall comparable between the VisuALL system and manual exam. There was a learning curve to overcome using the new technology for the patients and the operator.

P070

OCULAR OUTCOMES IN TOXIC EPIDERMAL NECROLYSIS (TENS): 13-YEAR REVIEW AT CHELSEA AND WESTMINSTER BURNS INTENSIVE UNIT

Corresponding Author: Manvi Sobti, orcid.org/0000-0002-5177-3180

Manvi Sobti¹, Olivia Li¹, Aditi Pandey¹, Richard Scawn¹, Declan Collins¹, Clare Bailey¹, Jorge Leon-Villapalos¹

¹Chelsea and Westminster Hospital, London, UK

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Introduction: Steven Johnson Syndrome (SJS) and its more severe form, Toxic Epidermal Necrolysis Syndrome (TENS) (UK incidence 1 per million/year) can lead to severe immune-mediated ocular surface destruction with chronic debilitating sequelae. Timely management is critical for minimizing inflammation, cicatrization and preserving vision. We present our 13-year experience with multidisciplinary management of this rare illness.

Method: Retrospective review of eye involvement in patients admitted with TENS at Chelsea and Westminster Hospital between 2010 and 2023.

Results: Among 24 TENS patients, 16 (66%) had ocular involvement, 13 % ($n=3$) had no ocular features and data for 21 % ($n=5$) was unavailable. Triggers included lamotrigine (25%, $n=6$), antibiotics (25%, $n=6$), diazepam (4%, $n=1$), NSAIDs (8%, $n=2$) and no identified trigger (4% $n=1$). Mild to moderate ocular disease was seen in 44% ($n=7$) and severe to extremely severe ocular in 56% ($n=9$). The mild-moderate group received intensive eye drops and systemic immunosuppression, while the severe-extremely severe group (except one patient who refused treatment) underwent cryopreserved amniotic membrane transplantation (AMT) in addition to medical management. In addition, 2 patients received dry amnion discs. Severe ocular disease correlated with higher Total Body Surface Area involvement (68% vs 81%), taking longer for skin re-epithelization (7.2 vs 9.6 days). No difference in the SCORTEN was observed between the groups. Ocular sequelae included visual impairment (31%, $n=5$), severe chronic dry eye (31%, $n=5$), no documented sequelae (31%, $n=5$). One patient died within 15 days of rash onset.

Conclusion: Ocular involvement occurred in 66% of TENS patients, with 60% exhibiting ocular sequelae. More severe ocular disease was seen in patients with severe skin involvement. Early and regular ophthalmic input is necessary to individualize ophthalmic treatment for TENS.

P071

VARIATION IN EYE CASUALTY ATTENDANCE DURING CHRISTMAS VERSUS SPRING PERIOD: A CROSS-SECTIONAL STUDY

Corresponding Author: Modupe Sodeinde, orcid.org/0000-0002-1618-6861

Modupe Sodeinde¹, Sohaib R Rufai^{1,2}, Steven Isherwood¹, Hamza Abdou¹, Chung Shen Chean¹, Basu Dawar¹, Jennifer Kwan¹, Swati Parida¹, Aaliya Nur¹, Akhil Mohindra¹, Ian De Silva¹, Tahir Islam¹, Antonella Berry-Brincat¹

¹Leicester Royal Infirmary, Leicester, UK; ²University of Leicester, Leicester, UK

Disclosures: no conflicts of interest and no funding

Introduction: No British study has explored the effect of the Christmas period on Eye Casualty attendance and casemix. This study aims to explore these phenomena at a tertiary hospital.

Method: This was a cross-sectional study conducted at Leicester Royal Infirmary's walk-in Eye Casualty. Data was collected over two-week periods in Christmas (19/12/2022-01/01/2023) and Spring (06/03/2023-19/03/2023). Eye casualty was closed on Christmas Day but open all other days. Main outcome measures included number of visits, total time in department, and discharge outcomes classified as follows: 1) discharge without follow-up; 2) discharge with routine follow-up; 3) discharge with urgent follow-up; 4) emergency admission. Casemix was divided as non-urgent (outcomes 1-2) and urgent (outcomes 3-4). The Chi-squared test was used to compare outcome measures across the two study periods.

Results: There were a total of 1323 Eye Casualty attendances (Christmas, n = 552; Spring, n = 771). When excluding Christmas day, mean daily attendances were 42 and 55 in Christmas versus Spring. There were 33% more attendances in Spring ($p < 0.001$). There was no significant difference in mean time in department per patient (Christmas; 116 minutes; Spring, 120 minutes; $p = 0.79$). There was no significant difference in overall casemix across the two periods ($p = 0.10$). However, subgroup analysis of the Christmas period revealed a 42% increase ($p = 0.002$) in urgent outcomes the week following Christmas (n = 87) versus the week leading to Christmas (n = 57).

Conclusion: Our study found significantly more eye casualty attendances in Spring compared to Christmas. Our subgroup analysis suggests that patients may delay attending eye casualty until the week after Christmas. These findings could help support patient education and optimise Eye Casualty staffing and strategy.

P072

FORECASTING A&E DEMAND: WEATHER CORRELATIONS IN OPHTHALMOLOGY EMERGENCY PATIENT ATTENDANCE

Corresponding Author: Matthew Stahl, orcid.org/0009-0004-5946-3736

Matthew Stahl¹, Salaam Botros², John Vekinis¹, Emma Jones¹, Gordon Hay¹, Tjebo Heeren¹

¹Moorfields Eye Hospital NHS Foundation Trust, London, UK; ²Luton and Dunstable University Hospital, Luton, UK

Disclosures: no conflicts of interest and no funding

Introduction: There is a strong belief amongst healthcare professionals that weather influences patient attendance. This retrospective study aimed to investigate if there was any

correlation between weather conditions and patient attendance at the Ophthalmology Accident and Emergency (A&E) department at Moorfields Eye Hospital, London.

Method: A&E attendances spanning from January 4, 2021, to August 31, 2023, were collected and analysed in conjunction with London weather data, including temperature, rainfall, wind, and sunlight. Statistical analysis was conducted using correlation coefficients and regression analyses employed to examine for the presence of a relationship between weather variables and A&E attendances.

Results: After excluding weekends and bank holidays, correlation coefficients demonstrated that there may be some correlation between weather conditions and A&E attendances. "Feels Like" and actual temperatures exhibited moderate positive correlations ($r = 0.256$ and $r = 0.270$, respectively) with statistical significance ($p < 0.0001$). Sun hours showed a modest positive correlation ($r = 0.195$) with statistical significance ($p < 0.0001$). Rainfall demonstrated a weak but statistically significant negative correlation ($r = -0.094$, $p = 0.0143$). Deeper statistical analysis demonstrated a complex relationship between weather variables and A&E attendance.

Conclusion: The findings suggest that there may be some correlation between warm conditions and higher A&E attendances. Conversely, rainier days may be associated with fewer A&E attendances. These results contribute insight into the potential influence of weather patterns on patient attendance at the Ophthalmology A&E department. In combination with analysing other factors, weather may have some use in the anticipation of high A&E attendance days and facilitating adequate staffing preparations. Such proactive measures, in combination with analysis of other factors, can contribute to the optimisation of resource allocation in the Ophthalmology A&E setting.

GLAUCOMA

P073. Abstract not included

P074

ONE YEAR OUTCOME OF PRESERFLO MICROSHUNT IN A DISTRICT GENERAL HOSPITAL

Corresponding author: Kashif Ali, orcid.org/0009-0002-0729-1894

Mumtaz Alam¹, Rabia Bashir¹, Alexandra Teclici¹, Makar Shenouda¹, Kashif Ali¹

¹Countess of Chester Hospital NHS Foundation Trust, Chester, UK

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Introduction: Trabeculectomy is considered the gold standard surgical treatment for glaucoma despite requiring significant postoperative management and some potentially sight threatening complications. Several Minimally Invasive Glaucoma Surgery (MIGS) procedures including the Preserflo MicroShunt have recently been introduced as safer and less traumatic options.

Method: A retrospective analysis of consecutive Preserflo MicroShunts performed between July 2021 and December 2022. Success was defined as IOP < 21 mm Hg with (qualified success) or without any medication (absolute success).

Results: A total of 81 Preserflo MicroShunts had twelve month post operative data and were included in the study.

The absolute success was 88% and the qualified success was 97.6%. Mean preoperative IOP was 25.36 mm Hg (range 14–58 mm Hg) and post operatively IOP was 11.37 mm Hg (range 3–24 mm Hg) at one year.

The mean number of glaucoma medications was 2.33 (range 1–5*) preoperatively and 0.17 (1–3**) post operatively (n = 8). (*Oral Acetazolamide was counted as medication) **One patient had postoperative IOP of 20 mm Hg on three medication and had revision surgery beyond twelve months.

The mean LogMAR visual acuity was 0.2 (-0.1 to 1.2) preoperatively and 0.2 (range -0.1 to 1.3) and post-operatively. Visual Field Test were available in 77 eyes with a mean deviation of -9.77 preoperatively and -9.58 postoperatively.

Five patients required revision surgery.

One patient had exposure of the Preserflo MicroShunt and this was removed.

Conclusion: Our results compare very favourably with published literature. This may be due to several factors e.g. patient ethnicity (97.6% caucasians), use of preservative free drops and discontinuation of blood thinners preoperatively, dose on Mitomycin C (0.4mg/ml) used and use of Prednisolone acetate 1% two hourly for a prolonged postoperative period.

P075

SAFETY AND EFFICACY OUTCOMES OF CATARACT SURGERY WITH HYDRUS MICROSTENT INSERTION

Corresponding author: Durray Nayab Ali, orcid.org/0009-0008-0920-2449

Durray Nayab Ali¹, Asma Adam¹, Chris Dineen¹, Fran Davies¹, Thomas Mackley¹, Gabriella Ghazi¹, Nabila Sheikh¹, Indira Madgula¹

¹Lancashire Teaching Hospitals NHS Foundation Trust, Preston, UK

Disclosures: No conflicts of interest and no funding

Introduction: Minimally invasive glaucoma surgery is increasingly used along with phacoemulsification procedure to treat glaucoma. Hydrus microstent works by creating a scaffold that dilates Schlemm's canal to bypass resistance of the trabecular meshwork and enhances aqueous flow through conventional outflow systems. We describe postoperative outcomes of combined phacoemulsification and Hydrus microstent for patients with Ocular Hypertension (OHT), Normal Tension Glaucoma (NTG) and Primary Open Angle Glaucoma (POAG).

Method: 95 patients included with POAG, OHT and NTG with co-existent cataract who underwent Hydrus microstent insertion between 2022 and 2023. Eyes with secondary glaucoma other than pseudoexfoliative or pigmentary glaucoma, angle closure and previous incisional glaucoma surgery were excluded. Main outcome measures included post-operative IOP, glaucoma medications post operatively, procedure related adverse events and repeat glaucoma surgery.

Results: Average age of patients was 81 years (range 49–92) with 58 males and 37 females. 26 eyes with OHT on 1.46 drops preoperatively achieved target IOP between 18–21 mmHg on 0.33 drops at 6 month follow up. 28 eyes with mild to moderate glaucoma had an average IOP of 16.72 mmHg on 1.82 drops preoperatively and at 6 months average IOP was 14.31 on 0.42 drops.

35 eyes with advanced POAG/ NTG had an average IOP of 16.2 mmHg on 2.34 medications preoperatively. Postoperatively, average IOP was 12.5 on 1 glaucoma drop. 1 patient in this group was listed for a trabeculectomy.

Adverse events included 1 retinal detachment and 2 device-related complications.

Conclusion: Patients with open angle glaucoma who underwent Hydrus microstent insertion and phacoemulsification displayed IOP reduction and attainment of target IOPs with low medication burden and few adverse events.

P076

OUTCOMES OF THE PRESERFLO MICROSHUNT: AN UPDATED SYSTEMATIC REVIEW AND META-ANALYSIS

Corresponding author: Ali Alkhabbaz, orcid.org/0000-0002-9931-5269

Ali Alkhabbaz¹, Mohammad Karam², Moath Baeshen², Ali Salimi², Waleed Al-Sarhani²

¹Kuwait University, Kuwait; ²McGill University, Montreal, Canada

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Introduction: Purpose: Assess the efficacy of the PreserFlo Microshunt (PMS) for treating primary open angle glaucoma

Method: A systematic review and meta-analysis were performed as per the Preferred Reporting Items for Systematic Reviews and Meta-analyses guidelines, and a search of electronic information was conducted to identify all studies of PMS Efficacy, PMS versus trabeculectomy, and PMS vs PMS + phacoemulsification. The primary outcome measures were the post-operative intra-ocular pressure (IOP) and post-operative anti-glaucoma medication (POAM). Secondary outcome measures included post-operative BCVA, adverse effects, post-operative interventions, and re-operation. Random-effects modelling was used for the analysis.

Results: 34 eligible studies that enrolled 3108 eyes were identified in the literature. There was a statistically significant reduction in IOP after PMS implantation up to 3 years post-operatively (SMD: 13.632, $P < 0.001$). There was a statistically significant reduction in POAM up to 5 years post-operatively (SMD: 1.287, $P = 0.018$). When comparing post-operative IOP between PMS and trabeculectomy, the latter showed superior outcomes up to 12 months post-operatively (MD = 1.54, $P = 0.01$); POAM were similar between the two groups at 12 months (MD = 0.10, $P = 0.38$). There was no statistically significant difference between PMS and PMS + phacoemulsification in terms of post-operative IOP at 12 months (MD = 0.82, $P = 0.54$).

Conclusion: The results of this meta-analysis confirm that PMS is effective in lowering both IOP and NOAM up to 3 and 5 years after surgery, respectively. Additionally, PMS is less effective in lowering IOP compared to trabeculectomy up to 12 months post-operatively; POAM are similar between the two groups. There is no statistically significant difference between PMS and PMS + phacoemulsification in terms of post-operative IOP at 12 months.

P077

AN ALTERNATIVE TO TRABECULECTOMY: A 2 YEAR RETROSPECTIVE STUDY COMPARING THE EFFICACY AND SAFETY OF THE PRESERFLO™ MICROSHUNT TO TRABECULECTOMY IN A TERTIARY CENTRE

Corresponding author: Toby Al-Mugheiry, orcid.org/0000-0002-2731-0266

Toby Al-Mugheiry¹, Disha Singhania¹, John Britton¹, Stylianos Georgoulas¹, Humma Shahid¹, Ian Tapply¹, Rupert Bourne¹

¹Cambridge University Hospitals, Cambridge, UK

Disclosures: No conflicts of interest and no funding

Introduction: Over recent years, there has been a rapid increase in development and use of Microinvasive Glaucoma Surgery devices (MIGS). This study was conducted to compare the long term efficacy and safety of the Preserflo™ microshunt as a surgical alternative to trabeculectomy in patients with moderate to advanced glaucoma at Cambridge University Hospitals, UK.

Method: Single centre retrospective case series of the first 46 patients who underwent Preserflo™ microshunt surgery (with 0.4mg/mL mitomycin C, MMC) alongside 27 patients who underwent trabeculectomy (with 0.2 mg/mL MMC) during the same time period (1/3/2021 – 10/12/2021). Visual acuity; intraocular pressure (IOP); number of glaucoma medications; incidence of complications and post-operative interventions were recorded with a follow up of 2 years.

Results: The mean ($\pm$ SD) follow up period was 23 ± 3.4 months for Preserflo™ and 23.1 ± 3.2 months for trabeculectomy patients. For Preserflo™, the mean IOP reduced from 21.6 ± 6.5 mmHg pre-operatively to 12.8 ± 5.2 mmHg ($p < 0.0001$) at 2 years and the mean number of medications reduced from 2.7 ± 1 to 0.5 ± 0.8 ($p < 0.0001$). Two patients required minimal post-operative intervention and one required a further procedure. For the trabeculectomy patients, the mean IOP reduced from 26.5 ± 12 mmHg pre-operatively to 11.7 ± 5.9 mmHg ($p < 0.0001$) at 2 years and the mean number of medications reduced from 3.8 ± 1 to 0.4 ± 0.9 ($p < 0.0001$). Six patients required post-operative an intervention or further procedure. The difference in average percentage IOP reduction between Preserflo™ and trabeculectomy was found to be non-significant (42.6% vs 53.1%, $p = 0.10$).

Conclusion: The Preserflo™ microshunt causes a significant reduction in IOP and medications comparable to trabeculectomy. Preserflo™ patients were less likely to require a post-operative intervention. Whilst we acknowledge potential surgical bias at pre-selection, the Preserflo™ microshunt seems to offer an effective and safe surgical alternative for glaucoma patients.

P078. Abstract not included

P079

THE IMPACT OF COVID-19-RELATED DELAYS ON VISUAL OUTCOMES IN GLAUCOMA PATIENTS: A RETROSPECTIVE COHORT STUDY ACROSS TWO TERTIARY CENTRES IN LONDON, UK

Corresponding author: Navya Arora, orcid.org/0009-0008-2018-3445

Navya Arora¹, Eduardo Normando¹

¹Imperial College Ophthalmology Research Group, London, UK

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Introduction: This study assessed the correlation between COVID-19-related appointment delays and visual outcomes in glaucoma patients.

Method: 13,744 patients seen under glaucoma services at the Western Eye and Charing Cross Hospitals were enrolled. 500 patients were randomly selected from this and screened. Data was extracted from the pre-COVID (prior to 11th March 2020) and post-COVID (4th July 2020) periods, including visual field mean

deviation (VFMD), intraocular pressure (IOP), best-corrected visual acuity (BCVA), demographic data and clinical background.

Results: 138 patients with various glaucoma types were included. The median patient age was 75 (IQR 15); 53% were male, 25% were British White, and 17% had diabetes. The median follow-up delay was 72 days (IQR 227). VFMD and delay correlation analysis showed a significant negative correlation for both eyes (right: $p = 0.0002$, left: $p = 0.0234$). Both IOP and BCVA showed no significant positive correlations across both eyes. Mean linear regression analysis found significant negative correlations between VFMD and delays across both eyes in African (right: $p = 0.0005$, left $p = 0.0004$) and Mixed White & Black Caribbean (right: $p = 0.035$, left: $p = 0.0396$) ethnicities, as well as a diagnosis of diabetes ($p = 0.0123$) in the left eye.

Conclusion: Our data shows that prolonged delays correlated with poorer visual field outcomes in this population. There is significant data to suggest that those with African, White & Black Caribbean ethnicities, and diabetes may have been more adversely affected than the rest of the population. We found an association between increased appointment delay over the acute COVID-19 period and worse visual field outcomes in glaucoma patients, disproportionately affecting some subgroups more.

P080

I-STENT INJECT IMPLANTATION COMBINED WITH CATARACT SURGERY-FOUR YEAR RESULTS

Corresponding author: Radomir Babovic, orcid.org/0000-0002-2331-4251

Radomir Babovic¹, Avinash Prabhu¹, Nimesha Alex¹

¹James Paget University Hospitals NHS Foundation Trust, Great Yarmouth, UK

Disclosures: No conflicts of interest and no funding

Introduction: The study aims to evaluate the safety and effectiveness of a second-generation trabecular meshwork micro by-pass stent (i-Stent Inject) combined with cataract surgery in patients with mild to moderate glaucoma. I-Stent inject implantation and cataract surgery were performed at the James Paget Hospitals, England.

Method: This study is a four year retrospective evaluation of 30 eyes of 30 patients who underwent trabecular i-Stent Inject implantation combined with phacoemulsification. Twenty-four eyes were diagnosed with primary open-angle glaucoma and one each with pseudo-exfoliative glaucoma and ocular hypertension. The rest of the patients have been diagnosed with normotensive glaucoma. The primary outcome measure was the intraocular pressure (IOP), success being defined as both an IOP less than 18 mmHg and a 20% reduction from preoperative baseline up to four years after the surgery. Safety was assessed by assessing the incidence of complications at routinely scheduled exams.

Results: The mean preoperative IOP was 21 mmHg. Post-operatively, the mean IOP was reduced to 14 mmHg (p less than 0.005). The number of ocular hypotensive medications remained unchanged. The success rate was 70%, with all the patients displaying a significant reduction of IOP. There were no causes of hyphaema, hypotony, endophthalmitis, or choroidal detachment. One eye developed cystoid macular oedema, which resolved in four weeks' time.

Conclusion: From our clinical experience, trabecular i-Stent inject implantation combined with cataract surgery appears to be effective at reducing IOP whilst carrying an exceptional safety profile.

P081. Abstract not included

P082

FOUR-ARM COMPARATIVE STUDY OF OUTCOMES FOR HYDRUS, ISTENT AND OMNI MIGS DEVICES WITH STANDALONE PHACOEMULSIFICATION IN GLAUCOMA: 6-MONTH FOLLOW-UP PRELIMINARY RESULTS

Corresponding author: Jeremy Chan, orcid.org/0000-0003-3147-7285

Jeremy J H Chan¹, Shaheryar Khan¹, Panagiotis Dervenis¹, Saad Khan², Mohammad Fazal¹, Chrysostomos Dimitriou¹

¹ Colchester Hospital, East Suffolk and North Essex NHS Foundation Trust, Colchester, UK; ²West Suffolk NHS Foundation Trust, Bury St Edmunds, UK

Disclosures: No conflicts of interest and no funding

Introduction: Aim of this study is to compare the Sclemms's canal based MIGS devices with standalone phacoemulsification in POAG and to assess their effect on intraocular pressure (IOP) and glaucoma progression and their safety.

Method: A retrospective, comparative study was carried out on POAG patients treated with iStent, Hydrus, OMNI and standalone phacoemulsification between 2nd August 2021 to 31st August 2023 at Colchester Hospital. Intergroup differences in IOP and visual acuity (VA) were compared preoperatively and postoperatively at first, second and third follow-up appointments at 6 months.

Results: 143 consecutive cases were analysed which included 50 Hydrus, 31 iStent, 36 OMNI, and 26 standalone phacoemulsification cases. There was no significant change in VA preoperatively and postoperatively at all intervals for all groups. At the last follow-up, IOP analysis showed iStent achieving 4.71 mmHg reduction, OMNI 3.26 mm Hg, Hydrus 2.67mmHg and standalone Phaco none (0.0mm Hg), ($p=0.05$ for Hydrus versus iStent; $p=0.14$ for Hydrus versus OMNI; $p=0.57$ for iStent versus OMNI). Mean medication reduction was also similar for Hydrus, iStent and OMNI, being 1.14, 1.09 and 1.0 respectively ($p=0.05$ Hydrus v/s iStent; $p=0.14$ Hydrus v/s OMNI; $p=0.09$ iStent v/s OMNI). Standalone phacoemulsification patients used 0.16 medications less ($p=0.009$ for Phaco v/s Hydrus). Complications were rare with no significant differences between groups.

Conclusion: iStent, Hydrus and OMNI had similar outcomes post-operatively at third follow up for good, sustained IOP reduction along with good safety profile and no significant difference between the groups, while standalone phacoemulsification did not reduce mean IOP. Mean medication reduction was also similar in all groups except standalone phacoemulsification which showed less reduction, but was only significantly different when compared to the Hydrus.

P083. Abstract not included

P084

THREE YEAR OUTCOMES OF PRESERFLO MICROSHUNT WITH MITOMYCIN C AT MANCHESTER ROYAL EYE HOSPITAL

Corresponding author: Clarissa Ern Hui Fang, orcid.org/0000-0002-7266-7117

Clarissa Ern Hui Fang¹, Ameer Ali¹, Pinky May Myat Noe Pwint¹, Cecelia Fenerty¹, Kenneth Yau¹, Jonathan Yu¹, Leon Au¹

¹Manchester Royal Eye Hospital, Manchester University NHS Foundation Trust, Manchester, UK

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Introduction: To report baseline characteristics and surgical outcomes of preserflo microshunt surgery in patients with glaucoma at a tertiary centre.

Method: Retrospective case series of consecutive patients who underwent preserflo microshunt surgery at Manchester Royal Eye Hospital with a minimum 1 year follow-up. Baseline characteristics, pre-operative and post-operative IOP, number of glaucoma medications, visual acuity, and adverse events were recorded.

Results: 162 patients who underwent preserflo microshunt surgery with 0.4-0.5mg/ml MMC were followed up at 1 year post-operation, 91 patients at 2 years and 69 patients at 3 years. The mean age was 66 ± 16 (21-91) years and 77 (47.5%) were female. 45.7% had POAG and 26% had uveitic glaucoma. The mean baseline IOP of 25.5 ± 7.6 (range 14-53) mmHg was reduced at each year post-operatively ($p < 0.0001$). The mean post-operative IOP at 1 year was 13.3 ± 5.0 , at 2 years 13.3 ± 5.7 and at 3 years 12.8 ± 5.1 mmHg. The mean number of pre-operative glaucoma medications was 2.8 ± 1.1 (range 0-5) compared with 0.5 at 1 year, 0.8 at 2 years and 0.7 at 3 years ($p < 0.0001$). Adverse events included hypotony (3 patients), early hyphaema (3 patients), corneal oedema (1 patient), CMO (1 patient), required revision of preserflo microshunt (10 patients) and required secondary tube surgery (3 patients). At last follow-up visit, complete success rate (IOP ≤ 21 mmHg without medication) was 63% and qualified success rate (IOP ≤ 21 mmHg, with medication) was 31%.

Conclusion: Our study has shown good IOP control with preserflo microshunt surgery at 3 years, which is comparable with studies in the literature. Appropriate patient selection was a key factor in success of surgery.

P085

RETROSPECTIVE ANALYSIS OF THE SAFETY AND EFFICACY OF THE MINJECT IMPLANT AT 12 MONTHS

Corresponding author: Mohammad Ihsan Fazal, orcid.org/0000-0002-5174-8372

Mohammad Ihsan Fazal¹, Hammad Husnain Malik², Luke Tanner³, Mahmoud Radwan¹

¹Colchester Hospital, Colchester, UK; ²Lister Hospital, Stevenage, UK; ³Ipswich Hospital, Ipswich, UK

Disclosures: No conflicts of interest and no funding

Introduction: The options for surgical management of glaucoma have greatly expanded with the provision of Minimally Invasive Glaucoma Surgery (MIGS). These procedures offer the modern surgeon an ability to augment the trajectory of disease with novel and emerging techniques. With an increasing number of MIGS devices being brought to market, the significance of measuring efficacy and safety should not be overlooked. One such device is the MINject implant, which targets the supraciliary space. This study presents the intraoperative and postoperative safety and efficacy profile from a single surgeon dataset in patients with 12 month follow up.

Method: Cases of standalone MINject insertion and MINject combined with phacoemulsification and intraocular lens insertion were identified via a retrospective analysis of records. Cases were performed by a single consultant surgeon from March 2022

onwards. Data on pre-operative and post-operative intraocular pressure (IOP) was collected, as well as number of drops used and adverse events. The primary outcome was reduction in IOP.

Results: 2 standalone procedures were identified and 26 combined cases. Average IOP was reduced for patients who underwent the procedure across all time points (3, 6 and 12 months). At 3 months the average IOP reduction was 23.5%. Furthermore, there was an overall average drop reduction at 3, 6 and 12 months for patients who had received the implant. There were no intra-operative adverse events. However, most common post-operative adverse events included a gross hyphaema (7.1%) and microhyphaema (10.7%).

Conclusion: 1 year after implantation the MINject can deliver and sustain IOP reduction and reduce drop dependence. This study adds further evidence for the modern surgeon to consider when managing patients with glaucoma.

P086

PORTABLE DEVICES FOR DIAGNOSIS OF GLAUCOMA IN COMMUNITY HEALTH CARE SETTINGS IN NIGERIA

Corresponding author: Farouk Garba, orcid.org/0000-0001-5341-0853

Farouk Garba^{1,2}, Fatima Kyari^{1,2}, Winifred Nolan^{1,4}, Matthew Burton¹, Victor Hu¹

¹London School of Hygiene and Tropical Medicine, London, UK; ²College of Medical Sciences, Ahmadu Bello University, Zaria, Nigeria; ³College of Health Sciences, University of Abuja, Abuja, Nigeria; ⁴NHIR Biomedical Research Centre, Moorfields and UCL Institute of Ophthalmology, London, UK

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Introduction: Glaucoma is an irreversible blinding disease with a complex diagnostic process which is mainly done by eye care specialist in tertiary and some secondary centres. Individuals with no access to such specialised centres present when significant vision is lost. The purpose of this study is to determine the performance of portable diagnostic tests for identifying people with possible glaucoma and evaluate their use in a community clinical setting to support earlier detection of glaucoma in adults in Abuja, Nigeria.

Method: This was a feasibility study which was carried out in two phases. Phase I was a hospital-based study where portable devices were compared to the corresponding standard hospital equipment. Phase II was a community-based study where these portable devices were evaluated for their use in the detection of glaucoma by health workers in community centres.

Results: Data was collected between January to September 2023. Three hundred and thirteen (312) participants were examined during Phase I of the study. Mean age was 44years (min age 18years and max 95years). Two thousand two hundred and fourteen (2214) participants were recruited in Phase II from (4) four community health care centres. Intraocular measurement using icare vs Goldman Applanation Tonometer and vertical cup disc ratio assessment using the Remedio fundus camera vs 78D slit lamp examination shows significant correlation with p-value < 0.001.

Conclusion: Portable devices have shown great prospects in the diagnosis of glaucoma. These devices have also been shown to be efficiently used by the community health care workers who had no training in eye care to detect and refer cases of glaucoma adequately. Thereby, contributing greatly in the fight against blindness due to glaucoma.

P087

THREE-YEAR SAFETY AND EFFICACY RESULTS OF A SUPRACILIARY DRAINAGE DEVICE IN OPEN ANGLE GLAUCOMA PATIENTS (STAR-GLOBAL)

Corresponding author: Gus Gazzard, orcid.org/0000-0003-1982-5005

Gus Gazzard¹, Antonio Fea², Philippe Denis³, Brian Flowers⁴, Inder Paul Singh⁵, Iqbal Ike K Ahmed⁶

¹Moorfields Eye Hospital, London, UK; ²Dipartimento di Scienze Chirurgiche-Clinica Oculistica, Torino, Italy; ³Department of Ophthalmology, Hôpital de la Croix-Rousse, Lyon, France; ⁴Ophthalmology Associates, Fort Worth, TX, USA; ⁵The Eye Centers of Racine and Kenosha, Wisconsin, USA; ⁶Department of Ophthalmology and Vision Sciences, University of Toronto, Toronto, Canada

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Funding: National Institute for Health and Care Research (NIHR) Biomedical Research Centre (BRC) at Moorfields Eye Hospital NHS Foundation Trust and UCL Institute of Ophthalmology

Introduction: We describe the 3-year safety and efficacy of a novel, minimally-invasive glaucoma surgery (MIGS) device (MINject[®]; iSTAR Medical, Belgium) implanted ab interno into the supraciliary space in patients with medically-uncontrolled open-angle glaucoma.

Method: The device was implanted in a standalone procedure in phakic and pseudophakic eyes with no medication washout in 3 prospective trials (STAR-I,II,III). The trials were completed at 2 years in 66 patients in 11 sites in Europe, Asia and Central America. Patients were invited to enrol in the STAR-GLOBAL study to continue follow-up annually from 3 until 5 years. Intraocular pressure (IOP) and IOP-lowering medication use were recorded annually, as well as a safety evaluation including adverse events and measurements of corneal endothelial cell density (ECD). Results from 48 patients who completed 3-year follow-up (72.7% of patients who completed the prior studies) are reported here

Results: Mean baseline diurnal IOP prior to implantation was 23.6 ± 3.4 mmHg using a mean of 2.2 ± 1.1 IOP-lowering medications (n = 48). At 2 and 3-year follow-up, mean diurnal IOP was 13.5 ± 3.8mmHg (-10.2mmHg, -42.2%; p < 0.0001) on 1.1 ± 1.3 medications and 15.1 ± 4.1mmHg (-8.5mmHg, -35.6%; p < 0.0001) on 1.4 ± 1.5 medications, respectively. At 3 years, 90% of patients achieved an IOP reduction of ≥ 20% from baseline, an IOP ≤ 18 mmHg was achieved in 85% of patients, and 42% of patients were medication-free. Adverse events since STAR-GLOBAL study enrolment were: 1 case of ECD loss and 1 cataract progression. The mean reduction in central ECD since baseline was 8.0% at the 3-year follow-up.

Conclusion: In an ethnically diverse cohort of patients, meaningful efficacy and a sustained reduction of hypotensive medications up to 3-years post-implantation was achieved by this supraciliary implant in a bleb-free procedure.

P088

PATTERN ELECTRORETINOGRAM AND VISUAL FIELD IN PATIENTS WITH PRIMARY OPEN-ANGLE GLAUCOMA

Corresponding author: Asaad Ghanem, orcid.org/0000-0001-5790-1553

Asaad Ghanem¹

¹Mansoura Ophthalmic Center, Mansoura University, Egypt

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Introduction: Glaucoma is a neurodegenerative disorder characterized by the loss of retinal ganglion cells and optic nerve fibres which result in loss of visual field. This study aimed to assess the amplitude and latency of pattern electroretinogram in patients with primary open angle glaucoma and to evaluate their correlation with visual field parameters.

Method: The current study included 50 patients with primary open-angle glaucoma and 20 healthy control subjects attending to outpatient clinic of Mansoura ophthalmic centre. The 50 glaucoma patients were divided into 3 groups (mild, moderate and severe). Ophthalmic examination included best corrected visual acuity, Gonioscopy, slit lamp biomicroscopy, intraocular pressure measurement using Goldmann applanation tonometry, visual field assessment by perimetry, and pattern electroretinogram was evaluated using RETI Port 21 (Electrophysiological Diagnostic System, Roland Consult; Brandenburg, Germany). Bivariate Correlations were assessed using Pearson's or Spearman's correlation coefficient depending on the nature of data.

Results: The mean age of the patients with POAG was 55.90 ± 8.839 years, while in the control group it was 52.75 ± 7.893 years. There were statistically significant differences between the two groups as regards the mean values of amplitude P50, latency P50, amplitude N95, latency N95, and latency N35. There were statistically significant positive correlations between cup/disc ratio and each of BCVA, UCVA, IOP, latency P50, latency N95, and latency N35 in glaucoma group. Statistically significant negative correlations were found between Cup/Disc ratio and each of amplitude P50, and amplitude N95 in glaucoma group

Conclusion: Pattern electroretinogram is a precise and objective detector of the electrical activity of the central retina's RGCs in patients with primary open-angle glaucoma.

P089

REAL-WORLD OUTCOMES OF SELECTIVE LASER TRABECULOPLASTY IN A TERTIARY REFERRAL GLAUCOMA SERVICE

Corresponding author: David Gildea, orcid.org/0000-0002-3011-2033

David Gildea¹, Jeremy O'Connor¹, Aoife Doyle¹

¹Royal Victoria Eye & Ear Hospital, Dublin, Ireland

Disclosures: No conflicts of interest and no funding

Introduction: Selective laser trabeculoplasty (SLT) is an effective treatment option in open-angle glaucoma (OAG) and ocular hypertension (OHT). The purpose of this study was to evaluate the real-world treatment outcomes of SLT in a tertiary referral glaucoma service.

Methods: We reviewed the medical records of consecutive patients who had undergone SLT in the glaucoma service at the Royal Victoria Eye and Ear Hospital, Dublin. All patients in this study were using topical IOP-lowering medications prior to SLT. The primary outcome measure was the mean change in intraocular pressure (IOP) at 6 to 8 weeks following the procedure.

Results: 71 eyes of 44 patients were included in this study. There was a mean reduction in IOP of 4.0mmHg (SD 3.6), from 19.1mmHg (SD 3.6) at baseline to 15.1mmHg (SD 4.0) at 6 to 8 weeks post-SLT ($P < 0.001$). The diagnosis was primary open-angle glaucoma (POAG) in 69.0% of eyes, pseudoexfoliation glaucoma (PXFG) in 5.6%, pigment dispersion glaucoma (PDG) in 2.8%, normal tension glaucoma (NTG) in 5.6%, ocular hypertension (OHT) in 15.5%, and pseudoexfoliation OHT in 1.4%. 100% of eyes were on topical IOP-

lowering medication (mean number of agents 2.7). In addition, 8.5% had one previous SLT procedure. No immediate (< 1 hour) post-SLT IOP spikes occurred in this study. Pearson correlation analysis revealed a moderate correlation ($r = -0.4021$, $p < 0.001$) between higher baseline IOP and greater IOP reduction.

Conclusion: SLT is an effective IOP-lowering treatment in open-angle glaucoma and ocular hypertension in a real-world setting.

P090

COMPARATIVE ANALYSIS OF PROMPTED AND UNPROMPTED CHATGPT RESPONSES TO FREQUENTLY ASKED QUESTIONS ABOUT PAEDIATRIC GLAUCOMA

Corresponding author: Deniz Goodman, orcid.org/0000-0002-7176-0694

Deniz Goodman¹, Alana L. Grajewski¹, Elena Bitrian¹, Eleonore J. Savatovsky¹, Ta Chen Peter Chang¹, on behalf of the SIGHT: Pediatric Glaucoma Study Group

¹Bascom Palmer Eye Institute, Miami, FL, USA

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Introduction: ChatGPT is a large language model capable of generating responses to queries, but its use in generating health information has not been well established. We evaluate the accuracy of ChatGPT responses to parents' frequently asked questions about paediatric glaucoma.

Methods: Ten frequently asked questions and responses published on the websites of national ophthalmology organisations and hospitals were compiled. The questions were individually entered into separate ChatGPT chat windows both with and without the following prompt: "Be specific and incorporate any applicable medical guidelines." Twenty-seven paediatric glaucoma specialists, 11 glaucoma specialists, and 4 paediatric ophthalmologists graded responses sentence by sentence as true or false. Descriptive statistics were used to compare the average accuracy between the three types of responses: unprompted ChatGPT, prompted ChatGPT, and reference website responses. Pearson's chi-square test was used to evaluate the association between response type and accuracy.

Results: Among all questions, unprompted ChatGPT responses had the highest proportion of statements graded as correct (0.96), followed by prompted ChatGPT responses (0.95) and website responses (0.92). The response source was significantly associated with the proportion of statements graded as correct ($p < .001$). Among all responses generated by GPT-3.5, prompted responses regarding the diagnosis, signs, and symptoms of paediatric glaucoma ($n = 2$) demonstrated the highest proportion of correct statements (0.98), while prompted responses regarding prognosis and treatment information ($n = 2$) had the lowest proportion of correct statements (0.94).

Conclusion: ChatGPT generates mostly accurate information to frequently asked questions about paediatric glaucoma, particularly about the diagnosis, signs, and symptoms of this condition. Future studies could investigate the quality of this information in terms of readability and comprehensiveness.

P091

A RETROSPECTIVE ANALYSIS OF EFFICACY AND SAFETY OUTCOMES IN PATIENTS UNDERGOING MICROSHUNT SURGERY FOR GLAUCOMA

Corresponding author: Timothy Lloyd, orcid.org/0009-0009-1210-4652

Disclosures: Pouya Alaghband has received lecture fees from Santen (manufacturer of the PreserFlo microshunt).

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Introduction: Minimally invasive bleb surgery (MIBS) has become increasingly popular due to its ability to achieve significant intraocular pressure (IOP) reduction with a good safety profile. This study aimed to assess the safety and efficacy of microshunt PreserFlo surgery at a single centre.

Method: This retrospective study analysed the medical records of 39 eyes that underwent microshunt PreserFlo surgery between March 2022 and August 2023. Data was collected from a combination of the electronic health record and paper notes. IOP measurements, adverse events and demographic data were recorded.

Complete success was defined as a $\geq 20\%$ reduction in IOP without any glaucoma medications whereas qualified success was defined as a $\geq 20\%$ reduction in IOP with additional glaucoma medications.

Results: Our demographic data was similar to those published in the literature with an average age of 69 years and mean pre-surgical IOP of 16.5mmHg. Of note 64.1% of the patients had moderate to severe glaucoma based on the Hodapp classification.

We found that 66.7% of eyes operated on met the definition of complete success by month 3 and a further 7.7% met the qualified success criteria. The incidence of early adverse events in this group was similar to the rates published previously with 28.2% of eyes experiencing hypotony before month 3.

Conclusion: This study suggests that microshunt PreserFlo surgery performed at our centre had similar safety and efficacy profiles compared to those published in the literature. This would support the case for increasing use of minimally invasive bleb surgery as an alternative to conventional glaucoma surgery.

P092

ENHANCED VIRTUAL GLAUCOMA SERVICE - A NOVEL MODEL OF CARE?

Corresponding author: Vidhi Naik, orcid.org/0000-0002-6508-1820

Vidhi Naik^{1,2}, Manjula Kumarasamy^{2,3}

¹University of Aberdeen, Aberdeen, UK; ²NHS Grampian, Aberdeen, UK; ³Aberdeen Royal Infirmary, Aberdeen, UK

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Introduction: An enhanced virtual glaucoma clinic (EVGC) was introduced to the glaucoma service post COVID, as an addendum to an already well-established virtual clinic service. The existing virtual service consists of a system whereby patients have one single visit to the hospital eye outpatient department and have IOP checks, RNFL OCT, and visual field testing. A glaucoma clinician reviews the results from these. The enhanced virtual glaucoma clinic provides an additional step in the clinic, alongside the tests carried out in the standard virtual glaucoma clinics (VGC), patients in the EVGC receive a slit lamp examination, gonioscopy, and Goldman applanation tonometry (GAT), carried out by an ophthalmic imaging practitioner. The EVGC acts as an intermediary between virtual and consultant-led face-to-face appointments.

The aim is to ascertain whether the introduction of EVGC is effective in reducing the requirement for face-to-face consultant-led glaucoma clinics.

Method: The review analyses data of 75 patients who attended EVGC over six months, beginning January to June 2023.

Results: 64% of patients were retained within the virtual glaucoma service, and out of that percentage 25% were kept in the EVGC service. Out of our patient group, only 25% were referred for consultant-led face-to-face appointments.

Conclusion: This is a useful model of care which reduces face-to-face appointment requirements by 75% from patients in this virtual cohort. This service is useful for monitoring long-term post-op trabeculectomy patients, and new patients for gonioscopy as slit lamp examination is utilised. Patients who are stable but unsuitable for VGC (e.g. wheelchair-bound) can be followed up in this clinic, reducing the requirement for a consultant-led clinic. This model is effective at reducing the burden on glaucoma services.

P093

EVALUATING POST-OPERATIVE FLARE IN GLAUCOMA SURGERY: MINIJECT, MICROSHUNT, BAERVELDT TUBE, CATARACT & ECP AND TRABECULECTOMY

Corresponding author: Ananth Ranjit, orcid.org/0009-0002-6624-1635

Ananth Ranjit¹, Bhavin Patel¹, Andrew Amon¹, Elizabeth Galvis¹, Kin Sheng Lim¹

¹Guy's and St Thomas' NHS Foundation Trust, London, UK

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Introduction: There is a paucity of published information on objective measurement of flare in the post-operative period for new approaches to managing glaucoma. Sub-clinical inflammation is not always apparent without objective measurements. This study highlights levels of long-term inflammation following interventions such as Miniject, Microshunt, Baerveldt Tube, Cataract & ECP and Trabeculectomy.

Method: Retrospective data collected from St Thomas' Hospital between July 2019 to September 2023. KOWA FM-600 Laser Flare Meter was used. Mean flare for each device was calculated at 1 week, 1 month, 3 months, 6 months and 1 year.

Results: PreserFlo mean flare at week 1, 42.64 (SD 49.43) and at 1 year 15.43 (SD 8.9). Baerveldt Tube mean flare at week 1, 28.65 (SD 26.35) and at 1 year, 44.65 (SD 40.63). Trabeculectomy mean flare at week 1, 61.27 (SD 68.81) and at 1 year, 33.45 (SD 30.97). Miniject mean flare at week 1, 92.25 (SD 81.10) and at 1 year, 11.97 (SD 6.10). Cataract & ECP mean flare at week 1, 60.75 (SD 49.78) and at 1 year, 26.43 (SD 24.32).

Conclusion: The flare meter provides an accurate, non-invasive assessment of aqueous flare which can be valuable in research and clinical decisions. A notable rise in Tube flare at 1 month may be linked to stent removal. Our data suggests sub-clinical inflammation persists in tubes and least with suprachoroidal devices (Miniject). The contrast in inflammation between IOP lowering devices (PreserFlo, Miniject and Baerveldt Tube) is likely influenced by the biomaterial. Sub-clinical inflammation seems to persist up to a year, if not longer, recognising this may direct appropriate long-term management. This study recognises the need for inclusion of objective flare assessment in glaucoma clinical trials.

P094

A MULTI-CENTRE STUDY OF A SUPRACILIARY GLAUCOMA DRAINAGE DEVICE IN PATIENTS WITH OPEN ANGLE GLAUCOMA: 1-YEAR FOLLOW-UP RESULTS

Corresponding author: Gokulan Ratnarajan, orcid.org/0000-0001-7990-5053

Gokulan Ratnarajan¹, Antonio Fea², Iqbal Ike K Ahmed³

¹Queen Victoria Hospital, East Grinstead, UK; ²Dipartimento di Scienze Chirurgiche-Clinica Oculistica, Torino, Italy; ³John Moran Eye Center, University of Utah, Salt Lake City, UT USA

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Introduction: To describe the safety and efficacy profile of a supraciliary, glaucoma drainage device, MINject® (iSTAR Medical, Belgium), in patients with medically uncontrolled open-angle glaucoma.

Method: The STAR-IV trial is a prospective, interventional, single-arm study at 2 sites in India and Panama. The MINject implant is a 5mm long network of hollow spheres made of soft, flexible silicone, and is implanted standalone and ab-interno into the supraciliary space to enhance outflow. The MINject implant was placed in 21 eyes in a stand-alone, *ab-interno* procedure. Intraocular pressure (IOP), medication use, and other ocular parameters were evaluated pre and post-operatively through 24 months. The primary endpoint was IOP reduction at 6 months compared to baseline, analysed with a paired t-test. Safety evaluation included the nature and frequency of adverse events. Interim study results at 12-months are reported here.

Results: Baseline mean diurnal IOP was 24.2 ± 3.2 mmHg using 2.5 ± 1.1 IOP-lowering medications. At 6 and 12-month follow-up, mean diurnal IOP was reduced by 10.3mmHg (-41.5%) to 14.2mmHg (n = 19) and by 9.2 mmHg (-37.0%) to 15.3 ± 4.1 mmHg (n = 17), respectively. Mean medication use at 12-months was 1.6 ± 1.2 , a mean reduction of 0.9 medications from baseline. An IOP ≤ 18 mmHg was achieved in 70.6% of eyes at 12 months. The most common adverse events were: anterior chamber inflammation, pupillary deformation, blurred vision, and visual acuity loss. There was no significant loss of endothelial cell density and no eye exceeded 30% loss from baseline.

Conclusion: This study confirms the potential efficacy and safety of standalone insertion of the MINject implant into the supraciliary space to reduce IOP and the dependency on glaucoma medications up to 12-month follow-up.

P095

INFLUENCE OF BASELINE INTRAOCULAR PRESSURE OR MEDICATION ON TWO-YEAR OUTCOMES OF A SUPRACILIARY DRAINAGE DEVICE IN THE STAR-I,II,III TRIALS

Corresponding author: Ian Rodrigues, orcid.org/0000-0002-4913-1306

Ian Rodrigues¹, Antonio Fea², Philippe Denis³, Brian Flowers⁴, Paul I Singh⁵, Ike K Ahmed⁶

¹Guy's and St Thomas' NHS Foundation Trust, London, UK; ²Università Degli Studi di Torino, Torino, Italy; ³Hôpital de la Croix-Rousse, Lyon, France; ⁴Ophthalmology Associates, Fort Worth, TX USA; ⁵Eye Centers of Racine & Kenosha, Racine, Wisconsin, USA; ⁶University of Toronto, Toronto, Canada

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Introduction: To describe the influence of baseline intraocular pressure (_bIOP) or number of medication classes used (_bMed) in patients with open-angle glaucoma on outcomes two years after implantation with a supraciliary, glaucoma drainage device, MINject® (iSTAR Medical, Belgium).

Method: The implant was delivered in a standalone, ab-interno procedure in phakic and pseudophakic eyes with no medication washout in three trials (STAR-I,II,III). Data from 66 patients who completed two-year follow-up were pooled.

Outcomes of patients with a high (>24mmHg) or moderate (21-24mmHg) _bIOP were compared as well as between patients with higher (3 or 4) or lower (1 or 2) _bMed use. Success was defined as IOP > 5 and ≤ 21mmHg with ≥ 20% reduction in IOP, without (complete) or regardless of (qualified) glaucoma medication (meds) use.

Results: At two-years, the high _bIOP group achieved a 41.8% IOP reduction (from mean _bIOP 28.1mmHg, n = 21) with 0.7 fewer medications compared to a 38.1% reduction (from mean 22.1mmHg, n = 45)(p = 0.448) and 1.0 fewer medications in the moderate _bIOP group.

The high _bMed group (mean _bMed 3.3, n = 28) had a 34.6% IOP reduction compared to 42.7% (p = 0.076) in the low _bMed group (_bMed 1.5, n = 38). Medication was similarly reduced by 1.3 and 0.7 (p = 0.143) in the high and low _bMed groups respectively.

Complete success was greater in the low μ Med group ($p = 0.003$), but both groups had a similar qualified success rate (84.2%, 67.9%; $p = 0.144$).

Conclusion: Comparable levels of relative IOP reductions at two years were seen in patients regardless of a high (mean 28.1mmHg) or moderate (mean 22.1mmHg) baseline IOP prior to implantation of this supraciliary drainage device. Likewise, a comparable reduction in IOP was achieved despite the number of medications used pre-operatively.

P096

AUTOMATED NON-CONTACT DIRECT SELECTIVE LASER TRABECULOPLASTY (DSLTL) AT THE NATIONAL TREATMENT CENTRE - HIGHLAND: INITIAL REAL WORLD RESULTS ON SAFETY AND EFFICACY

Corresponding author: Reham Shehada, orcid.org/0000-0002-7163-0793

Reham Shehada¹, Lai-Yeung Ngai¹, Thorfinn Leslie¹

¹National Treatment Centre, NHS Highland, Inverness, UK

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Introduction: The introduction of the Eagle Direct Selective Laser Trabeculoplasty (DSLTL) device presents a novel advancement in laser technology. In this study, we evaluate the safety profile and the capacity of DSLTL device to reduce intraocular pressure (IOP).

Method: This retrospective study analyses data from patients who underwent DSLTL between March and October 2023. A total of 51 patients (72 eyes) were included, encompassing 44 with primary open-angle glaucoma, 10 with normal tension glaucoma, 8 with pseudoexfoliation glaucoma, 5 with ocular hypertension, 3 with pigment dispersion syndrome, and 2 glaucoma suspects. Patients, whether treatment-naïve or after medication, underwent DSLTL. The procedure involved the application of 120 sequential noncontact laser shots (1.6-1.8 mJ), automatically delivered over 2 seconds on the limbus, guided by image analysis and eye tracking. IOPs were assessed at 30 minutes, 2 months, and 6 months post-laser treatment.

Results: The baseline mean IOP was 19.94 ± 5.21 mmHg. At 30 minutes post-laser treatment, the IOP was 17.94 ± 6.92 mmHg. Subsequent assessments at 2 months and 6 months revealed mean IOP values of 16.10 mmHg (± 3.64 mmHg) and 15.56 mmHg (± 3.79 mmHg), respectively. Remarkably, the reduction in IOP at 2 months from baseline showed a mean reduction of 3.41 mmHg (± 4.55 mmHg), demonstrating statistical significance with a p -value of $3.57 * 10^{-6}$. Similarly, the reduction in IOP at 6 months from baseline exhibited a mean reduction of 4.00 mmHg (± 6.16 mmHg), with a significant p -value of $8.96 * 10^{-4}$. Only four eyes experienced IOP spikes following the treatment.

Conclusion: DSLTL emerges as a promising and safe noncontact modality in reducing intraocular pressure IOP among glaucoma patients.

P097

SCOTTISH GLAUCOMA TRIAGE REDESIGN USING EXISTING COMMUNITY OPTOMETRY INFRASTRUCTURE (2020-2022)

Corresponding author: Matthew Sikorski, orcid.org/0009-0003-0207-8151

Matthew Sikorski¹, Roshni Sanders¹

¹NHS Fife, Dunfermline, UK

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Introduction: Glaucoma service redesign was implemented to tackle current patient pressures. An entire glaucoma service (2276 patients) was triaged to three levels in accordance with Glaucoma stratification guidance and local protocols. The patient information gathered included demographics, all glaucoma clinical findings, previous ocular surgery, ocular medication, and social deprivation scores.

Method: Level 1 patients ($n = 482$) were seen as part of Scottish general optometry service contract (GOS). Level 2 ($n = 714$) patients were seen in glaucoma accredited community optometry clinics (NHS Education Scotland glaucoma accreditation and training –NESGAT) with weekly input from glaucoma consultants through screen share. Level 3 ($n = 1080$) patients were retained in hospital. All outcomes were audited and analysed prospectively over 24 months.

Results: The study demonstrated statistically significant parameters between the three groups. Namely, Level 1 patients included more normal eyes, less pharmacological and surgical intervention, less mean deviation on visual fields and less social deprivation. After 24 months 40.2% of Level 1 patients ($n = 199$) returned to HES, with 62.8% of re-referrals being for cataract surgery or other eye related disease. 20.4% of re-referrals remained in HES or Level 2. For Level 2 patients, 9.24% ($n = 66$) returned to HES for SLT (33.3%), cataract surgery (22.7%), medication changes (21.2%), glaucoma surgery (9.1%). 2.7% of re-referrals remained in HES. Patient satisfaction was above 97%

Conclusion: This model shows safe triage of an entire glaucoma service and devolved care using the Scottish GOS contract and NESGAT qualified optometrists at community clinics. This frees hospital capacity with Queen Margaret Hospital seeing all new glaucoma patients within six weeks and review patients in allocated time with 25% leeway.

These results are encouraging and allow for other Scottish healthboards to follow suit thus aiming for national consistency.

P098

TO REPORT A CASE OF STURGE WEBER SYNDROME TYPE 2 IN A YOUNG FEMALE

Corresponding author: Bhupinderjeet Singh, orcid.org/0009-0005-8461-9808

Bhupinderjeet Singh¹, Reena Kumari¹, Dinesh Kumar Bhagat¹, Khushdeep Kaur¹

¹Mata Gujri Memorial Medical College and LSK Hospital, Kishanganj, Bihar, India

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Introduction: Sturge Weber Syndrome also known as encephalotrigeminal angiomas, belongs to a group of disorders known as phakomatosis. Incidence is 1 in 50,000-60,000 live births. It is caused by somatic mutations in GNAQ gene, on long arm of chromosome 9. According to ROACH scale classification, there are three types.

Method: An 18 year old female presented to Otorhinolaryngology department for oral ulcers and was referred to Ophthalmology department for progressive loss of vision in OS since 2 years. She also had dark red discoloration (port wine stain)

on left side of face extending to right mandibular region since birth. Unaided VA OD was 6/6 and OS was 6/18. Slit lamp examination OD anterior segment-iris-central heterochromia. Fundus examination OD-unremarkable. Slit lamp examination OS anterior segment-dilatation and tortuosity of episcleral vessels near limbus, iris-central heterochromia. Fundus examination OS showed CDR-0.9, Bayonetting sign-present. Gonioscopy-OU shows open angle in all quadrants with multiple iris processes. B-scan-OU-unremarkable. IOP by AT-OD-12mmHg, OS-22mmHg. HFA 24-2 threshold showed visual field defects corresponding to tunnel vision in left eye. OCT of OD- unremarkable. OCT of OS-CDR 0.9 and RNFL thinning in all quadrants. MRI showed swelling of subcutaneous tissue over left facial region.

Results: On slit lamp examination, OCT and based on MRI findings diagnosis of Sturge Weber Syndrome type 2 was made with chief ocular manifestation of glaucoma. Antiglaucoma medication was started and our patient was counselled for 3 monthly follow-up and was referred to Dermatology department.

Conclusion: Sturge Weber Syndrome is a rare disorder. Early detection of SWS is essential to detect development of complications like glaucoma that can lead to total blindness, choroidal haemangioma and other systemic manifestations.

P099

A COMPARATIVE STUDY OF SELECTIVE AND PASCAL LASER TRABECULOPLASTY IN THE TREATMENT OF OPEN ANGLE GLAUCOMA AND OCULAR HYPERTENSION

Corresponding author: Soly Somapalan, orcid.org/0000-0002-9196-0642

Soly Somapalan¹, Manoj Mathai¹

¹Wrightington, Wigan and Leigh Teaching Hospitals NHS Trust, Wigan, UK

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Introduction: PLT (pattern ALT)- conventional argon laser is applied in a patterned way around trabecular meshwork with the help of the PASCAL machine using specialised lens to avoid overlapping spot burn.

SLT (selective laser trabeculoplasty) green (532 nm), Q-switched, frequency doubled Nd:YAG laser applied to pigmented TM with less heat dissipation.

AIMS: Evaluate the efficacy of SLT in reducing IOP in patients of open angle glaucoma.

Efficacy of SLT in OHT, mild - moderate and advanced POAG. Assess the reduction in number of anti-glaucoma topicals used after SLT

Compare the results with PLT

Method: Data from the laser room logbook

Retrospectively analysed

Pre-laser IOP, IOP with 3 months, 6 months and 1 year were taken.

Treatment response was defined successful if there was a reduction of 3 mmHg or more (>20% from baseline) in IOP during the last visit when compared to pre- laser value.

Results: 96 eyes were further categorized into 3 groups:

- 12 eyes were advanced glaucoma
- 49 eyes were followup OHT and mild to moderate POAG
- 35 eyes were new patients diagnosed with either ocular hypertension or POAG

5 eyes with advanced glaucoma at 1 year follow up had baseline IOP -21.40 mmHg

11 eyes on followup OHT and POAG had average baseline IOP 24.90mmHg on maximal tolerated topicals

No reduction in the topicals both in advanced glaucoma and follow up OHT and POAG patients.

17/ 21 eyes -OHT (80.95%) and 7/14 -POAG (50%) treated with SLT are currently drop free.

Conclusion: SLT has shown promising results in lowering IOP in POAG and OHT

Results are definitely better than PLT

Treatment naïve eyes had better IOP lowering efficacy with SLT with drop-free eyes

P100

STUDYING THE RELATIONSHIP BETWEEN SOCIAL DEPRIVATION AND GLAUCOMA-RELATED QUALITY OF LIFE

Corresponding author: Dr Ryan Teo, orcid.org/0009-0006-2380-9912

Dr Ryan Teo¹, Kleonikos Tsakiris²

¹University Hospitals Plymouth NHS Trust, Plymouth, UK

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Introduction: As it is generally accepted that glaucoma is incurable, it is critical to maintain or improve the QOL of afflicted patients.

The purpose of this project is to investigate the relationship between social deprivation and its impact on glaucoma-related QOL, thereby advancing the European Glaucoma Society's aim of glaucoma treatment, which is to maintain QOL in affected patients at a sustainable cost.

Method: The methodology consists of the following:

- Systematic Literature Review
- The 25-item National Eye Institute Visual Functioning Questionnaire (VFQ-25) was administered in an interview format, and results correlated with the Scottish Index of Multiple Deprivation (SIMD), by postcode
- Statistical Analysis was carried out using IBM SPSS Statistics

Results: 150 patients, comprising 83 males and 67 females, undertook the questionnaire.

The mean age was 75 years and all patients were of Caucasian race.

Those with lower SIMD scores (more socially deprived) reported lower VFQ-25 scores (lower quality of life).

Statistical analysis established that Pearson correlation coefficient $r = 0.568$ with $p\text{-value} = 0.009$

Discussion: The results suggest there is a strong inverse relationship between social deprivation and glaucoma-related QOL.

This is extremely useful as more can and should be done to help socially deprived glaucoma patients.

Educational counselling, transport strategies, glaucoma support groups to facilitate peer monitoring should be considered.

Conclusion: Greater social deprivation is known to lead to poorer vision and presentation of glaucoma in its later stages.

However, this study has now established that greater social deprivation leads to a lower glaucoma-related QOL, encompassing the physical, psychological, emotional and social well-being of the patient.

More measures must be implemented to help improve the overall well-being of socially deprived glaucoma patients and maintain their QOL.

A VALIDATED WEB-APPLICATION FOR AUTOMATED CLASSIFICATION OF GLAUCOMATOUS VISUAL FIELD DEFECTS USING HODAPP-PARRISH-ANDERSON CRITERIA

Corresponding Author: Arun J Thirunavukarasu, orcid.org/0000-0001-8968-4768

Arun J Thirunavukarasu^{1,2,3}, Nikhil Jain^{3,4}, Helmut Yu³, Rohan Sanghera¹, Helmut C Y Yu^{3,5}, Rupert R A Bourne^{3,6}

¹University of Cambridge, School of Clinical Medicine, Cambridge, UK; ²Oxford University Clinical Academic Graduate School, Oxford, UK; ³Cambridge Eye Research Centre, Cambridge, UK; ⁴Bedfordshire Hospitals NHS Foundation Trust, Luton, UK; ⁵University of New South Wales, Sydney, Australia; ⁶Vision & Eye Research Institute, Anglia Ruskin University, Cambridge, UK

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Introduction: Perimetry analysis is an essential component of glaucoma diagnosis, prognostication, and management decisions. However, subjectivity and ambiguity of visual field classification limits accuracy and reliability. Standardised rules for classifying glaucomatous visual field defects exist, but these are labour-intensive and therefore impractical for day-to-day clinical work. An application for automatic application of Hodapp-Parrish-Anderson (HPA) criteria could mitigate these difficulties.

Method: In this cross-sectional study, 89 glaucoma patients with visual field defects had their perimetry data categorised according to HPA criteria by a web-application, Glaucoma Field Defect Classifier (GFDC), and by two independent researchers with disagreement resolved by a third researcher. In total, 168 perimetry results were analysed. To quantify the accuracy of GFDC, web-application output was compared to human researcher gradings which were used as ground truth. Specifically, a Kappa statistic was calculated and confusion matrix analysis employed to visualise agreement.

Results: GFDC exhibited perfect accuracy, reaching identical conclusions to human clinician gold standard decisions based on Hodapp-Parrish-Anderson criteria for every analysed perimetry result ($\kappa = 1.00$). 100% of mild, moderate, and severe glaucomatous field defects were correctly identified. Interpretability analysis reveals that GFDC correctly identifies plot boundaries and data points represented within perimetry result documents.

Conclusion: GFDC accurately classifies glaucomatous visual field defects according to Hodapp-Parrish-Anderson criteria. Subsequent work is underway applying GFDC in larger cohorts, and the application is well-suited to large-scale studies requiring visual field classification. GFDC and its code are freely available online, and may be customised to incorporate alternative perimetry data formats. Clinical applications of GFDC could reduce the subjectivity of decision-making in glaucoma, helping to provide patients with equitable and optimal care.

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EXPLORING THE POTENTIAL OF AUGMENTED REALITY AND VIRTUAL REALITY AS A DIAGNOSTIC TOOL FOR GLAUCOMA SCREENING IN OPHTHALMOLOGY

Corresponding author: Krishnika Vetrivel, orcid.org/0009-0001-6966-3801

Krishnika Vetrivel¹

¹Kings College London, London, UK

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Introduction: Glaucoma is one of the leading causes of irreversible blindness globally. Early detection is crucial. Yet, half of glaucoma cases remain undiagnosed. Whilst previous reviews have primarily focused on the application of VR/AR (virtual reality/augmented reality) in Ophthalmology for education, surgical training and treatment, the potential of this technology as a screening tool is limited in current literature. This review aims to provide an overview of the current state of research by exploring four key areas: limitations of existing screening tools, benefits and challenges of VR, the integration of VR with telemedicine and artificial intelligence (AI) for glaucoma diagnosis, and future directions.

Method: A comprehensive search for publications was conducted using the Medline and Embase databases. Original articles that adhered to the following criteria were included: regarding the use of VR or AR in diagnosis and screening in glaucoma, English language, full-text available and published in the last five years. The initial search yielded 2878 studies, from which eight relevant studies were identified. The included studies had a varying risk of bias and applicability, according to the QUADAS-2 evaluation.

Results: VR offers advantages such as enhancing accessibility, improving disease management and reducing wait times. Integrating telemedicine and VR with AI offers the potential for remote collection of patient data, early disease detection, clinical forecasting, disease progression tracking, and optimised treatment planning. Nevertheless, potential limitations include possible myopia progression and marketing-driven heterogeneity.

Conclusion: Future studies should investigate the accuracy and applicability of AR/VR in diagnosing glaucoma across diverse patient groups, including paediatric cases, encompassing different types and severity levels. In conclusion, VR holds potential for glaucoma diagnosis but necessitates collaboration and further research before integration into clinical practice.

P103

ORA VS GAT, DEVICES UNDER PRESSURE

Corresponding Author: Soon Chee Yap, orcid.org/0000-0002-2731-0266

Soon Chee Yap^{1,2}, Caroline Cobb^{1,2}, Stewart Gillan^{1,2}

¹NHS Tayside, Dundee, UK; ²University of Dundee, Dundee, UK

Disclosures: No conflicts of interest and no funding

Introduction: Consultant led glaucoma clinics have far exceeded capacity, leading to the widespread adoption of virtual glaucoma clinics. Ocular Response Analyzer (ORA) is a technician-friendly device that objectively measures intraocular pressure. Additionally, it measures corneal hysteresis (CH) which predicts susceptibility for glaucoma progression, aid in diagnosis and monitor response to glaucoma therapy.

Our study evaluates the agreement of IOP measurements between the Ocular Response Analyzer (ORA) and the Goldmann Applanation Tonometry (GAT). Furthermore, evaluate patients' preference between the two methods.

Method: Prospective study involving 50 patients attending glaucoma clinics at Ninewells Hospital, Dundee. IOP is measured in both eyes using both devices. Patients subsequently asked which method was preferred and why. Agreement between the

two methods is evaluated by calculating the 95% limit of agreement and evaluating the Bland-Altman plot.

Results: Primary angle closure glaucoma patients were found to be significant outliers and added to the exclusion criteria. The mean difference between GAT IOP and ORA IOPcc is -0.3 mmHg (95% CI -0.7 mmHg to $+0.1\text{ mmHg}$), while the 95% limit of agreement ranges from -4.2 mmHg to $+3.6\text{ mmHg}$. Bland-Altman plots are closely associated with smaller differences at lower IOP readings. 86% of IOPcc measurements are within 2 mmHg to GAT IOP.

Conclusion: ORA IOPcc showed great agreement with GAT IOP measurements with low proportion of clinically significant differences, especially in the lower IOP group. Marginally more patients prefer ORA over GAT.

ORA proves to be a great tool for IOP measurements in virtual glaucoma clinics, where majority of patients seen are of stable glaucoma.

Association between ORA IOP measurements and GAT IOP measurements in patients with PACG needs to be studied with a larger patient group and different study design.

P104. Abstract not included

MEDICAL RETINA

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REAL WORLD TREATMENT OUTCOMES IN POLYPOIDAL CHOROIDAL VASCULOPATHY IN A CAUCASIAN POPULATION OF BRITISH ETHNICITY

Corresponding author: Nishant Aggarwal, orcid.org/0000-0002-1931-390X

Nishant Aggarwal¹, Suresh Thulasidharan¹, Ramez Barbara¹, Alison Penwarden¹, Rino Frisina², Gabriella De Salvo¹

¹University Hospitals Southampton NHS Foundation Trust, Southampton, UK; ²University of Padova, Padova, Italy

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Introduction: Polypoidal choroidal vasculopathy (PCV) has been increasingly recognised in Caucasians recently. This study assesses treatment response of PCV in a Caucasian population of British ethnicity with intravitreal anti-vascular endothelial growth factor (anti-VEGF) injections alone and with a combination of anti-VEGF injections and photodynamic therapy (PDT).

Method: Retrospective review of 95 Caucasian patients in a single centre with diagnosis of PCV between 2013 and 2018 were included. Best corrected visual acuity (BCVA), central retinal thickness (CRT), indocyanine green angiography (ICGA) characteristics, numbers and type of treatment were analysed at baseline and at 1 year.

Results: One hundred and six eyes included from 95 patients received either anti-VEGF injections ($n = 72$, 71.3%) alone or combination therapy with anti-VEGF and PDT ($n = 13$, 12.9%). A third untreated group was also observed ($n = 16$, 15.8%). Five eyes were excluded from the study due to structural retinal damage. Mean number of injections was 16.4 in the monotherapy group and 19.9 in the combination group. Both treatment groups showed improvement in BCVA at 1 year and this was statistically

significant in the monotherapy group with a mean gain of 8.3 letters ($p < 0.001$). Mean CRT decreased in all treatment groups at twelve months and this change was significant, by 33.1 and $52.3\mu\text{m}$ for the monotherapy and combination groups respectively ($p < 0.01$ for both). In the observation group, vision and CRT remained stable.

Conclusion: This study demonstrates significant improvements in BCVA at one year in our Caucasian cohort of British ethnicity with PCV in the anti-VEGF monotherapy treatment group.

P106. Abstract not included

P107. Abstract not included

P108

USE OF ARTIFICIAL INTELLIGENCE IN CHOROIDAL NEOVASCULARISATION: A LITERATURE REVIEW

Corresponding author: Mohammad Anas, orcid.org/0000-0002-5818-5218

Mohammad Anas^{1,2}, Ahmed Ellabban¹, Muhammad Habbas¹, Taysab Hasan¹, Hashaam Choudhry¹, Sameer Khan¹

¹King's College London, Faculty of Life Sciences and Medicine, London, UK; ²Institute of Ophthalmology, University College London, London, UK

Disclosures: No conflicts of interest and no funding

Introduction: This literature review aims to synthesize the current state of AI applications in CNV and identify research gaps and future directions in this area. The objectives include understanding the capabilities of existing AI algorithms, exploring the datasets used in these studies and identifying any limitations in current approaches.

Method: A comprehensive literature search of OVID Embase, OVID Medline, Scopus, Web of Science and Cochrane Database of Systematic Reviews (CDSR) was conducted, extracting relevant studies and trials for critical appraisal.

Results: 50 papers, published from 2018 to present, were selected based on specific inclusion criteria. The reviewed studies showcased AI models, particularly convolutional neural networks, which demonstrated high performances in CNV diagnosis, classification, and segmentation. These AI models often achieved accuracy statistics comparable to or even surpassing those of expert ophthalmologists. However, finding a lack of research regarding AI's role in guiding CNV treatment decisions and follow-up, alongside discrepancies in dataset sizes and utilization, made it challenging to compare different deep learning models. Additionally, multi-modal integration was rarely explored, with many papers focusing solely on OCT images.

Conclusion: The findings from this review highlight the transformative potential of AI in CNV, emphasizing the importance of ongoing research and collaboration in this field. The integration holds a high promise, offering significant improvements in clinical decision-making and patient outcomes. However, several research gaps persist, including the need for standardized large datasets for external model validation and a greater focus on AI's role in guiding CNV treatment and follow-up. The findings from this review highlight the transformative potential of AI in CNV, emphasizing the importance of ongoing research and collaboration in this field.

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ENDOPHTHALMITIS POST-INTRAVITREAL TREATMENT: 15 YEARS EXPERIENCE AT A LARGE TERTIARY CENTRE

Corresponding author: Frederick Bailey, orcid.org/0000-0002-7206-3081

Frederick Bailey¹, Catherine Dardis¹, Muralidharan Upendran¹

¹Belfast Health and Social Care Trust, Belfast, UK

Disclosures: No conflicts of interest and no funding

Introduction: Intravitreal injections are an essential treatment for a variety of common medical retinal conditions (wAMD, DMO, RVO). Endophthalmitis is an ophthalmic emergency characterised by severe intraocular inflammation secondary to microbial entry. Previous studies have suggested a rate of post-intravitreal endophthalmitis of between 0.02-0.06 (1 in 1500 to 1 in 5000 cases). Standard treatment involves AC culture, intravitreal culture, and intravitreal antibiotic injection.

Method: The Fairview Unit of Mater Infirmorum Hospital, Belfast, is a large tertiary centre responsible for the medical retina treatment of a majority of patients in Northern Ireland. A range of demographic, clinical, and microbiological data was collected from cases of post-intravitreal endophthalmitis between 2008 and 2022 at the Fairview Unit. Cases were included if they had clinical signs of endophthalmitis along with standard treatment.

Results: 154,868 intravitreal injections took place and 46 cases of post-intravitreal endophthalmitis were reported between 2008 and 2022 at the Fairview Unit. The incidence rate was 0.290 (roughly 1 in 3500 injections) over this period, with a maximum of 14 cases in a given year. The median presentation time was 4 days. There were 28 culture positive and 18 culture negative cases. The commonest bacteria cultured was Staph epidermidis (14/28 cases), followed by Staph Warneri and Staph aureus (5/28 cases each). The median drop in VA for culture positive cases was significantly more than in culture negative cases (-26 letters culture positive; -2 culture negative; $p < 0.05$).

Conclusion: In summary, the incidence rate of post-intravitreal endophthalmitis at the Fairview Unit between 2008-2022 in Belfast is on par with other studies. Culture positive cases were associated with significant visual outcomes compared to culture negative cases.

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REAL-WORLD VOYAGER STUDY OF FARICIMAB IN NAMD AND DMO: BASELINE AND 3-MONTH DATA

Corresponding author: Clare Bailey, orcid.org/0000-0003-2141-4512

Clare Bailey¹, Monica Bengus², Voraporn Chaikitmongkol³, Varun Chaudhary⁴, Gloria C. Chi⁵, Robert P. Finger⁶, Roberto Gallego-Pinazo⁷, Frank Gilberg², Adrian Hock Chaun Koh⁸, Susumu Ishida⁹, Monica Lövestam-Adrian⁹, Mariacristina Parravano¹⁰, Jose Luna Pinto¹¹, Stefan Scheidl², Steffen Schmitz-Valckenberg^{12,13}, Eric H Souied¹⁴, Robyn Guymier¹⁵

¹Bristol Eye Hospital, University Hospitals Bristol, NHS Foundation Trust, Bristol, UK; ²F. Hoffmann-La Roche Ltd., Basel, Switzerland;

³Faculty of Medicine, Chiang Mai University, Chiang Mai, Thailand;

⁴Hamilton Regional Eye Institute, McMaster University, Hamilton, ON, Canada; ⁵Genentech Inc., San Francisco, CA, USA; ⁶University of Bonn, Venusberg-Campus, Bonn, Germany; ⁷Unit of Macula and

Clinical Trials, Oftalvist Clinic, Valencia, Spain; ⁸Camden Medical Centre, Singapore; Department of Ophthalmology, Faculty of Medicine and Graduate School of Medicine, Hokkaido University, Sapporo, Japan; ⁹Department of Clinical Sciences Lund, Lund University, Skane University Hospital, Lund, Sweden; ¹⁰IRCCS-Fondazione Bietti, Rome, Italy; ¹¹Centro Privado de Ojos Romagosa-Fundacion VER, Córdoba, Argentina; ¹²John A. Moran Eye Center, Department of Ophthalmology & Visual Sciences, University of Utah, Salt Lake City, UT, USA; ¹³University of Bonn, Venusberg-Campus, Bonn, Germany; ¹⁴Department of Ophthalmology, Centre Hospitalier Intercommunal de Créteil, Créteil, France; ¹⁵Centre for Eye Research Australia, Royal Victorian Eye and Ear Hospital, University of Melbourne, Melbourne, Australia

Disclosures: Clare Bailey - Financial Disclosures: Consultant: Alimera Sciences, Bayer, Boehringer Ingelheim, Janssen, Novartis, Roche. Monica Bengus - Financial Disclosures: Employee and Shareholder: F. Hoffmann-La Roche Ltd. Voraporn Chaikitmongkol - Financial Disclosures: Consultant: Bayer, Novartis, Roche; Grant Support: Bayer, Roche. Varun Chaudhary - Financial Disclosures: Consultant: Alcon, Bayer, Novartis, Roche; Grant Support: Bayer, Novartis. Gloria C. Chi - Financial Disclosures: Employee: Genentech, Inc. Robert P. Finger - Financial Disclosures: Consultant: Alimera, Apellis, Bayer, Boehringer Ingelheim, Chiesi, OD-OS, Opthea, ProQR, Roche/Genentech, Inc., Santen; Grant Support: Biogen, CenterVue, Novartis; Speaker: Alimera, Apellis, Bayer, Novartis. Roberto Gallego-Pinazo - Financial Disclosures: Consultant: Boehringer Ingelheim, Novartis, Ora, Roche, ZEISS; Grant Support: Bloss Group, Celltrion, Ionis, Iveric Bio, Novartis, NTC, Roche; Speaker: Allergan, Heidelberg Engineering, Novartis, Ora, Roche, ZEISS. Frank Gilberg - Financial Disclosures: Employee: F. Hoffmann-La Roche Ltd. Adrian Hock Chaun Koh - Financial Disclosures: Consultant: Alcon, Allergan/AbbVie, Apellis, Bayer, Boehringer Mannheim, Heidelberg Engineering, Novartis, Santen, Topcon, ZEISS. Susumu Ishida - Financial Disclosures: F. Hoffmann-La Roche Ltd. Alpha Communications, AMO Japan, Bayer, Bloom Technology, Bonac, Japan Archon, Nidec, Otsuka, Santen, Seed, Senju, Wakamoto, White Medical; [R] Advanced Medical, Alcon, Bayer, Bloom Technology, Chugai, HOYA, Japan Archon, Kowa, Kyowa Kirin, Mylan, Nidec, Nippon Boehringer Ingelheim, Nitto Medic, Novartis, Santen, Senju, Wakamoto. Monica Lövestam-Adrian - Financial Disclosures: Consultant: Allergan, Bayer, Novartis, Roche. Mariacristina Parravano - Financial Disclosures: Consultant: Allergan, Bayer, Novartis, Roche/Genentech, Inc. Jose D Luna Pinto - Financial Disclosures: Grant Support: F. Hoffmann-La Roche Ltd. Stefan Scheidl - Financial Disclosures: Employee: F. Hoffmann-La Roche Ltd. Consultant: AlphaRet, Apellis, bioeq, Katairo, Kubota Vision, Novartis, Oxurion, Pixium Vision, Roche, SparingVision; Grant Support: Apellis, Bayer, Heidelberg Engineering, Novartis, Roche, ZEISS; Owner: STZ GRADE Reading Center. Eric H Souied - Financial Disclosures: Consultant: Allergan/AbbVie, Bayer, Novartis, Roche, Teva. Robyn Guymier - Financial Disclosures: Consultant: Apellis, Bayer, Novartis, Roche/Genentech, Inc.

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Introduction: Vision outcomes with anti-vascular endothelial growth factor agents are often poorer in clinical practice than in clinical trials. The VOYAGER (NCT05476926) study will gather robust long-term clinical data among patients receiving faricimab (FAR) for neovascular age-related macular degeneration (nAMD) or diabetic macular oedema (DMO) in routine clinical practice globally to address this disparity.

Method: VOYAGER (NCT05476926) is a prospective observational study planning to enrol ~5000 patients in 28 countries using/initiating faricimab for nAMD or DMO in routine clinical practice. Patients will be observed for up to 5 years. The primary outcome is

VA change from baseline at month 12 per eye, by indication. Prior treatment and patient demographic data were collected at baseline. Ocular assessments were performed per routine clinical practice with analyses at baseline and month 3. Investigator tolerance to fluid and preferred treatment regimen were self-reported at baseline. Safety outcomes include the incidence, severity, duration and outcome of ocular and non-ocular adverse events.

Results: 98% of 321 eyes had received prior anti-VEGF or FAR treatment. In eyes switched from anti-VEGF to FAR at enrolment, mean visual acuity (nAMD [n = 56]: 59 to 65 letters; DMO [n = 25]: 70 to 72 letters) and central subfield thickness improved (nAMD [n = 44]: 314 to 264 µm; DMO [n = 19]: 311 to 269 µm). No safety events were reported. Physicians (N = 22) preferred treat-and-extend (nAMD: 83%; DMO 61%); most tolerated fluid for DMO (61%), but not nAMD (39%).

Conclusion: Interim results show benefit of FAR in switch patients in routine clinical practice and fluid tolerance variability among physicians.

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DRUG EXPOSURE AS A PREDICTOR IN DIABETIC RETINOPATHY RISK PREDICTION MODELS; A SYSTEMATIC REVIEW AND META-ANALYSIS

Corresponding author: Maria Anna Bantounou, orcid.org/0000-0002-0529-2845

Maria Anna Bantounou¹, Tulika Ashok Kumar Nahar², Niraj Kumar³, Mintu Nath⁴, Phyo K Myint⁴, Sam Philip^{1,5}

¹School of Medicine, University of Aberdeen, Aberdeen, UK; ²Queen's University Belfast School of Medicine, Belfast, UK; ³University College London, London, UK; ⁴Institute of Applied Health Sciences, University of Aberdeen, Aberdeen, UK; ⁵Grampian Diabetes Research Unit, Diabetes Centre, Aberdeen Royal Infirmary, Aberdeen, UK

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Introduction: Various diabetic retinopathy (DR) prediction models include drugs as predictors, however variation in how drug exposure is coded exists. This review aims to assess drug exposure handling, identify the drugs associated with DR by performing a network meta-analysis and determine which drugs enhance model performance by undertaking a meta-analysis.

Method: We searched EMBASE, MEDLINE and SCOPUS from inception to July 2022. We evaluated the quality of studies using the Prediction model Risk of Bias Assessment Tool (PROBAST). We conducted network meta-analysis and a meta-analysis to estimate the odds ratio (OR) and pooled C-statistic, respectively (PROSPERO: CRD42022349764).

Results: We included 19 records of 396,844 participants with type 1 or 2 diabetes, of which 27,374 (6.7%) had DR. Drugs included in the DR models were: insulin (n = 15), antihypertensive drugs (n = 3), oral antidiabetics (n = 6), lipid-lowering drugs (n = 5), antiplatelets (n = 1). Drug exposure was modelled primarily as a categorical variable (n = 14 studies). Insulin was associated with an increased risk of DR (OR = 3.09; 95%-CI: 1.66, 5.76). Models that included insulin (n = 5) had a higher pooled C-statistic (C-statistic = 0.85, 95%-CI: 0.78, 0.91), compared to models (n = 4) incorporating a combination of drugs alongside insulin (C-statistic = 0.82, 95%-CI: 0.75, 0.90), as

well as models (n = 2) not including insulin (C-statistic = 0.69, 95%-CI: 0.61, 0.77). These findings were consistent in the external validation meta-analysis (n = 3). Limitations include high risk of bias and heterogeneity.

Conclusion: This is the first review assessing DR prediction models according to drug exposure. Exposure to insulin was associated with a higher risk of DR and correlated with improved model performance. However, due to suboptimal drug handling, associations between other drugs and model performance may have been overlooked. We propose the routine evaluation of drugs as variables in DR prediction models.

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OCT FINDINGS IN OCULAR DECOMPRESSION RETINOPATHY. A REVIEW OF THE LITERATURE

Corresponding author: Ahmed Bilal, orcid.org/0000-0002-6846-5731

Ahmed Bilal¹, Ramya Sular Sundararajan², Mai Mair Nasser², Kevin Gallagher²

¹University Hospital of Wales, Cardiff & Vale University Health Board, UK; ²Cwm Taf Morgannwg University Health Board, Llantrisant, UK

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Introduction: Ocular decompression retinopathy (ODR) is a rare complication occurring after rapid intraocular pressure lowering interventions. A 43 year old woman developed ODR after medical management of acute angle closure. OCT scanning showed macular oedema with intraretinal fluid localised to Henle fibre layer, central subretinal fluid, a bright hyperreflective line at the level of the outer plexiform layer (so-called prominent middle-limiting membrane (pMLM) sign) and a central bacillary layer detachment, which prompted us to review the literature for the OCT signs in ODR.

Method: A Medline search for "ocular decompression retinopathy" returned 92 articles. Inclusion criteria for review were articles describing cases of ODR where OCT findings were reported or where the OCT image was included. 17 articles describing 19 eyes met the inclusion criteria and were reviewed. OCTs were examined and findings recorded.

Results: Intraretinal fluid was the most common finding in 53% of eyes. Subretinal fluid was found in 26% of eyes. Haemorrhages(sub-ILM, intraretinal, subretinal, Henle fibre layer) were noted in 6 eyes(32%), the most common location being the Henle Fibre layer(4 eyes, 21%). Hyper-reflectivity of retinal layers suggestive of retinal capillary ischaemia was noted in 3 eyes(including pMLM sign, paracentral acute middle maculopathy(-PAMM) and acute macular neuroretinopathy(AMN).

Conclusion: ODR is rare and most cases in the literature do not include OCT findings. OCT features include Henle fibre layer haemorrhage and pMLM sign, both associated with venous outflow impairment at the level of the deep capillary plexus. Other OCT signs(PAMM/AMN) indicate ischaemia at the level of the intermediate or deep capillary plexuses. OCT features can provide insight into the pathophysiology of ODR.

P113. Abstract not included

P114

THE USE OF ANTI-VEGF AGENTS IN PREGNANCY – WHICH AGENTS SHOULD WE USE?

Corresponding author: Ffion Brown, orcid.org/0000-0001-5818-9756

Ffion Brown¹, Ahmed Bilal², Ahmed Al-Janabi³, Kevin Gallagher²

¹Royal Gwent Hospital, Newport, UK; ²Royal Glamorgan Hospital, Llantrisant, UK; ³Betsi Cadwaladr University Health Board, Abergele, UK

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Introduction: VEGF plays an important role in embryogenesis, implantation and placentation. Disruption to VEGF pathways has been implicated in several obstetric complications including pre-eclampsia and miscarriage. Drug manufacturers and regulators advise to avoid anti-VEGF use in pregnancy unless the benefits outweigh the risks. There is limited human safety data for anti-VEGF agents in pregnancy.

Our aim is to apply pharmacokinetic and animal safety data to further understand the risk-benefit analysis of anti-VEGF in pregnancy and to inform agent selection.

Method: PubMed literature search.

Results:

Placental Transfer

The neonatal Fc receptor (FcRn) mediates transfer of antibodies from mother to foetus. In contrast to aflibercept and bevacizumab which possess an Fc fragment, ranibizumab does not and hence is relatively impermeable to placental transfer. Faricimab contains an Fc fragment with mutations to eliminate FcRn binding, theoretically reducing placental transfer.

Systemic half-life

FcRn binding prolongs the half-life of drugs in circulation. Hence, ranibizumab and faricimab have shorter half-lives (7 days) compared to aflibercept (11 days) and bevacizumab (20 days).

Systemic VEGF suppression

Suppressing maternal VEGF levels may also adversely affect pregnancy. Pharmacokinetic studies demonstrate aflibercept and bevacizumab significantly reduce systemic VEGF whereas ranibizumab and faricimab do not.

Animal safety studies

Aflibercept and bevacizumab were teratogenic in rabbits at all doses given. However, ranibizumab given in monkeys had no teratogenic effect. Faricimab given in monkeys has no foetal adverse effects but a higher rate of miscarriage compared to controls.

Conclusion: Pharmacokinetic and animal safety data should be considered when deciding to use anti-VEGF in pregnancy. Ranibizumab and faricimab may be preferable due to their reduced potential for placental transfer, shorter half-lives, minimal effects on systemic VEGF concentrations and better animal safety study outcomes.

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FARICIMAB REDUCES MACULAR LEAKAGE VS AFLIBERCEPT IN PATIENTS WITH DMO

Corresponding author: Ben Burton, orcid.org/0000-0001-9579-9078

Ben Burton¹, Kara Gibson², Florie Mar³, Tracey Wang⁴, Jeffrey Willis³

¹James Paget University Hospital NHS Foundation Trust, Great Yarmouth, UK; ²Roche Products Ltd., Welwyn Garden City, UK; ³Genentech, Inc., San Francisco, CA, USA; ⁴F. Hoffmann-La Roche Ltd., Mississauga, Canada

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advisor and been involved with commercial research with Bayer and Roche Kara Gibson Financial Disclosures: Employee: Roche Florie Mar Financial Disclosures: Employee: Genentech, Inc. Tracey Wang Financial Disclosures: Employee: Roche Jeffrey Willis Financial Disclosures: Employee: Genentech, Inc.

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Introduction: Increased vascular permeability is a hallmark feature of diabetic macular oedema (DMO). In preclinical mouse models, dual angiopoietin-2 (Ang-2)/vascular endothelial growth factor-A (VEGF-A) inhibition was associated with greater vascular leakage reductions versus Ang-2 or VEGF-A inhibition alone, suggesting synergistic actions of Ang-2 and VEGF-A. This analysis evaluated if dual Ang-2/VEGF-A inhibition with faricimab improves macular leakage over VEGF inhibition alone with aflibercept in patients with DMO.

Method: In the phase 3 YOSEMITE/RHINE trials (NCT03622580/03622593), patients with centre-involving DMO. were randomized 1:1:1 to 6.0mg faricimab every 8 weeks (Q8W) or according to a treat-and-extend (T&E) regimen, or 2.0mg aflibercept Q8W. This analysis included data from the first 16 weeks, in which all patients received assigned study drug Q4W. The faricimab Q8W and T&E arms were pooled as they received the same dosing regimen during this period. Outcomes included macular leakage area on fluorescein angiography and the proportion of patients with minimal to no macular leakage (0–1mm²).

Results: YOSEMITE/RHINE included 1216 faricimab patients and 593 aflibercept patients. Median baseline macular leakage area was similar between faricimab (24.58mm²) and aflibercept arms (25.64mm²). At week 16, median macular leakage area was lower in the faricimab versus aflibercept arms (3.59 vs 7.62mm²). A greater proportion of faricimab patients (28.4%) demonstrated minimal to no leakage at week 16 compared with aflibercept patients (15.2%).

Conclusion: Dual Ang-2/VEGF-A inhibition with faricimab resulted in a greater reduction in macular leakage and a larger proportion of patients with DMO achieving minimal to no leakage versus aflibercept. These findings suggest that dual Ang-2/VEGF-A inhibition provides greater vascular stability, which may contribute to faster fluid resolution and extended durability observed with faricimab versus aflibercept.

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THE EFFECT OF FOUR LOADING INTRAVITREAL AFLIBERCEPT INJECTIONS ON MACULAR FLUID IN TREATMENT-NAÏVE NEOVASCULAR AGE-RELATED MACULAR DEGENERATION

Corresponding author: Shruti Chandra, orcid.org/0000-0002-2634-9775

Shruti Chandra^{1,2}, Raffaele Raimondi², Alicia Lim¹, Amy Mohan⁴, Sneha Melmane⁴, Geeta Menon⁴, Manju Chandran⁴, Sobha Sivaprasad¹, Benjamin J L Burton⁶, Ajay Kotagiri³

¹Moorfields Eye Hospital NHS Foundation Trust, London, UK; ²University College London, London, UK; ³South Tyneside and Sunderland NHS Foundation Trust, UK; ⁴Frimley Health NHS Foundation Trust, Surrey, UK; ⁵James Paget University Hospitals NHS Foundation Trust, Great Yarmouth, UK

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Introduction: Purpose: To evaluate the effect of four versus three loading aflibercept injections on macular fluid resolution and visual acuity (VA) in exudative neovascular AMD (nAMD).

Method: Multicentre, retrospective cohort study of treatment naïve nAMD eyes undergoing 3 versus 4 loading doses pf aflibercept. Change in VA and fluid resolution on optical coherence tomography (OCT), were evaluated at 8 weeks post loading. Primary outcome was proportion of patients with no intraretinal (IRF) and/or subretinal (SRF) fluid IRF at central 1mm and whole macula at 8 weeks after loading. Data were summarised with mean $\pm$ SD for continuous variables, and n (%) for categorical variables.

Results: Data from 995 was analysed (355 patients - 4 loading doses and 640 - 3 loading doses). At 8 weeks post 4 loading doses proportion of eyes with no IRF/SRF, no IRF and no SRF were 62.8%, 88.7% and 79.2% at fovea versus 56.1%, 87.9% and 69.9% in the whole macula, respectively. Fluid resolution at both fovea and macula were significantly higher in eyes with 4 loading injections versus 3 ($p = 0.0001$). The mean VA change was $+4.0 (\pm 11.3)$ and $+5.4 (\pm 13.3)$ letters for 3 and 4 loading doses groups ($p = 0.09$).

Conclusion: Four loading dose injections of aflibercept results in higher proportion of eyes with total fluid resolution in the central subfield and total macular scan when compared to those receiving 3 loading dose injections at 8 weeks post loading phase. However, the better drying effect of 4th loading dose does not translate into better short-term VA outcomes.

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GREATER MASQUERADE: A RETROSPECTIVE ANALYSIS ON THE PRESENTATION OF OCULAR SYPHILIS

Corresponding author: Chibueze Chidi-Egeolu, orcid.org/0009-0006-2645-1047

Chibueze Enoch Divine-Gift Chidi-Egeolu¹, Annie See-Wing Ng¹, Thomas Nash¹, Matthew Stahl¹, Theresa Richardson¹

¹Imperial College London, London, UK

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Introduction: Ocular syphilis, a recognized manifestation of *Treponema pallidum*, emerges as a diagnostic challenge within ophthalmology. The diversity of its manifestations, including uveitis, retinitis, optic neuritis, and vasculitis, can potentially delay and skew diagnoses. Timely management is paramount to preventing severe sequelae. We aimed to better understand the

presenting features of ocular syphilis in patients presenting to the Western Eye Hospital.

Method: A retrospective audit from January 2010 to December 2023 was conducted to review electronic records of patients with documented ocular syphilis diagnoses at the Western Eye Hospital. Data on: demographics, ocular manifestations at presentation, visual acuity at presentation, initial clinical diagnoses, time from presentation onset to formal diagnosis, and documentation of any risky sexual behaviour was collected.

Results: 10 patients had a documented diagnosis of ocular syphilis (median age: 60 [48-87] years). Six out of the eight patients with documented initial diagnosis were misdiagnosed. The two remaining patients, correctly diagnosed, both had a previous syphilis diagnosis. The mean time to formal diagnosis was 55.1 days (± 75.8), with no significant differences seen between genders (Males: 48.5 ± 77.0 vs Females: 75.0 ± 97.6 , $p = 0.702$). Only two patients had any documented sexual history. Presenting symptoms included headache, visual disturbances, and painful eyes.

Conclusion: The heterogeneity of clinical phenotypes, such as uveitis, retinopathies, and optic nerve pathology, demands that ophthalmologists keep syphilis in mind as a differential diagnosis. This is particularly pertinent when considering presentations in older patients or those without perceived risky sexual behaviour and necessitates proactive assessment of patients' sexual histories. Further analysis regarding the consequences of misdiagnosis is warranted, as prompt diagnosis and treatment can prevent adverse outcomes.

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IMPACT OF FARICIMAB VS AFLIBERCEPT ON EPIRETINAL MEMBRANE FORMATION OVER 2 YEARS IN EYES WITH DMO IN THE YOSEMITE/RHINE PHASE 3 TRIALS

Corresponding author: Simona Degli Esposti, orcid.org/0000-0002-7476-1190

Simona Degli Esposti¹, Yuichiro Ogura², Gábor Deák³, Kara Gibson⁴, Ursula Schmidt-Erfurth⁵, Tracey Wang⁶, Peter Westenskow⁷, Jeffrey Willis²

¹Moorfields Eye Hospital NHS Foundation Trust, London, UK; ²Genentech, Inc., San Francisco, USA; ³Vienna Reading Center, Medical University of Vienna, Vienna, Austria; ⁴Roche Products Ltd., Welwyn Garden City, UK; ⁵Medical University of Vienna, Austria; ⁶F. Hoffmann-La Roche Ltd., Mississauga, Canada; ⁷F. Hoffmann-La Roche Ltd., Basel, Switzerland

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Yuichiro Ogura, MD, PhD Financial Disclosures: Consultant: Alcon Japan, Apellis, Astellas, Bayer, Boehringer Ingelheim, Chengdu Kanghong, Chugai Pharmaceutical Co., Ltd, HOYA, Iveric Bio, Kyoto Drug Discovery & Development, Novartis, Senju, and Wakamoto; Lecture fees: Bayer, KOWA, NIKON Healthcare, Novartis, Santen, Sanwa Kagaku, Topcon, and ZEISS; Employee: Genentech, Inc.

Gábor Deák, MD, PhD Financial Disclosures: None
Kara Gibson, PhD Financial Disclosures: Employee: Roche Products Ltd.

Ursula Schmidt-Erfurth, MD Financial Disclosures: None
Tracey Wang, MSc Financial Disclosures: Employee: F. Hoffmann-La Roche Ltd.

Peter Westenskow, PhD Financial Disclosures: F. Hoffmann-La Roche Ltd. (Employee)

Jeffrey Willis, MD, PhD Financial Disclosures: Employee: Genentech, Inc.

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Introduction: We compared the 2-year incidence of epiretinal membrane (ERM) formation in eyes with diabetic macular oedema (DMO) treated with faricimab vs aflibercept.

Method: In the phase 3 YOSEMITE/RHINE trials (NCT03622580/03622593), patients were randomized 1:1:1 to faricimab 6.0mg every 8 weeks (Q8W) after 6 initial Q4W doses, faricimab 6.0mg treat-and-extend (T&E, Q4W up to Q16W based on protocol-defined criteria) after 4 initial Q4W doses, or aflibercept 2.0mg Q8W after 5 initial Q4W doses. Presence of ERMs was an exclusion criterion for both trials. Masked, independent ERM grading was conducted at the Central Reading Centres (Duke, Vienna) at baseline and weeks 16, 48, 52, 56, 92, 96, and 100.

Results: Over 2 years, ERMs developed in 23/619 (3.7%), 31/618 (5.0%) and 45/604 (7.5%) of faricimab Q8W, T&E, and aflibercept Q8W eyes, respectively. Risk of ERM formation was lower for faricimab Q8W vs aflibercept (HR 0.48; 95%CI 0.29-0.80; nominal $P=0.0039$). The corresponding data for T&E patients vs aflibercept were HR 0.65 (95%CI 0.41-1.03; nominal $P=0.0644$). Using a treatment agnostic approach, at year 2, eyes that developed ERMs had worse BCVA and CST than those without ERMs. Eyes that developed ERMs also had higher rates of intra- (63/83; 75.9% vs 649/1336; 48.6%) and sub-retinal fluid (9/85; 10.6% vs 47/1372; 3.4%) at 2 years than those without ERMs. Among faricimab-treated eyes in the T&E arm, dosing was extended to Q16W in a larger proportion of eyes without ERMs (332/516; 64.3%) than those with ERMs (7/28; 25%).

Conclusion: We demonstrated a potential anti-fibrotic effect of faricimab vs aflibercept in DMO eyes. ERMs were associated with worse visual acuity and anatomic outcomes, and a lower rate of dose extension to Q16W.

P119

TREATMENT OUTCOMES WITH ANTI-VEGF THERAPY FOR RETINAL VEIN OCCLUSION: AN ANALYSIS OF REAL-WORLD DATA FROM 16 UK CENTRES

Corresponding author: Christiana Dinah, orcid.org/0000-0002-0815-4771

Christiana Dinah¹, Melanie Dodds², Pablo Arrisi³, Liliana P Paris⁴, Alice V R Lake², Patrick Nielsen², John Holdsworth², Insaf Saffar³, Richard Pohler³, Jeannette Raine³, Gloria C Chi⁴

¹London North West University Healthcare NHS Trust, London, UK;

²Medisoft Ltd, Leeds, UK; ³Roche Products Ltd, Welwyn Garden City, UK; ⁴Genentech, Inc., San Francisco, CA, USA; ⁵F. Hoffmann-La Roche Ltd., Basel, Switzerland

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Introduction: To characterize treatment patterns and long-term visual acuity (VA) outcomes in patients with macular oedema secondary to retinal vein occlusion (RVO) treated in routine clinical practice in the UK.

Method: This was a retrospective observational study using Medisoft electronic health record data from 2013-2023 contributed by 16 UK National Health Service trusts. Patients with treatment-naïve eyes with macular oedema secondary to branch RVO (BRVO), central RVO (CRVO), or hemi-RVO (HRVO) that received ≥ 1 intravitreal anti-vascular endothelial growth factor (VEGF) injection on or after diagnosis were included.

Results: 3511 eyes from 3465 BRVO patients and 3568 eyes from 3514 C/HRVO patients were included. Of all RVO eyes, 57.4% were followed-up for ≥ 24 months and 19.1% for 60 months after index. BRVO and C/HRVO eyes received 4.3 and 4.1 anti-VEGF injections from months 0-6 and 2.1 and 2.0 injections from months 7-12. Mean injection numbers in BRVO and C/HRVO eyes decreased further from 3.4 and 3.2 in year 2 to 2.5 and 2.4 in year 5. Mean VA change (95%CI) from index injection at month 3 was 11.1 (10.6,11.6) for BRVO and 13.8 (13.1,14.4) ETDRS letters for C/HRVO. By month 60, mean VA change (95%CI) decreased to 8.3 (6.7,10.0) for BRVO and 7.0 (4.8,9.2) letters for C/HRVO. VA changes were lower in eyes with low vs high injection numbers. Mean VA change from index injection to 12 months in BRVO and C/HRVO eyes with low (<4), medium (4-8), and high (>8) injection numbers in year 1 were 7.5 and 6.1, 11.9 and 11.3, and 13.4 and 15.5 letters.

Conclusion: These data from the UK show that anti-VEGF injections for macular oedema secondary to RVO improve vision.

P120. Abstract not included

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DUAL PATHWAY INHIBITION WITH FARICIMAB FOR PREVIOUSLY TREATED NEOVASCULAR AGE RELATED MACULAR DEGENERATION AND DIABETIC MACULAR OEDEMA: GUIDANCE FROM A UK PANEL OF RETINA SPECIALISTS

Corresponding author: Louise Downey, orcid.org/0000-0001-5944-3437

Louise Downey¹, Sobha Sivaprasad², Romi R Chhabra^{3,4}, Clare Bailey⁵, Soma Chakrabarti⁶, Andrew Lotery⁷, Jignesh Patel⁸, Julie Silvestri⁹, Sarah-Lucie Watson¹⁰, Gwyn Williams¹¹, Samer Elsherbiny¹², Saima Khokhar¹³, Antony Parker¹³

¹Hull Royal Infirmary, Hull, UK; ²Moorfields Eye Hospital NHS Foundation Trust, London, UK; ³Manchester Royal Eye Hospital, Manchester, UK; ⁴University of Manchester, Manchester, UK; ⁵University Hospitals Bristol and Weston NHS Foundation Trust, Bristol, UK; ⁶NHS Greater Glasgow & Clyde, Glasgow, UK; ⁷Southampton Eye Unit and Faculty of Medicine, University of Southampton, Southampton, UK; ⁸Colchester District General Hospital, Colchester, UK; ⁹Royal Victoria Hospital, Belfast, UK; ¹⁰Royal Berkshire NHS Foundation Trust, Berkshire, UK; ¹¹Swansea Bay University Health Board, Swansea, UK; ¹²South Warwickshire University NHS Foundation Trust, Warwick, UK; ¹³Roche Products Ltd., Welwyn Garden City, UK

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Samer Elsherbiny: Financial Disclosures: Honoraria and Travel Support: AbbVie, Alimera Sciences, Bayer, Novartis, Topcon, Roche.

Jignesh Patel: Financial Disclosures: Consultancy: Bayer, Roche, Travel Support: Bayer, Roche.

Giuliana Silvestri: Financial Disclosures: Consultancy: Bayer, Roche.

Sarah-Lucie Watson: Financial Disclosures: Consultancy and Honoraria: Roche, Novartis, Bayer.

Gwyn Williams: Financial Disclosures: Honoraria: Roche.

Antony Parker: Financial Disclosures: Employee: Roche.

Saima Khokhar: Financial Disclosures: Employee: Roche.

Andrew Lotery: Financial Disclosures: Travel support: Roche, Equity in Eyebio.

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Introduction: Some eyes with neovascular age-related macular degeneration (nAMD) and centre-involving diabetic macular oedema (DMO) fail to respond sufficiently or lose response over time to standard of care anti-vascular endothelial growth factor (anti-VEGF) therapy. We explore clinical scenarios for switching to dual angiopoietin-2/VEGF-A inhibition with faricimab (Vabysmo, Roche Products Ltd).

Method: A national steering group meeting of UK medical retina specialists was held in London on 27 October 2023. Clinician practices and experience were reviewed together with pivotal clinical trial data and early findings from real world settings. The meeting was facilitated and funded by Roche Products Ltd.

Results: While there is no unequivocal standardised protocol for identifying suboptimal response, the authors review relevant clinical biomarkers of disease activity commonly used in clinical practice to determine patient response and guide treatment decisions. Common reasons identified for considering a change of treatment were lack of efficacy, suboptimal anatomic or visual improvement, and insufficient durability. The authors outline switch pathways, with monthly or interval-matched initiation, that may be integrated into current faricimab treat-and-extend clinical pathways for treating patients with nAMD and DMO. Switch options for suboptimal response to ranibizumab biosimilar, as well as approaches for bilateral involvement and patient counselling and consent are also reviewed.

Conclusion: Clinicians may consider a treatment switch to faricimab in nAMD and DMO patients who have suboptimal disease control or insufficient durability on initial anti-VEGF monotherapy.

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FARICIMAB IMPROVES FLUID PARAMETERS IN PATIENTS WITH NAMO

Corresponding author: Richard Gale, orcid.org/0000-0002-8847-044X

Richard Gale^{1,2}, Aachal Kotecha³, Jeffrey R Willis⁴, Audrey Souverain⁵, Yevgeniy (Eugene) Shildkrot⁴, Philippe Margaron⁵

¹Hull York Medical School, University of York, York, UK; ²York and Scarborough Teaching Hospitals NHS Foundation Trust, York, UK;

³Roche Products Ltd, Welwyn Garden City, UK; ⁴Genentech Inc, San Francisco, CA, USA; ⁵F. Hoffmann-La Roche Ltd, Basel, Switzerland

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Aachal Kotecha - Financial Disclosures: Employee: Roche Products Ltd.

Jeffrey R. Willis - Financial Disclosures: Employee: Genentech, Inc.

Audrey Souverain - Financial Disclosures: Employee: F. Hoffmann-La Roche Ltd.

Yevgeniy (Eugene) Shildkrot - Financial Disclosures: Employee: Genentech, Inc.

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Introduction: The phase 3 TENAYA/LUCERNE (NCT03823287/NCT03823300) trials demonstrated that faricimab, a dual angiopoietin-2 (Ang-2)/vascular endothelial growth factor-A (VEGF-A) inhibitor, maintained vision with extended treatment durability and fewer injections versus aflibercept in patients with neovascular age-related macular degeneration (nAMD). We evaluated fluid outcomes, including time to first absence of fluid, in TENAYA/LUCERNE.

Method: Treatment-naïve patients in TENAYA/LUCERNE (N = 1329) were randomised 1:1 to faricimab 6.0mg up to every 16 weeks (Q16W; n = 665), with fixed up to Q16W dosing, based on protocol-defined criteria, in the first year and a treat-and-extend regimen in the second, or aflibercept 2.0mg Q8W (n = 664). Change in CST and absence of SRF and IRF during the matched-dosing period (weeks 0–12), and time to first absence of SRF and IRF up to week 48 were assessed in the faricimab versus aflibercept arms.

Results: During the matched-dosing period, faricimab resulted in a greater reduction in CST from baseline versus aflibercept (week 12: -145 vs -133 µm; nominal $P \leq 0.0001$). Likewise, a larger proportion of patients achieved absence of SRF and IRF with faricimab versus aflibercept (week 12: 77% vs 67%; nominal $P \leq 0.0001$). In patients with SRF and IRF at baseline, first absence of IRF and SRF was achieved faster and with fewer injections with faricimab versus aflibercept (75th percentile of time to first absence of IRF and SRF was week 8 with faricimab versus week 12 with aflibercept; corresponding median number of injections: 2 vs 3).

Conclusion: In patients with nAMD in TENAYA/LUCERNE, dual inhibition of Ang-2 and VEGF-A with faricimab resulted in a greater improvement in anatomical outcomes during the matched-dosing period, as well as more rapid anatomical improvements with fewer injections.

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VISUAL AND ANATOMICAL OUTCOME OF PATIENTS TREATED WITH SUBTHRESHOLD MICROPULSE (577NM) LASER FOR CHRONIC CENTRAL SEROUS CHORIORETINOPATHY

Corresponding author: Mallika Goswami, orcid.org/0009-0000-7954-5043

Mallika Goswami¹, Oluwabunmi Ashano¹

¹Ipswich Hospital, East Suffolk and North Essex Foundation Trust, Ipswich, UK

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Introduction: CSCR (Central Serous Chorio Retinopathy) is a multifactorial disease without a known mechanism of action. It is hypothesized to occur because of choroidal hyperpermeability and secondary RPE dysfunction leading to serous retinal detachment. Subthreshold micropulse laser (SML) uses subthreshold energy that selectively targets RPE cells without inducing chorioretinal damage or scarring and has been suggested as a treatment option.

Method: A one year retrospective observational study of visual and anatomical outcomes in 12 eyes of 12 patients who received subthreshold 577nm micropulse laser treatment for chronic CSR.

Inclusion criteria:

CSR for a minimum of 6 months

No previous treatment

Presence of subretinal fluid secondary to chronic CSR when the treatment was performed.

Follow up for visual acuity gain, central retinal thickness and side effects at month 6, 12 and 18.

Results: 9 eyes (75.0%) had one session of laser and 3(25.0%) received two sessions. 2 eyes received anti-VEGF injections for associated CNV.

Mean CRT was noted to be reduced from 343 μ at baseline to 314 μ at month 6 and 261 μ (-59.33 $\pm$ 11.20) at month 12.

6 (50%) patients had complete resolution of SRF at the last follow-up, 2(16%) had qualified resolution of SRF whilst 4(33%) had persistent SRF at last follow-up.

6 (50%), 4 (44%) and 4(66%) patients gained at least one line of VA at 6, 12 and 18-months follow-up.

No side effects were reported following the treatment.

Conclusion: Acute CSCR can be observed and is self-resolving in most cases, but progression to chronic CSCR warrants treatment. Although studies show photodynamic therapy to have better results, our study shows that subthreshold laser can be considered as an option in the management of chronic CSCR with good visual and anatomical outcome.

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COMPARISON OF 90° FIELD CLARUS 700 IN STAGED-MYDRIATIC MODE AND MYDRIATIC TWO 45° FIELD DIGITAL IMAGING IN THE NHS DIABETIC EYE SCREENING

Corresponding author: Marta H Gruszka-Goh, orcid.org/0009-0002-8742-9135

Marta Gruszka-Goh^{1,2}, Anthony Vukic¹, Paul Galsworthy¹, Scott Vallance¹, Peter H Scanlon^{3,4}

¹Gloucestershire Hospitals NHS Foundation Trust, Cheltenham, UK;

²The Royal College of Ophthalmologists' National Ophthalmology Audit, London, UK; ³Nuffield Department of Clinical Neuroscience, University of Oxford, Oxford, UK; ⁴University of Gloucestershire, Gloucester, UK

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Introduction: The purpose of the study was to determine if the Clarus 700 (Clarus) was safe to use in the staged-mydriatic mode in the Diabetic Eye Screening Programme (DESP).

Method: Trial participants were 1497 individuals (2993 eyes) aged 15 to 94 with diabetes mellitus participating in DESP or virtual eye clinics for delayed hospital follow up.

Clarus photographs were taken before dilation and compared to two-field 45 degrees mydriatic digital photography. When image quality was poor or pupil size less than 3mm, photos were taken with Clarus before and after dilation. If the non-mydriatic images were gradable, they were used in comparison to the mydriatic digital images. To compare the same areas, Clarus photos were graded with two 45° field overlay using DESP grading criteria.

Results: Out of 2993 photos taken using the Clarus, 3.3% with 95% CI(2.7%-4.0%) were ungradable pre-dilation and 1.7% (95% CI: 1.3%-2.2%) with staged mydriasis.

Sensitivity and specificity in the central area comparing Clarus with the two 45-degree field mydriatic digital reference were calculated for 1,486 patients (2,905 eyes). The sensitivity and specificity for any diabetic retinopathy (DR) were 94.2% (95% CI: 92.9%-95.3%) and 87.3% (95% CI: 85.4%-89.0%) respectively. The sensitivity and specificity for referable DR were 86.0% (95% CI: 82.9%-88.8%) and 92.8% (95% CI: 91.7%-93.8%) respectively.

Conclusion: The ungradable image rate of 3.3% for Clarus without mydriasis in a predominantly White Caucasian population was lower than reports for digital cameras and it performed accurately in the detection of diabetic retinopathy in the central area.

There was a steep learning curve for the grading team with some novel presentations of acquisition artefacts and colour differences in DR features.

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REAL WORLD EFFICACY AND DURABILITY OF FARICIMAB IN SUB-OPTIMAL RESPONDER PATIENTS WITH NEOVASCULAR AGE-RELATED MACULAR DEGENERATION

Corresponding author: Jimena Soto Hernaez, orcid.org/0009-0001-4787-0666

Jimena Soto Hernaez¹, Christine Goodchild^{2,3}, Clare Bailey², Eslam Ahmed², Serena salvatore²

¹University of Bristol, Bristol, UK; ²Bristol Eye Hospital, University Hospitals Bristol and Weston NHS Foundation Trust, Bristol, UK;

³Royal Victoria Eye and Ear Hospital, Dublin, Ireland

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Introduction: Faricimab is the first and only bispecific molecule with targets Anti-VEGF and Ang-2. It gain approval by the MHRA in May 2022 and our unit started using it by October 2022 for nAMD. The purpose of this study is to evaluate the efficacy and treatment interval extension of faricimab in this patient cohort.

Method: This is a single centre retrospective study. Patients with neo-vascular AMD who were sub-optimal responders and received at least 1 injection of faricimab were included.

Results: 266 eyes of 215 patients were examined. Mean age was 81 years old and 57% of our patients were women. Prior to

switching, all our patients were on aflibercept and 80% were on less than 6 weekly injections. Mean follow up period was 8.5 months (range 7–10 months)/ average of 6.3 injections (range 1–9). Best recorded visual acuity (BRVA) was maintained throughout at 68 letters. A maximum reduction in central subfield thickness was noted after third loading dose at 248 μm ($\pm 79\mu\text{m}$, $p = 0.000$). Intraretinal (IRF) and subretinal fluid (SRF) was seen initially in 33% and 78% of patients respectively. Maximum resolution of fluid was seen after 2 injections of faricimab where only 17% of patients had IRF and 28% had SRF ($p = 0.06$, $p0.000$). At the end of the loading course only 30% of patients required injection every 6 weeks or less. Out of the 1680 injections given, ocular adverse event was seen in 4 patients.

Conclusion: Our study shows that faricimab maintain visual acuity, improve anatomical parameters and extension of treatment interval. The safety of faricimab is consistent published data. We will have 1 year data for presentation in the near future.

P126

HOW PATIENTS CAN DESIGN THEIR OWN INFORMATION LEAFLET! A QUALITY IMPROVEMENT PROJECT ON RETINAL VEIN OCCLUSION

Corresponding author: Sarah Hignett, orcid.org/0000-0002-1340-9193

Sarah Hignett¹, Michael Williams²

¹Northern Ireland Medical and Dental Training Agency (NIUMDTA), Belfast, UK; ²Centre for Medical Education, Queens University Belfast, Belfast, UK; ³Belfast Health and Social Care Trust, Belfast, UK

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Introduction: Retinal vein occlusion (RVO) is a common cause of sudden vision loss. Previous studies have shown there is a misunderstanding about the condition amongst patients. This misunderstanding affects quality of life, leading to increased anxieties about prognosis, available treatments and how this condition will affect activities of daily life such as ability to work/drive. This quality improvement project was designed to measure and improve patients understanding of RVO.

Method: A questionnaire was completed by patients receiving injections for RVO at the Macular Clinic in the Mater Hospital, Belfast. The questionnaire involved three knowledge-based questions, satisfaction scores based on current information they had been provided on RVO, and finally a free text section for any specific questions patients had. Thematic analysis was conducted on the free text, grouping questions under three main themes (prognosis, treatment logistics and self-management). These themes were used to write a new patient information leaflet, incorporating all the questions listed by patients. The new leaflet was distributed to patients and the same questionnaire was repeated.

Results: In the first cycle, a total of 30 patients filled in the questionnaire, after the leaflet was introduced, a further 18 patients filled in the questionnaire. Initially, participants scored a mean of 56% on MCQ questions on RVO, which improved to 87% after the leaflet was introduced. Initially, 83% of participants had unanswered questions about RVO, however after the leaflet was distributed, this reduced to 33%.

Conclusion: This type of data collection can be used to better understand patients' anxieties/questions regarding their own health conditions. Furthermore, in this way patients can be involved in writing their own information leaflets, ensuring they are patient centred.

P127

VISUAL OUTCOMES OF RETINAL VEIN OCCLUSION IN PATIENTS YOUNGER THAN 50 YEARS OF AGE – A CLINICAL AUDIT

Corresponding author: Joshua Himsworth, orcid.org/0009-0005-8401-4761

Joshua Himsworth¹, Balasubramanian Ramasamy¹, Rose Herbert¹

¹Wirral University Teaching Hospital NHS Foundation Trust, Liverpool, UK

Disclosures: No conflicts of interest and no funding

Introduction: Aim: To review the visual outcomes and incidence of complications in patients younger than 50 years, presenting with Retinal Vein Occlusion (RVO).

Method: Electronic patient records were retrospectively reviewed. Visual acuity (VA) at presentation and final follow-up and complications including macular oedema, retinal ischaemia, and secondary glaucoma were recorded and analysed.

Results were compared to The Royal College of Ophthalmologists (RCOphth) 2022 RVO guidelines.

Results: Eighteen patients under the age of 50 were identified with RVO in the audit period of January 2016 to present. 8 were Branch RVO (BRVO) and 10 Central RVO (CRVO).

Mean baseline VA was 0.94 ± 0.52 logMAR in the CRVO and 0.68 ± 0.38 logMAR in the BRVO group. Mean final VA was 0.32 ± 0.27 logMAR in the CRVO and 0.69 ± 0.67 logMAR in BRVO group.

In the BRVO group, four patients required sectoral pan-retinal photocoagulation (PRP) for ischaemia and one required Ozurdex implant, whilst two patients with CRVO required sectoral PRP and two required Ozurdex implants. One CRVO patient experienced raised intra-ocular pressure secondary to steroid response. There were no cases of neovascular glaucoma.

Conclusion: The RCOphth RVO guideline states that 20% of patients under 50 years with RVO develop poor visual outcomes with severe neovascular complications, whilst most CRVO eyes recover to better than 0.3 logMAR with anti-VEGF injections. 78% maintain or improve vision at follow-up.

Our data showed that VA improved to 0.3 logMAR or better in 70% of CRVO affected eyes and 63% of patients with BRVO. 90% of CRVO patients improved or maintained their vision at final follow-up, compared to 75% of the BRVO group. No patients experienced severe neovascular complications. Limitations include small case number, however we will continue to prospectively audit cases.

P128

EVALUATING FARICIMAB EFFICACY IN TREATMENT-RESISTANT DIABETIC MACULAR OEDEMA PATIENTS: A REAL-WORLD PERSPECTIVE

Corresponding author: Rehab Ismail, orcid.org/0000-0002-3936-7825

Rehab Ismail¹, Kamal El Badawy², Ben Scrivens³, Serena Salvatore³, Lina Kobayter³

¹NHS Grampian, Aberdeen, UK; ²Queen Elizabeth The Queen Mother Hospital, Margate, UK; ³Bristol Eye Hospital, Bristol, UK

Disclosures: No conflicts of interest and no funding

Introduction: Diabetic Macular Oedema (DMO) significantly contributes to vision impairment in diabetic patients. Traditional treatments include anti-vascular endothelial growth factor (anti-VEGF) injections, corticosteroids, and laser photocoagulation. In May 2022, Faricimab received NICE approval for DMO treatment. This study assesses Faricimab's effectiveness and durability in Bristol Eye Hospital.

Method: We conducted a retrospective analysis from January 2023 to November 2023. The study encompassed DMO patients switching to Faricimab due to inadequate response to previous therapies. We evaluated demographic details, best-recorded visual acuity (BRVA), central subfield thickness (CST), optical coherence tomography (OCT) central subfield thickness (CST), and re-treatment intervals. Outcome measures were reported at 6 months post loading.

Results: Our results included 9 patients (62 eyes), with 20 undergoing bilateral treatment. The average patient age was 64, predominantly with type 2 diabetes and Caucasian. Upon switching to Faricimab, patients received four-weekly loading doses, followed by a treat-and-extend regime. The mean BRVA pre- and post-Faricimab remained comparable at 67 letters. However, there was a significant reduction in CST by 24.3%, from 470.5 μm to 356 μm , following Faricimab loading (p -value 0.0015). Approximately 15% of patients achieved complete fluid resolution. Of the 62 eyes evaluated, 43% extended treatment intervals up to or beyond eight weeks, 41% demonstrated a response but required four- to six-week intervals, and 16% did not respond, with 90% of this latter group switching to intravitreal corticosteroids.

Conclusion: Faricimab demonstrates efficacy in DMO management, particularly for patients inadequately responding to other treatments requiring frequent re-treatments. Notably, 43% of patients extended their treatment up to or beyond 8 weeks; a significant improvement from prior therapies. Future directions include expanding the study to treatment-naïve patients and extending follow-up duration for comprehensive evaluation.

P129. Abstract not included

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CENTRAL SEROUS CHORIORETINOPATHY AND CHOROIDAL THICKNESS IN UK BIOBANK

Corresponding author: Catherine Jamison, orcid.org/0000-0003-0511-8000

Catherine Jamison¹, Barbra Hamill¹, Michael J Quinn^{1,2}, Alyson Muldrew¹, Alan Sproule¹, Peter Blows¹, UK Biobank Eye and Vision Consortium, Savita Madhusudhan³, Konstantinos Balaskas⁴, Andrew Lotery⁵, Tunde Peto^{1,2}, Foster Paul⁶

¹Queen's University Belfast, Belfast, UK; ²Belfast Health and Social Care Trust, Belfast, UK; ³University of Liverpool, Liverpool, UK; ⁴Moorfields Eye Hospital, London, UK; ⁵University of Southampton, Southampton, UK; ⁶University College London, London, UK

Disclosures: No conflicts of interest and no funding

Introduction: Central serous chorioretinopathy (CSR) is part of the pachychoroid spectrum. Imaging characteristics are presented of eyes graded as CSR, versus controls in UK Biobank (UKBB).

Method: UKBB included 500,000 UK participants aged 40-69. Colour and spectral domain optical coherence tomography images of 68517 participants were obtained using Topcon-1000 and graded masked.

Eyes graded as CSR were assessed for number of fluid areas, fluid height and diameter, serous pigment epithelial detachment (SPED) presence, sub-foveal choroidal thickness (SFCT), sub-fluid choroidal thickness (SFLCT), maximum choroidal thickness (MCT), greatest (vertical) choroidal vessel diameter (CVD) on foveal scan and sub-fluid (CVD_{SF}). These were compared to 109 pathology-free UKBB eyes. T-tests were used for comparisons, using SPSS v26.

Results: 273 eyes (227 subjects) were graded CSR; 38 eyes were excluded as non-CSR, resulting in 235 eyes of 195 subjects (40 bilateral).

Most (88%) had one fluid-area, 8.5% had 2, and 3% had ≥ 3 . In 64%, fluid was extrafoveal, 21% subfoveal and 15% juxtafoveal. Nasal fluid (within the arcades) was most frequent (60%) followed by 22% central and 18% temporal location. Mean fluid-height was 89 μm (0-309) and diameter 1073 μm (130-5298). 85% had no SPEDs, 8.6% had 1, 5.6% 2 or more.

CSR-eyes had significantly thicker mean SFCT and MCT compared to controls (SFCT: 348 μm (188-502 μm) versus 265 μm (111-443 μm), $p < 0.001$; MCT 394 μm (225-536 μm), 320 μm (176-458 μm) ($p < 0.001$) respectively). CVD was significantly higher in CSR eyes (181 μm (88-438 μm) versus 165 μm (93-267 μm) for controls ($p = 0.003$)).

Conclusion: CFT and MCT were significantly higher in those graded as CSR than controls. Additionally, CVD of the largest gradable vessel (foveal scan) was significantly greater in those with CSR, signifying choroidal vascular changes.

P131

INCIDENCE AND PREVALENCE OF COMPLETE RPE AND OUTER RETINAL ATROPHY SECONDARY TO AMD ASSESSED USING OCT IN A POPULATION IN GLOUCESTERSHIRE, UK

Corresponding author: Ushna Javed, orcid.org/0009-0002-3160-0052

Ushna Qureshi Javed¹, Charlotte F E Norridge¹, Rebecca Jones¹, Shaswati Bhowmik¹, Shweta Pandey¹, Paul Galsworthy¹, Yan Li¹, James McFadyen¹, Evelyn Qian¹, Esther Samuel¹, Peter Scanlon^{1,2}, Quresh Mohamed¹

¹Gloucestershire NHS Foundation Trust, Cheltenham, UK; ²University of Oxford, Oxford, UK

Disclosures: None

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Introduction: There is limited information on the number of patients with foveal and non-foveal geographic atrophy (GA) who may benefit from treatment. We conducted a retrospective, study to assess all patients in Gloucestershire with a diagnosis of AMD using CAM criteria to estimate incidence and prevalence of GA.

Method: We analysed all eyes in persons ≥ 55 years with diagnosis of AMD on Medisoft electronic medical record seen in Gloucestershire NHS hospitals between 01/07/18-30/06/23.

Graders assessed OCT and infrared images for Complete RPE and Outer Retinal Atrophy (cRORA), foveal involvement and lesion size.

Results: There were 2495 patients who had a GA diagnosis in at least one eye. The median age of the patient at first eye diagnosis was 85.2 years (IQR: 79.2 to 89.8).

Estimated prevalence of GA in at least one eye was 1.0% for population of Gloucestershire ≥ 55 years, varying with age.

The incidence of foveal involving GA in at least one eye was 3.6 per 10000 person-years and the incidence of non-foveal GA was 3.3 per 10000 person-years over study period.

Of those with GA, 43.2% had non-foveal involvement with better LogMAR visual acuity (VA) of 0.26 (IQR: 0.14 to 0.38) compared to 0.61 (IQR 0.38 to 0.94) in eyes with foveal involvement.

32.1% patients with GA would have been eligible for the Pegcetaplan OAKS/DERBY study.

Conclusion: Almost 25% of foveal involving GA eyes had a VA at the end of the study period worse than 1.0 LogMAR. Around 20% of eyes with foveal-involving GA maintained VA better than 0.3 LogMAR.

P132. Abstract not included

P133. Abstract not included

P134

OUTCOMES IN THE SECOND EYES IN NEOVASCULAR AGE-RELATED MACULAR DEGENERATION WHILST ON TREATMENT WITH INTRA-VITREAL ANTI-VEGF FOR THE FIRST EYE

Corresponding author: Venkat Kotamarthi, orcid.org/0000-0003-3748-9279

Venkat Kotamarthi¹, Raly Job Vellaniperumbal¹, Adnan Hafeez¹, Omara Arshad¹, Summaya Zaffar¹

¹Mid Cheshire Hospitals NHS Trust, Leighton Hospital, Crewe, UK

Declarations: No conflicts of interest and no funding

Introduction: The anatomical and functional outcomes were compared in the first eye and the second eye in case of sequential development of neovascular AMD at the end of 18 months.

Method: In a retrospective observational study, 60 patients had their first and second eyes compared when they developed sequentially neovascular AMD. All the patients' eyes went through a phase of loading dose followed by a fixed dosing and a phase of treat and extend with intravitreal Anti-VEGF of the same type. Baseline visual acuity, the morphological features of the macula on the OCT were taken into consideration. Both the normally distributed variables and non-normally distributed variables were compared between the two groups using statistical tests and p-values below 0.05 were considered statistically significant.

Results: The visual acuity in the second eye was better at the time of diagnosis and again at the end of 18 months in all the eyes, with a mean of 62 letters in the second eye compared to a mean of 51 letters in the first eye ($P = .003$). Morphological features on OCT were better. 88% of the first eye patients were symptomatic but only 27% of the second eyes had any visual symptoms.

Conclusion: Routine screening of the non-involved eye at every visit in the rapid access one-stop AMD clinics and the patient's awareness of the nature of the neovascular AMD entity showed a better outcome in the second eye as it resulted in early diagnosis and treatment compared to the first eye. It also gave an insight that algorithm to the second eye involved cannot be altered while trying to synchronise both eyes, by extending the treatment intervals very quickly.

P135

VISION IN FOCUS: EXPLORING POSSIBLE SERTRALINE-RELATED MACULOPATHY THROUGH A CASE SERIES ANALYSIS

Corresponding author: Venkat Kotamarthi, orcid.org/0000-0003-3748-9279

Venkat Ramana Rao Kotamarthi¹, Paras Agarwal¹, Mathew Fenech¹, Simon Peterson¹

¹Mid Cheshire Hospitals NHS Trust, Leighton Hospital, Crewe, UK

Disclosures: No conflicts of interest and no funding

Introduction: To report a case series of maculopathy in 9 patients after initiating sertraline therapy (selective serotonin reuptake inhibitor) which ranged from retinal pigment epithelial (RPE) changes and dry macular degeneration to neovascular age related macular degeneration (nAMD), cystoid changes and central serous chorioretinopathy (CSCR) like picture, some of which were bilateral and some of which were accelerated macular changes to pre-existing changes.

Method: Patients selected ranging from age group 40 to 64 years, 7 male Caucasians and 2 female *Caucasians*, had a history of seeing a central scotoma with substantial evidence of onset of visual symptoms following commencement with Sertraline. Complete ocular examination with colour vision testing and investigation with multimodal imaging which included colour fundus photography, fundus fluorescein angiography (FFA), autofluorescence (AF), optical coherence tomography (OCT) and optical coherence tomography angiography (OCTA).

Results: Clinical examination, fundus photos, AF and OCT showed changes ranging from extensive macular scarring with retinal pigment epitheliopathy, disruption of the outer retinal layers with intra-retinal fluid in cases presenting as Age related macular degeneration (AMD), to the extent of development of a choroidal neovascular membrane needing treatment with intravitreal Anti-VEGF. Some cases developed sub retinal fluid (SRF), with RPE changes similar to CSCR. Stopping the Sertraline medication showed in some cases reversal of symptoms reversal of some changes like intra-retinal fluid and sub-retinal fluid (especially in CSCR) but the RPE changes remained.

Conclusion: Sertraline is most commonly prescribed for depressive (83%) and anxiety disorders (11%), the rest including obsessive-compulsive disorders, post-traumatic stress disorders, panic and eating disorders. Patients should be warned before commencing Sertraline and should be monitored regularly. Needs more extensive study.

P136

FUNCTIONAL AND STRUCTURAL DEFICITS IN PRECLINICAL AND EARLY DIABETIC RETINOPATHY POINT TO DIABETIC RETINAL NEUROPATHY

Corresponding author: Jae Yee Ku, orcid.org/0000-0002-5758-7346

Jae Yee Ku^{1,2}, Simon Harding^{1,2}, Gabriela Czanner³, David Parry², Paul Knox¹

¹Department of Eye and Vision Science, University of Liverpool, Liverpool, UK; ²St. Paul's Eye Unit, Liverpool University Hospitals NHS Foundation Trust, Liverpool; ³Liverpool John Moores University, Liverpool, UK

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Introduction: Increasing evidence shows that diabetic retinopathy (DR) is not only a microvascular but also a neurodegenerative disease. Many studies of diabetic retinal neurodegeneration (DRN) only examined the ganglion cell complex (GCC). As far as we are aware, this is the first longitudinal

study that compared the macular function and structure of healthy participants (HP) with participants with diabetes (PWD) using the handheld radial shape discrimination (hRSD) test and evaluates retinal thickness of all layers in early DR.

Method: Macular function was measured using hRSD, distance and near visual acuity (VA). Macular structure was assessed using Heidelberg Spectralis OCT.

Results: 50 HP (Group 1, 55 ± 14y, 26 male), 26 PWD with no DR (Group 2, 55 ± 14y, 14 male) and 46 PWD with minimal DR (Group 3, 57 ± 17y, 23 male) were recruited. hRSD threshold (mean ± SD: Group 1 -0.77 ± 0.11 logMAR, Group 2 -0.68 ± 0.18, Group 3 -0.61 ± 0.25), distance (Group 1 -0.08 ± 0.12, Group 2 0.03 ± 0.15, Group 3 0.06 ± 0.16) and near VA (Group 1 0.06 ± 0.16, Group 2 0.21 ± 0.25, Group 3 0.19 ± 0.22) were generally worse with increasing disease severity (all $p \leq 0.001$). Compared to HP, total retinal thickness was reduced by 2.7–4.9% in PWD across all subfields. PWD had retinal nerve fibre layer (RNFL; 0.8–12.3%), ganglion cell layer (GCL; 4.6–9.9%), inner plexiform layer (IPL; 4.0–9%) and outer nuclear layer (ONL; 5.1–9.7%) thinning in all subfields. Longitudinal data for 23 PWD showed no change in macular function but a decrease in GCL, IPL and the inner nuclear layer (INL) after 205 ± 109 days (all $p < 0.05$).

Conclusion: Functional and structural changes are detectable in early DR, consistent with DRN developing before observable microvascular abnormalities. It is important to examine all retinal layers and not only the GCC in DRN.

P137

EARLY EFFECT OF INTRAVITREAL FARICIMAB ON HARD EXUDATES IN DIABETIC MACULAR OEDEMA

Corresponding author: Ameeta Kumar, orcid.org/0000-0002-8683-2443

Ameeta Kumar¹, Ali Al-Gilgawi², Arevik Ghulakhshian¹, Giovanni Ometto³, Christiana Dinah¹

¹Central Middlesex Hospital, London, UK; ²Royal London Hospital, London, UK; ³Moorfields Eye Hospital, London, UK

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Introduction: Intravitreal Faricimab is the first bispecific antibody to target vascular endothelial growth factor A (VEGF-A) and angiopoietin-2 (Ang-2). We examine the early effect of intravitreal Faricimab on hard exudates (HE) in patients with diabetic macular oedema (DMO).

Method: Patients initiated on Faricimab for centre-involving DMO associated with HE, between August 2022 and January 2023, were reviewed. Areas of HE were confirmed and measured using ImageJ on colour fundus photographs taken at initiation of Faricimab ('baseline') and 4–6 months later ('follow-up'). Demographic data, including age, gender and ethnicity, as well as best-corrected visual acuity (BCVA) and central retinal thickness (CRT) were recorded.

Results: Fifteen eyes from 15 patients were included in this analysis. The mean age of the cohort was 59 years (range 33–67); 8 patients (53%) were male, and 53% were of South Asian ethnicity. 12 eyes (80%) were treatment-naïve. The mean follow-up was 4.8 months, with eyes receiving a mean of 3.7 injections.

At follow-up, the median total HE area had reduced from $0.12 \pm 0.28 \text{ mm}^2$ to $0.10 \pm 0.62 \text{ mm}^2$. 10 eyes (67%) demonstrated a

visible reduction in total HE area, of which 6 eyes (40%) had a reduction within the 1mm foveal subfield. Mean BCVA improved from LogMar 0.23 ± 0.19 to 0.14 ± 0.15 ($p < 0.01$), whilst mean CRT reduced from $392 \pm 78 \mu\text{m}$ to $303 \pm 53 \mu\text{m}$ ($p < 0.001$).

Conclusion: Intravitreal Faricimab was shown to reduce HE area in DMO within 6 months of initiation in two-thirds of the cohort. Previous studies have noted reduction of HE area with anti-VEGF agents much later in the treatment course. We hypothesise that early reduction with Faricimab is due to the anti-inflammatory effects of Ang-2 and will present a chronological pictorial analysis of the reduction.

P138

LIGHT-ADAPTED ELECTRORETINOGRAMS OBTAINED WITH A PORTABLE DEVICE IN PATIENTS WITH ABCA4-RETINOPATHY: ASSOCIATIONS WITH GENOTYPE

Sancia X H Lam, orcid.org/0009-0000-5437-417X

Sancia X H Lam^{1,2}, Elena Schiff^{1,2}, Serene Lin^{1,2}, Mark Simcoe^{1,2}, Zaina Bouzia^{1,2}, Gavin Arno^{1,2}, Michel Michaelides^{1,2}, Andrew Webster^{1,2}, Omar Mahroo^{1,2}

¹UCL Institute of Ophthalmology, London, UK; ²Moorfields Eye Hospital, London, UK

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Introduction: Electroretinogram (ERG) parameters can quantify severity in Stargardt disease (ABCA4-retinopathy) and provide prognostic information. Standard tests take up to an hour and often require a separate patient visit. A portable device can record light-adapted ERGs within minutes in the outpatient clinic. We explored whether parameters obtained with the device associate with severity by genotype.

Method: Patients with genetically confirmed ABCA4-retinopathy underwent light-adapted ERGs with the portable device (RETeval system, LKC Technologies) and specialised skin electrodes (Sensor Strips, LKC). Variants were classified by severity (4 categories: mild; mild/moderate; moderate; severe) according to a recent preprint (Cornelis et al., medRxiv 2023.04.24.23288782). Each patient was classified by severity of the milder of the two alleles. Light-adapted ERG b-wave and flicker amplitudes were compared between groups. Parameters were averaged for both eyes where available.

Results: Fifty-five patients (29 females) were included: mean (SD) age was 51 (16) years. Of these, 30, 9, 8 and 8 were classified genotypically as mild, mild/moderate, moderate and severe respectively. Mean (SD) b-wave amplitudes were 23.05 (30.49), 13.83 (12.71), 7.40 (6.89), 0.81 (1.11) microvolts respectively. Flicker ERG amplitudes were 21.51 (12.74), 15.58 (11.71), 14.52 (10.55), 2.71 (3.54) microvolts respectively. For both parameters, significant differences ($p < 0.05$) were observed between the severe group and each of the other groups; in addition, for b-wave amplitudes, the mild group differed significantly from the moderate group ($p = 0.014$).

Conclusion: Portable recordings yield an assessment of retinal function that correlates with genotype. Amplitudes of both parameters fell with increased severity of genotype. The test may help predict severity of as-yet unclassified variants as well as predicting severity in patients who cannot access genetic testing.

BEYOND BASAL LAMINAR DRUSEN: EXPANDED RETINAL PHENOTYPES IN MEMBRANOPROLIFERATIVE GLOMERULONEPHRITIS

Corresponding author: Claudia Liang Peng, orcid.org/0009-0008-8751-5598

Claudia Liang Peng¹, Abraham Olvera-Barrios¹, Roy Schwartz¹, Gabriela Grimaldi², Catherine Egan¹, Adnan Tufail¹

¹Moorfields Eye Hospital NHS Foundation Trust, London, UK;

²Neurocenter of the Southern Switzerland, Lugano, Switzerland

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Introduction: To describe various retinal manifestations of a rare, complement-driven disease known as membranoproliferative glomerulonephritis.

Method: Retrospective single-centre observational case series of patients with ocular pathology and renal biopsy-confirmed membranoproliferative glomerulonephritis, seen between 2006-2023. Multimodal imaging and electronic health records were reviewed for demographic parameters and disease course, including any changes in visual acuity or ocular pathology.

Results: A total of 6 patients (12 eyes) with a mean age at presentation of 32.8 (SD = 14) years and a mean follow-up of 4.5 years were included in the study. Visual acuity remained unchanged in 67% of the patients, and all patients remained with vision better than 6/12 by the end of follow-up. All patients presented with bilateral drusen at initial visit, primarily in the temporal or macular regions. Fundus autofluorescence revealed these to be hyperfluorescent. Pigmentary changes were observed in 4/12 eyes, with 2/12 eyes having over 12 years of follow up and demonstrable development and progression of retinal pigment epithelium atrophy without foveal involvement. Additionally, optical coherence tomography revealed vertical hyperreflective lesions/outer retinal columnar abnormalities (ORCA), as well as cystoid macular oedema in 2 eyes.

Conclusion: Basal laminar drusen were seen bilaterally in both type I and type II membranoproliferative glomerulonephritis. Yet it is of interest to observe the phenotypic variation in this rare complement-driven disease, with additional findings such as outer retinal columnar abnormalities, cystoid macular oedema and progressive retinal pigment epithelium atrophy that may be seen in longitudinal follow-up. Future studies genotyping the patient cohort would be of interest to understand any underlying molecular mechanisms.

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THE REAL-WORLD EFFICACY AND SAFETY OF FARICIMAB IN BOTH TREATMENT NAÏVE AND TREATMENT RESISTANT PATIENTS WITH NEOVASCULAR AGE-RELATED MACULAR DEGENERATION

Corresponding author: Aimee Lloyd, orcid.org/0000-0001-5317-1194

Aimee Lloyd^{1,2}, Salwa Abugreen¹, Miguel Kurc², Mahmoud Elbalawi³

¹Manchester Royal Eye Hospital, Manchester, UK; ²East Lancashire NHS Foundation Trust, Blackburn, UK; ³University of Central Lancashire, Preston, UK

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Funding: None

Introduction: The aim of the study was to collect real-world data on the efficacy of Faricimab in naïve and switch patients diagnosed with nAMD.

Method: All patients within the study period treated with Faricimab for nAMD were included. Patients were divided into groups. Group 1 (G1) was made up of switch patients with minor sub retinal fluid (SRF) or intra-retina fluid (IRF) on optical coherence tomography (OCT). Group 2 (G2) was made up of switch patients with significant activity on OCT. Group 3 (G3) was treatment naïve patients. The East Lancashire NHS Trust Faricimab protocol was created and followed. The outcome was based on VA and optical coherence tomography (OCT) showing central retinal thickness (CRT), presence or absence of SRF, IRF, and structural retinal changes.

Results: Ninety-nine eyes were included in the study. A total of 350 Faricimab injections were given with a mean of 3.54 ± 1.05 (range 1- 6). For G1 & G2 patients the mean change in CRT after Faricimab was $-88.53\mu\text{m}$ (p value < 0.0001, 95% CI 58.58 to 118.49). For G3 patients the mean change in CRT was $-127.1\mu\text{m}$ (p < 0.0001, 95% CI 95.51 to 158.65). The mean vision acuity (VA) change was, $+0.041 \pm 0.340 \log\text{MAR}$ (p = .904). In G1 & G2: 56.7% had a total resolution of fluid. In G3, 79.5% had a total resolution of fluid. 1 patient had bilateral intraocular inflammation. There was no occlusive vasculitis.

Conclusion: Faricimab was associated with an improvement in CRT in both treatment naïve and treatment resistant nAMD. One patient developed intraocular inflammation, but no evidence of vasculitis or vein occlusion. Further work on the long-term outcomes of Faricimab will allow us to identify treatment intervals for our cohort.

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GREATER REDUCTION IN PIGMENT EPITHELIAL DETACHMENT SIZE WITH FARICIMAB VS AFLIBERCEPT DURING HEAD-TO-HEAD DOSING IN PATIENTS WITH NAMD

Corresponding author: Andrew Lotery, orcid.org/0000-0001-5541-4305

Andrew Lotery¹, Philippe Margaron², Audrey Souverain², Ming Yang³, Yevgeniy Shildkrot³, Aachal Kotecha⁴, Jeffrey Willis³, Shriji Patel³

¹University of Southampton, Southampton, UK; ²F. Hoffmann-La Roche Ltd., Basel, Switzerland; ³Genentech Inc., San Francisco, CA, USA; ⁴Roche Products Ltd., Welwyn Garden City, UK

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Roche Products Ltd. Jeffrey R. Willis: Employee: Genentech, Inc. Shriji Patel: Employee: Genentech, Inc.

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Introduction: In TENAYA/LUCERNE (NCT03823287/NCT03823300), dual inhibition of angiopoietin-2 (Ang-2) and vascular endothelial growth factor receptor (VEGF)-A with faricimab resulted in greater improvement in anatomical outcomes during the head-to-head dosing period vs aflibercept, and more rapid drying of the retina with fewer injections. This analysis evaluates changes in pigment epithelial detachment (PED) over time in TENAYA (N = 671)/LUCERNE (N = 658).

Method: Treatment-naïve patients were initially randomized to faricimab 6.0mg up to every 16 weeks (Q16W) after 4 initial Q4W doses or aflibercept 2.0mg Q8W after 3 initial Q4W doses. We performed a post hoc analysis evaluating PED baseline characteristics and anatomic changes during the head-to-head dosing period (weeks 0–12). PED was defined as retinal pigment epithelium elevation with width $\geq 350\mu\text{m}$; graded as serous PED or fibrovascular PED.

Results: Baseline PED characteristics were similar between treatment arms: foveal centre involvement (faricimab, 80.7%; aflibercept, 77.5%); fibrovascular type (faricimab, 79.9%; aflibercept, 82.1%), serous type (faricimab, 20.1%; aflibercept, 17.9%); mean(SD) maximum thickness within 6-mm Early Treatment Diabetic Retinopathy Study grid (faricimab, 253 μm [189 μm]; aflibercept, 240 μm [186 μm]). At week 12, the adjusted mean decrease from baseline in maximum PED thickness was greater with faricimab than aflibercept (all PED, -87.9 vs -74.5 μm [nominal $P = 0.0067$]; serous PED, -136.1 vs -108.2 μm [nominal $P = 0.0147$]). Similarly, at week 12, among eyes with serous PED at baseline, a greater proportion of patients experienced resolution of serous PED with faricimab vs aflibercept (95.3% vs 86.6%; nominal $P = 0.0258$).

Conclusion: Faricimab was associated with greater improvements in PED outcomes vs aflibercept in the head-to-head dosing period of TENAYA/LUCERNE. These findings are consistent with the greater drying of retinal fluid seen with faricimab.

P142

EVALUATION OF THE CARE PATHWAY FOR PATIENTS REFERRED FROM PRIMARY CARE WITH SUSPECTED NEOVASCULAR AGE-RELATED MACULAR DEGENERATION

Corresponding author: Georgina Mack Garcia, orcid.org/0009-0006-0348-4591

Georgina Mack Garcia¹, Martin McKibbin², Aditi Mohla²

¹University of Leeds, Leeds, UK; ²Leeds Teaching Hospitals Trust, Leeds, UK

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Funding: None

Introduction: Following changes to the local secondary care pathway, this study investigated the true positive rate for suspected NvAMD referrals and times to start treatment and complete the initial 3 injections. Identifiable reasons for any delays were recorded.

Method: All patients starting treatment for NvAMD in at least one eye between April and September 2023 were included. Data was collected retrospectively from mediSIGHT, Trust electronic records and the dedicated email Inbox for suspected NvAMD referrals from primary care. The true positive referral rate was estimated and the intervals between referral, start of treatment and the 3rd dose of intra-vitreous therapy were recorded. Identifiable delays were grouped into patient and hospital categories.

Results: During the 6-month period, treatment for NvAMD was initiated in 95 eyes. The true positive referral rate was 22%. Treatment was started within 14 and 28 days of referral in 50% and 83% of eyes respectively. The first 3 intra-vitreous injections were given within 10 and 12 weeks in 91% and 100% of eyes. The ability to meet both targets varied from month to month. For the 49 eyes for which treatment was not started within 14 days of referral, patient and hospital factors were the cause in 24% and 76% of eyes respectively.

Conclusion: Changes to the local care pathway have improved the delivery of the loading phase of treatment, compared to prior performance documented in the UK AMD Audit. Hospital factors, especially lack of capacity, mean that some patients still experience delays with initial assessment and treatment. Prospective review of clinic capacity and more effective triage of primary care referrals are required to improve access to initial treatment.

P143

OUR INITIAL REAL WORLD OUTCOME OF FARICIMAB (VABYSMO) FOR NAIVE NAMD AT CHFT

Corresponding author: Safaa Mahmoud, orcid.org/0009-0007-2634-8944

Safaa Mahmoud¹, Indra Dias¹

¹Calderdale & Huddersfield NHS Foundation Trust, Huddersfield, UK

Disclosures: The first author was sponsored to present this study at the Yorkshire Vabysmo meeting on November 2023

Funding: none

Introduction: Faricimab, a recently NICE-approved anti-VEGF for wet AMD, was introduced at CHFT in January 2023. A retrospective audit of 99 naïve wet AMD eyes treated with Faricimab from January 1 to September 1, 2023, was conducted to assess outcomes and guide local guidelines.

(Declaration: The first author was sponsored to present this study at the Yorkshire Vabysmo meeting.)

Method: Utilizing data from Medisight Audit Suit, clinical notes, and OCT scans, the study focused on injection patterns, OCT and clinical morphologic changes, and visual acuity at Baseline, Week 16, and the last visit. The treatment protocol included four loading injections at 4-week intervals, followed by a week 16 review to determine further plans.

Results: Results at week 16 revealed 75% of eyes with dry macular OCT, a 24% reduction in mean CRT (347 to 263 microns post-loading), and a positive trend in PED improvement. Last visit outcomes demonstrated a mean ETDRS VA increase of 4 letters, with 77% experiencing vision gain or avoidance of loss. Notably, 50% maintained good vision up/near to driving standards.

The mean post-loading injection interval was 8 weeks, and 66% transitioned to T&E. Although three eyes experienced atrophy-related vision loss, and three developed RPE rip, stability was maintained. One case switched to an alternative anti-VEGF due to unexplained pain post-injection

Conclusion: In conclusion, CHFT's initial results suggest Farcimab's favourable tolerability and efficacy in managing wet AMD. This retrospective audit contributes valuable insights for refining local guidelines and optimizing patient outcomes.

P144

OUR INITIAL REAL WORLD OUTCOME OF FARICIMAB (VABYSMO) IN TREATMENT EXPERIENCED WET AMD AT CHFT

Corresponding author: Safaa Mahmoud, orcid.org/0009-0007-2634-8944

Safaa Mahmoud¹, Indra Dias¹

¹Calderdale & Huddersfield Foundation Trust, Huddersfield, UK

Disclosures: The first author was sponsored to present this study at the Yorkshire Vabysmo meeting on November 2023

Funding: No applicable funding

Introduction: A retrospective audit of 165 treatment experienced wet AMD eyes treated with Farcimab from January 1 to September 1, 2023, was conducted to assess outcomes and guide local guidelines. (Declaration: The first author was sponsored to present this study at the Yorkshire Vabysmo meeting)

Method: Utilizing data from Medisight and OCT scans, the study focused on injection patterns, OCT and clinical morphologic changes, and visual acuity at Baseline, Week 4 and 24. The treatment protocol included switching previously treated wet AMD eyes who are unstable on their current anti VEGF or requiring less than 8 weekly interval to stabilize, four loading injections were considered in eyes with more aggressive diseases. When stable T&E considered.

Results: At week 4 revealed 62% of eyes had dry macular OCT, 17% reduction in mean CRT (323 to 265 microns at week 4 and 24), and a positive trend in PED improvement. Last visit outcomes demonstrated a mean ETDRS VA stable at 62 letters, with 68% experiencing vision gain or avoidance of loss. Notably, 46% maintained good vision up/near to driving standards. The mean injection interval was 7 weeks, and 62% transitioned to T&E. 61% are now on or over 8 weekly intervals. One patient had CVA, three developed atrophy with subsequent visual loss, two eyes showed poor response and switched to another anti VEGF, and one eye developed RPE rip.

Conclusion: In contrast to our audit on naïve wet AMD patients treated with Farcimab, the group with prior treatment exhibited lower vision gains, as anticipated given their more challenging and resistant nature. Nevertheless, more than 60% of these eyes successfully underwent T&E. A subsequent project is essential to assess the long-term outcomes and stability.

P145

LONG-TERM VISUAL OUTCOMES FOLLOWING ANTI-VEGF TREATMENT IN NEOVASCULAR AGE-RELATED MACULAR DEGENERATION; REAL-WORLD EVIDENCE FROM A UK DISTRICT HOSPITAL

Corresponding author: Nyaish Mansoor-Ali, orcid.org/0000-0002-8872-3435

Nyaish Mansoor-Ali^{1,2}, Shakti Thakur¹

¹Royal Bolton Hospital, Bolton, UK; ²Manchester Royal Eye Hospital, Manchester, UK

Declarations: No conflicts of interest and no funding

Introduction: To report 2-, 3- and 4-year visual outcomes of patients undergoing anti-VEGF therapy for neovascular age-related macular degeneration 'nAMD' in a real-world setting.

Method: This single-centred, retrospective interventional case series evaluation obtained data from treatment naïve patients with nAMD commenced on treatment from January to December 2018. Electronic notes (OpenEyes™) were reviewed to obtain information on visual acuity, type and number of anti-VEGF injections.

Results: Overall, 160 eyes had records completing 2-years follow up, 112 had 3-years and 64 completed 4-years: 64%, 59% and 56% eyes had ≤4 letter loss at follow-up respectively. Eyes with functional vision (≥70 letters) increased or were unchanged in every category (39% to 44% at 2-years, 43% to 43% at 3-years and 33% to 42% at 4-years). Average baseline VA was 59 ± 20 letters. Median change in VA was unchanged from baseline at 2-years, -2 letters at 3-years and +3 letters at 4-years. In total, 54% had exclusive aflibercept treatment, 26% exclusive ranibizumab and the remainder had combination. Mean number of injections per year was 3.6 ± 1.6 in 2018, 4.3 ± 2.0 in 2019, 4.6 ± 2.4 in 2020, 5.3 ± 2.6 in 2021 and 5.2 ± 2.7 in 2022.

Conclusion: More than half eyes had stable vision at each year of follow-up. Our study had a notably higher proportion of patients with functional vision at baseline compared with other studies which was well retained in all groups despite the COVID pandemic.

P146

ANTI-VEGF TREATMENT OF RETINAL VEIN OCCLUSION IN THE WESTERN TRUST, NORTHERN IRELAND

Corresponding author: Orlaith McGrath, orcid.org/0000-0002-4302-7081

Orlaith McGrath¹, Ivan Drinan¹, Asif Orakzai¹

¹Western Health and Social Care Trust, Derry, UK

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Introduction: This is a study of anti-VEGF treatment for Retinal Vein Occlusion (RVO) in the Western Trust, Northern Ireland. Since the COVID-19 pandemic, nurse-led injector clinics and virtual clinics have resulted in a significant increase in the quantity of anti-VEGF injections administered.

Method: Patient data from RVO clinic visits are recorded in Medisoft software. Medisoft data was analysed for 150 eyes with a baseline injection start date from 01/01/2020 to 01/01/2022. Age at first injection, diabetic status, risk factors, Visual Acuity (VA) at each visit and number of intravitreal injections administered were recorded. Excel was used to analyse the data.

Results: In 2019, a total of 363 intravitreal injections were given for RVO. In 2022, 848 injections were administered.

27 (18.0%) eyes could read <35 ETDRS letters at baseline. One month post-treatment, they gained a mean of 14.2 ETDRS letters. At 18 months they had a 21.6 letter gain. 50 (33.3%) eyes could read 35 to 55 ETDRS letters at baseline. A mean of 9.9 letters were gained after one month and 20.3 ETDRS letters at 18 months. 45 (30.0%) eyes could read 56 to 70 ETDRS letters at

baseline. After two months, they had a mean gain of 5.7 letters, the maximum VA they achieved. 28 (18.7%) eyes could read >70ETDRS letters at baseline. They had a mean gain of 1.7 letters at 13 months.

Conclusion: Nurse-led injection clinics and virtual clinics have enabled earlier intervention and reduced waiting lists ensuring patients with RVO in Northern Ireland get timely treatment and thus, the best possible chance of a good visual outcome.

P147

CLINICAL EVALUATION OF DEM DX OPHTHALMOLOGY TRIAGE SYSTEM (DOTS) FOR A&E REFERRALS: AN ARTIFICIAL INTELLIGENCE MODEL

Corresponding author: Mariane Melo, orcid.org/0000-0003-4014-954X

Mariane Melo^{1,2}, Priyanka Sanghi¹, Camilo Brandao-de-Resende^{1,2}, Anish Jindal¹, Yan Ning Neo¹, Elsa Lee¹, Pearse Keane¹, Gordon Hay¹, Alex Day¹

¹Moorfields Eye Hospital NHS Foundation Trust, London, UK; ²DemDX Ltd, London, UK

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Introduction: As demands for emergency eye care escalate, effective triaging of accident and emergency (A&E) referrals is essential to optimise allocation of resources. The Dem Dx ophthalmology triage system (DOTS) utilising artificial intelligence (AI) has been developed and validated from over 11,000 A&E attendances, showing potential to appropriately redirect attending patients to elective/urgent care. This study investigates DOTS in a real-world ophthalmic A&E setting by triage nurses.

Method: This single-site, quasi-experimental study at a tertiary A&E ran from July 11, 2023 to October 6, 2023. Triage nurses inputted structured clinical data into DOTS and received the AI recommendation, including "see in A&E", "see in urgent care clinic," and "Advice at Triage Only/refer to GP". Nurses maintained the final decision and could override the AI recommendation. Discharge diagnoses was used as source of truth for triage accuracy. DOTS + nurses was compared with data collected from July 1, 2022 to September 16, 2022 collected by triage nurses alone, and demographic balance was achieved through propensity matching.

Results: DOTS + nurses (1426 A&E attendances) showed higher sensitivity (98.2% [95% CI 96.8-99.9]) versus nurses alone (1425 attendances) (95.2% [95% CI 93.3-96.6], $p = 0.005$) in identifying emergencies. It also showed superior specificity (40.7% [95% CI 37.3-44.2] vs. 23.9% [95% CI 21.0-27.1], $p = 0.0006$), reducing same-day A&E appointments by 9%.

Conclusion: To our knowledge, this is the first clinical trial in the UK to test an AI model in emergency eye care. DOTS + nurses showed superior sensitivity and specificity to triage nurses alone in an ophthalmic A&E setting. The use of DOTS has the potential to reduce unnecessary A&E referrals and cost to the health service, whilst maintaining patient safety.

P148. Abstract not included

P149

COST EFFECTIVENESS OF BIOSIMILAR ONGAVIA IN NEW AND STABLE TREATED EYES IN VARIOUS RETINAL CONDITIONS AS COMPARED TO LUCENTIS

Corresponding author, Hafiz Muhammad Abdullah, orcid.org/0009-0009-0910-1581

Hafiz Muhammad Abdullah¹, Waqar Ahmed¹, Raza Ali Zaidi¹, Craig Goldsmith¹, Benjamin Burton¹, Muhammad Raja¹

¹James Paget University Hospitals, Great Yarmouth, UK

Disclosures: No conflicts of interest and no funding

Introduction: Biosimilars are the compounds that have small structural differences from the original licensed product but it does not alter the medicine safety, efficacy or reaction with the body's immune system however they are much cheaper than the original licenced products. Ranibizumab has long been proven efficient in treating various retinal pathologies associated with retinal leakage. Since these conditions require treatments over long periods of time, these have a profound cost impact on NHS. In our study we compared the efficacy and safety of Biosimilar (ONGAVIA) in terms of cost as compared to Lucentis

Method: Study Design: Retrospective cross sectional.

Sample size: 123 eyes, 102 were new eyes, 21 were switched

Assessment Criteria: Vision, OCT changes

End point criteria: Improved, no improved, worse

Results: Commencing the treatment with ONGAVIA or switching to ONGAVIA from Lucentis in stable treated eyes is approximately 20% cost effective as compared to Lucentis with same number of eyes.

Conclusion: Biosimilars are effective alternative to original licenced product in terms of efficacy and safety. They can be safely switched once retinal pathology remains stable as this is a viable choice due to being cost effective. Cost effectiveness depends on number of factors including total number of injections administered over a period of time and local trust purchasing contract with the supplier.

P150

KEY STEPS IN THE OPTIMISATION OF THE PORT DELIVERY SYSTEM WITH RANIBIZUMAB (PDS) PROCEDURES AND THE IMPACT ON PATIENT OUTCOMES

Corresponding author: Mahi Muqit, orcid.org/0000-0003-1161-3956

Mahi Muqit¹, André Vicente², Philip Jaycock^{3,4}, Carlos Quezada-Ruiz^{4,5}, Natasha Singh⁵

¹Moorfields Eye Hospital NHS Foundation Trust and UCL Institute of Ophthalmology, London, UK; ²F. Hoffmann-La Roche AG, Basel, Switzerland; ³Roche Products Limited, Welwyn Garden City, UK; ⁴Clinica de Ojos Garza Viejo, San Pedro Garza Garcia, Nuevo Leon, Mexico; ⁵Genentech Inc., San Francisco, CA, USA

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André Vicente - Financial Disclosures: Employee: Hoffmann-La Roche AG, Basel, Switzerland

Philip Jaycock - Financial Disclosures: Employee: Roche Products Limited

Carlos Quezada-Ruiz - Financial Disclosures: Employee and Stocks or Stock Options: Genentech, Inc.

Natasha Singh - Financial Disclosures: Employee and Stocks or Stock Options: Genentech, Inc.

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Introduction: The PDS is a unique drug delivery system for continuous delivery of a customized formulation of ranibizumab into the vitreous. The PDS implant insertion procedure has 7 critical steps, which have evolved during the clinical trial programme. Here we highlight some key steps of the PDS implant insertion procedure and adaptations implemented since the start of the trial programme and their impact on patient outcomes.

Method: PDS trials include: LADDER (NCT02510794), ARCHWAY (NCT03677934), and PORTAL (NCT03683251) in neovascular age-related macular degeneration; PAGODA (NCT04108156) in diabetic macular oedema; and PAVILION (NCT04503551) in diabetic retinopathy.

Results: The PDS Instructions for Use (IFU) was optimised in LADDER, reducing the rate of postoperative vitreous haemorrhage from 50.0% to 5.1%. During ARCHWAY/PORTAL, the frequency of implant dislocations was 1.1%; 5/6 patients had a longer-than-specified scleral incision length (>3.7 mm). Notably, a 3.5 mm incision length was mandated with suturing for longer incisions to help mitigate implant dislocations. In PAGODA and PAVILION, the frequency of implant dislocations was 0.3% and 0%, respectively. The IFU was updated to reduce the risk of conjunctival events. Subsequently, a trend towards reduction in the frequency of these events has been observed across PDS trials (4.5% in Ladder [n = 179]; 6.4% in ARCHWAY [n = 248]; 3.2% in PAGODA [n = 320]; 2.9% in PAVILION [n = 105]). Further optimisation of the PDS is ongoing.

Conclusion: Attention to the different key steps and meticulous adherence to the IFU are critical for surgical success in the PDS implant insertion procedure. Learnings from different clinical trials have led to a continuous evolution of PDS surgical methodologies, with the goal of further optimising patient outcomes.

P151

EVALUATING THE ACCURACY OF REFERRALS TO THE SOUTHAMPTON FAST TRACK AMD CLINIC

Corresponding author: Abdi Malik Musa, orcid.org/0000-0002-1304-5457

Abdi Malik Musa¹, Andrew Lotery²

¹University of Southampton, Southampton, UK

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Funding: None

Introduction: This study aimed to assess the accuracy of referrals to the fast-track clinic for suspected neovascular age-related macular degeneration (nAMD). The objectives included determining the proportion of referrals that resulted in a diagnosis of nAMD, and exploring the variables within referral letters that correlated with a positive nAMD diagnosis.

Method: In this retrospective study, data from 137 eyes of 117 patients referred to the emergency nAMD clinic were analysed. Pertinent data was extracted from referral letters and Medisoft.

Results: Optometrists accounted for 87.1% (n = 115) of referrals, while internal referrals from eye casualty and other ophthalmic speciality clinics comprised 8.3% (n = 11), and external ophthalmologists constituted 4.6% (n = 6). Of the 132 eyes referred, 25.8% (n = 34) were diagnosed with nAMD. The majority of the remaining 74.3% (n = 98) received a diagnosis of dry AMD (54.1%, n = 53). Subretinal fluid was significantly more reported in referrals from patients who received a final diagnosis of nAMD (73.5%, p = 0.003). Reduced visual acuity, distortions, scotomas, and retinal haemorrhages were equally reported in both groups. The nAMD group were significantly more likely to have had an OCT prior to referral (73.5%, n = 25, p = 0.0431) compared to the non-nAMD group (52%, n = 51). The median time from referral to appointment was 19 days (IQR 15-29), while the median time from referral to injection was 38 days (IQR 27.25-51).

Conclusion: Three out of four patients referred with suspected nAMD receive an alternative diagnosis. A positive diagnosis of nAMD is associated with an OCT before referral and a report of subretinal fluid. Inappropriate referrals delay treatment of genuine nAMD cases. Greater use of community OCT would facilitate more accurate rapid access referrals of nAMD.

P152. Abstract not included

P153

INTERGRADER VARIABILITY IN THE GEOGRAPHIC ATROPHY INCIDENCE STUDY (GAINS) IN GLOUCESTERSHIRE, UNITED KINGDOM

Corresponding author: Shweta Pandey, orcid.org/0009-0007-7343-2153

Shweta Pandey¹, Ushna Javed¹, Charlotte Norridge¹, Rebecca Jones¹, Shaswati Bhowmik¹, Paul Galsworthy¹, Li Yan¹, James McFadyen¹, Evelyn Qian¹, Esther Samuel¹, Peter Scanlon¹, Quresh Mohamed¹

¹Gloucestershire NHS Foundation Trust, Cheltenham, UK

Declarations: No conflicts of interest and no funding.

Introduction: A 5-year retrospective audit using Medisoft to identify all patients with Geographic Atrophy (GA) in Gloucestershire showed high intergrader variability. A team of clinicians were trained by an experienced medical retina consultant to grade Optical Coherence Tomography (OCT) images for features of GA as defined by the Consensus on Atrophy Meeting (CAM) group criteria.

Method: The 10 members of the grading team consisted of 50% ophthalmologists and 50% other allied health professionals. They undertook a one-hour training session by an experienced medical retina consultant prior to grading. A random sample of 50 patients (5% of the sample) were secondary graded by the same medical retina consultant to check reproducibility and intergrader variability. The areas assessed were the 1) diagnosis of GA, 2) foveal involvement and 3) presence of cRORA (complete RPE and outer retinal atrophy). Cohen's Kappa values were used to measure the inter grader variability and a value of >0.70 was considered to be good agreement.

Results: There was only slight agreement (kappa = 0.20) in the classification of cRORA and iRORA making it the biggest variation in grading. There was moderate agreement when grading more

broadly as GA vs non-GA ($\kappa = 0.42$). There was good agreement on the classification of foveal involving GA or not. ($\kappa = 0.78$).

Conclusion: Although a small sample was used for secondary grading, it highlighted the high inter-grader variability and the need for further training. Consequently, further training was conducted to identify features of GA, data collection forms were streamlined and all data was routinely secondary graded. An arbitration process with an experienced medical retina consultant was implemented for any discrepancies and any cases where an agreement was not reached were excluded.

P154

TACKLING THE BURDEN OF REFERRALS FROM A DIABETIC EYE SCREENING PROGRAMME IN DORSET

Corresponding author: Anjali Patel, orcid.org/0009-0005-0790-9318

Anjali R Patel¹, Owen Anderson¹

¹Royal Bournemouth Hospital, Bournemouth, UK

Disclosures: No conflicts of interest and no funding

Introduction: Diabetic Eye Screening Programmes (DESP) should refer patients, and the Hospital Eye Services (HES) should offer appointments, in a timely manner. These time periods are set by NHS England. This enables treatment to be initiated to reduce complications of diabetic retinopathy. An audit evaluated whether patients referred to HES from DESP met NHS standards.

Method: An audit reviewed patients referred as R2 or M1 by DESP to HES between 1st April - 30th June 2022 and R3A referrals made between 1st April - 31st December 2022. A clinician generated DESP referral log and HES patient records were reviewed. Time from DESP screening to DESP referral, and HES consultation, were compared against NHS England standards.

Results: 47 of 48 (98%) R2/M1 patients and 27 of 36 (75%) R3A patients were referred by DESP in a timely manner. However, only 19% of R2/M1 referrals and 22% of R3A referrals had a timely consultation. Our virtual grading review of referral images found that 9% of R3A patients were misclassified by DESP. Dorset DESP do not have OCT referral refinement.

Conclusion: HES capacity is limited due to budgetary, staffing and real estate constraints. Capacity can be increased through introducing virtual clinic reviews of routine M1 referrals and expanding one-stop laser clinics for R3A referrals. Training and feedback between HES and DESP may improve referral quality. Dorset DESP refer based on photographic M1 criteria without OCT referral refinement; introduction of OCT into DESP would reduce M1 referrals and release clinic slots for more urgent patients. Ultimately, HES in Dorset are not meeting Dorset DESP demand; increased resources and more inventive ways of working are required to address this.

P155. Abstract not included

P156

AUTOMATED SEGMENTATION OF HYPERREFLECTIVE FOCI IN DMO SHOWS GREATER VOLUME REDUCTION WITH FARICIMAB VS AFLIBERCEPT IN THE PHASE 3 YOSEMITE AND RHINE TRIALS

Corresponding author: Tunde Peto, orcid.org/0000-0001-6265-0381

Tunde Peto¹, Andreas Maunz², Esther von Schulthess², Katie Patel², Isabel Bachmeier², Ian Lloyd Jones², Yaniv Cohen², Kara Gibson³, Jeffrey R Willis⁴, Carl Glittenberg², Sasha Fauser²

¹Queen's University of Belfast, Royal Victoria Hospital, Belfast, UK; ²F. Hoffmann-La Roche Ltd., Basel, Switzerland; ³Roche Products Ltd, Welwyn Garden City, UK; ⁴Genentech Inc., San Francisco, CA, USA

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Andreas Maunz - Financial Disclosures: Employee: F. Hoffmann-La Roche Ltd.

Esther von Schulthess - Financial Disclosures: Employee: F. Hoffmann-La Roche Ltd.

Katie Patel - Financial Disclosures: Employee: F. Hoffmann-La Roche Ltd.

Isabel Bachmeier - Financial Disclosures: Employee: F. Hoffmann-La Roche Ltd.

Ian Lloyd Jones - Financial Disclosures: Employee: F. Hoffmann-La Roche Ltd.

Yaniv Cohen - Financial Disclosures: Employee: F. Hoffmann-La Roche Ltd.

Kara Gibson - Financial Disclosures: Employee: Roche Products Ltd.

Jeffrey R. Willis - Financial Disclosures: Employee: Genentech, Inc.

Carl Glittenberg - Financial Disclosures: Employee: F. Hoffmann-La Roche Ltd.

Sasha Fauser - Financial Disclosures: Employee: F. Hoffmann-La Roche Ltd.

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Introduction: In diabetic macular oedema (DMO), hyperreflective foci (HRF) are considered to represent retinal inflammation and may be an imaging biomarker for disease progression. Aim was to test if the anti-inflammatory properties of faricimab, a dual angiopoietin-2 (Ang-2)/vascular endothelial growth factor (VEGF)-A inhibitor, would translate into improved resolution of retinal HRF in DMO, vs aflibercept.

Method: We used a deep learning-based segmentation model to automatically segment HRF in SD-OCT volumes. The U-Net, a convolutional neural network for biomedical-image segmentation, was trained for 250 epochs using manually annotated B-scans from phase 2 BOULEVARD (NCT02699450). These B-scans were annotated by placing ellipses over HRF $\leq 50\mu\text{m}$ in diameter, and manually outlining larger bright objects. Ellipses were shrunk to the most hyperreflective centre using adaptive thresholding. HRF were segmented in OCT volumes from 1527 patients in phase 3 YOSEMITE/RHINE (NCT03622580/03622593; treated with faricimab Q8W or treat-and-extend [T&E], or aflibercept Q8W). Objects $\leq 50\mu\text{m}$ in diameter were classified as HRF. Volumes were assessed within the 1.0/3.0mm diameter ETDRS rings, separated into inner and outer retina.

Results: Mean (SE) baseline HRF volumes in the inner retina at 1mm diameter were 222.41(11.92)pL, 240.16(14.53)pL and 238.95(14.05)pL for faricimab Q8W, faricimab T&E and aflibercept, respectively. At week 16, adjusted mean HRF volumes were 251.74(15.68)pL, 224.69(15.55)pL and 287.76(15.92)pL, respectively. By week 48, HRF volumes reduced by -118.29pL, -130.05pL and -58.67pL for faricimab Q8W, faricimab T&E and aflibercept, respectively, compared with baseline (nominal $P = 0.0006$ and $P < 0.0001$ for faricimab Q8W and T&E vs aflibercept averaged across all visits, respectively). Similar results were obtained for the inner retina at 3mm diameter and the outer retina.

Conclusion: Greater retinal HRF reductions were observed in faricimab-treated eyes compared with aflibercept.

P157

INADVERTENT ANTI-VASCULAR ENDOTHELIAL GROWTH FACTOR INJECTION INTERVAL INCREASE, DUE TO THE DISRUPTION FROM THE COVID-19 PANDEMIC, WAS INSIGHTFUL

Corresponding author: Ankita Abhinandan Punit, orcid.org/0009-0006-1745-8896

Ankita Abhinandan Punit¹, Fernando J. Huelin¹, Lesley Brandes¹, Tafadzwa Young-Zvandasara¹

¹The Newcastle Upon Tyne Hospitals NHS Foundation Trust

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Introduction: The COVID-19 pandemic presented unique challenges to health systems, globally. Guidelines from the Royal College of Ophthalmologists suggested maintaining nAMD patients on 8-weekly injections, if no visual deterioration.

Method: Inclusion criteria included patients presenting with neovascular age related macular degeneration (nAMD); deemed to require baseline treatment between 01 September 2019 – 24 February 2020.

Outcomes for those affected by the beginning and peak of the pandemic are included in the study. Data was collected prospectively in 2020 and early 2021, giving all subjects a minimum of 18 months since their baseline injection.

Results: Data is available for 135 eyes. The mean BCVA before treatment was 58.16± 15.48 letters. At 12 months, this was 60.51± 17.19 letters. Eyes with > 70 letters at baseline were 37/134 (27%) and at 12 months this was 43/132 (32%) [P = 0.005]

Anticipated dates for 8-week injection were determined. The 8-week delay time was a median of 65± 16 days. Further analysis showed at 10, 15, 30, 60, 90 days delay, no statistical significance was found for letter gain of 5 (P = 0.572), 10 (P = 0.481), 15 (P = 0.415) letters. No statistical significance was found for letter loss at 5 days (P = 0.366), 10 (P = 0.414) and 15 days (P = 0.621). When eyes were not loaded as planned, visual acuity gain wasn't statistically significant at 5+ letters [P = 0.283], 10+ letters [P = 0.315], 15+ letters [P = 0.599]. Similarly, loss of 5 letters [P = 0.322], 10 letters (P = 0.186) and 15 letters [P = 0.099] wasn't statistically significant.

Conclusion: This data emphasizes the need to reconsider the loading schedule of Anti-VEGF injections. Missing some injections wasn't deleterious and the 8-week injection being inadvertently delayed, did not affect visual outcomes.

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INTRAOCULAR INFLAMMATION FOLLOWING INTRAVITREAL FARICIMAB FOR NEOVASCULAR AGE-RELATED MACULAR DEGENERATION (NAMD)

Corresponding author: Ankita Abhinandan Punit, orcid.org/0009-0006-1745-8896

Ankita Abhinandan Punit¹, Hani Hasan¹, Joanna Da Costa¹

¹The Newcastle Upon Tyne Hospitals NHS Foundation Trust, Newcastle Upon Tyne, UK

Declarations: No conflicts of interest and no funding

Introduction: Intraocular inflammation is an uncommon, potentially vision-threatening adverse event, following intravitreal injection therapy. It is of increasing importance due to escalating numbers of injections performed and concerns about inflammation with the use of newer anti-VEGF agents.

Method: Interventional case series (n = 3)

Results: Three patients are presented. The first patient presented with anterior uveitis, vitritis, choroidal effusion and a shallow retinal detachment, 28 days following an eighth intravitreal Faricimab injection. Intraocular pressure was normal. Previous treatment included ten uneventful aflibercept injections. Topical and oral steroid treatment was ineffective. Vitreoretinal surgical intervention was required. The second patient presented with anterior uveitis and vitritis, 61 days following a fifth Faricimab intravitreal injection. Past treatment included eleven uneventful aflibercept injections. Topical steroid treatment was instituted for anterior uveitis and intraocular inflammation resolved. The third patient presented with anterior uveitis, vitritis and a choroidal effusion, 32 days following right intravitreal Faricimab.

Conclusion: Clinical presentation in all three cases was not compatible with endophthalmitis. Causes of intraocular inflammation post intravitreal injection may be divided into patient, medication, and drug-delivery-specific factors. No patient in our series was using pro-inflammatory drugs such as prostaglandins. Molecular structural differences between drugs and manufacturing processes are postulated to be causes of differential rates of intraocular inflammation. A small proportion of patients switched from Aflibercept to Faricimab may have developed anti-drug antibodies which may have contributed to intraocular inflammation. Aflibercept is administered from pre-filled syringes. Faricimab is physically drawn from the vial with a filter needle into a syringe prior to administration through a 30G needle. Syringes and needles are a source of silicone oil which may induce an immunogenic response contributing to intraocular inflammation.

P159

THE OCULAR PHENOTYPE OF IRD SECONDARY TO VARIANTS IN EYS IS INFLUENCED MORE BY AGE THAN GENOTYPE

Corresponding author: Riddhi Singh, orcid.org/0009-0000-2958-3682

Riddhi A Singh¹, Dong Sun¹, Chris Inglehearn¹, Martin McKibbin²

¹University of Leeds, Leeds, UK; ²Leeds Teaching NHS Hospitals Trust, Leeds, UK

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Introduction: Variants in the EYS gene account for 5% of cases of inherited retinal degeneration (IRD). Progression of the ocular phenotype is seen with age but the impact of genotype is not well known.

Aim: To compare the impact of age and genotype severity on the ocular phenotype of EYS-associated IRD.

Method: Colour, autofluorescence and spectral domain OCT images were used to record the retinal phenotype and collect quantitative data on retinal structure. The impact of age and predicted genotype severity was made by comparing the

phenotype by age (<40, 40-50 and >50 years) and by genotype severity (most severe, moderate, least severe).

Results: Data was available for 27 patients from 22 families. The median age was 50.5 years and 63% were male. The onset of visual impairment was in early adulthood (aged 20-40 years) in 63% of cases, with no association in decade of onset and predicted genotype severity. Intra-retinal white dots were present in 41% of all eyes and were most common in the fifth decade but were rare in older patients. Bone spiculation was present in 90% of all eyes, was universal from aged 40, and often extended beyond the mid-peripheral retina. Well-defined chorio-retinal atrophy was present in 32% of all eyes but was most common in the eyes of the oldest patients. Visual acuity, central retinal thickness and ellipsoid zone length all showed negative associations with increasing age. No similar associations between phenotype, visual acuity or retinal structure were observed with predicted genotype severity.

Conclusion: Age is more strongly associated with function and phenotype in EYS-associated IRD than genotype.

P160

REAL-WORLD TREATMENT PATTERNS AND OUTCOMES AMONG EYES WITH DIABETIC MACULAR OEDEMA (DMO) THAT RECEIVED >6 MONTHS OF FARICIMAB TREATMENT IN THE UK: FARWIDE-DMO

Corresponding author: Sobha Sivaprasad, orcid.org/0000-0001-8952-0659

Sobha Sivaprasad¹, Rhianon Reynolds², Tunde Peto³, Clare Bailey⁴, Louise Downey⁵, Christine Kiire⁶, Ian Pearce⁷, Gloria Chi⁸, Natalee James⁹, Alice Rothwell⁹, Neil Shah¹⁰, Amanda Downey¹⁰, Melanie Dodds¹¹, Parul Dayal⁸

¹Moorfields Eye Hospital NHS Foundation Trust, London, UK; ²Aneurin Bevan University Health Board and Cardiff University School of Vision Science, Cardiff, UK; ³Queen's University Belfast and Belfast Health and Social Care Trust, Belfast, UK; ⁴University Hospitals Bristol NHS Foundation Trust, Bristol, UK; ⁵Hull University Teaching Hospitals NHS Trust, Hull, UK; ⁶Oxford Eye Hospital, Oxford University Hospitals NHS Foundation Trust, Oxford, UK; ⁷Liverpool University Hospitals NHS Foundation Trust, Liverpool, UK; ⁸Genentech Inc, San Francisco, CA, USA; ⁹Roche Products Ltd, Welwyn Garden City, UK; ¹⁰Hoffmann-La Roche Ltd, Basel, Switzerland; ¹¹Medisoft Ltd, Leeds, UK

Disclosures:

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Tunde Peto - Financial Disclosures: Consultant: Alimera, Apellis, Bayer, Genentech, Inc., Heidelberg, Novartis, Oxurion

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Louise Downey - Advisory Board/Speakers' Honoraria: Alimera, Allergan, Bayer, Biogen, Novartis, Roche; Travel Grants: Allergan, Bayer, Novartis

Christine Kiire - Advisory Board, Consulting, Honoraria or Travel Expenses: Roche, Boehringer Ingelheim, Bayer, Alimera Sciences

Ian Pearce - Nothing to disclose

Gloria Chi - Employee: Genentech, Inc.

Natalee James - Employee: Roche Products Ltd.

Alice Rothwell - Employee: Roche Products Ltd.

Neil Shah - Employee: Hoffmann-La Roche Ltd.

Amanda Downey - Employee: Hoffmann-La Roche Ltd.

Melanie Dodds - Employee: Medisoft Ltd.

Parul Dayal - Employee: Genentech, Inc

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Introduction: Faricimab, the only bispecific antibody inhibiting angiopoietin-2 (Ang-2) and vascular endothelial growth factor (VEGF)-A, was approved for DMO in the UK in May 2022. We report results from a multicentre real-world data study evaluating patient characteristics, treatment patterns and outcomes among eyes with 6 months of follow-up on faricimab.

Method: The ongoing, retrospective, FARWIDE study includes patients (n = 1921) with a documented diagnosis of DMO (n = 2673 eyes) who initiated faricimab (≥1 injection) between June 2022-June 2023 at participating National Health Service sites (n = 21) that use the Medisoft ophthalmic electronic medical record system.

Results: 36% of patient-eyes were naïve, 64% were previously-treated (86% aflibercept). 71% of naïve eyes and 58% of previously-treated eyes received four initial injections per UK label. 17% of naïve eyes and 21% of previously-treated eyes had ≥6 months follow-up. Previously-treated eyes had received anti-VEGF treatments for a mean (SD) of 2.8 (2.5) years before initiating faricimab. Among eyes with ≥6 injections, mean (SD) treatment interval between the 5th and 6th faricimab injection was 9.0 (2.6) weeks in naïve eyes and 7.4 (2.6) weeks in previously-treated eyes. Mean (SD) gain in vision from baseline by injection 6 was +8.1 (1.6) letters (nominal p < 0.001) among naïve eyes and +1.1 (0.8) letters among previously-treated eyes. 303 eyes (33% naïve) had ≥12-months of follow-up on faricimab. Naïve eyes received a mean (SD) of 4.5 (0.9) and 6.3 (1.7) injections in the first 6 and 12-months of faricimab treatment, and previously-treated eyes received 4.6 (1.1) and 7.1 (2.2) faricimab injections, suggesting a rapid extension in intervals after the initial doses.

Conclusion: Initial results support the real-world durability of faricimab in treatment naïve and previously treated eyes.

P161

RETINOPATHY SEVERITY AND PRESENTATIONS IN TEENAGERS WITH SICKLE HAEMOGLOBINOPATHIES

Corresponding author: Evdokia Sourla, orcid.org/0009-0008-3084-3355

Evdokia Sourla¹, Peck-Lin Lip¹

¹The Birmingham & Midland Eye Centre, Sandwell & West Birmingham Hospitals, NHS Trust, Birmingham, UK

Disclosures: No conflicts of interest and no funding

Introduction: Sickle haemoglobinopathy is a common but specific disease entity limited to certain ethnic groups. Disease complications include severe proliferative retinopathy, vitreous haemorrhage, tractional retinal detachment and vision loss.

Method: Retrospective identification of patients attended single specialist sickle eye clinic in 2022-23, with first eye presented age ≤19yrs.

Participants: 35 teenagers (70 eyes) with confirmed sickle haemoglobinopathies.

Purpose: To investigate the disease prevalence and severity correlations with genotype and age onset in teenagers.

Results: Mean presenting age 16.9yrs (SD:2.07, range 10–19yrs), 54% females. 23/35 (65.7%) patients had HbSC genotype, 9 (25.7%) HbSS, 3 (8.6%) HbSthalassaemia. Mean follow-up period is 7.6yrs (SD:5.9, range 0.5–19.2yrs).

Screening referrals from haematologist/paeds were 28/35 (80.0%), 4/35 (11.4%) from opticians, 3/35 (8.6%) from ophthalmologists. Asymptomatic presentation was 58/70 eyes (82.9%).

Presentations: vision 6/9 or better in 64/70 (91.4%) eyes; 5/70 (7.1%) eyes were 6/12–6/18, 1/70 (1.4%) was 3/60 vision. 31/70 (44.3%) eyes had no retinopathy, 24/70 (34.3%) eyes presented with established sickle stage3 disease, 8/70 (11.4%) were severe with vitreous haemorrhage, 6/70 (8.6%) had salmon patch; 37/70 (52.8%) eyes had some features of non-proliferative signs.

Of 23 HbSC patients, 20/46 (43.5%) eyes had sickle stage3 disease, 6/46 (13%) had stage4. Of 9 HbSS patients, only 1 eye (5.6%) had sickle stage3, no stage4. Of 3 HbSthalassaemia patients, 3/6 (50%) eyes had stage3 and 2/6 (33.3%) had stage 4.

Follow-up(FU): 5/35 patients had no FU visit. 25/35 (71.4%) had DNA > 1.

Three eyes achieved vision 6/12, 1 eye 6/60. Progression to stage3 were found in 8/38 (21%) HbSC eyes, 4/16 HbSS eyes (25%), no disease progression in HbSthalassaemia. 10/60 (16.7%) eyes experienced recurrence vitreous haemorrhage, 2/60 (3.3%) developed localised retinal-detachment were SC patients.

Conclusion: Our cohort showed teenagers with sickle haemoglobinopathy can present with severe retinopathy, higher frequency and severity in HbSC genotype. Retinopathy screening could be beneficial from age 10 to detect early treatable retinopathy.

P162. Abstract not included

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REAL-WORLD TREATMENT PATTERNS AND OUTCOMES AMONG EYES WITH NAMD THAT RECEIVED >6 MONTHS OF FARICIMAB TREATMENT IN THE UK: FARWIDE-NAMD

Corresponding author: James Talks, orcid.org/0000-0001-6126-6476

James Talks¹, Deepali Varma², Gabriella de Salvo³, Praveen Patel⁴, Samantha de Silva⁵, Richard Gale⁶, Martin McKibbin⁷, Gloria Chi⁸, Natalie James⁹, Alice Rothwell⁹, Neil Shah¹⁰, Amanda Downey¹⁰, Melanie Dodds¹¹, Parul Dayal⁸

¹Newcastle Upon Tyne Hospitals NHS Foundation Trust, Newcastle, UK; ²South Tyneside and Sunderland NHS Foundation Trust, Sunderland, UK; ³Southampton University Hospital, Southampton, UK; ⁴Moorfields Eye Hospital NHS Foundation Trust and UCL Institute of Ophthalmology, London, UK; ⁵Oxford University Hospitals, Oxford, UK; ⁶University of York and York and Scarborough Teaching Hospitals NHS Foundation Trust, York, UK; ⁷St James's University Hospital, Leeds, UK; ⁸Genentech, Inc, San Francisco, CA, USA; ⁹Roche Products Ltd, Welwyn Garden City, UK; ¹⁰Hoffmann-La Roche Ltd, Basel, Switzerland, UK; ¹¹Mediso Ltd, Leeds, UK

Disclosures: James Talks - European advisory board, speaker, travel grant; Allergan - Advisory board, speaker, travel grant; Apellis - Advisory board; Roche - Consultancy, advisory board, travel grant; Bayer - Advisory board, speaker, travel grant, educational grant; Heidelberg Engineering - Speaker, travel grant; Boehringer Ingelheim - Consultancy; Teva - Advisory board

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Gabriella De Salvo - Novartis - European advisory board, speaker, travel grant; Allergan - Advisory board, speaker, travel grant; Apellis - Advisory board; Roche - Consultancy, advisory board, travel grant; Bayer - Advisory board, speaker, travel grant, educational grant; Heidelberg Engineering - Speaker, travel grant; Boehringer Ingelheim - Consultancy; Teva - Advisory board.

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Samantha De Silva - Consultancy: Roche, AbbVie. Travel: Bayer Richard Gale - AbbVie, Apellis, Bayer, Biogen, Boehringer Ingelheim, Novartis, Roche

Martin McKibbin - Consultancy fees, educational travel grant and research funding: Roche

Gloria Chi - Employee: Genentech, Inc.

Natalie James - Employee: Roche Products Ltd.

Alice Rothwell - Employee: Roche Products Ltd.

Neil Shah - Employee: Hoffmann-La Roche Ltd

Amanda Downey - Employee: Hoffmann-La Roche Ltd

Melanie Dodds - Employee: Mediso Ltd

Parul Dayal - Employee: Genentech, Inc.

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Introduction: Faricimab, the only bispecific antibody inhibiting angiopoietin-2 (Ang-2) and vascular endothelial growth factor (VEGF)-A, was approved for nAMD in the UK in May 2022. We report results from a multicentre real-world data study evaluating patient characteristics, treatment patterns and outcomes among eyes with 6 months of follow-up on faricimab.

Method: The ongoing, retrospective, FARWIDE study includes patients (n = 5804) with a documented diagnosis of nAMD (n = 6978 eyes) who initiated faricimab (≥1 injection) between June 2022–June 2023 at participating National Health Service sites (n = 21) that use the Mediso ophthalmic electronic medical record system.

Results: 28% of patient-eyes were naïve, 72% were previously treated (aflibercept 84.3%). 81% of naïve eyes and 59% of previously-treated eyes received four initial injections per UK label. 18.4% of naïve eyes and 25.3% of previously-treated eyes had ≥6 months follow-up. Previously-treated eyes had received anti-VEGF treatments for a mean of 3.2 years (SD2.8) before initiating faricimab. Among eyes with ≥6 injections, the treatment interval between the 5th and 6th faricimab injection was 9.7 (SD2.6) weeks in naïve eyes and 7.6 (SD2.1) weeks in previously-treated eyes. Mean (SD) gain in vision from baseline by injection 6 was +5.3 (SD0.9) ETDRS letters (p < 0.001) among naïve eyes and +0.6 (SD0.4) letters among previously-treated eyes. 742 eyes (24% naïve) had ≥12-months of follow-up on faricimab. Naïve eyes received a mean (SD) of 4.7 (0.6) and 6.7 (1.4) injections in the first 6 and 12-months of faricimab treatment, and previously-treated eyes received 4.3 (1.0) and 7.1 (2.0) faricimab injections, suggesting a rapid extension in intervals after the initial doses.

Conclusion: Initial results support the real-world durability of faricimab in treatment naïve and previously treated eyes.

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TOWARDS AUTOMATED OPTICAL COHERENCE TOMOGRAPHY (OCT) DATA ANALYSIS FOR A REAL-WORLD DIABETIC MACULAR OEDEMA (DMO) TREATMENT COHORT

Corresponding author: James Talks, orcid.org/0000-0001-6126-6476

James Talks¹, Tunde Peto², Xiaoyong Wang³, Anda Gershon³, Geeta Menon⁴, Faruque Ghanchi⁵, Xin Chen³, Vivian Look³, Huanxiang Lu³, Michael H Chen³, Daniela Ferrara³, Gloria Chi³, Qi Yang³

¹Newcastle Upon Tyne Hospitals NHS Foundation Trust; ²Centre for Public Health, Queen's University Belfast, Belfast, UK; ³Genentech, Inc., San Francisco, CA, USA; ⁴Frimley Health NHS Foundation Trust, Frimley, UK; ⁵Bradford Teaching Hospitals NHS Foundation Trust, Bradford, UK

Disclosures: Talks and Peto: advisory boards Roche. Other authors employed by Genentec

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Introduction: To facilitate automated and fast interpretation of large real-world datasets incorporating imaging and enabling in-depth scientific and clinical investigations, an OCT image analysis pipeline was built for retrospective quantitative image analysis. This was tested on a large real-world data DMO cohort from the National Health Service in the UK.

Method: An anonymized cohort of 858 patients with DMO and anti-vascular endothelial growth factor (VEGF) treatment was extracted from the Medisoft electronic health record. Corresponding anonymized Heidelberg OCT images were organized into a standard DICOM format. Full retinal thickness layer and fluids were segmented by an in-house automated algorithm on extracted volumes. 80 randomly selected baseline OCT volumes were manually graded for image quality, fovea centration, internal limiting membrane (ILM), Bruch's membrane (BM), intraretinal fluid (IRF), and subretinal fluid (SRF) segmentations to confirm the robustness of the automated segmentation.

Results: 858 patients (one eye per patient) received 59,481 scans and 11,742 visits. 11,544 visits included OCT images; most protocols were 25 B-scans. After matching images at first treatment and follow-ups, 9042 macular OCT volumes were segmented. 85% of OCT volumes were graded as fair/good image quality. 91% of images were rated as acceptable for fovea centration. Among images with fair/good quality, 88% of ILM and BM segmentations and 97% IRF and SRF segmentations were graded as fair/good. With univariate Fisher's exact tests, segmentation quality was significantly related to image quality.

Conclusion: A fit-for-purpose OCT image analysis pipeline was feasible for this patient cohort and yielded fair/good quality segmentations for most images. Manual reviews on representative subsets of the data added confidence on the trustworthiness of the pipeline.

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WET AMD EMERGENCY SLOT AT EYE CASUALTY PILOT PROJECT

Corresponding author: Georgios Tsokolas, orcid.org/0000-0003-0935-4844

Georgios Tsokolas¹, Owen Anderson¹, Non Matthews¹, Charalambos Tossounis¹

¹Royal Bournemouth Hospital, University Hospitals Dorset NHS Foundation Trust, Bournemouth, UK

Disclosures: No conflicts of interest and no funding

Introduction: Due to COVID-19 pandemic and the frequent review/treatment nature of the Wet-AMD disease, there is significant pressure on NHS Trusts. This pilot project aims to

relieve these pressures and provide a fast-track access to known already treated Wet-AMD patients with sudden changes in their vision.

Method: The project was introduced in February 2023 and is a collaboration between the Medical Retina (MR) Service and the Eye Emergency Department (Eye-ED). An emergency Wet-AMD slot allocation was introduced to the morning weekday Eye-ED booking diary. It focuses on Wet-AMD patients already known to the service with sudden deterioration in the vision of the treated or the fellow eye.

The booking is done through the Eye-ED telephone service. Vision, IOP, dilation and imaging occurs on arrival. The Eye-ED doctor reviews the patient, the scans and liaises with the MR team. A decision is made whether an urgent injection/further follow-up is needed. The injection is performed on the same day and a management plan is formulated and documented.

Results: 8-months of data were presented during the departmental audit meeting in October 2023. 65 patients were seen at the eye-ED Wet-AMD slot. 45 were known Wet-AMD patients and 20 were newly diagnosed Wet-AMD patients. 34 of the 65 patients (52%) were injected the same day.

Conclusion: The project highlighted an efficient management of a significant number of Wet-AMD patients in distress with sight-threatening changes. The prevention of sight loss and reduction of stress levels of the Eye-ED & MR staff were noted. Additionally, a cost-effective management of these patients meant that no extra injection/one-stop clinics were needed. This service could be easily implemented on every Eye-ED.

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EFFICACY AND SAFETY OF FARICIMAB IN ALTIMETER: A TRIAL EXPLORING BIOMARKERS OF ANG-2 INHIBITION IN PATIENTS WITH DMO

Corresponding author: Deepali Varma orcid.org/0000-0002-6769-2967

Deepali Varma¹, Florie Mar², Manuel Amador², Carl Glittenberg³, Kara Gibson⁴, Alex Kotak⁴, Audrey Souverain³

¹South Tyneside and Sunderland NHS Foundation Trust, Sunderland, UK; ²Genentech Inc., San Francisco, CA, USA; ³F. Hoffmann-La Roche Ltd., Basel, Switzerland; ⁴Roche Products Ltd., Welwyn Garden City, UK

Disclosures:

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Introduction: Dual angiopoietin-2 (Ang-2) and vascular endothelial growth factor (VEGF)-A pathway inhibition with faricimab promotes vascular stability and extends treatment

durability for diabetic macular oedema (DMO). The phase 2b ALTIMETER biomarker hypothesis-generating study will explore the associations between clinical endpoints, multimodal imaging assessments, and aqueous humour biomarker patterns in patients with DMO treated with faricimab.

Method: ALTIMETER (NCT04597918) was a 6-month, open-label, single-arm, exploratory study that enrolled treatment-naïve patients (age ≥ 18 years) with a diagnosis of DMO ($n = 99$). Patients received faricimab 6mg every 4 weeks (Q4W) until week 24. Exploratory endpoints include changes from baseline over time in BCVA, CST, IRF, macular leakage, multimodal imaging, and aqueous humour biomarker patterns. All exploratory outcomes will be analysed in the modified intent-to-treat (mITT) population (90.9%, $n = 90$).

Results: Mean (SD) age was 59.5 (9.8) years and most patients were male ($n = 61$, 61.6%) and white ($n = 86$, 86.9%). At week 24, adjusted mean (95%CI) change from baseline in BCVA and CST was +9.2 letters (7.5, 10.9) and $-200.2 \mu\text{m}$ (-214.1 , -186.2), respectively. The proportion of patients with absence of IRF at week 24 was 26.1% (17.3%, 36.6%). Macular leakage area was reduced following treatment with faricimab, from a median (range) of 28.6 mm^2 (1.7–40.4) at baseline to 2.8 mm^2 (0–38.6) at week 20. High macular leakage was associated with elevated Ang-2-related proteins and immune regulators at baseline, which were reduced following treatment with faricimab. Faricimab was generally well-tolerated, and safety was consistent with the known safety profile of faricimab.

Conclusion: At 6 months, Q4W faricimab injections in ALTIMETER demonstrated meaningful improvements in vision and anatomic outcomes consistent with those in the phase 3 YOSEMITE/RHINE trials.

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REAL WORLD USE OF FARICIMAB IN TREATING NAMD PATIENTS PREVIOUSLY TREATED WITH OTHER ANTI VEGF IN CTMUHB

Corresponding author: Shuchita Verma, orcid.org/0009-0004-8668-8367

Shuchita Verma¹, Amit Gaur¹

¹Royal Glamorgan Hospital, Cwm Taf Morgannwg University Health Board, Llantrisant, UK

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Introduction:

Purpose

To assess efficacy and safety of intravitreal Faricimab in patients previously treated with Eylea or Lucentis

Method: Retrospective study of clinical notes of 32 eyes diagnosed previously with nAMD.

Visual acuity, OCT parameters (CMT plus presence of SRF/ IRF) were assessed at the beginning of switching treatment to Faricimab and at each visit. Interval between injections and complications were also noted.

Results: 32 eyes previously treated with Eylea/ Lucentis received Faricimab.

Study period was September 2022 to August 2023.

31 eyes had nAMD (7 were occult at diagnosis), 1 had RAP/ CSCR.

Cohort included patients diagnosed and receiving treatment from 2013 onwards, 9 previously treated with Lucentis plus Eylea and 23 with Eylea.

Reasons for switching treatment included:

1. unable to extend interval between injections to beyond four weeks (16)
2. unable to extend beyond six weeks (7)
3. persistent SRF (3)
4. not responding to treatment (4)
5. chronic SRF (1)
6. recurrence (1)

20 eyes were dry after 4 injections.

19 eyes were dry by end of August 2023 of which 7 were dry after two injections.

We were able to extend interval between treatment from four to six weeks in 27 patients.

No complications were noted.

Conclusion: Switching eyes to Faricimab had a positive effect on vision, improving it by an average of 3 ETDRS letters with OCT showing improvement in CMT. This outcome was following treatment with four injections. Follow up between injections were able to be increased leading to reduced hospital appointments. The patients are being followed up for longer term outcome analysis.

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COMPARISON OF 135° FIELD SCANNING CONFOCAL OPHTHALMOSCOPE IN STAGED-MYDRIATIC MODE AND MYDRIATIC TWO 45° FIELD DIGITAL IMAGING IN NHS DIABETIC EYE SCREENING

Corresponding author: Anthony Vukic, orcid.org/0000-0001-8935-622X

Anthony Vukic¹, Marta Gruszka-Goh^{1,2}, Paul Galsworthy¹, Scott Vallance¹, Peter H Scanlon^{3,4}

¹Gloucestershire Hospitals NHS Foundation Trust, Cheltenham, UK;

²The Royal College of Ophthalmologists' National Ophthalmology Audit, London, UK; ³Nuffield Department of Clinical Neuroscience, University of Oxford, Oxford, UK; ⁴University of Gloucestershire, Gloucester, UK

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Introduction: The purpose of the study was to determine if the Optos California (Optos) was safe to use in the staged-mydriatic mode in the Diabetic Eye Screening Programme (DESP).

Method: Trial participants were 1497 individuals (2,993 eyes) aged 15 to 94 with diabetes mellitus participating in DESP or virtual eye clinics for delayed hospital follow up.

Scanning Confocal Ophthalmoscope (SCO) photographs were taken before dilation and compared to two-field 45 degrees mydriatic digital photography. When image quality was poor or pupil size less than 3mm, photos were taken with Optos before and after dilation. If the non-mydriatic images were gradable,

they were used in comparison to the mydriatic digital images. To compare the same areas, SCO photos were graded with two 45° field overlay using DESP grading criteria.

Results: Out of 2993 photos taken using the Optos, 5.1% with 95% CI(4.3%-6.0%) were ungradable pre-dilation and 3.4% (95% CI: 2.8%-4.2%) with staged mydriasis.

Sensitivity and specificity in the central area comparing the Optos with two 45-degree field mydriatic digital reference were calculated for 1477 patients (2856 eyes). The sensitivity and specificity for any diabetic retinopathy (DR) were 91.9% (95% CI: 90.5%-93.2%) and 78.1% (95% CI: 75.7%-80.3%) respectively. The sensitivity and specificity for referable DR were 77.6% (95% CI: 73.9%-80.9%) and 95.4% (95% CI: 94.5%-96.2%) respectively.

Conclusion: The ungradable image rate of 5.1% for Optos without dilation in a predominantly White Caucasian population was lower than reports for digital cameras and it performed accurately in the detection of any retinopathy in the central area.

There was a steep learning curve for the grading team with some novel presentations of acquisition artefacts and colour differences in features of DR.

P169

THE EFFECT OF LIPOPOLYSACCHARIDES ON INDUCIBLE NITRIC OXIDE SYNTHASE IN HUMAN MICROVASCULAR ENDOTHELIAL CELLS (HMEC-1) AND HUMAN CHOROIDAL ENDOTHELIAL CELLS (HCECs)

Corresponding author: Ruoxin Wei, orcid.org/0009-0006-4207-8895

Ruoxin Wei¹, Elizabeth A Stewart¹, Winfried M Amoaku¹

¹The University of Nottingham, Nottingham, UK

Disclosures: WA is on the Eye editorial board

Introduction: Toll-like receptors (TLRs) are a family of proteins that initiate the innate immune response in reaction to invading microbes. It has been shown that TLR4 is expressed significantly in both human microvascular endothelial cells (HMEC-1) and human choroidal endothelial cells (hCECs). Lipopolysaccharide (LPS), recognized by TLR4, is the significant component of the outer cell membrane of Gram-negative bacteria. The TLR4-LPS complex activates MAPK and NF-κB pathways, which will induce the expression of nitric oxide synthase (NOS), a known mediator of angiogenesis that is involved in the pathogenesis of choroidal and retinal diseases. This research aims to investigate the expression of iNOS in HMEC-1 and hCECs following LPS stimulation.

Method: hCECs were isolated and cultured as per the method previously developed and published in our laboratory in Nottingham. The protein expression of iNOS were examined by western blotting.

Results: The expression of iNOS protein increased in a time-dependent manner in response to LPS stimulation in HMEC-1. The protein expression reached a maximum following 24-hour treatment. In hCECs, the expression of iNOS was up-regulated significantly after 12-hour incubation with LPS. The rise was then followed by a fall at 24 hour.

Conclusion: This study identified a more prominent expression of iNOS in primary cells (hCEC), and elucidated the mechanism of TLR4-LPS for initiating angiogenic factors which associate with and modulate progression of intraocular vascular disease. This study will help improve our understanding of different choroidal vascular diseases, as well as the development of future treatments of intraocular vascular diseases.

P170

EFFICACY, SAFETY, AND DURABILITY OF FARICIMAB IN MACULAR OEDEMA DUE TO RETINAL VEIN OCCLUSION: 72-WEEK RESULTS FROM THE PHASE 3 BALATON AND COMINO TRIALS

Corresponding author: Michael Williams, orcid.org/0000-0002-5051-5921

Michael Williams¹, Francis R Abreu², Pablo Arrisi³, Aachal Kotecha³, Ying Liu², Liliana Paris², Anne-Cecile Retiere³, Jeffrey R Willis²

¹Queen's University of Belfast, Belfast, UK; ²Genentech Inc, San Francisco, CA, USA; ³Roche Products Ltd, Welwyn Garden City, UK

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Liliana P. Paris - Financial disclosures: employee Genentech Inc.

Anne-Cecile Retiere - Financial disclosures: employee Roche Products Ltd.

Jeffrey R. Willis - Financial disclosures: employee Genentech Inc.

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Introduction: Week 24 results from the phase 3 BALATON/COMINO trials support the efficacy and safety of faricimab, a dual angiopoietin-2/vascular endothelial growth factor (VEGF)-A inhibitor, for the treatment of macular oedema due to retinal vein occlusion (RVO). Week 72 results from these trials will include new information on the long-term efficacy, safety, and durability of faricimab in this patient population.

Method: Anti-VEGF treatment-naïve patients with centre-involving macular oedema secondary to branch RVO (BALATON [NCT04740905], N = 553) or central/hemiretinal RVO (COMINO [NCT04740931], N = 729) were enrolled in phase 3, multicentre, randomized, double-masked trials of faricimab. From day 1 through week 20, patients received 6 monthly intravitreal injections of faricimab 6.0 mg or aflibercept 2.0 mg; the primary endpoint was at week 24. From week 24 through 72, all patients received faricimab 6.0 mg at up to every 16-week (Q16W) intervals per a modified treat-and-extend (T&E)-based dosing regimen (using CST and BCVA).

Results: BCVA gains from baseline at week 24 with faricimab were noninferior to aflibercept in BALATON (adjusted mean[95% CI] change: +16.9[15.7,18.1] letters vs +17.5[16.3,18.6] letters) and COMINO (+16.9[15.4, 18.3] letters vs +17.3[15.9, 18.8] letters). CST reductions from baseline were similar across treatment arms. Faricimab was well tolerated through week 24, with a safety profile similar to aflibercept.

Conclusion: Following positive 24-week results, 72-week results from BALATON/COMINO will provide important information on the longer-term efficacy and safety of faricimab in patients with RVO, in addition to the durability of faricimab treatment.

MISCELLANEOUS

P171

DIABETIC RETINOPATHY CHALLENGES IN DEVELOPING COUNTRIES: AWARENESS AND SEVERITY AT PRESENTATION OF DIABETIC RETINOPATHY AND BARRIERS TO ITS TREATMENT

Corresponding author: Nazli Gul, orcid.org/0009-0008-8224-4100

Imran Ahmad¹, Nazli Gul¹

¹Khyber Medical College/Khyber Teaching Hospital Peshawar, Peshwar, Pakistan

Declarations: No conflicts of interest and no funding

Introduction: Diabetic retinopathy is a common cause of visual loss in diabetic patients. Diabetic population is increasing throughout the world and Pakistan is on number third in terms of diabetic population. Pakistan is having the highest prevalence of diabetes 26.7%. Due to low socio-economic status of the people, poor health education and resources for treatment our patients present with severe disease. In this study awareness level of diabetic patients about diabetic retinopathy, risk factors for progression, referral system from physician/ endocrinologist, stage of retinopathy at presentation and barriers to treatment were acquired.

Method: Descriptive study carried out in Ophthalmology Department of Khyber Teaching Hospital Peshawar and Qazi Hussain Ahmad Hospital Nowshera from March 2023 to September 2023. Diabetic patients presenting for first time were included in the study. Total 1056 eyes of 528 patients were enrolled in the study.

Results: Diabetic macular edema (DME) was present in 53.88% with center-involving DME on OCT in 21.94% eyes whereas non-center involving DME in 32.38%. In 167 (15.81%) eyes were having mild NPDR, 216 (20.45%) moderate NPDR, 117 (11.07%) severe NPDR, 145 (13.73%) PDR and 43 (4.07%) eyes presented with ADED. Endocrinologist referred 92 (17.42%) patients and 23 (4.35%) by general physician. 56 (10.60%) knew that diabetic retinopathy is associated with visual loss. The common barrier to treatment of diabetic retinopathy was lack of awareness (83.14%), affordability (74.62%), dependency (42.80%), transportation (23.86%).

Conclusion: Health education about diabetic retinopathy is very important in developing countries in order to decrease burden on health system which will directly affect the screening system. Along with patients, treating physician and endocrinologist need to be made aware about this blinding disease for proper referral and timely management.

P172. Abstract not included

P173

ARE SOCIAL MEDIA PROMOTING OCULAR PROTECTION IN THE COMMUNITY? CONTENT ANALYSIS OF HOME-IMPROVEMENT DO-IT-YOURSELF VIDEOS ON YOUTUBE

Corresponding author: Chung Shen Cean, orcid.org/0000-0002-1625-4078

Chung Shen Cean¹, UK Midlands Ophthalmology Trainees Research Collaborative¹, Ian De Silva¹

¹University Hospitals of Leicester NHS Trust, Leicester, UK

UK Midlands Ophthalmology Trainees Research Collaborative (UMOTRC) Writing group: Chung Shen Cean, Christina Lim, Daisy Duke, Hamza Abdou, Loic Sermeus, Alicia Poyser, Hetvi Bhatt, Kerolos Bassilious, Maninee Purohit, Jennifer Kwan, Modupe Sodeine, Mervyn Thomas, Alexander Stubbing-Moore, Helen Kuht, Vivek Tank, Sabbur Anwar, Sara R Sabur, Anjali Gohil, Ian De Silva

Declarations: No conflicts of interest and no funding

Introduction: During the COVID-19 pandemic, many countries worldwide introduced lockdown measures to control its spread and people are spending more time at home. There were reports of increasing incidence of Do-it-Yourself (DIY)-related trauma. As YouTube is becoming more popular among people to search for practical video demonstrations of home DIY projects, we would like to evaluate whether ocular safety is demonstrated in these videos.

Method: A cross-sectional evaluation of YouTube videos was performed using keywords of "DIY home" determined using Google Trends. Search parameters were restricted to the period of the pandemic. As YouTube users usually click the first 3 pages of search results, only the first 300 videos were included. Data collected included baseline video characteristics, DIY activities shown, ocular safety hazard involved and whether ocular safety was demonstrated. UK Health and Safety Executive (HSE) regulations on eye protection were used as a reference guidance to determine whether eye protection was adequate.

Results: After excluding 9 videos which were not available for analysis, 148 videos met the inclusion criteria. Most videos (67.6%) demonstrated DIY projects involving dust, projectiles or power-driven tools. Only 6 videos mentioned ocular safety in video description and 11 videos discussed ocular safety in the video. 109(73.6%) DIY videos did not demonstrate any ocular protection. Based on reference guidance, 126(85.1%) DIY videos showed inadequate ocular protection. Worryingly, no video provides any advice in case of eye injury.

Conclusion: Portrayal of ocular protection in social media platforms is still inadequate. It is imperative to increase public awareness of the importance of wearing appropriate ocular protection when doing DIY work. A campaign to raise awareness about this should be organised, especially in the UK national eye health week.

P174

THE FATE OF OPHTHALMOLOGY TRAINEES IN THE UK—CCT HOLDERS 2017 TO 2021

Corresponding author: Zhihang Cheng, orcid.org/0000-0002-9390-0942

Zhihang Cheng¹, Muhammad Zuhair Mustafa²

¹Ophthalmologists in Training Group, Royal College of Ophthalmologists, London, UK

Declarations: No conflicts of interest and no funding

Introduction: The Ophthalmologist in Training group (OTG) has been conducting surveys to explore the outcomes of Certificate of Completion of Training (CCT) holders. This aimed to assess the impact of the Ophthalmic training programme and monitor changes with the upcoming curriculum transition. Both the OTG and the College are interested in understanding trainees' choices of Specialist Interest Areas (SIAs) and long-term practice locations for workforce planning.

Method: The Royal College of Ophthalmologists provided a list of CCT holders (2017-2021). Data on fellowship and consultant

posts, including SIAs and locations, were collected. Analysis focused on the proportion of consultant posts by SIAs, trends over the period, and the geographic alignment of consultant posts with training and fellowship regions.

Results: Out of 401 trainees attaining CCT, 82% were included in the analysis. Compared to previous surveys, there's a consistent uptake of consultant posts within the training region (61%), out of region (31%) and abroad (8%). Detailed analysis for 2019–2021 suggests a trend of post-CCT holders staying in the region where they undertook fellowship (49%). Some undertook fellowships elsewhere but returned to their original region for consultancy (16%). Specialist interest areas distribution remains even, but a notable reduction in uptake of medical retina posts is observed from 22% in the last survey 2011–16 to 11% over the last 4 years.

Conclusion: This data sheds light on the current state of ophthalmic training towards consultant practice. The study anticipates changes with the new curriculum, particularly in consultant post uptake within regions and the distribution of fellowships. The findings provide valuable insights for ongoing workforce planning in ophthalmology.

P175

INVESTIGATING SLEEP DEPRIVATION IN INDIVIDUALS WITH VISUAL IMPAIRMENT

Corresponding author: Katie Curran, orcid.org/0000-0002-7071-109X

Katie Curran¹, Laura Cushley¹, Lajos Csincsik¹, Guiliiana Silvestri², Ruth Hogg¹, Alyson Muldrew¹, Tunde Peto^{1,2}

¹Queen's University Belfast, Belfast, UK; ²Belfast Health and Social Care Trust, Belfast, UK

Disclosures: Giuliana Silvestri: Consultant Advisor Roche and Bayer. The other authors declare no conflicts of interest.

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Introduction: This study focused on gathering opinions from the visually impaired community on the impact of sleep deprivation.

Method: This patient and public (PPI) scoping exercise employed patient questionnaires at a Patient Engagement Day in Belfast to assess sleep issues among adults with VI. The Patient Engagement Day provided an interactive platform for open discussions and shared experiences, enabling a deeper understanding of the impacts of sleep difficulties on daily life. The tailored questionnaire investigated sleep patterns and quality, offering insights into the impact of sleep problems among people with VI.

Results: The questionnaire was completed by 16 participants, with 5/16 (31.3%) identifying as female. In total, 8/16 individuals (50%) reported experiencing notably low sleep quality, ranging from very poor to poor. Five participants (31.3%) reported using sleeping aids or medications, including, talking books, positive affirmations or sleeping tablets, with 2/5 (40.0%) reporting moderate-good sleep. One participant tried melatonin medication previously but found it to be ineffective. Meanwhile, another individual experienced distressing nightmares while using this medication, prompting them to discontinue its usage. One individual expressed that "lack of sleep is more distressing than sight loss", while another admitted to experiencing as little as two hours of sleep at night, leading to persistent fatigue.

Conclusion: Our findings highlight a research gap in understanding poor sleep quality among individuals with VI; however, it's important to note that sleep problems are widespread,

affecting people irrespective of blindness. The study's results underscore the importance of recognising and addressing the patients' issues stemming from loss of sight. Devising effective interventions for individuals with VI, involving patients/public in shaping future research is crucial to ensure alignment with the needs and experiences of those affected.

P176

GENDER DISTRIBUTION OF SPEAKERS AT MAJOR OPHTHALMOLOGY CONFERENCES

Corresponding author: Maria Georgi, orcid.org/0000-0001-6509-4691

Maria Georgi^{1,2}, Kristie Hing Chi Leung³, Arslan Raja⁴, Sara Raja⁵, Nain Tara Raja⁵, Rashmi G Mathew⁶

¹Imperial College London, London, UK; ²West Middlesex University Hospital, London, UK; ³Brighton and Sussex University Hospitals NHS Trust, Brighton, UK; ⁴University of Edinburgh, Edinburgh, UK; ⁵University of Aberdeen, Aberdeen, UK; ⁶Moorfields Eye Hospital, London, UK

Declarations: No conflicts of interest and no funding

Introduction: Females are disproportionately under-represented in ophthalmology despite the upward trend in medical school entry for girls. Visibility of female clinicians at academic meetings gives a cross-sectional view of inclusion within the field, and previous study looking at American ophthalmology conferences have revealed a relatively low percentage of women speakers. As most data originated from the United States (US), we aimed to quantify the prevalence of female faculty and all-male panels ("manels") at major international ophthalmology meetings in Oceania, Europe as well as the US.

Method: We retrospectively analysed the most recent annual meetings held by major international ophthalmology organisations in 2016 to 2021. Demographic and academic metrics were collected for each invited lecturer, panellist and chair/moderator. Data was obtained through Web of Science, online biography and Gender Balance Assessment Tool.

Results: Conferences held by four international ophthalmology organisations were identified, including 202 sessions delivered by 1145 speakers. On average across all speakers, 27.9% (n = 317) were female. Of the 191 panel sessions, 29.8% (n = 57) were manels. There was a statistically significant difference in the proportion of manels over time (p = 0.01), at its lowest in 2018 and highest in 2021. Male speakers had statistically higher mean H-index (p = 0.01) and number of publications (p = 0.01) compared to their female counterparts.

Conclusion: Despite the encouraging findings that manels are a minority, female speaker representation remains poor at international conferences. Conference organisers should ensure the distribution of speakers reflects the diversity of the profession, which would provide different thoughts and perspectives in advancing the field.

P177

WELLBEING AND BURNOUT AMONGST OPHTHALMOLOGY DOCTORS IN TRAINING (DIT) IN THE NORTHERN REGION: OUR PERSPECTIVE AND PLAN

Corresponding author: William Hodgkinson, orcid.org/0009-0000-1768-3870

William Hodgkinson¹, Rajen Gupta¹, Lucy Clarke¹

¹Royal Victoria Infirmary, Newcastle upon Tyne, UK

Declarations: No conflicts of interest and no funding

Introduction: In 2023 the GMC reported that increasing workload, reduced satisfaction and heightened risk of burnout were reported as key factors affecting doctors' wellbeing in the UK. This follows on from a 2021 GMC report stating that ophthalmology had the greatest increase in risk of burnout compared to other specialties. This raises the question of the causation of these findings.

Method: We sought to understand and quantify the levels of burnout amongst ophthalmology DiT in the Northern Region. We sent out two simultaneous questionnaires - the first a validated single question to quantify the severity of burnout and the second was the Burnout Clinical Subtypes Questionnaire (BCS-12), aiming to characterise the burnout being experienced. We also included qualitative questions to gain further background.

Results: Over 60% of DiT in the Northern Region responded to the questionnaire. 100% of them reported having symptoms of burnout with 63% reporting definite burnout. The BCS-12 characterised the burnout being experienced as 'Frenetic/Overloaded' with a large proportion also experiencing 'Underchallenged/Lack of Development'. Qualitative answers included themes of lack of support (personal and clinical), difficulties with short rotations and reduced training opportunities during clinical duties due to service pressures.

Conclusion: Our next steps sought to make realistic change through discussions aiming for longer rotations, encouraging the social aspect by returning to face-to-face teaching and implementing a buddy system to improve cohesion and provide dedicated learning opportunities. Our study has highlighted that specific and targeted solutions should be used to tackle wellbeing and burnout issues rather than a blanket approach. We encourage other regions to undertake similar projects with an aim of reducing national risk of burnout in ophthalmology DiT.

P178

SPEAKING UP: DOES GENDER BALANCE OF SPEAKERS AT THE RCOPTH ANNUAL CONGRESS REFLECT ITS AIMS?

Corresponding Author: Rebecca Jones, orcid.org/0000-0002-4104-9949

Rebecca Jones¹

¹Cheltenham General Hospital, Cheltenham, UK

Declarations: No conflicts of interest and no funding

Introduction: In its role of as the voice of UK Ophthalmology, the RCOphth has a clear set of values encouraging equity and diversity, to ensure the College is representative of its membership and the wider general public. As the flagship event organized by the RCOphth, the Congress acts as a way to prove its commitment to goals of diversity. There is known to be an underrepresentation of female doctors in ophthalmology (33.8% of members are female) compared to the wider medical community (51.9% of GMC registered doctors are female). It is therefore important to recognise if the outward-facing image of the College, shown by the gender divide in speakers at the Congress, propagates this inequality.

Method: Using the programme of talks in addition to the digital forum for recorded talks, speakers were identified and their gender determined by their publicly available profiles.

Results: Of the 321 invited speakers, 58% (n = 186) identified as male and 42% (n = 135) identified as female. This varied between subspecialties: in Cataract sessions, 14% of speakers were female, compared to 67% in Oculoplastics. When looking at the chairs of each session (n = 76), the gender divide was greater, with 68% (n = 52) of sessions chairs being male and 32% (n = 24) female.

Conclusion: In order to demonstrate the aspirations of the RCOphth, the Congress should strive to foster representation, by ensuring the make-up of the chairs and speakers reflects a desire to promote gender equality in the ophthalmic workplace.

P179

TEBENTAFUSP TREATMENT SLOWS DISEASE PROGRESSION IN PREVIOUSLY TREATED MALIGNANT UVEAL MELANOMA

Corresponding author: Muhammad Khursheed Ullah Khan Marwat, orcid.org/0009-0003-7906-3038

Muhammad Khursheed Ullah Khan Marwat^{1*}, Dulani Ranatunge^{1*}

¹Hull University Teaching Hospital NHS Trust, Cottingham, UK
*co-first authors

Disclosures: No conflicts of interest and no funding

Introduction: Survival rates are low in patients with metastatic uveal melanoma, and no treatments have been proven to benefit survival. However, in the recent trial by Nathan P et al Tebentafusp has shown overall survival of 73% at 1 year vs 59% by the Investigator's Choice.

Ocular melanoma is the most common ocular, malignancy with five years metastasis incidence rate of 25% with liver being the most common site of metastasis (89%). [1] Metastasis is a poor prognostic factor with a median survival of less than a year. [1,2]

Method: A prospective study of a disease status in a 70-year old patient receiving tebentafusp.

Results: A 72-year-old female with known metastatic Uveal melanoma was initially diagnosed in 2017. She initially received 2 lines of treatment with disease progression.

Tebentafusp was used as 3rd line treatment under early access programme by Immunocore.

Treatment started Dec 2021 Initially, a partial response to treatment was observed, which has since been followed by the maintenance of stable disease up to the most recent staging scan conducted in September 2023. She has received a total of 97 weeks of treatment till 30th October 2023 with good tolerance apart from mild rash during 2nd cycle and occasional sickness.

Conclusion: Tebentafusp can be considered as a viable treatment option, warranting further research and potential consideration as a 2nd or 3rd line treatment for previously treated metastatic uveal melanoma (mUM). Given the limited options for treatment of metastatic uveal melanoma, Tebentafusp stands as a hopeful addition to the treatment armamentarium, providing insights into its efficacy and paving the way for enhanced understanding of its mechanisms for the broader benefit of patients facing this challenging malignancy.

P180

DOES THE NUMBER OF PRE-OPERATIVE ASSESSMENTS OF STRABISMUS PATIENTS INFLUENCE THE SURGICAL OUTCOME?

Corresponding author: Idrhys Zaman-Khan, orcid.org/0009-0008-2390-0964

Declarations: No conflicts of interest and no funding

Introduction: Surgical strabismus traditionally attended for several face-to-face (FTF) measurements before being listed for surgery. During the Covid-19 pandemic, a streamlined care-pathway was adapted to reduce FTF contact time to protect staff and manage floor-flow. Patients were listed for surgery after just one face-to-face orthoptic assessment, followed by a further measurement at pre-assessment.

Method: This is a case cohort to look at surgical outcomes for consecutive exotropia patients who underwent re-do horizontal muscle surgery over the last 14 years, from a surgeon's prospective database.

Results: 65 patients had surgery before the pandemic (Group-A) and 28 since the pandemic (Group-B). The mean age was 34.9 and 39.1 years respectively. The mean pre-operative near-angle was 37.5 prism dioptres (PD) and distance-angle of 34.8PD for group-A. Group B was 39.3PD for near and 36.3PD for distance. 89.1% in group-A and 89.2% in group-B had suppression. Postoperative alignment within 15PD was achieved, after a combination of fixed and adjustable procedures, for 75.4% in group-A and 88.5% in group-B with predominant adjustable procedure. The mean 2-weeks post-operative alignment was -1PD for group-A and -1.3PD for group-B. Group-A had an average of 3.6 measurements, with 3.3 doctor consultant prior to surgery and group-B had 2 FTF orthoptics and 1 FTF doctor contact.

Conclusion: 44.4% reduction in face-to-face orthoptics contact-time and 69.7% less consultant contact-time had not negatively affected the surgical outcome.

Strabismus surgery can be considered in selected cases with less FTF pre-operatively and improve cost-efficiency of our adult strabismus service with 'block-contract' status.

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ARE THERE PREDICTORS OF SUCCESSFUL BIOPSY TESTS IN PATIENTS TREATED WITH RUTHENIUM PLAQUE BRACHYTHERAPY FOR CHOROIDAL MELANOMA?

Corresponding author: Keri McLean, orcid.org/0000-0002-8907-1176

Keri McLean^{1,2}, Hassaan Amer², Rumana Hussain^{2,3}

¹St Paul's Eye Unit, Liverpool University Hospitals Foundation Trust, Liverpool, UK; ²Institute of Life Course and Medical Science, University of Liverpool, Liverpool, UK; ³Liverpool Ocular Oncology Centre, Liverpool University Hospitals Foundation Trust, Liverpool, UK

Declarations: No conflicts of interest and no funding

Introduction: Metastatic disease in choroidal melanoma develops almost exclusively in patients with monosomy 3. Genetic analysis is offered to establish prognosis and inform patient monitoring. Patients having ruthenium plaque brachytherapy (RPB) usually have their biopsy taken the same day as plaque surgery however some are taken months later. Little is known about the temporal effect RPB has on these biopsies. We aimed to establish if there are risk factors for cytogenetic test failure following RPB.

Method: Data was extracted from the Liverpool Ocular Oncology Centre (LOOC) database and the NHS Cancer registry on the 10 November 2023. Data included patient demographics, clinical and tumour characteristics, treatment and genetic

diagnosis, as well as occurrence of metastasises and death. Descriptive statistics, binary logistic regression and survival analysis was performed on SPSSv.28.

Results: Five hundred and eighty-one patients were included. Most biopsies were taken on the day of plaque insertion (median = 0 days, range = 115 day before to 7611 days after RPB). Overall failure rate of histology tests was 13.2% compared to 21.7% of cytogenetic tests. Binary logistic regression demonstrated that cytogenetic test failure was not impacted by time from RPB to biopsy, patient age, sex, tumour height, biopsy modality or type of genetic technique performed (Nagelkerke $R^2 = 0.042$). Cox proportional hazard modelling confirmed monosomy 3 was a significant risk factor death from metastatic disease compared to disomy 3 (HR 12.1, $p < 0.001$).

Conclusion: Cytogenetic test failure in RPB patients is not impacted by time from RPB to biopsy, age, sex, tumour height, biopsy modality and genetic technique. Caution should be taken however as 70.9% (412/581) of patients had their biopsy within four weeks of RPB.

P182

HEALTH SEEKING BEHAVIOURS OF THOSE ACCESSING PRIMARY EYE CARE SERVICES IN TAMIL NADU, INDIA

Corresponding author: Navaneeth Natarajan, orcid.org/0009-0001-3152-7820

Navaneeth Natarajan^{1,2}, Priya Morjaria¹, Bharath Balasubramaniam³, Shaffa Hameed¹

¹London School of Hygiene and Tropical Medicine, London, UK; ²King's College London, London, UK; ³Sankara Eye Foundation and Eye Institute, India

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Introduction: Approximately 270 million Indians are visually impaired. Currently, a global effort is being made to improve primary eye care services as they are the frontline for the health needs of most populations. The goal of this study was to ascertain the health-seeking behaviours of patients attending vision centres (VCs) part of the Sankara Eye Group in Tamil Nadu, India.

Method: A mixed methods study was conducted to explore health-seeking behaviours of 158 patients attending seven VCs across three Tamil Nadu districts. Quantitatively, patients' sex, age, monthly household income, travel behaviours, and clinical patterns were recorded for all patients. Qualitatively, some (53) were interviewed concerning their views on their own eye health, their knowledge and importance of the subject and their ideas on increasing VC attendance. Analysis involved deducing medians, interquartile ranges and associations for quantitative and inductive thematic analysis for qualitative data.

Results: Quantitatively, three significant associations were found. Women were more likely to attend the VCs with a companion and older patients/those with a higher educational attainment were more likely to pay to travel to a VC. Qualitatively, themes found were: views on hinderances and motivators to attendance, methods of gaining eye health knowledge, and methods to increase VC attendance.

Conclusion: Key behaviours found were patients being mistaken about the foreseen costs of attending a VC and not attending until their lifestyles were dramatically hindered. Some also highly valued the importance of eye health. Positive recommendations were a key deciding factor for VC attendance and was a major advertising tool. Hypotheses to test in the future

are women bringing companions to clinics and whether there is an intersection between age and cost of travel to a VC.

P183

PATIENT PARTICIPATION IN CLINICAL TRIALS: BARRIERS FACING PATIENTS AND THEIR EFFECT ON FUTURE OPHTHALMOLOGY TRIALS

Corresponding author: Maryam Noeman, orcid.org/0009-0000-8911-9320

Maryam Noeman¹, Gurpreet Bansal¹, Gibran Butt¹, Graham Wallace², Saaeha Rauz^{1,2}

¹Birmingham and Midland Eye Centre, Birmingham, UK; ²Institute of Inflammation and Ageing, University of Birmingham, Birmingham, UK

Declarations: No conflicts of interest and no funding

Introduction: Patient participation in clinical research is critical, but can be limited by barriers such as finances, lack of information and lack of time. This study examines public attitudes and barriers to patient participation, as well as identifying existing workplace policies focused on trial participation and public understanding of these policies.

Method: An online survey was conducted amongst healthcare staff and members of the public to assess willingness to participate in clinical trials and understanding of relevant workplace policies. Freedom of information requests were made to city councils and other government organisations on employer policies regarding trial participation.

Results: 501 public responses and 132 healthcare professional (HP) responses were collected. Overall, members of the public were more willing to participate in trials compared to HPs (59.1% vs 47.4%, $p < 0.001$). Respondents were more willing to participate as healthy volunteers rather than patients (65.2% vs 49.3%, $p < 0.001$) and preferred to take annual leave rather than unpaid leave (66.3% vs 48.3%, $p < 0.001$). The main barriers identified were financial concerns (47.5%), desire for incentivisation (19.2%) and lack of understanding of workplace policies (12.0%).

48.0% of respondents believed their employer did not have policies relevant to trial participation in place and 33.6% of respondents were uncertain. 88.9% of government organisations did not have specific policies in place, with 30/336 suggesting alternative options such as annual leave or sick leave.

Conclusion: Overall, there are many barriers to participation in trials. This study found that most respondents have a poor understanding of workplace policies, while public organisations do not have policies which promote trial participation. By addressing barriers and implementing employer policies which ease patient volunteering, we can encourage participation in ophthalmology and general clinical trials.

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THE IMPACT OF COVID-19 ON MEDICAL EDUCATION: A LOST GENERATION OF OPHTHALMOLOGISTS?

Corresponding author: Swapnil Parmar, orcid.org/0009-0008-5666-3133

Swapnil Parmar¹, Ishika Bansal¹, Janvi Karia²

¹North West Anglia NHS Trust, Huntingdon, UK; ²Cambridge University Hospital NHS Foundation Trust, Cambridge, UK

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Introduction: COVID-19 had a significant impact on undergraduate medical education. There has been extensive analysis of the generic impact on medical education, however, the individual impact on specialities, particularly ophthalmology, has not been widely researched. We explored the impact of COVID-19 on the undergraduate ophthalmology experience in the UK, characterising the effect on medical education and considering positive outcomes that could be implemented in future undergraduate curricula.

Method: An online-cross sectional study was performed using a 13-item questionnaire, completed by 101 participants from UK medical schools. The three main topics covered by the questions included demographics, details of placement and the perceived impact of COVID-19.

Results: Statistical and qualitative analysis revealed a significant reduction in clinical exposure during the pandemic with an almost 100% shift to complete online lecture-based teaching. This teaching style has been adapted post-pandemic to a hybrid style of both online and in-person sessions. The impact of this has affected student confidence in dealing with ophthalmic conditions and deterred them from considering ophthalmology as a career.

Conclusion: COVID-19 has led to an increase in the proportion of undergraduate medical education taking place online, even post-pandemic. This has exacerbated a pre-existing gap in training medical trainees to deal with ophthalmic presentations. Despite online teaching increasing accessibility to education for students, efforts should be made to ensure students have hands-on, clinical exposure, especially in practical placements such as ophthalmology.

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THE IMPACT OF AN EDUCATIONAL FELLOW IN OPHTHALMOLOGY: A COMPARATIVE STUDY

Corresponding author: Nicole Quah, orcid.org/0000-0002-2212-4843

Nicole QX Quah¹, Thomas Nash¹, Mostafa Dowidar¹, Rahila Zakir¹

¹Western Eye Hospital, London, UK

Declarations: No conflicts of interest and no funding

Introduction: With NHS Ophthalmology waitlists at near unprecedented levels post-pandemic, balancing ophthalmic training with service provision requirements has become a significant challenge. To counter this, Health Education England has funded a 12-month Ophthalmology Educational Fellow post based at Western Eye Hospital (WEH) to support training recovery for pan-London trainees, including 1:1 clinical supervision in outpatients & Eye Casualty and surgical simulation courses. We aimed to quantify the impact of the Educational Fellow on trainees' teaching and on patient flow in Eye Casualty.

Method: We retrospectively surveyed four Year 1 Specialty Trainees (ST1s) at WEH and four ST1s at other London hospital sites to compare successfully completed ePortfolio sign-offs during the first four months of training. For the WEH ST1s, we also compared the average time needed for a WEH ST1 to manage a posterior vitreous detachment (PVD) case in Eye Casualty with the Educational Fellow present versus without.

Results: We found that, on average, the WEH ST1s had a higher number of ePortfolio sign-offs compared to other London ST1s (case-based discussions: 5 vs 3; direct observation of procedural skills: 4 vs 2; clinical rating scales: 6 vs 2), and felt more supported to manage eye emergencies on a Likert scale (4.8 vs 3.5). Furthermore, trainees completed PVD cases faster with the Educational Fellow present versus without (17 mins vs 29 mins).

Conclusion: Investment into an Ophthalmology Educational Fellow post has been associated with a positive impact on the training, safety, and efficiency of WEH ST1s, in comparison with their London peers, enabling increased training opportunities and service provision. Further research should be performed on a larger trainee cohort to better understand the cost-benefit economics of this investment.

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EVALUATION OF A REMOTE CONSENT PROJECT USING THE NASSS FRAMEWORK

Corresponding author: Rishi Ramessur, orcid.org/0000-0003-3420-6050

Rishi Ramessur^{1,2}, Andrea Viteri¹, Claire Lovegrove¹, Sharma Ganesanathan¹, Jeremy Chan¹, Safwana Abdulrasid¹, Alexander Sherlock¹, Pei-Fen Lin¹, Peter Thomas¹, Swan Kang¹

¹Moorfields Eye Hospital, London, UK; ²Imperial College London NHS Trust, London, UK

Declarations: No conflicts of interest and no funding

Introduction: Traditional paper consent processes for medical procedures can be time-consuming and stressful for patients and lead to operational impacts, theatre delays or cancellations through lost forms.

Method: We sought to evaluate a pilot initiative to introduce remote consenting to Moorfields Eye Hospital using mixed methodology with the established NASSS (Nonadoption, Abandonment, and Challenges to Scale-Up, Spread, and Sustainability) framework.

Results: 288 patients were sent consent forms between May 2022 and January 2023. 63% were seen virtually and 98% of those patients did not require face-to-face appointments until the day of surgery, saving the average patient a 7.5-mile, 45-minute, £7 commute emitting 0.89 KgCO₂e.

Other strengths identified included ease of technology use (100% of clinicians and 83% of patients reported that the platform was easy to use), efficiency (surgeons explained risks and benefits of surgery in 92 seconds on average, and patients signed in 58 seconds with remote consenting), strong Trust-Supplier co-design, and good alignment with organisational vision and the socio-political landscape.

Barriers to scaling and adoption included interdependent technologies as possible points of failure and the need to integrate with more than one electronic health record. Age was identified as a statistically significant risk factor for patients not signing the remote consent form (relative risk 1.85, $p < 0.001$); visual acuity of 0.3 LogMAR or worse and white ethnicity were not statistically significant risk factors.

Most patient comments were positive. Themes for improvement included providing more information (by text messages and letters) and more language options.

Conclusion: Evaluating this pilot using the NASSS framework has helped identify potential sources of difficulty as the Trust seeks to maximise the chance of successful implementation during the post-pilot deployment and scaling phase.

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A COMPARISON OF HIGH RISK PATHOLOGICAL FEATURES BETWEEN PRIMARY AND SECONDARY ENUCLEATION FOR RETINOBLASTOMA

Corresponding author: Reham Shehada, orcid.org/0000-0002-7163-0793

Reham Shehada¹, Mona Mohammad², Yacoub Yousef²

¹National Treatment Center, NHS Highland, UK; ²King Hussein Cancer Center, Amman, Jordan

Declarations: No conflicts of interest and no funding

Introduction: The International Intraocular Retinoblastoma Classification (IIRC) indicated that 47% of group D eyes might be saved. Treating the eyes in group D posed a challenge for medical professionals because a conservative course of management may promote metastasis and death. In this study, we investigated High-Risk Pathologic Features (HRPF) rates in eyes undergoing secondary versus primary enucleation.

Method: A retrospective analysis of 121 eyes from 118 patients who underwent enucleation at the King Hussein Cancer Center (KHCC) Amman, Jordan, between November 2009 and January 2020.

Results: Patients in the secondary group (49/121 eyes, 40%) were considerably younger at diagnosis ($p = 0.0014$), had bilateral disease ($p = 0.0001$), and had less-progressed disease at presentation ($p = 0.016$) compared to the primary enucleation. Primarily enucleated eyes were more-likely to have massive choroidal invasion ($p = 0.0315$) and post-laminar optic nerve invasion ($p = 0.027$), in spite of the finding that the overall prevalence of HRPF was similar between the two groups (35.5 percent vs. 37.5 percent; $p = 0.585$). The likelihood of anterior chamber invasion, was considerably higher in secondary enucleated eyes ($p = 0.013$). We evaluated primary and secondary enucleation for each subgroup (D and E) of the IIRC and found the prevalence of HRPF was comparable ($p = 0.58$, 1.0, respectively). The difference in time between diagnosis-to-enucleation in secondary enucleation did not predict HRPF ($p = 0.50$). There was no discernible difference between primary and secondary enucleated eyes in terms of metastasis or survival ($p = 0.156$ and 0.44, respectively).

Conclusion: Systemic chemotherapy has the ability to reduce the extent of tumour expansion that has been pathologically identified. Primary and secondary enucleated eyes are comparable in low metastatic risk only when strict examination and management guidelines are followed.

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ACCURACY AND RELIABILITY OF SELF-ADMINISTERED VISUAL ACUITY TESTS: SYSTEMATIC REVIEW OF PRAGMATIC TRIALS

Corresponding author: Arun J. Thirunavukarasu, orcid.org/0000-0001-8968-4768

Arun J. Thirunavukarasu¹, Refaat Hassan¹, Aaron Limonard¹, Viral Gudiwala¹, Shalom V. Savant¹

¹University of Cambridge School of Clinical Medicine, Cambridge, UK

Declarations: No conflicts of interest and no funding

Introduction: Remote self-administered visual acuity tests could allow patients and non-specialists to assess vision without eye health professional input. Validation in pragmatic trials is necessary to demonstrate the accuracy and reliability of tests in relevant settings to justify deployment. Here, published pragmatic trials of these tests were synthesised to summarise the effectiveness of available options and appraise the quality of their supporting evidence.

Method: A systematic review was undertaken in accordance with a preregistered protocol (CRD42022385045). The Cochrane Library, Embase, MEDLINE, and Scopus were searched. Screening

was conducted according to the following criteria: (1) English language; (2) primary research article; (3) visual acuity test conducted out of eye clinic; (4) no clinical administration of remote test; (5) accuracy or reliability of remote test analysed. There were no restrictions on trial participants. Quality assessment was conducted with QUADAS-2.

Results: Of 1227 reports identified, 10 studies were included. One study was at high risk of bias and two studies exhibited concerning features of bias; all studies were applicable. Two studies—trialling DigiVis, iSight Professional, and Peek Acuity—exhibited comparable accuracy between remote and clinical tests. All other trials exhibited inferior accuracy, including conflicting results involving iSight Professional and Peek Acuity. Two studies evaluated test-retest agreement; one trial showing that DigiVis was as reliable as clinical assessment. The three most accurate tests required access to digital devices. Reporting was inconsistent and often incomplete, particularly with regards to describing methods and conducting statistical analysis.

Conclusion: Digital self-administered visual acuity tests exhibit promise for facilitating teleophthalmology, non-specialist eye assessment, pre-consultation triage, and autonomous long-term monitoring of vision. Further pragmatic trials are indicated to justify deployment in carefully defined contexts to facilitate patient or non-specialist led assessment.

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LARGE LANGUAGE MODELS APPROACH EXPERT-LEVEL CLINICAL KNOWLEDGE AND REASONING IN OPHTHALMOLOGY: A HEAD-TO-HEAD CROSS-SECTIONAL STUDY

Corresponding author: Arun J. Thirunavukarasu, orcid.org/0000-0001-8968-4768

Arun James Thirunavukarasu^{1,2}, Shathar Mahmood¹, Andrew Malem³, William Paul Foster^{1,4}, Rohan Sanghera¹, Refaat Hassan¹, Sean Zhou⁵, Shiao Wei Wong⁶, Yee Ling Wong⁶, Yu Jeat Chong⁷, Abdullah Shakeel¹, Yin-Hsi Chang⁸, Benjamin Kye Jyn Tan⁹, Nikhil Jain¹⁰, Ting Fang Tan¹¹, Saaeha Rauz^{7,12}, Daniel Shu Wei Ting^{11,13,14}, Darren Shu Jeng Ting^{7,12,15}

¹University of Cambridge School of Clinical Medicine, Cambridge, UK; ²Oxford University Clinical Academic Graduate School, University of Oxford, Oxford, UK; ³Eye Institute, Cleveland Clinic Abu Dhabi, Abu Dhabi, United Arab Emirates; ⁴Department of Physiology, Development and Neuroscience, University of Cambridge, Cambridge, UK; ⁵West Suffolk NHS Foundation Trust, Bury St Edmunds, UK; ⁶Manchester Royal Eye Hospital, Manchester University NHS Foundation Trust, Manchester, UK; ⁷Birmingham and Midland Eye Centre, Sandwell and West Birmingham NHS Foundation Trust, Birmingham, UK; ⁸Department of Ophthalmology, Chang Gung Memorial Hospital, Linkou Medical Center, Taoyuan, Taiwan; ⁹Yong Loo Lin School of Medicine, National University of Singapore, Singapore; ¹⁰Bedfordshire Hospitals NHS Foundation Trust, Luton and Dunstable, UK; ¹¹Singapore Eye Research Institute, Singapore National Eye Centre, Singapore, Singapore; ¹²Academic Unit of Ophthalmology, Institute of Inflammation and Ageing, University of Birmingham, Birmingham, UK; ¹³Duke-NUS Medical School, Singapore, Singapore; ¹⁴Byers Eye Institute, Stanford University, Palo Alto, CA, USA; ¹⁵Academic Ophthalmology, School of Medicine, University of Nottingham, Nottingham, UK

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system to detect retinal disease. DSJT authored the book used in the study and receives royalty from its sales. AJT and RS have received funding from HealthSense for unrelated research work involving large language models. The other authors have no competing interests to declare.

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Introduction: Large language models (LLMs) are generating great excitement in medicine for their emergent abilities to successfully complete tasks with no or minimal prior training or exposure. A plethora of ophthalmological studies have suggested that large language models may be capable of providing useful advice and guidance to eye health professionals. However, these studies generally fail to provide a clinical benchmark to contextualise the results used to justify their conclusions.

Method: GPT-3.5 and GPT-4 were trialled on 347 FRCOphth Part 2 questions before GPT-3.5, GPT-4, PaLM 2, LLaMA, expert ophthalmologists, and doctors in training were trialled on a mock examination of 87 questions. Performance was analysed with respect to question subject and type (first order recall and higher order reasoning). Masked ophthalmologists graded the accuracy, relevance, and overall preference of GPT-3.5 and GPT-4 responses to the same questions.

Results: The performance of GPT-4 (69%) was superior to GPT-3.5 (48%), LLaMA (32%), and PaLM 2 (56%). GPT-4 compared favourably with expert ophthalmologists (median 76%, range 64–90%), ophthalmology trainees (median 59%, range 57–63%), and unspecialised junior doctors (median 43%, range 41–44%). Low agreement between LLMs and doctors reflected idiosyncratic differences in knowledge and reasoning with overall consistency across subjects and types ($p > 0.05$). All ophthalmologists preferred GPT-4 responses over GPT-3.5 and rated the accuracy and relevance of GPT-4 as higher ($p < 0.05$).

Conclusion: LLMs are approaching expert-level knowledge and reasoning skills in ophthalmology. State-of-the-art LLMs such as GPT-4 may provide useful medical advice and assistance where access to expert ophthalmologists is limited, as they offer comparable or superior performance to non-specialists. Clinical benchmarks provide useful assays of LLM capabilities in ophthalmology before clinical trials are planned to evaluate benefits to eye health outcomes.

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PARTNERING WITH LOCAL CHARITIES AND SOCIAL SERVICES TO PROVIDE A COMPREHENSIVE SIGHT LOSS SUPPORT SERVICE IN THE EYE CLINIC

Corresponding author: Christina Rennie, orcid.org/0000-0001-9895-4758

Wenquian Wu¹, Kris Gibson², Christina Rennie¹

¹University Hospital Southampton NHS Foundation Trust, Southampton, UK; ²Southampton Sight, Southampton, UK

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Introduction: Over 300 adult patients a year are registered as either sight impaired (SI) or severely sight impaired (SSI) at this busy eye unit. Following loss of dedicated support for these patients in 2019 a joint initiative between the NHS, local sight loss charities and social services, has re-imagined the support, rehabilitation and empowerment of patients living with visual impairments. Initial study of patients and staff to understand barriers to support led to the design of a sight loss advisor (SLA) course and awareness training for all staff. There are 7 embedded sight loss advisors, including within paediatric ophthalmology and neurology. Ongoing oversight is provided by a stakeholder group from involved parties.

Method: Assessment of the impact of the new SLA service and training, including surveys of staff and patients.

Results: 683 patients seen by SLA since September 2021, with 668 referred to local charities for support. Staff noting good or excellent support offered to patients increased from 37% to 74% post implementation. 84% of staff feel the SLAs are very helpful. Patient satisfaction with support offered increased from 36% to 73% after SLA input. After intervention patients had improved understanding of their condition, reduced anxiety, and better knowledge of support services.

Conclusion: The SLA's provide an invaluable resource for staff and patients, empowering patient choice and control, moving away from the medical model focused on the disease. This mirrors the core move towards shared decision making, not only listening to the patient but understanding their needs and signposting where appropriate. The model of clinic staff becoming SLA's provides career opportunities and ensures close links with clinical staff, maximising patient referrals, including early intervention before SI certification.

P191. Abstract not included

NEURO OPHTHALMOLOGY

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ASSESSING CHATGPT, BING, BARD'S COMPETENCY IN NEURO-OPHTHALMOLOGY: A TEXTBOOK REVIEW QUESTIONS TRIAL

Corresponding author: Eray Atalay, orcid.org/0000-0002-2536-4279

Mert Egemen Caliskan¹, Semih Ceylan², Betin Bilkan Karaman¹, Musacan Yalcin¹, Eray Atalay¹

¹Eskisehir Osmangazi University, Eskisehir, Turkey; ²Eskisehir Mahmudiye State Hospital, Eskisehir, Turkey

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Introduction: This study aims to assess the competency of three popular open access models (LLM) Chatbots, namely ChatGPT 3.5 (Open AI), Bing (Microsoft) and Bard (Google), in answering 64 "Review Questions" from the Neuro-ophthalmology section of the Review of Ophthalmology textbook (Elsevier, 4th edition).

Method: All questions were multiple choice with one choice being correct. Questions were asked individually on a new chat window each time. The prompt structure was "The Question with Choices" + "What is the correct answer?". The answers were contrasted with the answer sheet of the textbook and labelled as correct or incorrect. Questions were categorized as "neuroanatomy", "physiology" and "clinical" as per the content in the book.

Analyses were performed using Chi-square test and statistical significance was set at $p < 0.05$.

Results: Among 64 questions, ChatGPT 3.5 performed best answering 35 questions correct (54.68%), followed by Bard and Bing both answering 33 questions correct (51.56%). However, the difference was not statistically significant ($p = 0.91$). For neuroanatomy, both ChatGPT 3.5 and Bing answered 5 questions correct out of 6, Bard answered 4 out of 6 questions correct ($p = 0.72$). For physiology ChatGPT 3.5 performed the worst answering 4 questions correct out of 8, Bard and Bing performed better both answering 5 out of 8 questions correct ($p = 0.84$). For clinical, ChatGPT 3.5 performed the best answering 26 out of 50 questions correct followed by Bard answering 24 out of 50 correct and lastly Bing answering 23 out of 50 questions correct ($p = 0.82$).

Conclusion: LLM Chatbots were only able to answer approximately half of the questions correctly. The chatbots performed similarly for different question categories.

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CORRELATION BETWEEN THE WHITE MATTER HYPERINTENSITIES ON MRI AND OPTICAL COHERENCE TOMOGRAPHY ANGIOGRAPHY IN ALZHEIMER'S DISEASE PATIENTS

Corresponding author: Eray Atalay, orcid.org/0000-0002-2536-4279

Semih Ceylan¹, Mert Egemen Caliskan², Uğur Toprak³, Nazlı Çelik³, Eray Atalay³, Abdullah Küçükhüseyin³

¹Eskisehir Mahmudiye State Hospital, Eskisehir, Turkey; ²Eskisehir Osmangazi University Medical School, Eskisehir, Turkey; ³Eskisehir Osmangazi University Faculty of Medicine, Eskisehir, Turkey

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Introduction: Due to shared embryological origins of retinal and cerebral microcirculation, evaluation of retinal microcirculation using optical coherence tomography angiography (OCT-A) holds promise for unravelling cerebral microcirculation dynamics in Alzheimer's disease (AD).

This study aims to assess correlations between metrics derived from retinal OCT-A and white matter hyperintensity volumes (WMH) on magnetic resonance imaging (MRI) in AD patients.

Method: Patients were prospectively recruited and exclusion criteria were significant cataract, co-existing retinopathy and optic neuropathy, intraocular pressure >21 mmHg, spherical equivalent ± 4 D. Foveal and average perifoveal vessel densities (VD) of the superficial and deep capillary plexus (SCP and DCP, respectively) were recorded from macular OCT-A scans (Triton, Topcon Inc.). T1 and T2 FLAIR MRI images within 3-year timeframe were used for quantifying WMH via fully automated toolbox (UBO-Detector, Jiang et al. 2018). Volume assessments were performed for periventricular-WMH (PVWMH), deep-WMH (DWMH) and whole brain-WMH (WBWMH). Pearson correlation was used, and statistical significance was set at $p < 0.05$.

Results: A total of 14 patients with mean age of 72.6 years (range: 65 – 82) were recruited. There was a moderate positive correlation between DWMH volume and foveal DCP ($p = 0.029$, coefficient: 0.583). Although VD levels at the DCP layer showed similar correlation trends with WBWMH (foveal DCP coefficient: 0.49, $p = 0.07$; perifoveal DCP coefficient: 0.48, $p = 0.08$) and

PVWMH (foveal DCP coefficient: 0.46, $p = 0.09$; perifoveal DCP coefficient: 0.47, $p = 0.09$), statistical significance was not reached.

Conclusion: The volume occupied by WMH was positively correlated with VD at the DCP layer in patients with AD. This may have potential associations with the recent discovery of upregulated cerebrospinal fluid vascular endothelial growth factor levels in relation to cerebral amyloid angiopathy and WMH accumulation in AD.

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SERVICE EVALUATION: ORTHOPTIST-LED CLINIC FOR SUSPICIOUS OPTIC DISCS

Corresponding author: Zhihang Cheng, orcid.org/0000-0002-9390-0942

Zhihang Cheng¹, Robert Cheeseman¹

¹St Paul's Eye Unit, Liverpool University Hospitals, Liverpool, UK

Declarations: No conflicts of interest and no funding

Introduction: A significant number of optic discs raising suspicion of papilloedema are referred through the GP e-Referral service based on optometrists' recommendations. We devised and implemented an orthoptist-led suspicious optic disc clinic with specialist oversight to assess these patients in a specialised clinical environment.

Method: This retrospective analysis involved patients aged 16 and above who were referred and triaged to the suspicious disc clinic. Initial assessments included the collection of demographic data and symptoms such as postural headache, blurred vision, diplopia, pulsatile tinnitus, and transient visual obscuration. Final diagnoses and the aetiology of optic nerve swelling appearance were documented during follow-up visits.

Results: Between January 2022 and June 2023, 75 patients underwent evaluation, with an average age of 32.5 years (range 16–62). Remarkably, 34 patients (45%) presented with no symptoms. While one patient was referred to the medical team for assessment, no cases of raised intracranial pressure were diagnosed at the final follow-up. Of the total, 49 patients (65%) were discharged after the initial visit, and 64 patients (85%) were discharged within six months of their initial assessment. Clinical assessment revealed that 43 patients had no disc swelling, and the remaining 31 patients were diagnosed with pseudopapilloedema, causes included peripapillary hyperreflective ovoid mass-like structures (22), optic disc drusen (4), tilted disc (4), and crowded disc (1).

Conclusion: In our catchment area, cases referred by optometrists via the GP referral pathway are unlikely to yield a diagnosis of raised intracranial pressure. The outcomes of our orthoptist-led suspicious disc clinic highlight its potential for managing low-risk patients through comprehensive assessment in a dedicated setting, utilising allied eye care professionals in extended roles.

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ANTINUCLEAR AND ASSOCIATED AUTO-ANTIBODIES IN PATIENTS WITH OPTIC NEURITIS SECONDARY TO NEUROMYELITIS OPTICA SPECTRUM DISORDER

Corresponding author: Pulkit Gupta, orcid.org/0009-0008-5273-2373

Pulkit Gupta¹

¹Salisbury NHS Foundation Trust, Salisbury, UK

Declarations: No conflicts of interest and no funding

Introduction: First identified in 1894, 'neuro-myélite optique aiguë'/Devic's disease was renamed as NeuroMyelitis Optica Spectrum Disorder (NMOSD) in 2015. It is well established that optic neuritis (ON) is one of the three absolute diagnostic criteria for NMOSD; with Aquaporin-4 (AQP4) antibodies being its biomarkers (carrying 91–95% specificity), due to the characteristic loss of astrocytic AQP4. Alternately, cases of optic neuritis secondary to connective tissue disorders (CTD) with positive autoantibodies have also been reported, with or without associated positive AQP4.

Method: We came across three patients (A, B, C) who presented with unilateral visual loss secondary to retrobulbar optic neuritis. The ages of these two females and one male ranged between 26 to 43 years, and visual acuities at presentation were perception of light (A), 1.2 logMAR (B), and no perception of light (C). Blood assessment indicated positive AQP4 antibodies in all.

Additional flagged blood results:

Patient A: Syphilis Total Antibody, positive Anti-nuclear (ANA), Anti-Ro and Anti-RNP antibodies. The CSF analysis was unremarkable and there was no clinical evidence to indicate asymptomatic, meningeal, meningo-vascular, or parenchymal Neurosyphilis.

Patient B: Positive ANA, Anti-Ro antibodies.

Patient C: Elevated IgG-DS DNA, positive ANA antibodies. Unremarkable lupus, anti-cardiolipin and vasculitis screen.

Results: All three patients received Intravenous methylprednisolone 1 gm pulsed over 3 days followed by oral prednisolone. Significant visual recovery was noted between 3–6 weeks follow-up.

Conclusion: A literature review indicates that patients with CNS involvement associated with CTDs exhibit atypical ON with mucocutaneous, musculoskeletal, haematologic or reticuloendothelial manifestations, and limited response to conventional therapies. Uncharacteristically, although all our patients had positive CTD antibodies, they responded well to IV steroid therapy, and neither manifested any mucocutaneous involvement.

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RNFL THINNING IN PATIENTS AFTER TRAUMATIC BRAIN INJURY

Corresponding author: Youstina Metry, orcid.org/0009-0005-5697-9363

Youstina Metry^{1,2,3}, Elinor Laws^{1,2,3,4}, Noor Haziq Bin Saliman⁵, Antonio Belli^{1,2,3}, Richard J Blanch^{1,3,6}

¹University Hospitals Birmingham NHS Foundation Trust, Birmingham, UK; ²Institute of Inflammation and Ageing, College of Medical and Dental Sciences, University of Birmingham, Birmingham, UK;

³NIHR Surgical Reconstruction and Microbiology Centre, University Hospitals Birmingham NHS Foundation Trust, Birmingham, UK;

⁴NIHR Birmingham Biomedical Research Centre, University of Birmingham, Birmingham, UK; ⁵Centre for Optometry Studies, Faculty of Health Sciences, Universiti Teknologi MARA Cawangan

Selangor, Bandar Puncak Alam Selangor, Malaysia; ⁶Academic Department of Military Surgery and Trauma, Royal Centre for Defence Medicine, Birmingham, UK

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author(s) and not necessarily those of the NIHR or the Department of Health and Social Care.

Introduction: Optical Coherence Tomography(OCT) is more effective in the early detection of traumatic optic neuropathy(-TON) retinal changes compared to optic disc examination. This study investigates retinal changes in patients sustaining traumatic brain injury(TBI) and their relationship to TBI severity.

Method: We conducted a retrospective study analysing OCT images from patients who sustained mild to severe TBI between March 2020 and May 2022. Macula ganglion cell layer(GCL) and circumpapillary retinal fibre layer(RNFL) thickness changes were assessed on serial OCT. OCT changes were associated with TBI severity using the DoD classification and CT head findings using Marshall scoring system. RNFL changes were subcategorised as single-sectoral, multi-sectoral thinning, TON or thickening.

Results: One-hundred-and-thirty-five patients were included in the study, with a mean age of 31 years and 109(81%) were male patients. Median visual acuity was -0.06 on initial presentation and -0.05 on follow-up, with 34%(46) patients reporting visual symptoms on presentation. Sixty percent (81) had mild TBI, 26.7% (36) had moderate TBI, and 13.3%(18) had severe TBI. Of the 135 patients, 62(45.9%) had follow-up OCT imaging, where 2/62 (3.2%) had TON changes, 4(6.5%) had single-sectoral RNFL thinning, 13(21%) had multi-sectoral RNFL thinning, 6(10%) RNFL thickening, and 6(10%) had isolated GCL changes. TBI severity and Marshall scoring were strongly associated with RNFL thickness changes ($p < 0.001$ for all). There was no association between TBI severity or Marshall scoring with GCL thickness changes ($p = 0.29$, $p = 0.1$ respectively).

Conclusion: We report a strong association between TBI severity, Marshall scoring and the longitudinal RNFL thickness changes, where subclinical reduced RNFL thickness was the most common detectable OCT change, suggesting that OCT changes may have future potential to help assess TBI severity by acting as an oculomic imaging biomarker.

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INCIDENCE AND DEMOGRAPHICS OF ACUTE OPTIC NEURITIS PRESENTING TO EYE CASUALTY IN BELFAST

Corresponding author: James Parker, orcid.org/0009-0005-4893-0199

James Parker¹, James Parker¹, Monica Hrabovsky¹, Emma Donaldson¹

¹Royal Victoria Hospital Belfast, Belfast, UK

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Introduction: The rate of optic neuritis in the UK general population is quoted in a 2020 study as 3.7 per 100,000 per person-years. The prevalence of multiple sclerosis (MS) in Northern Ireland is higher than the UK average. We therefore have looked at the incidence and characteristics of acute optic neuritis (ON) in Northern Ireland, and to our knowledge, there is little published literature on this.

Method: Belfast Eye Casualty (serving just over 2/3 of the Northern Ireland population) and electronic records were used to retrospectively identify patients who presented in 2022 with presumed new acute optic neuritis. Demographic data was collected, in addition to management implemented (investigations and treatment) and ongoing follow-up with ophthalmology and neurology services. A final diagnosis and visual acuity were recorded from the most recent clinical encounter.

Results: 16,826 patients attended Belfast Eye Casualty in 2022. 49 patients were identified as likely acute ON (71% female, 29% male), with average age 35.4 years old. Almost 2/3 of the cases could be described as having symptoms and findings of "typical" ON with 1/3 "atypical". On more detailed analysis, 39 of these patients were confirmed optic neuritis at their most recent clinic encounter (45% subsequently had a diagnosis of MS and 2 anti MOG disease). Of note, 11 (28%) of these patients were treated with corticosteroids. There was no neuromyelitis optica associated ON. Taking into account the 2021 census for our catchment area, the annual incidence of acute optic neuritis is 2.78 per 100,000, 1.57 per 100,000 MS-associated ON and 0.14 per 100,000 antiMOG associated ON.

Conclusion: The incidence of acute optic neuritis in Northern Ireland is a little lower than the UK average.

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AN ASSESSMENT OF THE RELATIONSHIP BETWEEN INTRACRANIAL PRESSURE AND VISUAL FIELD LOSS IN PAEDIATRIC IDIOPATHIC INTRACRANIAL HYPERTENSION PATIENTS AT A TERTIARY UNIT

Corresponding author: Robert Brady, robertthomas.brady@nhs.net

Rahul Shah¹, Robert Brady², Pooja Harijan², Deepa Krishnakumar², Brinda Muthusamy²

¹University of Cambridge, Cambridge, UK; ²Cambridge University Hospitals NHS Trust, Cambridge, UK

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Introduction: This work aims to characterise the relationship between quantifiable proxy measures of intracranial pressure (ICP) and visual field loss (VFL) in paediatric idiopathic intracranial hypertension.

Method: Reliable (fixation losses < 20% or false positives < 33% or false negatives < 33%) Humphrey Field Test (HFT) parameters (Visual Field Index-VFI, Mean Deviation-MD and Pattern Standard Deviation-PSD) of 60 eyes of 37 paediatric IIH patients were retrospectively reviewed. CSF infusion study or tap opening pressures, Optical Coherence Tomography (OCT) retinal nerve fibre layer (RNFL) thickness and papilloedema Frisén grading at presentation were analysed.

Results: The 37 patients reviewed were 11:26 male:female, had a mean age of 13.3 ± 2.2 and BMI of 26.9, and 73% were of White British ethnicity. Frisén grading was available for 42 eyes; mean grade 2.4, OCT mean RNFL thickness for 57 eyes; mean $194.3 \mu\text{m}$, and CSF opening pressure for 46 eyes of 28 patients; mean opening pressure 25mmHg. Weak but statistically significant correlation was found between Frisén Grade and each of VFI, MD and PSD ($r = -0.4773$, -0.5025 and 0.3953 , $p < 0.01$), between RNFL thickness and the same parameters ($r = -0.3803$, -0.4126 and 0.3774 , $p < 0.01$) and between CSF opening pressure and the same parameters ($r = -0.3986$, -0.3985 and 0.3943 , $p < 0.01$).

Conclusion: Statistically significant, fair correlations exist between Frisén grade, RNFL thickness or CSF opening pressure and reliably-measured VFI, MD and PSD in children with IIH. This suggests the extent of ICP as measured by degree of papilloedema on slit lamp examination, OCT or CSF manometry is predictive of the extent of VFL, and vice versa: the extent of VFL is predictive of ICP. Further subgroup analysis to stratify cases by papilloedema and VFL severity will help guide intervention to prevent lasting VFL.

ORBIT AND OCULOPLASTICS

P199

LEVOBUPIVACAINE VERSUS BUPIVACAINE IN PERIBULBAR BLOCK FOR OPHTHALMIC SURGERIES: A SYSTEMATIC REVIEW AND META-ANALYSIS

Corresponding author: Ali A M A M Alkhabbaz, orcid.org/0000-0002-9931-5269

Ali A M A M Alkhabbaz¹, Mohammad Karam², Marwa Al Ghafri², Adam Samet², Christian El-Hadad³

¹Kuwait University, Kuwait; ²McGill University, Montreal, Canada; ³University of Toronto, Montreal, Canada

Disclosures: No conflicts of interest and no funding

Introduction:

Purpose: Compare the efficacy of levobupivacaine versus bupivacaine in peribulbar block for ophthalmic surgery.

Method: A systematic review and meta-analysis were performed as per the PRISMA guidelines, and a search of electronic information was conducted to identify all comparative studies of levobupivacaine versus bupivacaine in peribulbar block for ophthalmic surgery. The primary outcome measures were akinesia score, time to onset of effect, post-operative diplopia, verbal pain scales, and moderate hypotension. Secondary outcome measures included surgeon satisfaction, patient satisfaction, requirement of supplementary topical anaesthesia intra-operatively, requirement of supplementary injection, and duration of block. Our analyses were summarized by calculating mean differences (MD) using fixed-effects meta-analysis.

Results: 9 eligible studies that enrolled 800 patients were identified in the literature. There was no statistically significant difference in akinesia score at 2 minutes (MD = 0.17, P = 0.12) or 10 minutes post-injection (MD = 0.02, P = 0.89). Similarly, there was no statistically significant difference between the two groups in terms of time of onset of the effect of the anaesthetic (MD = 0.23, P = 0.27), post-operative diplopia (MD = 0.62, P = 0.17), and hypotension (MD = 1.00, P = 1.00). There was also no significant difference between the two anaesthetic agents in terms of verbal pain scales during the block (MD = 0.04, P = 0.66), at the end of surgery (MD = -0.01, P = 0.94) or 4 hours post-operatively (MD = -0.01, P = 0.73).

Conclusion: The results of this meta-analysis conclude that there is no significant difference between levobupivacaine and bupivacaine in terms of akinesia score at 2- and 10-minutes post-injection, time to onset of anaesthetic effect, post-operative diplopia, hypotension, or verbal pain scales.

P200. Abstract not included

P201

A REVIEW OF THE QUALITY OF INFORMATION ON EXTERNAL HORDEOLA ON TIKTOK

Corresponding author: Alice Ditchfield, orcid.org/0009-0003-8331-8793

Alice Ditchfield¹

¹Imperial College Healthcare NHS Trust, London, UK

Disclosures: No conflicts of interest and no funding

Introduction: The rise of video-sharing platforms such as TikTok has seen an increase in the availability and reach of health-related videos. While this could pose an opportunity for healthcare education, some studies suggest that videos are often of poor quality and/or contain misinformation. This study sought to characterise the quality of information on external hordeola on TikTok.

Method: The TikTok app was searched using the term "stye". Of the top 50 results, 23 videos were excluded as they did not contain informational content on external hordeola. Basic statistics were collected for the remaining 27: number of views, likes, and comments; and video duration. Creator type (eye health professional, non-eye health professional, non-health professional) was recorded. The DISCERN tool was used to assess the quality and accuracy of the information provided. The numerical score was correlated to a quality category: excellent, good, satisfactory, poor, very poor.

Results: The 27 videos assessed had a combined total of 210 million views. The mean number of views was highest for non-health professionals (12.05 million), and lowest for eye health professionals (1.47 million). The mean DISCERN score was highest for non-eye health professionals (33.5). Eye health professionals scored a mean of 31.4, and non-health professionals 27.4. One video was rated 'satisfactory' based on the DISCERN score; the remainder were 'poor' or 'very poor'. The management advice given included rubbing garlic on the stye, using a gold ring, and a hard-boiled egg in a sock.

Conclusion: Videos on external hordeola have an enormous reach on TikTok, but frequently contain misinformation, which has the potential to cause harm. Knowledge of this can help ophthalmologists tailor their advice to patients, to mitigate these risks in future.

P202

5-YEAR REVIEW OF BASAL CELL CARCINOMA (BCC) – PRE (2017 – 19) AND COVID (2020 -21) ERAS

Corresponding author: Mariam Doctor, mariamhd@gmail.com

Mariam Doctor¹, Bridget Hemmant¹

¹James Paget University Hospitals NHS Foundation Trust, Great Yarmouth, UK

Disclosures: No conflicts of interest and no funding

Introduction: We determine incidence, effects of incomplete lesion excision, analyse overall recurrences of BCC and plan follow-up recommendations at hospital offering services to a sun exposed beach county in England during the pre (2017-19) and covid (2020-21) eras.

Method: Retrospective analysis of 160 patients with ocular BCC.

Results: 160 patients (pre:~36/year | covid:~25/year) with median age 75 years and male/female ratio ~0.85 during both era's. Average reference time ratio (pre/covid) from GP to clinic was ~0.45 days and from clinic to biopsy was ~0.71 days. Location of occurrences (pre%|covid%) were Lower Lid (47% | 37%), Upper Lid (16% | 22%), M-Canthus (25% | 19%), L-Canthus (2 % | 4 %), Eyebrow (2 % | 2 %) and Multiple locations (8 % | 16 %). Histology findings (pre% | covid%) were nodular BCC (65 % | 61 %), superficial (6 % | 5 %), Morpheaform (11 % | 10 %), Micronodular (4 % | 1 %) and mixed pattern (14 % | 23 %). Complete excision was in 80 % and majority of the repeat biopsy were required for

morpheaform (6 (18 %)) patients. Overall true recurrence was 1.83 % and that of morpheaform histology was 9 %.

Conclusion: Higher incidence in our study is mainly attributed to a beach town having more sun exposure activities in the serving population. True recurrence is extremely low in BCC and occurs mainly in morpheaform and infiltrative histology.

- r-BCC, morpheaform, infiltrative histologies need to be followed for at least 5 years.
- Primary nodular BCC can be discharged for self-monitoring after satisfactory 6 monthly post-op review.
- Recommendations from this study will help to utilize limited and precious health care resources efficiently without compromising on patient safety.

P203

DOES THYROIDECTOMY IMPROVE OUTCOMES IN THYROID EYE DISEASE? EXPERIENCE IN A MULTI-ETHNIC METROPOLITAN COHORT TREATED IN A MULTIDISCIPLINARY TED (MDTED) SERVICE

Corresponding author: Nour Houbby, orcid.org/0000-0001-6865-4682

Nour Houbby¹, Daisy Metcalfe², Natalie Man², Malik Moledina¹, Vickie Lee¹

¹Imperial College Healthcare Trust, London, UK; ²NIHR Imperial Clinical Research Facility, London, UK

Disclosures: No conflicts of interest and no funding

Introduction: 90% of Thyroid Eye Disease (TED) is associated with Grave's Hyperthyroidism (GH). The benefit of thyroidectomy on TED remains controversial in the peer-reviewed literature. Our aims are to evaluate the indications, and endocrine and TED outcomes for thyroidectomy in our TED cohort.

Method: Retrospective collection of demographic data, endocrine diagnosis, TSH receptor (TRAb) antibody levels, treatment and TED parameters (Clinical Activity Score and immunosuppression treatment), indications for thyroidectomy, intra-operative data, complications and outcomes for patients >18 years attending the MDTED service. Statistical analysis was carried out using SPSS.

Results: 13% patients (75/568) underwent total thyroidectomy of which 51% (38/75) received immunosuppression treatment for TED. 46 patients had complete medical records. 96% (44/46) had GH and 2 Hashimoto's disease. 98% were female with mean thyroid onset age of 39 years (range 21-72 years). 52% (14/27) were current or ex-smokers. Indications for thyroidectomy: 30% (14) unstable control on medical treatment, 17% (8) to improve TED control; 35% (16) post-GH relapse, 11% (5) facilitation of pregnancy; 17% (8) declined/allergy/relapse radio-iodine treatment, suspected tumour (n = 1) and co-existing primary hyperparathyroidism (n = 1). N = 11 unknown indication. 37% (17/46) experienced no post-operative complications. T4 values recorded in 19 patients significantly increased post-operatively (12.8 vs 14.5, *p = 0.016). TRAb levels significantly decreased post-operatively (11.45 vs 7.32, *p = 0.036) in the 6 patients with pre and post-operative TRAb levels. CAS and TSH did not significantly change between pre-thyroidectomy, immediately post-thyroidectomy and in the six months post-thyroidectomy (p = 0.320 and p = 0.339 respectively).

Conclusion: 17% of our cohort were recommended for thyroidectomy to improve their TED control. Our study demonstrates a clinically significant reduction in TRAb and increase in T4 levels post-operatively but no significant change in TED parameters.

P204

A STUDY EVALUATING A 3D-PRINTED EYELID MODEL FOR OCULOPLASTIC TRAINING

Corresponding author: Ali Istanaksai, orcid.org/0009-0001-4132-4509

Ali Istanaksai¹, Mohsan Malik²

¹King's College London, London, UK; ²Moorfields Eye Hospital, London, UK

Disclosures: No conflicts of interest and no funding

Introduction: This study evaluated the validity of a novel 3D-printed eyelid tissue model, aiming to determine its potential as an alternative to cadaveric tissue.

Method: 3D printing was used to produce a simplified eye model with four tissue planes: skin, orbicularis muscle, tarsus, and conjunctiva. A survey, modified from a validated design, assessed face validity, content validity, utility, and teaching potential. The model was rated on a 5-point Likert scale by a total of 9 participants, including oculoplastic fellows (n = 4) and consultants (n = 5). A score of 1 represented strong disagreement and a score of 5 suggested strong agreement.

Results: Results from our study indicate positive face validity, with participants particularly impressed with the realism of the skin and orbicularis layers of the eyelid model (median: 4.00 and IQR: 1.00). In terms of content validity, participants felt the eyelid model provided adequate simulation of human cadaveric tissue (mean: 3.38 and IQR: 1.00). Furthermore, the model demonstrated excellent teaching potential for basic eyelid suturing and wedge resection but less potential for other procedures such as canalicular repair. In terms of utility, participants favoured the eyelid model over standard tissue pads for layer identification, showing potential as a valuable teaching resource. The model's cost-effectiveness and sustainability were also recognised. In general, senior oculoplastic fellows rated the model higher than consultants across all domains.

Conclusion: Overall, our 3D-printed model scored positively across all domains and participants recommended the model as a widely accessible, low-cost alternative to cadaveric tissue in oculoplastic training.

P205. Abstract not included

P206. Abstract not included

P207

BOTULINUM TOXIN INJECTION FOR HEMIFACIAL SPASM AND BENIGN ESSENTIAL BLEPHAROSPASM: A META-ANALYSIS AND REVIEW COMPARING PRETARSAL AND PRESEPTAL INJECTION TECHNIQUES

Corresponding author: Mohammad Saleki, orcid.org/0000-0002-7418-2690

Mohammad Saleki¹, Muhammad Ashhad Noor², Ahmad Abul³, Ali Alkhabbaz⁴, Ali Abel³, Mohammad Karam⁵, Sarah Al-Youha³

¹Barts Health NHS Trust, London, UK; ²Manchester University NHS Foundation Trust, Manchester, UK; ³Jaber AlAhmad Hospital, Kuwait; ⁴Amiri Hospital, Kuwait; ⁵McGill University, Montreal, Canada

Disclosures: No conflicts of interest and no funding

Introduction: This meta-analysis examined two botulinum toxin injection techniques, pre-tarsal and pre-septal for hemifacial spasm (HFS) and benign essential blepharospasm (BEB), aiming to clarify superiority in injection approach to inform optimised treatment strategies for these conditions.

Method: A systematic review and meta-analysis were performed as per the Preferred Reporting Items for Systematic Reviews and Meta-analyses guidelines with a search of electronic databases conducted to identify all Randomised Controlled Trials, case control studies and cohort studies comparing the outcomes between the two techniques. The change in Jankovic Rating Scale (JRS) was the primary outcome. Secondary outcome measures included adverse events, duration to maximum response and latency to first response. Fixed and random effects modelling were used for the analysis.

Results: 384 patients with 2657 injections for HFS patients and 1033 injections for BEB patients was reported across six studies. The mean difference analysis for the change in JRS scores in HFS patients reveals a statistically significant difference favouring the pre-tarsal group for improved JRS ratings (MD = 0.47, CI = 0.19 to 0.75, $P = 0.001$) and change in JRS scores in BEB patients favouring the pre-tarsal group (MD = 0.92, CI = 0.48 to 1.36, $P = 0.001$). There was a significant difference favouring the pre-tarsal group in duration of maximum response in HFS ($P = 0.0005$) and fewer occurrences of ptosis ($P = 0.0009$) with low amount of heterogeneity ($I^2 = 0\%$, $P = 0.65$). For all other secondary outcomes including post-procedure tearing, irritation and lagophthalmos both groups had similar results.

Conclusion: When compared to the pre-septal approach, pre-tarsal botulinum toxin injection is a preferable alternative for BEB and HFS, minimising the extent and frequency of spasms, less occurrences of ptosis, and a greater duration of maximal response.

PAEDIATRIC OPHTHALMOLOGY

P208

AUDIT OF THE MANAGEMENT OF PAEDIATRIC ORBITAL AND PRE-SEPTAL CELLULITIS AND THE ROLE OF SPECIALIST INPUT IN LINE WITH PUBLISHED GUIDELINES

Corresponding author: Yasmin Mahamed Cumar Cabdi, orcid.org/0000-0001-5746-2132

Yasmin Mahamed Cumar Cabdi¹, Youssra Al Hajji¹, Tina Kipioti², Varsha Kadaba²

¹University of Birmingham, Birmingham, UK; ²University Hospitals Birmingham NHS Trust, Birmingham, UK

(*joint 1st authors).

Disclosures: No conflicts of interest and no funding

Introduction: Orbital cellulitis is a rare but serious bacterial infection of the eye; it is an ophthalmological emergency and has the potential for serious long-term consequences if managed inappropriately. Pre-septal cellulitis is a less serious infection of the eyelid that may present similarly to orbital cellulitis. Management of orbital cellulitis can be highly variable, highlighting non-optimal treatment.

This audit aimed to assess the compliance of management of orbital and pre-septal cellulitis with local and national guidelines (BIPOSA, The Royal College of Ophthalmologists).

Method: A retrospective analysis was undertaken of the data of 110 paediatric patients. Data was collected on timepoints of specialist input (ENT/Ophthalmology), treatment regime (including antibiotic choice and duration), length of stay and clinical outcome. Management was then assessed for compliance.

Results: Management of orbital cellulitis deviated from the local guidelines, with only 71% of patients receiving the recommended first-line IV antibiotic. 29/44 patients were admitted for inpatient management. There was a positive correlation between time to first administration of IV antibiotic and overall length of stay. 22% of inpatients did not receive any specialist input (from ENT or ophthalmology). 9% of patients required surgical abscess drainage. Only 59% of patients were discharged with 7-14 days of oral co-amoxiclav, and less than 50% of patients were followed up.

Conclusion: Orbital and pre-septal cellulitis in the paediatric population needs close monitoring and timely intervention with the appropriate management. Specialist teams need to be involved early in these patient's care, where deemed necessary. The audit has highlighted gaps in existing guidelines that should be addressed. We have proposed a new, revised guideline for optimal management of these conditions to enhance patient outcomes.

P209

EVALUATION OF A VIRTUAL MULTIDISCIPLINARY APPROACH TO DECISION-MAKING IN ROP MANAGEMENT

Corresponding author: Barnaby Carr, orcid.org/0009-0006-3820-5812

Barnaby P B Carr¹, Eibhlin McLoone², on behalf of the TeleROP-NI MDT team

¹Queens University Belfast, Belfast, UK; ²Belfast Trust, Belfast, UK

Disclosures: No conflicts of interest and no funding

Introduction: The sustained provision of a robust Retinopathy of Prematurity (ROP) screening service in Northern Ireland has become more challenging in recent times due to the increasing survival rate of very premature infants, resulting in an increased workload and pressure on individual ROP screening Ophthalmologists. A TeleROP-NI regional model for ROP screening was implemented to meet this challenge, enabling imaging of infants across neonatal units in Northern Ireland by non-Ophthalmologist ROP imagers, supported by a dwindling workforce of Speciality Doctor and Consultant Ophthalmologists, with subsequent virtual Multidisciplinary Team (MDT) collective decision-making for infants developing ROP.

Method: This study used qualitative methodology; specifically, thematic analysis of focus group data, through a series of questions answered by members of the MDT, comprising of Treaters/ Screeners/ Imagers/ Trainees/ Administrators. Various parameters were quantitatively analysed to indicate the value of the MDT (1-10, 1 least and 10 most).

Results: Thematically, assessment of the educational value of the MDT (average 9.3/10) highlighted better management of cases, improved feedback on decisions and continued learning right up to senior roles. Collective decision-making (average 9.4/10) shared burden on practitioners, reducing risk and ensuring optimal care of infants. Job satisfaction (average 9.2/10) improved across all roles due to reduction in stress, a clarity in individual roles and collective approach. MDT increased the likelihood of trainees undertaking future ROP screening (average 9/10) due to excellent support and less stressful experience.

Conclusion: Overall, the MDT approach ensured safer and more effective decision-making, enhancing quality of job role whilst encouraging trainees to consider ROP screening and treatment in their future careers to help create a sustained and populated workforce.

P210

TELEROP-NI: A REGIONAL MULTIDISCIPLINARY APPROACH TO ROP SCREENING & DECISION MAKING

Corresponding author: Barnaby Carr, orcid.org/0009-0006-3820-5812

Barnaby P B Carr¹, Eibhlin McLoone²

¹Queens University Belfast, Belfast, UK; ²Belfast Trust, Belfast, UK

Disclosures: No conflicts of interest and no funding

Introduction: The sustained provision of a robust Retinopathy of Prematurity (ROP) screening service in Northern Ireland has become more challenging in recent times due to the increasing survival rate of very premature infants, resulting in an increased workload, whilst simultaneously a dwindling workforce of ophthalmologists experienced in ROP management. A TeleROP-NI regional model for ROP screening was developed to meet this challenge, enabling imaging of infants across neonatal units in Northern Ireland by non-Ophthalmologist ROP imagers with subsequent Multidisciplinary Team (MDT) collective decision-making for infants developing ROP.

Method: The telemedicine approach uses a team of ROP imagers to image infants with the Forus-Neo Camera and upload images for review at the weekly MDT virtual meeting. Between 2nd May – 24th October 2023, 431 screens (n = 127 infants) were undertaken: 250 by Imagers, 40 by Imagers and Ophthalmologists using Binocular Indirect Ophthalmoscopy (BIO), and 141 by Ophthalmologists. In line with the 2022 RCOphth ROP screening recommendations, BIO was performed on all infants by an Ophthalmologist, to confirm vascularisation into Zone 3, before final discharge from the ROP screening programme.

Results: Overall, 58% of screens were completed by ROP Imagers alone; with 98% sensitivity in picking up any ROP, 100% sensitivity in detecting treatment-warranted ROP and 100% specificity of determining absence of ROP (as verified by Ophthalmologist on BIO pre-discharge. 98% of all imaging screens were well tolerated by infants.

Conclusion: This study confirms that the tele-medicine model significantly reduced Ophthalmologist workload and streamlined organisational barriers. Imaging infants by non-clinicians is a safe and valuable practice which allows an MDT approach to collective decision-making when managing patients in an area of medicine which has serious clinical and medicolegal implications if disease is missed.

P211. Abstract not included

P212

EARLY SPECTACLE INTERVENTION FOR BABIES WITH DELAYED VISUAL MATURATION

Corresponding author: Niku Dhillon, orcid.org/0000-0001-7846-4575

Niku Dhillon¹, Jane MacKinnon¹

¹NHS Highland, Inverness, UK

Disclosures: No conflicts of interest and no funding

Introduction: A blind baby is a neurodevelopmental emergency. A 2-month-old infant unable to fix and follow, but who subsequently improves by 6 months, is described as having delayed visual maturation (DVM). We propose early intervention with spectacle overcorrection.

Method: Retrospective study of babies under 6 months of age identified as being visually impaired between 2015 and 2022 in NHS Highland. A subset of infants with a retrospective diagnosis of DVM was identified. Benefit from early intervention with glasses in DVM was assessed.

Results: 37 babies identified as being visually impaired before 6 months of age. 24 (65%) were male. Socio-economic status was not normally distributed ($p = 0.007$) and skewed towards more affluent areas. 16 had a primary diagnosis of neurological abnormalities, 8 ocular abnormalities, 5 intrauterine drug exposure, 2 a genetic syndrome and 6 isolated DVM. 20 had a secondary diagnosis of global developmental delay, 5 were premature and 12 had seizures. 19 infants (51%) had a retrospective diagnosis of DVM and 16 (84%) of these babies were prescribed over-plus glasses. 6 had isolated DVM (type 1), 8 had DVM related to systemic disease (type 2) and 5 had DVM associated with ocular disease (type 3). All babies demonstrated an improvement in visual behaviour. Where parental comment was recorded the majority reported a positive effect with glasses, often shortly after commencing wear.

Conclusion: The findings support active intervention with near overcorrection to “kick start” vision in DVM infants. As vision is paramount to global development, we advocate taking advantage of early neuroplasticity with refractive intervention in order to improve long-term neurodevelopmental outcome.

P213

EFFECTIVENESS OF INTERVENTIONS FOR CONVERGENCE INSUFFICIENCY: A PROSPECTIVE STUDY EVALUATING CLINICAL OUTCOMES AND HEALTH-RELATED QUALITY OF LIFE USING AS20

Corresponding author: Joel Lee Zher Jong, orcid.org/0000-0002-8339-2083

Joel Lee Zher Jong^{1,2}, Zainab Saleem², Joshua Simmons², Martin Rhodes², Jessy Choi^{1,2}

¹Medical School, The University of Sheffield, Sheffield, UK; ²Sheffield Teaching Hospitals NHS Trust, Sheffield, UK

Disclosures: No conflicts of interest and no funding

Introduction: Convergence insufficiency (CI) is a common condition that can impair visual performance and comfort during close visual work. This prospective study evaluated the effectiveness of interventions on clinical outcomes and health-related quality-of-life using the adult strabismus quality-of-life questionnaire (AS20) in patients with CI.

Method: Data was extracted from a database collected at first consultation from 2015 to 2022. Demographics, interventions and outcomes of 80 patients with CI (mean age 46.5 ± 24.6 years) were analysed.

Results: Orthoptic exercises were prescribed to 58% of patients, 31% received prisms, 11% received no treatment, with 3 discharged on the same day. At latest follow-up review, 22.5% were recommended to continue exercises, 20% had prisms, 1 underwent bimedial resection and 2 had botox. The median follow-up was 5.0(0-55) months, 97.5% were discharged with 27.5% following failure to attend and 10% deceased. At the latest follow-up, the

attendance failure rate was higher for exercises (36%) than for prisms (15%). Both near prism cover test and near-point of convergence (NPC) improved from a median of 12(95%CI 10-14)D to 8(95%CI 5-10)D, and 17.5(95%CI 15-20)cm to 10.0(95%CI 8.3-12.0)cm respectively ($p < 0.05$). Near prism fusion range also improved from 10(95%CI 8-14)D to 19(95%CI 16-20)D. The median AS20 score at presentation were 100(30-100) and 47.5(0-100), and post-intervention were 100(80-100) and 77.5(12.5-97.5) for psychosocial and functional components, respectively. Improvement was noted in NPC (33%) and mean AS20 scores was 9% higher psychosocially and 32.8% functionally, highlighting the benefits of intervention on patients' quality-of-life.

Conclusion: This cohort provides valuable insights into the clinical management of CI, as evidenced by improvements in NPC and AS20 scores. However, the study also found that long-term compliance with treatment is intrinsically challenging, emphasising the importance of disease education.

P214. Abstract not included

P215

INCOMPLETE CONGENITAL STATIONARY NIGHT BLINDNESS ASSOCIATED WITH A NOVEL MUTATION IN THE *CACNA1F* GENE

Corresponding author: Shi Pei Loo, orcid.org/0009-0003-8680-7218

Shi Pei Loo¹, Chloe Shipton¹, Mark Hamilton¹, Andrew Brown¹, Eoghan Millar¹, Iqbal Malik¹, Madeleine Craig¹, Ruth Hamilton¹

¹NHS Greater Glasgow and Clyde, Glasgow, UK

Disclosures: No conflicts of interest and no funding

Introduction: Congenital stationary night blindness (CSNB) refers to a group of inherited, non-progressive retinal diseases. A subtype is incomplete CSNB (iCSNB). Most cases of iCSNB result from mutations in the X-linked gene *CACNA1F*. Herein, we describe iCSNB in two male siblings, both carrying a novel mutation c.4041+5G > T in the *CACNA1F* gene inherited from their mother.

Method: Electroretinography (ERG) testing was carried out. Genetic testing with next generation sequencing, in-silico analysis and functional analysis were performed to further characterise the novel variant.

Results: A 3-year-old male had high myopia and reduced visual acuities. Ophthalmic examination was normal and he underwent ERG testing which demonstrated findings resembling that of iCSNB, with a markedly reduced amplitude to dim flashes and an electronegative waveform to bright flashes in dark-adapted ERGs, leading to its diagnosis. Next generation sequencing of genes linked to retinal disorders identified a hemizygous c.4041+5G > T variant in the *CACNA1F* gene. Follow-up at seven years demonstrated stable visual acuities. His younger brother at 2 years old also had reduced visual acuities and myopia. ERG demonstrated findings resembling iCSNB and genetic testing identified the same c.4041+5G > T variant in *CACNA1F*. Their mother had mostly normal ERGs and was found to have the same variant.

Conclusion: We report a novel *CACNA1F* variant in three patients not previously identified in the literature. Although functional analyses were unable to confirm pathogenicity of this variant, in-silico tools predicted that it would disrupt normal splicing of *CACNA1F*, leading to a null effect, consistent with the pathogenesis of iCSNB. Reporting of this family further widens the genotypic spectrum of iCSNB.

P216

QUALITATIVE SURVEY OF PARENT EXPERIENCE AT THE STEPPING HILL SQUINT SERVICE

Corresponding author: Nyaish Mansoor-Ali, orcid.org/0000-0002-8872-3435

Nyaish Mansoor-Ali^{1,2}, Anna Maino²

¹Stepping Hill Hospital, Stockport, UK; ²Manchester Royal Eye Hospital, Manchester, UK

Disclosures: No conflicts of interest and no funding

Introduction: To explore the perception and experience of parents whose children underwent squint surgery at Stepping Hill Hospital, Stockport to ultimately enhance the service for an improved future experience.

Method: Qualitative data was collected via online anonymous surveys from families whose children underwent squint surgery between February and July 2023. The focus was on key themes such as understanding healthcare roles, information resources, and delivery preferences.

Results: Fifteen responses were obtained, with 80% expressing a keen interest in understanding the different healthcare roles. Two-thirds felt adequately informed, emphasising a desire to particularly know the skill set and role in the care pathway. Two-thirds of respondents conducted their own research before appointments primarily using online resources, followed by leaflet materials. Parents particularly prioritised understanding squint causes and recovery details.

Regarding information about the condition, individuals are generally content receiving it from any doctor. However, when it comes to details about the surgical procedure, the majority express a preference for information to be delivered by the Consultant specifically. Satisfaction was highest for the surgery day (100%), followed by the post-op appointment (87%) and pre-clinic appointment (86%). In this re-audit, a greater number of participants expressed a desire for more tangible information before their first appointment, such as in the form of a leaflet.

Conclusion: Patients desire a comprehensive understanding of healthcare roles with emphasis on role clarification and involvement. The preference for more information before the first appointment, especially on squint causes and post-op recovery, suggests a need for improved pre-appointment resources. Participants favoured Consultants for delivering information on surgery. Overall, positive satisfaction was reported at all stages, with effective communication cited as a key factor.

P217

EXPERIENCE OF UTILISING 0.1% CYCLOSPORINE AS A LONG-TERM TREATMENT IN CHILDREN WITH SEVERE VERNAL KERATOCONJUNCTIVITIS

Corresponding author: Daniel Morris, orcid.org/0009-0007-3425-3312

Daniel Morris¹, Kuan-yin Wu², Abdul-Jabbar Ghauri³

¹Sandwell and West Birmingham Hospital Trust, Birmingham, UK; ²Aston University, Birmingham, UK; ³Birmingham & Midlands Eye Centre, City Hospital NHS Trust, Birmingham, UK

Disclosures: No conflicts of interest and no funding

Introduction: Vernal keratoconjunctivitis (VKC) is a rare, chronic and severe form of allergic conjunctivitis characterised by inflammation of the ocular surface and conjunctiva. Although the condition is often self-limiting and tends to resolve by the end of puberty, a subset of patients can develop sight-threatening complications if not treated properly. Topical cyclosporine A (CsA) 0.1% is an effective steroid sparing agent in the management of severe VKC.

Method: This study aims to investigate if CsA 0.1% eye drops are being utilised correctly in severe or sight threatening VKC and if its early use reduces future clinic attendances or flare ups. We have designed a retrospective, non-randomised case series with analysis of 30 patients (mean age 8.4 ± 3.0 years) with severe VKC treated with CsA 0.1% eye drops. The mean treatment period during this study was 38.5 months.

Results: Our results show 4 out of 30 (13%) patients were started on CsA at initial attendance to clinic or eye casualty. 17 out of 30 (56%) were subsequently started on CsA within the first 3 attendances and 88% of these patients were shown to have clinical improvement in their VKC following initiation of CsA.

Conclusion: Cyclosporin eye drops are an effective and safe long term treatment modality for severe VKC. Early proactive use of CsA is a viable option as a steroid sparing agent. Fine adjustment of dosage is essential to avoid flare ups requiring attendance to eye casualty. Proactive use of CsA is a viable option as a steroid sparing agent for long term treatment of VKC. The importance of patient compliance is crucial for optimal clinical improvement.

P218

PARENTAL AND HEALTH CARE PROFESSIONAL PERCEPTION OF NURSE LED VIDEO CONSULTATIONS IN PAEDIATRIC OPHTHALMOLOGY CYST AND WATERY EYE CLINICS: A QUALITATIVE EVALUATION

Corresponding author: Roisin Mulholland, orcid.org/0009-0002-6488-7715

Roisin Mulholland¹

¹Belfast Health and Social Care Trust, Belfast, UK

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Introduction: The Paediatric Ophthalmology Service, Royal Victoria Hospital Belfast, is a regional centre of excellence. A significant number of referrals received are in relation to cysts and watery eyes. A large proportion, based on the referral, did not actually require any ophthalmic equipment to assess and plan their care. With the Covid-19 pandemic acting as a catalyst, a virtual nurse led cyst and watery eye clinic was developed for what are considered low risk referrals. The objective is to clinically assess the patient through video consultation, and either discharge with advice on management of symptoms, board for surgical intervention, or refer to a face to face clinic appointment if further assessment is required. The aim was to make more face to face clinic appointments available for children who require specialist ophthalmic examinations. The virtual clinic has proved efficient in reducing waiting lists due to the ease of scheduling ad-hoc clinics, as no physical clinic rooms are required.

Method: The objective of this ongoing quality improvement project is to evaluate the satisfaction levels of the parents and nurses over a 6 month period through questionnaires.

Results: To date, there have been 61 virtual appointments scheduled:

54% of these were either discharge with advice or DNA discharge

58 face to face clinic appointments have been saved
The waiting list for this clinic has been reduced by half over the period of only 3 months due to the ease of scheduling ad-hoc clinics.

92% of parents who attended state they would be happy to attend another virtual appointment based on their experience.

Conclusion: These figures provide overwhelming evidence of clinic efficiency and how the Paediatric Ophthalmology department is successfully modernising its service.

P219. Abstract not included

P220

PREVALENCE OF MYOPIA IN THE PAEDIATRIC POPULATION OF EUROPE: A SYSTEMATIC REVIEW AND META-ANALYSIS

Corresponding author: Aoife O Donnell, o_donnell-a4@ulster.ac.uk

Aoife O Donnell^{1,2}, Megan Doyle^{2,3}, Siofra Harrington^{2,3}, Michael Moore², Veronica O'Dwyer^{2,3}, Michael Moore^{2,3}

¹Ulster University, Ulster, UK; ²Technological University Dublin, Dublin, UK; ³Centre for Eye Research Ireland, Dublin, UK

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Introduction: Myopia is a global health issue of epidemic proportions. Its impact extends beyond optical inconvenience to ocular pathology with the potential for irreversible visual impairment. However, without comprehensive prevalence levels, the gravity and urgency of this issue cannot be evidenced. Therefore, the purpose of this meta-analysis was to quantify myopia prevalence in the paediatric population of Europe. Additional objectives to explore variations in myopia rates based on age, sex, geographical location, and urbanisation levels were also addressed.

Methods: Studies were sourced through a systemic search of research databases (PubMed and Cochrane) using refractive error terminology and European nations with distinctions made between countries for data analysis purposes. Suitable studies were identified by employing the inclusion criteria and data collated. The pooled prevalence of myopia and high myopia were provided with 95% confidence intervals.

Results: The cumulative prevalence of myopia was 11.48% (95% CI 9.20; 13.97). Subgroup analysis showed increasing myopia prevalence in urban environments and as a factor of age. Publication year and sex had minimal influence on the prevalence of myopia.

Conclusion: This is the first meta-analysis to explore the overall prevalence of paediatric myopia in European regions. The findings indicate that myopia is moderately prevalent among children in Europe and increases as a factor of age and urbanisation. However, there is a significant lack of data therefore further research is required in this field to offer a more comprehensive overview of paediatric European myopia prevalence.

P221

A MIXED-METHODS STUDY INVESTIGATING MULTI-LEVEL FACTORS ASSOCIATED WITH PSYCHOSOCIAL ADJUSTMENT OUTCOMES OF CHILDREN AND YOUNG PEOPLE WITH VISUAL IMPAIRMENT

Corresponding author: Ana Šemrov, orcid.org/0000-0001-5447-1295

Ana Šemrov^{1,2}, Mario Cortina Borja², Rachel Knowles², Roz Shafran^{2,4}, Ameenat Lola Solebo^{2,4}, Valerija Tadić³, Jugnoo S. Rahi^{2,4,5}

¹University of Ljubljana, Ljubljana, Slovenia; ²UCL Great Ormond Street Institute of Child Health, London, UK; ³University of Greenwich, London, UK; ⁴Great Ormond Street Hospital NHS Foundation Trust, London, UK; ⁵UCL Institute of Ophthalmology, London, UK

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Introduction: Children and young people with visual impairment (CYP-VI) usually face increased psychosocial challenges. This mixed-methods study examined the relationship between multiple factors shaping psychosocial adjustment outcomes in CYP-VI, representing a first step towards multimodal intervention development.

Method: Recruitment from two UK tertiary-care ophthalmology departments and three vision loss charities. Seventy-eight CYP-VI aged 8–18 years and their caregivers participated in a quantitative postal survey. Outcomes were assessed using (i) Vision-Related Quality of Life (VQoL) for Children and Young People and (ii) Strengths and Difficulties Questionnaires. Semi-structured interviews were conducted with 18 of these families. Linear multi-variable regression modelling and thematic analysis were undertaken.

Results: Better VQoL ($\text{Adj.}R^2 = .533$) was significantly ($p < .05$) predicted by greater peer support, fewer functional vision difficulties, better connection in school, and higher carer's satisfaction with life. Fewer internalising problems ($\text{Adj.}R^2 = .511$) were predicted by greater peer support, an absence of additional chronic health conditions, and fewer carer's anxiety symptoms. Fewer externalising problems ($\text{Adj.}R^2 = .298$) were predicted by greater peer support, later onset of VI, and more prosocial behaviour. The thematic analysis further underscored, from the children's and carer's perspective, the crucial role of positive educational experience, family, and peer support in CYP-VI's psychosocial adjustment.

Conclusion: Our findings highlight significant associations between individual, family, and school characteristics and mental health and VQoL in CYP-VI, offering potential targets for family-centred and school-based psychosocial interventions. These findings should aid healthcare professionals in identifying 'at-risk' patients requiring additional support. Collaborative efforts across multiple services, integrating formal and non-formal support from VI diagnosis onward, are imperative for optimal care and improving outcomes for CYP-VI and their families.

P222

COMPARABILITY OF OPTIC NERVE HEAD ASSESSMENT USING TWO OPTICAL COHERENCE TOMOGRAPHY MACHINES IN CHILDREN WITH CRANIOSYNOSTOSIS

Corresponding author: Riddhi Shenoy, orcid.org/0000-0001-8118-9009

Riddhi Shenoy¹, Dermot Roche², Richard Bowman², Mervyn G Thomas¹, Vasiliki Panteli², Robert H Henderson², Catey Bunce³, Irene

Gottlob¹, Noor ul Owase Jeelani², Frank A Proudlock¹, Sohaib R Rufai^{1,2}

¹University of Leicester, Ulverscroft Eye Unit, Leicester, UK; ²Great Ormond Street Hospital, London, UK; ³The Royal Marsden, London, UK

Disclosures: Dr Thomas and Dr Proudlock, consultants, Leica Microsystems

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Introduction: Ophthalmological monitoring is important in craniosynostosis to exclude papilloedema. No study has compared handheld versus table-mounted optical coherence tomography (OCT) assessment in craniosynostosis. This study assessed comparability between machines and clinicians. (Competing interests: Thomas+Proudlock, Consultants, Leica Microsystems.)

Method: This cross-sectional study took place at Great Ormond Street Hospital, London. Twenty children with craniosynostosis were recruited from 21/10/2021–24/03/2022. The Envisu C2300 handheld OCT (Leica Microsystems) and Spectralis (Heidelberg Engineering) were used for bilateral optic nerve head scans in all participants. Masked, randomised optic nerve tomograms were graded qualitatively by an expert grader (Sohaib Rufai), and novice grader (Riddhi Shenoy) before and after brief training. The following criteria were graded as yes/no: anterior displacement of Bruch's membrane, raised cup and rim, cup obliteration, severe optic atrophy. Intermachine and intergrader agreement was assessed.

Results: Bilateral OCT scans were successful in all children using both machines. Ten participants (50%) were female. Median age was 6 years (range: 4–16; IQR: 5–8). Seven patients had syndromic craniosynostosis and thirteen patients had non syndromic craniosynostosis. Expert grading delivered 100% intermachine agreement (40/40). Untrained novice grading delivered 96% intergrader agreement (77/80) and 93% inter-machine agreement (37/40); following training, 100% intergrader (80/80) and intermachine agreement (40/40) was achieved. Of the 20 participants, one demonstrated bilateral anterior displacement of Bruch's membrane, seven displayed bilateral raised rim and cup, two displayed unilateral cup obliteration and none displayed severe optic atrophy.

Conclusion: This qualitative optic nerve head grading system was robust across both machines. The novice grader achieved 100% intergrader and intermachine agreement with brief training. Both machines can be used to qualitatively assess the optic nerve head in craniosynostosis.

P223

ASSOCIATIONS BETWEEN GENES ASSOCIATED WITH SYNDROMIC CEREBRAL VISUAL IMPAIRMENT (CVI), WITH CVI-RELATED VISION TESTS IN A COMMUNITY-BASED SAMPLE OF CHILDREN

Corresponding author: Cathy Williams, orcid.org/0000-0002-9133-2021

Cathy Williams^{1,2}, Jeremy Guggenheim³, David Odd⁴, Nora Mohd¹, Panos Maghsoudlou¹, Silvia Pregolato², Anna O'Connor⁵, Ilona Kovacs^{6,7}, Yi-Zhong Wang^{8,9}, Eileen Birch^{8,9}, Karen Luyt²

¹Bristol Eye Hospital, Bristol, UK; ²Bristol Medical School, Bristol, UK; ³Cardiff School of Optometry & Vision Sciences, Cardiff University, UK; ⁴Cardiff University, Cardiff, UK; ⁵Liverpool University, Liverpool, UK; ⁶Institute of Psychology, Faculty of Education and Psychology,

Eötvös Loránd University, Budapest, Hungary; ⁷HUN-REN-ELTE-PPKE Adolescent Development Research Group, Budapest, Hungary; ⁸Crystal Charity Ball Pediatric Vision Laboratory, Retina Foundation of the Southwest, Dallas, TX, USA; ⁹University of Texas Southwestern Medical Center, Dallas, TX, USA

Disclosures: No conflicts of interest

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Introduction: A recent report identified novel genetic variants associated with syndromic cerebral visual impairment (CVI). Few genetic data are available regarding non-syndromic CVI. We opportunistically looked at single nucleotide polymorphisms (SNPs) in some of the recently-reported variants and compared these with CVI-related vision test results, in a community-based cohort of children aged 7yr - 15yr.

Method: We compared genotypes for 5 SNPs: one in DLG4 and four in GRIN genes (1, 2A, 2B, 3A), with the results of CVI-related vision tests: pursuit eye movements (7yr); contour detection threshold (11yr, 12yr); radial deformation hyperacuity: RDH (11yr, 12yr); resolution and recognition vernier acuity (15yr); total score and three derived subscales from a CVI-questionnaire (13yr). We conducted non-parametric univariate tests to identify any associations $p < 0.05$ and examined these in univariable regression analyses. We analysed all children with data and a subgroup admitted to hospital as neonates (at-risk).

Results: Results were available for 54 - 3133 children and 8 non-parametric comparisons were $p < 0.05$. Three SNPs were associated with RDH at 12 years: GRIN2A for the whole sample (AA vs AC, $p = 0.048$) and GRIN1 (GA vs GG, $p = 0.045$) and DLG4 (GA vs GG, $p = 0.045$) in the at-risk subgroup. A further SNP (GRIN3A) was associated with contour detection threshold at 11 years, again in the at-risk subgroup (CT vs CC, $p = 0.023$). There was little evidence of an association between GRIN2B and any outcome.

Conclusion: These results provide weak but suggestive evidence that these SNPs may contribute to CVI-related visual functions in a community-based cohort of children as well as in children with genetic syndromes. This may help towards understanding the pathophysiology involved in CVI and/or design future treatments for CVI.

UVEITIS

P224

EVALUATING THE USE OF AN INVESTIGATIVE PROTOCOL FOR ACUTE ANTERIOR UVEITIS (AAU) IN THE EMERGENCY EYE CLINIC USING MOST AGREED GUIDELINES

Corresponding author: Durray Nayab Ali, orcid.org/0009-0008-0920-2449

Durray Nayab Ali¹, Nyaish Mansoor¹, Abdelsattar Farrag¹

¹Bolton NHS Trust, Bolton, UK

Disclosures: No conflicts of interest and no funding

Introduction: Acute anterior uveitis (AAU) is inflammation of the uveal tract commonly found to be an extra-articular manifestation of spondyloarthropathies. Currently there is no universally agreed investigative protocol for AAU. Local guidelines were implemented by the Uveitis Lead for patients

presenting with AAU in the acute referral clinic (ARC). We assessed if patients were investigated accordingly and reviewed subsequent health outcomes.

The authors of the report have no financial interests/funding to declare.

Method: Adults presenting to ARC between May and June 2022 with AAU were included. Information collected as part of the investigative protocol included: history of systemic disease associated with anterior uveitis, previous attacks and unilateral or bilateral attack. Patients were reviewed to assess if they followed the protocol according to local guidelines. Inclusion of a complete metabolic panel, chest x-ray, syphilis serology inflammatory causes and markers was recorded.

Results: 31 patients were identified. 77.4% of patients correctly followed the investigative protocol. 6.5% of patients were unnecessarily investigated further. 5 patients (16.1%) did not have further investigations requested despite guidelines stating they would benefit from further tests. 2 of these patients had a flare of AAU in less than 12 months and 1 was found to be HLAB27 positive on subsequent review. 19% of patients had a specialist investigation requested. Of these patients, two-thirds had no clear indication documented.

Conclusion: Results show that only 77.4% of patients are being investigated according to guidelines in ARC. A flowchart illustrating the investigative pathway has been made and displayed in clinic rooms. Overall health outcomes may be improved if patients are adequately investigated on acute presentation with early identification of systemic diseases in order to reduce visits with a recurrent attack.

P225. Abstract not included

P226

ADALIMUMAB IN THE TREATMENT OF REFRACTORY NON-INFECTIOUS SCLERITIS: 18-MONTH OUTCOMES

Corresponding author: Ian Yeung, orcid.org/0000-0001-6497-8866

Emilia Bober^{1,2}, Majid Fotuhi¹, Kristina Frain^{1,3}, Joshua Luis¹, Jonathan Virgo⁴, Edward Hindle¹, Safiyya Diwany¹, Jianfei Ma⁵, Mark Westcott¹, Angela Rees¹, William Tucker¹, Peter Addison¹, Carlos Pavesio¹, Narciss Okhravi¹, Ian Yeung¹

¹Moorfields Eye Hospital NHS Foundation Trust, London, UK; ²The Royal Marsden Foundation Trust, London, UK; ³Imperial College Healthcare NHS Trust, London, UK; ⁴Guy's and St Thomas' NHS Foundation Trust, London, UK; ⁵Whittington Hospital, London, UK

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Introduction: Adalimumab, a tumour necrosis alpha inhibitor, has become a globally well-recognised treatment for refractory uveitis. Evidence generated from a recent case series provided an argument for expanding its use to patients with refractory non-infectious scleritis. Long-term outcomes are still underreported in this cohort of patients which makes conclusions on its overall efficacy challenging. Hence, we aim to report 18-month outcomes of patients with non-infectious, refractory scleritis treated with adalimumab.

Method: We included 14 adult patients with refractory non-infectious scleritis initiated on adalimumab between September 2014 and October 2021 at our tertiary referral centre. Data was collected retrospectively from electronic patient records. Patients' 18-month outcomes were analysed to determine the glucocorticoid-sparing effect of adalimumab, rate of disease reactivation, reasons for stopping the treatment and presence of any significant adverse events.

Results: Out of 14 patients, 11 continued their treatment for the duration of 18 months. Three patients stopped due to ineffectiveness secondary to developing anti-adalimumab antibodies ($n = 2$) and primary ineffectiveness, not associated with development of anti-adalimumab antibodies ($n = 1$). On initiation of adalimumab, 28.6% of patients were on ≤ 5 mg of daily prednisolone as compared to 90.9% at 18 months of treatment ($P < 0.004$, Fisher exact test). The rate of disease reactivation was calculated as 1.01 flare-ups/patient-year. Adverse events included pain on injecting adalimumab, a single episode of bleeding gums, headache and nausea, peripheral neuropathy and infection.

Conclusion: This case series demonstrates a good response to adalimumab in the majority of patients after 18 month of treatment. Development of anti-adalimumab antibodies was an obstacle that stopped two patients from continuing long-term therapy. We hope this data will contribute to further understanding of the role of adalimumab in the treatment of refractory scleritis.

P227

PHENOTYPES OF TUBERCULOUS UVEITIS IN QUANTIFERON-GOLD POSITIVE PATIENTS AND OUTCOMES OF ANTI-TUBERCULOUS TREATMENT: RETROSPECTIVE OBSERVATIONAL STUDY

Corresponding author: Chung Shen Chean, orcid.org/0000-0002-1625-4078

Chung Shen Chean¹, Priti Kulkarni¹, Modupe Sodeinde¹, George Hills¹, Martin Wiselka¹, Periyasamy Kumar¹, Bharat Kapoor¹

¹University Hospitals of Leicester NHS Trust, Leicester, UK

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Introduction: To evaluate the features of tuberculous uveitis (TBU) in QuantiFERON-Gold (Q-Gold) positive patients and to report the outcomes of anti-tuberculous treatment (ATT) in this cohort.

Method: A retrospective observational study was performed among Q-Gold positive patients who had features of TBU who presented in 2009-2021. Demographic and baseline characteristics including age and ethnicity, ocular features of TBU, results of other diagnostic tests, details of ATT and their outcomes and effect on TBU were evaluated. These outcomes were compared to COTS-1 Trial.

Results: Among 94 patients, the commonest manifestation was anterior uveitis (62.2%, 117/188 eyes) followed by intermediate uveitis (44.1%, 83/188 eyes). 74 patients (78.7%) were of Asian or Asian British origin. Mean age was 53.4 years (SD 16.2). There were 49 (52.1%) male and 45 (47.9%) female. 7 eyes (3.7%) had choroidal granuloma, 30 eyes (16.0%) had vasculitis and 32 eyes (17.0%) had macular oedema. 83% (78/94) received topical steroids, 18.1% (17/94) received peri-ocular steroids and 8.5% (8/94) intra-ocular steroids. 40.4% (38/94 patients) needed systemic oral steroids and 5.3% (5/94) needed steroid-sparing immunosuppressant. Most patients had bilateral involvement (62.8%). 17 patients (18.1%) had systemic manifestations of TB. 40 patients (42.6%) were identified to have active TB and 46 patients (48.9%) had latent or previously treated TB. 79 patients received ATT, out of which 92.4% completed ATT. 3 patients did not complete ATT due to side effects. 2 patients' ATT were ongoing, and 1 patient was lost to follow-up. As per the COTS-1 Trial criteria, 19 patients (26%) had treatment failure. Majority (74%) had a favourable outcome to systemic ATT and 78% had no recurrence of TBU.

Conclusion: Our evaluation of TBU phenotypes and ATT outcomes demonstrated a favourable outcome of ATT on recurrence of TBU. The findings were limited by the retrospective data collection methodologies.

P228

IL-6 INHIBITION WITH VAMIKIBART IN PATIENTS WITH UVEITIC MACULAR OEDEMA: PHASE 3 MEERKAT AND SANDCAT TRIALS

Corresponding author: Helen Devonport, orcid.org/0000-0003-1113-1869

Helen Devonport¹, Laura Steeples², Markus C. Elze², Lachlan Macgregor², David Silverman², Zdenka Haskova³

¹Bradford Royal Infirmary, Bradford, UK; ²F. Hoffmann-La Roche AG, Basel, Switzerland; ³Genentech Inc., San Francisco, CA, USA

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 Markus C. Elze - Financial Disclosures: Employee: F. Hoffmann-La Roche AG; Stocks/Stock Options: Roche
 Lachlan Macgregor - Financial Disclosures: Employee: F. Hoffmann-La Roche AG; Stocks/Stock Options: Roche
 David Silverman - Financial Disclosures: Employee: F. Hoffmann-La Roche AG; Stocks/Stock Options: Roche
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Introduction: Uveitic macular oedema (UMO) is a vision-threatening complication of noninfectious uveitis (NIU). UMO pathophysiology is characterized by increased intraocular levels of proinflammatory cytokines, including interleukin-6 (IL-6). Vamikibart (RG6179) is a recombinant monoclonal antibody that inhibits IL-6 activity; it was studied in patients with UMO in a Phase 1 study (DOVETAIL). We report the study designs of MEERKAT and SANDCAT, two ongoing Phase 3 trials of vamikibart in UMO.

Method: Two identical, global, randomised, double-masked, sham comparator-controlled Phase 3 trials, MEERKAT (study GR44277, NCT05642312) and SANDCAT (study GR44278, NCT05642325), are investigating efficacy, safety, PK, and PD of intravitreal vamikibart in patients with UMO ($N = 225$ per study; 26 countries). Eligibility criteria include macular oedema, defined by macular thickening, associated with NIU, and best corrected visual acuity (BCVA) scores of 73–19 EDTRS letters. Patients with active or inactive, acute, or chronic NIU (any etiology or anatomical type) are eligible. Patients are randomised into 3 groups: vamikibart 0.25-mg, vamikibart 1-mg, and sham. Study treatment is administered 4 times every 4 weeks through week 12, followed by as needed dosing from week 20–48. Predefined rescue criteria allow standard-of-care treatment for worsening BCVA, UMO, or inflammation.

Results: Primary outcome measure is proportion of patients with ≥ 15 letter BCVA improvement from baseline at week 16. Secondary endpoints include changes in BCVA and central subfield thickness, rescue medications requirements, adverse events, and concentrations of IL-6 and vamikibart in aqueous humor and serum.

Conclusion: The ongoing global Phase 3 MEERKAT and SANDCAT program will evaluate vamikibart and its potential to address the unmet need for effective non-corticosteroid treatments in UMO.

P229

ADALIMUMAB IN THE TREATMENT OF NON-INFECTIOUS UVEITIS: A STUDY FROM THE UNITED KINGDOM: PART 1 EFFECTIVENESS

Corresponding author: Kristina Frain, orcid.org/0000-0002-5690-4098

Kristina Frain^{1,2}, Emilia Bober^{1,3}, Majid Fotuhi¹, Edward Hindle¹, Mahima Jhingan¹, Joshua Luis¹, Rajna Rasheed¹, Safiyya Diwany¹, Mark Westcott¹, Angela Rees¹, William Tucker¹, Peter Addison¹, Carlos Pavesio¹, Ian Yeung¹, Narciss Okhravi¹

¹Moorfields Eye Hospital NHS Foundation Trust, London, UK;

²Imperial College London, London, UK; ³The Royal Marsden NHS Foundation Trust, London, UK

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Introduction: Adalimumab, an anti-tumour necrosis factor- α inhibitor, is an effective treatment licensed for use in non-infectious uveitis affecting the posterior segment unresponsive to conventional disease-modifying anti-rheumatic drugs (cDMARDs) or immunosuppressants. There is significant value in reporting real world outcomes for the use of this agent since the uses and outcomes may differ from those seen during pivotal trials.

Method: Retrospective cohort study including 202 adult patients with non-infectious uveitis initiated on adalimumab between September 2013 and May 2021 at our tertiary centre. Those established on adalimumab for at least 12 months were analysed. Aim was to determine the glucocorticoid-sparing effect of adalimumab, choice of cDMARD and adalimumab biosimilars used.

Results: 125(61.9%) female and 77(39.1%) male patients with an average age of 47 ± 15 met criteria for analysis. Birdshot retinochoroiditis was the most frequent aetiology (54(26.7%)) with idiopathic uveitis the next most common ($n = 50$ (24.8%)). Most patients were initiated on Hyrimoz® ($n = 111$ (55.0%)) with the remainder commenced on Humira® ($n = 90$ (44.6%)) or Amgevita®($n = 2$ (1%)). Prior to adalimumab commencement, 195(92.9%) patients were on cDMARDs compared to 139(66.2%) at 12 months. On initiation of adalimumab, 134(66.3%) patients were on ≤ 10 mg per day of oral Prednisolone, 86 (43%) of which were on ≤ 5 mg a day. At 12 months, 167(66.3%) patients were on ≤ 10 mg per day of prednisolone with 126(62.3%) on ≤ 5 mg a day. The steroid sparing effect of adalimumab as determined by event rate per patient years of ≤ 10 mg and ≤ 5 mg prednisolone per day was 0.87 and 0.66 respectively. Adverse events reported by 24 months include bacterial infection ($n = 38$ (18.8%)), COVID-19 infection ($n = 19$ (9.4%)) and neoplasia ($n = 7$ (3.5%)).

Conclusion: Adalimumab is an effective treatment for uveitis. At 12 months our study concluded that our adalimumab patients required less oral prednisolone.

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DEVELOPMENT AND VALIDATION OF A NOVEL QUALITY OF LIFE ONTOLOGY (QOLO) AND QUESTIONNAIRES TO LOWER EXISTING BARRIERS TO DEVELOPING PATIENT REPORTED OUTCOME MEASURES

Corresponding author: Charles O'Donovan, orcid.org/0000-0003-4416-5037

Charles O'Donovan^{1,2}, Samantha Pendleton³, Genista Curry¹, Sam Norton¹, Heidi Lempp¹, Alastair Denniston⁴, George Gkoutos⁴, Malika Prem Senthil⁵, Konrad Pesudovs⁶, Tasanee Braithwaite^{1,2}

¹King's College London, London, UK; ²Guy's and St Thomas' NHS Foundation Trust, London, UK; ³University of Oxford Nuffield Department of Medicine, Oxford, UK; ⁴University of Birmingham, Birmingham, UK; ⁵Flinders University College of Nursing and Health Sciences, Adelaide, Australia; ⁶University of New South Wales, Sydney, Australia

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Introduction: Traditional PROM item identification entails qualitative research and thematic content analysis (e.g. NVivo), typically of single-centre focus group transcripts. The resources and expertise required create barriers to PROM development, and the PROM may lack external generalisability. Our study aimed to a) develop a new ontology for ophthalmic Quality of Life(QoL) to enable annotation of unstructured text data; b) Explore online questionnaires as an alternative text data source; and c) to validate these approaches to PROM item identification, through comparison to a 'Gold Standard' (focus group/interview transcript data, $n = 41$), taking uveitis as our exemplar.

Method: We developed QoLo(v1.0) using Protégé, using items in 11 QoL domains from 6 Rasch-validated, published ophthalmic PROMs. We recruited participants via 10 uveitis charities and 7 Facebook groups, to complete online questionnaires. These explored impacts of uveitis and treatment, by QoL domain. We text-mined this dataset to extend QoLo, adding new uveitis-relevant items, and patient-preferred synonyms, using a natural language processing approach and statistical ranking. We validated this approach versus the Gold Standard.

Results: QoLo(v1.0) contained 765 unique classes within 11 QoL domains. Participants($n = 58$) resided in 5 continents(81% UK-based), with mean age 55(sd 11.2,range 31-80)years. They reported anterior(47%), intermediate(9%), and posterior/pan-uveitis(44%). Two authors reviewed 1641 text items identified through statistical ranking, selecting new classes and synonyms. The extended QoLo annotated 504/796(63%) classes in the questionnaire data, and 399/796(50%)in the transcript data, revealing significant overlap in QoL items across eye diseases.

Conclusion: QoLo, a novel ontology, is hosted and open-access on GitHub, and offers an accelerated, low-cost approach for annotating text data. We illustrate application to identify new uveitis PROM content from online questionnaires, addressing key barriers and generalisability issues for PROM development across ophthalmology.

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DOES THE HUMAN UVEA CONTAIN A PREVIOUSLY UNCHARACTERISED POPULATION OF T RESIDENT MEMORY CELLS?

Corresponding author: Ian Reekie, orcid.org/0000-0003-2542-7329

Ian Reekie¹, Lakshanie Wickramasinghe¹, Andrew D Foers¹, Tom Thomas¹, Ananya Bhalla¹, Dylan Windell¹, Robert Hedley², David Copland³, Colin Chu³, Sarah Coupland⁴, Andrew Dick³, Srilakshmi Sharma¹, Jonathan Sherlock⁵, Mark C Coles¹, Christopher D Buckley¹

¹Kennedy Institute of Rheumatology, Oxford University, Oxford, UK;

²Sir William Dunn School of Pathology, University of Oxford, Oxford,

UK; ³Bristol Medical School, Translational Health Sciences, University of Bristol, Bristol, UK; ⁴Department of Cellular and Molecular Pathology, University of Liverpool, Liverpool, UK; ⁵Janssen Research and Development, LLC, Spring House, PA, USA

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Introduction: The eye is well known to possess a degree of immune privilege, and has classically been thought to be devoid of resident cells of the lymphocyte lineages. Tissue resident T-cells have now been identified in numerous tissues around the human body, and play a vital role in the adaptive immune response. Here we identify a novel population of T-cells within the human uvea which express markers of resident memory T cell status.

Method: Uveal tract was dissected from post-mortem adult human eyes with no history of eye disease ($n = 3$) and split into iris, ciliary body and choroid. Uveal tract was also dissected from foetal globes at two stages of development ($n = 8$). Single cell suspensions of these tissues were prepared for single cell RNA sequencing (scRNAseq) using the 10X Genomics Chromium platform. T-cells were also identified by flow cytometry, and spatial location was assessed on formalin-fixed paraffin-embedded eye sections by multiplexed immune-fluorescent imaging on the Cell DIVE platform (Leica).

Results: scRNAseq identified discrete T cell clusters within all adult uveal tissues. These clusters are shown to express markers of T resident memory status, and not of naïve T-cell status. By contrast, T-cells within foetal uveal tissues express markers of naïve status. Immune-fluorescent imaging confirmed the presence of CD3+, CD69+, CD45RO+ cells suggestive of a resident memory phenotype outside of CD31+ blood vessels in adult human eyes.

Conclusion: We demonstrate that resident T cells populate the healthy adult human uvea, and are not found in foetal eyes. This work is of relevance to the understanding of the uveal immune system, and potentially to autoimmune uveitis.

P232

SEEING THROUGH OCULAR SYPHILIS. A CASE SERIES OF OCULAR SYPHILIS IN NORTHERN IRELAND

Corresponding author: Clare Shute, orcid.org/0000-0001-8831-7421

Clare Shute¹, Sean McGinley¹, Clara McAvoy¹

¹Belfast Health and Social Care Trust, Belfast, UK

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Introduction: The incidence of syphilis infection is dramatically increasing in the UK with a 159% rise in syphilis diagnosis over the past 10 years. Infection with *Treponema Pallidum* can affect many body systems and can therefore present to multiple different specialties, including ophthalmology, before a diagnosis is made. Ocular involvement can occur at any stage of syphilis.

This case series aims to identify and describe various presentations of ocular syphilis to increase awareness among ophthalmologists.

Method: A Northern Ireland wide trawl of patients presenting to an ophthalmology department with ocular syphilis was carried out. Patients were identified from a database kept by the regional uveitis service from 2015 – 2023. Demographics and case information was obtained from electronic patient records, paper records and direct communication with the genitourinary medicine (GUM) department.

Results: Fourteen patients, 12 male and 2 female, presented with ocular syphilis from 2015 to 2023. Of those who were male, a third (4 patients) were men who have sex with men. In all cases, patients presented primarily to ophthalmology who made the diagnosis of syphilis. Forty-three percent (6 patients) had ocular symptoms alone and of those with symptoms, only 2 had genital ulcers. Uveitis of varying types was the most common presentation of ocular syphilis, followed by syphilitic optic neuropathy.

Conclusion: Syphilis is on the rise and the ophthalmology department may be where it first presents. Ophthalmologists need to be vigilant for ocular signs of syphilis and keep an awareness that it may masquerade as other ophthalmic conditions, especially uveitis. An established relationship with the GUM team is vital to aid in patient management.

P233

WHAT IS THE ROLE OF A UVEITIS VIRTUAL CLINIC IN A POST PANDEMIC WORLD?

Corresponding author: Hannah Whelan, orcid.org/0000-0002-8446-3146

Hannah Whelan¹, Kiran Ahmed¹, Kanchan Bhan¹

¹Leeds Teaching Hospitals NHS Trust, Leeds, UK

Disclosures: No conflicts of interest and no funding

Introduction: During the Covid-19 pandemic, shielding patients did not attend face to face (F2F) clinics. A virtual service was set up to allow these vulnerable patients to be reviewed by a clinician to aid management of chronic uveitis. Patients are no longer shielding but the success of the virtual clinic has led to continuation of this service. We audited the role of this clinic and how it works.

Method: Data collected for 50 patients between 30/01/2023 and 30/12/2023. Parameters observed included diagnosis, whether attended uveitis investigations clinic, management decision and follow up.

Results: For 24% of patients, medication was reviewed; 42% of which had changes made to medication. 14% had a general review and 24% discussed the investigation results. Nil adverse consequences were found for patients attending virtual clinics instead of F2F; zero patients had increased intraocular pressure or clinically significant changes on imaging. One patient had a drop in visual acuity and was subsequently seen F2F which demonstrated a cataract. Follow up was arranged face to face for 48%, 6% were discharged from uveitis clinic and 16% were advised to have further imaging.

Conclusion: Virtual clinic has a role for all uveitis which can be monitored via imaging and does not require slit lamp examination. Anterior uveitis and vitritis should still be seen F2F. All patients have a safety net of being seen in face-to-face clinic annually, with the six-month review being virtual. This service enables patients to have undilated investigations so they can drive to and from their appointment and have results of these investigations discussed via telephone, supporting the working population.

VITREA RETINAL DISEASES AND SURGERY

P234. Abstract not included

P235

EVALUATION AND OUTCOME OF SLIT-LAMP PROPHYLACTIC LASER-RETINOPEXY FOR RETINAL TEAR IN A DISTRICT GENERAL HOSPITAL IN WALES

Corresponding author: Somnath Datta, orcid.org/0009-0004-4121-4125

Dr. Somnath Datta¹, Ramya Sulur Sundararajan¹

Royal Glamorgan Hospital, Cwm Taf Morgannwg University Health Board NHS Wales, Llantrisant, UK

Disclosures: No conflicts of interest and no funding

Introduction: Authors report an outcome of initial slit-lamp prophylactic-laser-retinopexy to treat retinal-break with relation to number of breaks treated successfully at first-instance, number of breaks needed subsequent additional-laser-photocoagulation, number of breaks needed vitreo-retinal surgical-intervention, most frequent location of tear, and their demographics.

Method: A retrospective-study of 51(=n) eyes with single retinal-tear, undergone laser-retinopexy for prevention of retinal-detachment, in a DGH in Wales, from January-2020 to September-2021, were performed. Main outcome measures were position of retinal-tear, if primary laser-retinopexy was adequate, whether further to-up retinopexy was required or been referred to vitreo-retinal surgical procedure along with their demography-variation.

Results: Primary laser-retinopexy were found to be adequate in 35 eyes(69%), out of 51(=n) of total treated eyes. Additional top-up retinopexy were performed in 10 eyes(20%). Vitreo-retinal procedures after initial retinopexy were performed in 6 eyes(12%). Most common location of retinal-tear was supero-temporal, of them, 64% was found to be adequate after initial retinopexy. Infero-temporal, supero-nasal and infero-nasal retinal-tear retinopexy were found to be 100% adequate without necessitating any further vitreo-retinal surgical intervention. Patients were aged between 40 and 80. Female patients were found to have most adequate retinopexy at first instance. A Left-tailed-proportion-hypothesis test was performed to validate authors' proposal of 68% of retinopexy would be adequate at first application, yielding **P-value** = 0.54(>Significance level α = 0.005(0.5%)) where Null Hypothesis: H_0 : $P = 0.68$, Alternate Hypothesis: H_a : $P < 0.68$, Sample size (n)=51, Number of success (x)=35, Proportion of success ($\hat{p}$) = 0.68, Proportion of failure (q) = 0.32 with decision rule being **P-value** > α meaning failure to reject Null Hypothesis.

Conclusion: Initial laser-retinopexy is adequate and effective means of treatment and prevention of retinal-detachment in vast majority of cases. However, there is high-rate of requirement of additional laser-top-up or vitreo-retinal surgical intervention.

P236

AN ASSESSMENT OF THE IMPACT OF DISEASE ACTIVITY CRITERIA ON DOSING INTERVAL ASSIGNMENT IN CLINICAL TRIAL PATIENTS WITH NAMD

Corresponding author: Faruque Ghanchi, orcid.org/0000-0002-4448-8162

Faruque Ghanchi¹, Christina Y Weng², Audrey Souverain³, Ivaylo Stoilov⁴, Philippe Margaron³

¹Bradford Teaching Hospitals NHS Foundation Trust, Bradford, UK;

²Baylor College of Medicine, Cullen Eye Institute, Houston, TX, USA;

³F. Hoffmann-La Roche, Basel, Switzerland; ⁴Genentech, Inc., San Francisco, CA, USA

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Introduction: Clinical treatment decisions for patients with neovascular age-related macular degeneration (nAMD) are based on disease activity. Likewise, in clinical trials, disease activity criteria guide dosing regimen allocation and dosing interval adjustment. This analysis evaluated how applying different disease activity criteria may have impacted faricimab (a dual angiopoietin-2/vascular endothelial growth factor-A inhibitor) dosing interval assignment in phase 3 trials (TENAYA/LUCERNE).

Method: In TENAYA/LUCERNE (NCT03823287/NCT03823300), patients (N = 1329) were randomised 1:1 to faricimab 6.0mg or aflibercept 2.0mg Q8W for 112 weeks. In the faricimab arm, patients were assigned to treatment intervals up to Q16W (n = 665) based on protocol-defined disease activity criteria: BCVA loss of ≥ 5 (vs mean over 2 previous visits) or ≥ 10 letters (vs highest BCVA over 2 previous visits); CST $> 50\mu\text{m}$ (vs mean over 2 previous visits) or $\geq 75\mu\text{m}$ (vs lowest CST over 2 previous visits); new macular haemorrhage. Application of these criteria at week 20 was compared with application of alternative disease criteria (adapted from PULSAR; NCT04423718) at week 20: BCVA loss > 5 letters (vs week 16 BCVA; CST increase $> 25\mu\text{m}$ (vs week 16 CST); new macular haemorrhage.

Results: When disease activity was based on protocol-defined disease activity criteria, 78% of faricimab patients in TENAYA/LUCERNE were assigned to $\geq Q12W$ dosing at week 20. In contrast, in a hypothetical scenario where disease activity was based on alternative criteria, 96% of faricimab patients in TENAYA/LUCERNE would have been assigned to $\geq Q12W$ dosing at week 20.

Conclusion: Application of disease activity criteria in clinical trials can have a significant impact on dosing interval. Attention should be paid to which criteria best reflect real-world practice to understand the likelihood of trial results translating to treatment burden reductions.

P237. Abstract not included

P238

IS PARS PLANA VITRECTOMY THE ANSWER FOR ALL POSTERIOR CAPSULE TEARS? A FIVE-YEAR RETINAL DETACHMENT PERSPECTIVE

Corresponding author: Charlotte Ho, orcid.org/0000-0003-1131-1448

Charlotte Ho¹, Matthew Stahl¹, Thomas Nash¹, Korina Theodorako¹, Graham Dugiud¹

¹The Western Eye Hospital, London, UK

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Introduction: Posterior capsule rupture (PCR) in cataract surgery is one of the most recognised complications. PCR with vitreous loss but without dropped nucleus lens fragment (DNLF) is commonly managed with anterior vitrectomy (AV) alone. On the other hand, cases that are complicated by DNLF are usually managed with subsequent secondary pars plana vitrectomy (PPV) and removal of the retained fragments. Studies have shown that cataract surgery complicated by PCR are at increased risk of rhegmatogenous retinal detachment (RRD). The aim of this study is to analyse the risk of RRD following AV alone versus AV with subsequent PPV at the Western Eye Hospital, London.

Method: Electronic notes of patients who underwent cataract surgery which was complicated by PCR, requiring either AV or AV and subsequent PPV from 2013 to 2018 at the Western Eye Hospital were reviewed retrospectively to allow for a 5-year follow up period. Data was analysed using SPSS. A chi-square test was performed to test for statistical difference at the 5% level.

Results: In this study, 113 patients were managed with AV alone, whilst 60 patients were managed with AV and PPV. Six patients in the AV group and 0 patients in the PPV group developed RRD. However, chi-square test did not show statistical significance at the defined level.

Conclusion: This study demonstrated a clear trend indicating a reduced incidence of RRD in patients who underwent a PPV versus an AV following complicated cataract surgery with PCR. A statistically significant difference between the two groups was not established at the 5% level in this study. This is likely due to the sparsity of DNF during this time period. A prospective study at our centre is warranted.

P239. Abstract not included

P240

THE POTENTIAL FOR Q20W DOSING WITH FARICIMAB AND EXTENDED TREATMENT OUTCOMES IN NAMD: A POST HOC ANALYSIS OF THE TENAYA/LUCERNE TRIALS

Corresponding author: Praveen Patel, orcid.org/0000-0001-8682-4067

Praveen Patel¹, Lauren Hill², Audrey Souverain³, Aachal Kotecha⁴, Philippe Margaron³

¹Moorfields Eye Hospital, London, UK; ²Genentech, Inc., San Francisco, CA, USA; ³F. Hoffmann-La Roche Ltd., Basel, Switzerland; ⁴Roche Products Ltd., Welwyn Garden City, UK

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Introduction: In the phase 3 TENAYA/LUCERNE (NCT03823287/NCT03823300) trials, faricimab gave durable vision gains (non-inferior to aflibercept) in patients with neovascular age-related macular degeneration (nAMD). We evaluated how many patients could have extended to Q20W.

Method: Treatment-naïve patients were randomised to faricimab up to every 16 weeks (Q16W) (n = 665) or aflibercept every 8 weeks (Q8W) (n = 664). After week 24, faricimab-treated patients received fixed dosing up to Q16W until week 60 and then entered a treat-and-extend-based (T&E) dosing regimen. In this post hoc analysis, we evaluated the change in best corrected visual acuity (BCVA) and central subfield thickness (CST) at year 1 (averaged over weeks 40, 44 and 48) and year 2 (averaged over weeks 104, 108 and 112) of faricimab-treated patients who were always-on $\geq$ Q12W (~80%) and always-on Q16W (>60%) for the duration of the 2-year trials. Protocol-defined disease criteria (BCVA loss of ≥ 5 or ≥ 10 letters; CST $> 50\mu\text{m}$ or $\geq 75\mu\text{m}$; new macular haemorrhage) were applied to patients on Q16W dosing during the T&E phase to see whether they met the criteria for extension to Q20W.

Results: At year 1, faricimab-treated patients always-on $\geq$ Q12W (n = 372) and always-on Q16W (n = 204) demonstrated robust BCVA gains (mean[SD]; always-on $\geq$ Q12W, +7.0[10.8]letters; always-on Q16W, +7.9[10.7]letters) that were maintained through year 2 (always-on $\geq$ Q12W, +6.6[12.9]letters; always-on Q16W, +7.5[11.8]letters). Patients on extended faricimab treatment had meaningful reductions in CST from baseline at year 1 (always-on $\geq$ Q12W, -135.7[112.5] μm ; always-on Q16W, -139.0[108.1] μm) that were maintained through year 2 (always-on $\geq$ Q12W, -142.8[112.7] μm ; always-on Q16W, -145.3[110.8] μm). 56% of faricimab-treated patients could have extended to Q20W dosing.

Conclusion: Stable outcomes achieved by faricimab-treated patients always-on extended treatment intervals could allow for potential extension to Q20W.